Microwave Driven Transport in Mesoscopic Superconducting Junctions from Tunnel Barriers to Atomic Contacts

Doctoral thesis for obtaining

the academic degree

Doctor of Natural Sciences

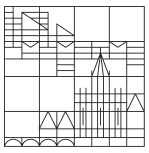
(Dr. rer. nat.)

submitted by

Aschenbrenner, Oliver

at the

Universität Konstanz



Faculty of Science

Department of Physics

Konstanz, 2026

Zusammenfassung

Bestrahlt man einen supraleitenden Kontakt mit Mikrowellen, so tragen mehrere Transportprozesse gleichzeitig zum Strom bei. Quasiteilchen tunneln, Cooper-Paare werden übertragen, und Ladungsträger durchlaufen mehrfache Andreev-Reflexionen (MAR). Alle drei erzeugen Strukturen in derselben Kennlinie. Aus der Lage einer Struktur allein lässt sich nicht mit Gewissheit sagen, welcher Prozess sie hervorruft.

Um die Prozesse dennoch zu trennen, messe ich jeden Kontakt zweimal. Zuerst charakterisiere ich ihn ohne Mikrowelle. Der statische Transport legt fest, bei welchen Energien Strukturen einsetzen und welche Kanäle zum Strom beitragen. Danach nehme ich vollständige Serien von Kennlinien bei wachsender Mikrowellenamplitude auf. In diesen Serien gibt der Abstand der Replika an, wie viel Ladung ein Prozess überträgt. Wie stark die Replika mit der Amplitude anwachsen und wieder verschwinden, unterscheidet inkohärentes Tunneln von kohärenter Phasenkopplung.

Ich untersuche Aluminiumkontakte bei Temperaturen von wenigen Millikelvin, von der Oxidtunnelbarriere bis zum einzelnen atomaren Kontakt in einem mechanisch kontrollierbaren Bruchkontakt. Die Mikrowellenamplitude, die tatsächlich am Kontakt ankommt, bestimme ich aus den Messdaten selbst.

Am ersten von zwei Tunnelkontakten beschreibt das Tien-Gordon-Modell die Messung vollständig. Es sagt voraus, dass die Mikrowelle Kopien der statischen Kennlinie erzeugt, verschoben um ganzzahlige Vielfache der Photonenenergie. Diese Replika bleiben bis mindestens zur 25. Photonenordnung einzeln sichtbar. Meines Wissens wurden einzelne Replika an einem supraleitenden Tunnelkontakt bisher in keiner vollständigen Amplitudenserie durchgehend bis zu einer vergleichbaren Ordnung aufgelöst.

Der zweite, nominell baugleiche Kontakt verhält sich anders. Unter Mikrowellen richtet er den Strom gleich, die Kennlinie wird unsymmetrisch, lokal um bis zu 25 %. Gleichspannung, Mikrowellenamplitude und Frequenz stellen dabei gemeinsam ein, wie stark und in welche Richtung die Gleichrichtung wirkt. Das Tien-Gordon-Modell kann das nicht beschreiben, denn es ist in der Spannung symmetrisch. Die Frequenz, bei der der Effekt auftritt, stimmt mit einer Mode der Probenumgebung überein, die unabhängig davon nachgewiesen wurde. Ein resonant angeregter schwacher Josephson-Kontakt erklärt die Einsatzschwelle, die Frequenzabhängigkeit und den Vorzeichenwechsel plausibel. Der mikroskopische Mechanismus bleibt offen.

Bei atomaren Kontakten tragen nur wenige Kanäle zum Transport bei, jeder mit einer eigenen Transmission. Diesen Satz von Transmissionen, den Pincode des Kontakts, gewinne ich aus dem statischen Verlauf der Andreev-Strukturen. Verfolgt man ihn beim langsamen Öffnen des Kontakts, so ändern sich die Transmissionen innerhalb der Leitwertplateaus stetig und ordnen sich an den Sprüngen neu. Der Normalleitwert gibt nur die Summe der Transmissionen wieder. Die

Sprünge sind darin zu sehen, die stetige Entwicklung innerhalb eines Plateaus und die Umordnung der Kanäle jedoch nicht.

Der Pincode geht anschließend als feste Vorgabe in die Beschreibung der Mikrowellenmessungen ein. Dazu zerlege ich den berechneten statischen Leitwert an den Schwellen der einzelnen MAR-Ordnungen. Jede Ordnung überträgt eine andere Ladung und erhält deshalb ihr eigenes Muster an Replika. So entsteht eine Beschreibung der MAR-Prozesse unter Photonenanregung (PAMAR). Ergänzt um inkohärentes Cooper-Paar-Tunneln unter Photonenaustausch (PAICPT) gibt sie die wichtigsten Linien der Messung und ihre Gewichte wieder, von mittlerer Transmission bis zu einem nahezu ballistischen Kanal. Kein Parameter wird dabei an die Mikrowellenmessung selbst angepasst.

Die Replika bei Nullspannung folgen dem Quadrat der Besselfunktion. Das spricht für inkohärentes Cooper-Paar-Tunneln und gegen kohärenten Shapiro-Transport. Beim höchsten untersuchten Leitwert stößt die Beschreibung an ihre Grenze. Dort treten zusätzliche Replika im halben Cooper-Paar-Abstand auf, die meines Wissens bisher nicht berichtet wurden.

Die Mikrowellenspektroskopie verbindet damit zwei Grenzfälle. Im Tunnellimit ist der Transport quantitativ verstanden. Bei wenigen Kanälen liefert die Kombination aus PAMAR und PAICPT ein praktikables Bild. An den Grenzen dieses Bildes werden die beiden neuen Effekte sichtbar, die mikrowelleninduzierte Gleichrichtung und die Replika im halben Cooper-Paar-Abstand.

Abstract

Under microwave irradiation, quasiparticle, Cooper pair, and multiple Andreev reflection (MAR) transport overlap in the same spectrum, where position alone no longer identifies a process. I separate them with independent static characterization and complete microwave amplitude maps. Static transport fixes the energy thresholds and the channel composition, the replica spacing sets the transferred charge, and the amplitude evolution distinguishes incoherent tunneling from coherent phase locking. I apply this to aluminum junctions at millikelvin temperatures, from oxide tunnel barriers to atomic contacts in a mechanically controllable break junction, with the local ac drive calibrated in situ.

In the first of two tunnel junctions, the Tien-Gordon model reproduces the complete driven evolution, with individual photon assisted quasiparticle replicas resolved to at least photon order 25. To my knowledge, no previous amplitude map of a superconducting tunnel junction resolves individual replicas continuously to comparable order. A second, nominally similar junction shows microwave induced rectification reaching a local current asymmetry of 25 %, tuned in magnitude and polarity by dc bias, microwave amplitude, and frequency. The response lies beyond the bias symmetric Tien-Gordon description. Its drive frequency matches an independently identified mode of the sample environment, and a resonantly driven Josephson weak link plausibly explains the threshold, frequency sensitivity, and polarity reversal, although the mechanism remains open.

For atomic contacts, channel resolved fits to the static MAR spectra yield the mesoscopic pincode. Along an opening trace the transmissions evolve within plateaus and reorganize at abrupt jumps. The total normal-state conductance shows the jumps but not the continuous evolution within a plateau or the internal channel reorganization behind them. Using the pincode as independent input, I partition the static MAR conductance around its thresholds and dress each order by its charge, giving threshold weighted photon assisted MAR (PAMAR). Combined with photon assisted incoherent Cooper pair tunneling (PAICPT), it reproduces the dominant trajectories and their spectral weight from moderate transmission to a nearly ballistic leading channel, without fitting individual driven features. The squared-Bessel amplitude evolution of the zero-bias replicas supports PAICPT over coherent Shapiro transport. At the highest conductance the description reaches its limit and a subharmonic PAICPT response appears, with replicas at half the conventional Cooper pair spacing that to my knowledge have not been reported before.

Microwave amplitude spectroscopy therefore links quantitative photon assisted tunneling in the tunnel limit to a practical PAMAR and PAICPT description of driven few-channel transport, and exposes two responses at its limits, a tunable rectification and a subharmonic PAICPT response.

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1 Introduction

Computation now underlies science, industry, and everyday life. As the scope and scale of computation expand, so does the demand for processing power. Meeting that demand requires more energy [1]. For decades, computing advanced to the rhythm of Moore's law. Smaller transistors placed ever more functionality on each chip, while reductions in operating voltage limited the accompanying increase in power consumption [2, 3]. This strategy no longer guarantees proportional improvements in performance and energy efficiency because power dissipation, fabrication complexity, material properties, and device physics make further scaling increasingly difficult [4].

As conventional scaling approaches its practical limits, quantum mechanics offers a way out, an entirely new basis for computation. A sufficiently powerful quantum computer can already solve selected problems beyond the practical reach of classical machines, as demonstrated by a verifiable simulation [5]. Such simulations could transform how molecules, pharmaceuticals, catalysts, and other quantum materials are understood and designed [6, 7]. Other algorithms could break widely used public-key cryptosystems, forcing a fundamental change in how digital communication is secured [8]. These machines may not replace classical computers, but their impact on science, industry, and everyday life could be revolutionary.

Superconducting circuits form the hardware basis of several of the world's most advanced quantum processors and mark a principal route toward scalable quantum computing [9 - 11]. Their fast operation, lithographic fabrication, and compatibility with integrated microwave control have enabled increasingly complex processors and major advances in quantum error correction [12]. These processors rely on Josephson junctions, typically formed by two superconducting electrodes separated by an ultrathin insulating barrier. Their nonlinearity enables qubit operation, while their dissipation and microwave response originate from microscopic charge transport. The same scalability that makes superconducting circuits attractive also exposes their microscopic limitations. Nonequilibrium quasiparticles tunneling across a Josephson junction can relax the qubit, while defects in amorphous tunnel barriers absorb energy and produce fluctuations in relaxation time [13 - 15]. As processors grow, understanding and controlling these local loss channels becomes essential to their reproducibility and performance.

Reaching that microscopic understanding requires studying a single junction in isolation, apart from the full qubit circuit whose collective response can hide individual transport processes. I compare two complementary junction regimes for that purpose. Oxide tunnel barriers set the low-transmission limit, while atomic contacts expose transport through discrete channels of finite and high transmission [16, 17]. This controlled progression tests where an effective circuit description remains sufficient and where individual charge transfer processes must be resolved. Microwave irradiation extends this comparison into the driven regime. The spacing and amplitude evolution of the resulting structures reveal transferred charges and overlapping transport processes, directly testing transport descriptions under drive.

In this thesis, I investigate aluminum junctions at millikelvin temperatures under dc bias and microwave irradiation. Oxide tunnel barriers serve as the low-transmission reference, while a mechanically controllable break junction forms atomic contacts with different channel compositions. Static spectra establish the superconducting energy scales and channel transmissions. Driven spectra then reveal how quasiparticle, multiple Andreev reflection (MAR), and Cooper pair contributions evolve with microwave amplitude and transmission.

State of the Art and Open Questions

Static transport offers a well-established reference for driven measurements. In the tunnel limit, single-particle tunneling is elastic, and the current arises from electrons tunneling between the two BCS densities of states, weighted by their thermal occupation. Because each density of states diverges at the gap edge, the *I*–*V* characteristic develops a sharp current onset once the bias voltage reaches the sum of the two superconducting gaps, and this quasiparticle peak directly maps the size of the gap [18, 19]. As the barrier transmission increases, an incident quasiparticle can undergo multiple Andreev reflections at the two interfaces before finally crossing the junction. Each reflection converts the quasiparticle between electron and hole character and lets it pick up an additional increment of bias energy, so that a transport process of order m turns on once the bias voltage reaches $eV = 2\Delta/m$. This mechanism drives current at subgap bias and produces a series of conductance features known as the subharmonic gap structure [20, 21]. Because the amplitude of each order depends sensitively on the channel transmission, fitting the complete subgap *I*–*V* characteristic with multichannel scattering theory extracts the discrete channel transmissions of an atomic contact [16, 17].

Coherent Cooper pair transport gives a complementary macroscopic description of the same junction. Cooper pairs tunnel without dissipation, and the resulting supercurrent follows the Josephson relations, which set the current by the sine of the superconducting phase difference and set the time evolution of that phase by the bias voltage. Embedding this nonlinear, phase dependent el-

ement in a parallel resistance and capacitance, the RCSJ model reproduces the classical current–voltage dynamics of a real junction, including the hysteresis between its switching and retrapping branches [22 - 24]. A junction cannot generally be treated in isolation from this circuit environment, even under static bias. Each tunneling event, whether it transfers a quasiparticle or a Cooper pair, can exchange energy with the electromagnetic modes of the surrounding capacitance and impedance, and this exchange reshapes the measured I–V response near the gap and near zero bias. In the tunnel limit, $P(E)$ Theory captures this effect by convolving the bare tunneling rate with the probability that the environment absorbs or supplies a given energy during the tunneling event [25, 26].

Microwave driving adds coherent and incoherent responses to this environment dependent baseline. In the tunnel limit, the oscillating drive periodically shifts the electrochemical potential of the electrodes, so a tunneling quasiparticle can absorb or emit one or more photon quanta during the tunneling process. The Tien-Gordon model describes the resulting current as a sum of energy shifted copies of the static quasiparticle current, weighted by the probability of exchanging a given number of photons, and these copies appear as replicas displaced from the static onset by integer multiples of the photon energy [27]. Coherent Cooper pair tunneling responds differently. Under dc bias, the Josephson relations already predict an oscillating supercurrent at a frequency set by the bias voltage, and when this oscillation locks to the microwave drive, the phase advances by a fixed number of cycles per drive period. This phase locking pins the dc voltage to a comb of values set by the drive frequency, which appear as Shapiro steps in the I–V curve [28]. At finite transmission, the same photon exchange also acts on the Andreev reflection process, so each MAR order can additionally absorb or emit photons, and photon assisted multiple Andreev reflection (PAMAR) shifts the static charge order thresholds into a comb of finite-bias sidebands [29]. Cooper pairs can also tunnel incoherently under drive, absorbing photons without the running phase that Shapiro steps require, and this photon assisted incoherent Cooper pair tunneling (PAICPT) produces its own replica pattern near zero bias, distinct because it does not depend on phase locking [30]. These mechanisms can overlap in the same spectrum, making their separation a central interpretive problem.

Voltage biased aluminum atomic contacts provide the closest experimental match. Static MAR yields the channel transmissions, while microwave spectra show coherent integer and fractional Shapiro resonances on a PAMAR background. The low-impedance environment supports this phase locking, and a microscopic driven MAR theory reproduces representative finite-bias conductance traces using the measured transmissions and calibrated drive without adjustable parameters [31]. These measurements establish quantitative PAMAR, but do not address its amplitude resolved separation from incoherent Cooper pair transport.

Josephson STM experiments identify PAICPT through its replica spacing and squared-Bessel amplitude dependence [30]. Across different STM conductance regimes, quasiparticle tunneling, PAICPT, and driven MAR appear in the same

spectra. Direct charge dependent dressing of the static full counting statistics components fails to reproduce the irradiated MAR response, while a microscopic calculation performs better but retains substantial deviations [32]. Related measurements of Yu-Shiba-Rusinov states further show that separate electron and hole thresholds can make the visible sideband spacing differ from the net transferred charge [33]. A different platform, gate tunable semiconductor–superconductor nanowire Josephson junctions, reveals charge dependent MAR sidebands in microwave amplitude and frequency maps. A phenomenological comparison dresses the measured static current under the piecewise assumption that one MAR order dominates near each threshold [34].

Taken together, previous studies leave an experimental as well as a modeling gap. They demonstrate individual driven transport mechanisms in selected devices and spectra, but do not systematically map coexisting PAMAR and PAICPT across complete microwave amplitude sweeps and independently characterized contact configurations. The microscopic calculation and the phenomenological comparison above illustrate the resulting trade-off. Full Floquet treatments capture coherent microwave dynamics but are computationally demanding, while phenomenological constructions are easier to apply but rely on measured line shapes and piecewise assignments [32, 34]. What is missing is therefore both a systematic comparison across the few-channel regime and a practical description that predicts the driven response from independently characterized static transport.

Overview

In this thesis, I systematically measure microwave amplitude sweeps across atomic contacts with substantially different channel compositions. Each sweep is a series of I–V curves recorded at increasing microwave amplitude. The data reveal coexisting PAMAR and PAICPT throughout the few-channel regime and establish how their trajectories and spectral weights evolve with transmission and drive. To explain these maps, I develop a deterministic threshold decomposition. It partitions the simulated static MAR conductance around its known thresholds and reconstructs an effective current for each charge order. Charge dependent photon dressing and an independently constrained PAICPT contribution then reproduce the dominant trajectories and spectral weight redistribution without selecting structures from the driven measurement. In Chapter 2, I develop the theory behind this analysis, from BCS quasiparticle tunneling and Josephson phase dynamics to channel resolved MAR and environmental energy exchange. In Chapter 3, I then detail the underlying experiment, beginning with sample fabrication. I also cover the cryogenic MCBJ setup, filtering, and the electrical measurement circuit, along with the resulting acquisition, processing, and fitting.

In Chapter 4, I present experiments on tunnel barriers and atomic contacts. First, in Sec. 4.1, I observe a new microwave induced rectification effect in a

tunnel barrier. The local current asymmetry reaches 25 %, and this strong bias asymmetric response lies beyond the established bias symmetric Tien-Gordon model. The effect occurs within a specific operating range set jointly by voltage bias, microwave amplitude, and frequency, and together these three parameters tune the magnitude and polarity of the rectification. Then, in Sec. 4.2, I show that the static environmental analysis and the zero-bias replicas support incoherent Cooper pair transport rather than a coherent Shapiro response. Measurements spanning moderate transmission up to a nearly ballistic leading channel show that the combined framework captures the principal few-channel microwave signal. The framework acts as a practical model for testing charge assignments and separating overlapping mechanisms. At high conductance, the same data expose an additional subharmonic photon assisted feature, a genuinely new observation whose microscopic origin remains an open question.

In Chapter 5, I draw these results together into a unified picture of driven charge transport in aluminum junctions, and the same quasiparticle tunneling, MAR, and Cooper pair processes resolved throughout this thesis govern the dissipation and decoherence that ultimately limit superconducting quantum circuits.

2 Superconducting Transport

This chapter first introduces the transport processes supported by a superconducting junction, then resolves their channel dependence at finite transmission, and finally describes how charging and the electromagnetic environment modify their dynamics. This sequence builds the description required for the experimental systems studied in this thesis, which combine mesoscopic transport with stochastic environmental effects.

The discussion begins in the weak transmission regime. Bardeen-Cooper-Schrieffer theory provides the quasiparticle spectrum and the basis for quasiparticle tunneling, while the Josephson description captures Cooper pair transport and phase dynamics. Both descriptions arise from the same superconducting foundation and expose complementary aspects of transport through a superconducting tunnel junction.

At finite transmission, the individual conduction channels and their transmission probabilities must be retained explicitly. This channel resolved treatment captures Andreev reflection, Andreev bound states, and multiple Andreev reflection. Finally, we embed the junction in its electromagnetic environment. Energy exchange with this environment and the energetic cost of charging determine the probability of individual charge transfer events. Together, these theoretical layers provide the framework for interpreting the measured *I–V*, *dI–dV*, and microwave driven transport features.

All numerical simulations in this chapter use aluminum parameters, zero temperature, and zero Dynes broadening, $T = \gamma = 0$. Irradiated simulations use a drive frequency of 15 GHz.

2.1 BCS Theory and Tunnel Spectroscopy

We develop the minimal weak-coupling Bardeen-Cooper-Schrieffer (BCS) description needed for superconducting transport. We first define the normal-state spectrum on which pairing acts. BCS mean field theory then introduces the pair potential and gives the Bogoliubov quasiparticles whose spectrum determines the superconducting density of states (DOS). We use this spectrum and its occupation to formulate the static tunnel current in the low-transmission limit. Finally, we extend this description to photon assisted tunneling (PAT) under harmonic driving. Phase dynamics and finite-transmission transport are treated in later sections. [35, 36]

We consider a spin singlet *s*-wave superconductor with one effective isotropic band. This provides the low-energy description used for the aluminum electrodes in this thesis but does not reproduce their detailed band structure. It excludes anisotropic, strongly correlated, and multiband superconductors.

2.1.1 BCS Mean Field and Bogoliubov Quasiparticles

The normal-state problem supplies the reference DOS and the diagonal part of the Bogoliubov-de Gennes (BdG) Hamiltonian. Pairing then couples its electron and hole sectors.

At the energies relevant for superconducting transport, the electronic structure is represented by the effective single-particle Hamiltonian

$$\hat{H}_{\mathrm{N}}(\vec{r}) = -\frac{\hbar^2}{2m^*}\nabla^2 + U(\vec{r}) - \mu\,. \tag{2.1}$$

Here m^* describes the band curvature, while $U(\vec{r})$ includes the effective lattice background, confinement, barriers, and disorder. Energies are measured relative to the chemical potential μ. This normal-state operator later forms the diagonal part of the BdG Hamiltonian. [37]

In a homogeneous system, the effective band introduced above is approximated as parabolic,

$$\varepsilon(\vec{k}) \approx \frac{\hbar^2|\vec{k}|^2}{2m^*}\,, \qquad E_{\mathrm{F}} = \frac{\hbar^2 k_{\mathrm{F}}^2}{2m^*}\,, \qquad v_{\mathrm{F}} = \frac{\hbar k_{\mathrm{F}}}{m^*}\,. \tag{2.2}$$

The corresponding three-dimensional DOS per unit volume, including both spin directions, is

$$N(\varepsilon) = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{\varepsilon}\,. \tag{2.3}$$

Here, ε is measured from the band edge. Superconductivity involves only a narrow energy shell around E_{F}. We therefore use

$$N(\varepsilon) \approx N_0 \equiv N(E_{\mathrm{F}})\,. \tag{2.4}$$

In the superconducting description below, ξ and E are instead measured relative to the chemical potential, which satisfies $\mu \simeq E_{\mathrm{F}}$ for a conventional metal.

Within the free-electron approximation with three conduction electrons per atom, bulk aluminum has $m^* \approx m$, $E_{\mathrm{F}} \approx 11.7\,\mathrm{eV}$, $v_{\mathrm{F}} \approx 2.0 \cdot 10^6\,\mathrm{m\,s^{-1}}$, and $\lambda_{\mathrm{F}} = 0.36\,\mathrm{nm}$. We use these as representative material scales, not as parameters extracted from the devices studied in this thesis. [38]

The normal-state electron and hole excitations are independent. BCS pairing couples these sectors and reorganizes them into Bogoliubov quasiparticles.

BCS Mean Field Approximation

In BCS theory, an effective attraction pairs electrons in time reversed states near the Fermi surface. The mean field approximation replaces this interaction by the complex local pair potential

$$\widetilde{\Delta}(\vec{r}) = -V\left\langle \hat{\psi}_{\downarrow}(\vec{r})\hat{\psi}_{\uparrow}(\vec{r})\right\rangle = \Delta(\vec{r})e^{i\mathring{\phi}(\vec{r})}\,, \qquad \Delta(\vec{r}) := |\widetilde{\Delta}(\vec{r})|\,. \tag{2.5}$$

In the homogeneous bulk system considered below, the nonnegative magnitude Δ equals the excitation gap, while ϕ describes condensate coherence. The attractive coupling $V < 0$ acts within the weak-coupling pairing shell. The pair potential must be determined self-consistently from the quasiparticle spectrum derived below. [35]

Bogoliubov-de Gennes Eigenproblem

For spin singlet pairing without spin dependent fields, the mean field problem separates into two equivalent 2×2 electron–hole blocks. We define the reduced Nambu spinor and BdG Hamiltonian as

$$\Psi_k(\vec{r}) = \begin{pmatrix} u_k(\vec{r}) \\ v_k(\vec{r}) \end{pmatrix}\,, \qquad \hat{H}_{\mathrm{BdG}}(\vec{r}) = \begin{pmatrix} \hat{H}_{\mathrm{N}}(\vec{r}) & \widetilde{\Delta}(\vec{r}) \\ \widetilde{\Delta}^*(\vec{r}) & -\hat{H}^*_{\mathrm{N}}(\vec{r}) \end{pmatrix}\,. \tag{2.6}$$

The amplitudes u_k and v_k describe the electron and hole components of one quasiparticle mode. The diagonal blocks carry both components according to the normal-state problem, while the off-diagonal pair potential converts an electron into its time reversed hole partner and vice versa. A superconducting excitation is therefore generally a coherent mixture of both components.

The allowed quasiparticle modes follow from the eigenvalue problem

$$\hat{H}_{\mathrm{BdG}}(\vec{r})\Psi_k(\vec{r}) = E_k\Psi_k(\vec{r})\,. \tag{2.7}$$

Particle–hole symmetry relates every solution at $+E_k$ to a partner at $-E_k$ with exchanged electron and hole components. In the normal-state limit $\widetilde{\Delta} \to 0$, the off-diagonal coupling vanishes and both sectors decouple. [39 - 41]

Homogeneous Limit

Above, k is just a generic mode index, not necessarily a momentum. A uniform system restores translational invariance, and the eigenmodes become plane waves. We then switch to $\vec{q}$ to mark that the label is now an actual momentum. For uniform $\widetilde{\Delta} = \Delta e^{\mathring{i}\phi}$, the normal-state energy $\xi_{\vec{q}} = \hbar^2 q^2/(2m^*) - \mu$ gives

$$\begin{pmatrix} \xi_{\vec{q}} & \widetilde{\Delta} \\ \widetilde{\Delta}^* & -\xi_{\vec{q}} \end{pmatrix} \begin{pmatrix} u_{\vec{q}} \\ v_{\vec{q}} \end{pmatrix} = E_{\vec{q}} \begin{pmatrix} u_{\vec{q}} \\ v_{\vec{q}} \end{pmatrix}, \qquad E_{\vec{q}} = \sqrt{\xi_{\vec{q}}^2 + \Delta^2}\,. \tag{2.8}$$

The quasiparticle spectrum is separated from the Fermi level by the gap Δ. The homogeneous spectrum has two immediate consequences. It closes the BCS self-consistency equation for $\Delta(T)$, and it maps the normal-state DOS onto the superconducting DOS.

Superconducting Gap

Writing the homogeneous dispersion from Eq. (2.8) as $E(\xi)$, evaluation of the pair expectation value in Eq. (2.5) gives the weak-coupling self-consistency equation

$$1 = \frac{|V|N_0}{2} \int_0^{\hbar\omega_{\mathrm{D}}} \frac{1}{E(\xi)} \tanh\!\left(\frac{E(\xi)}{2k_{\mathrm{B}}T}\right) \mathrm{d}\xi\,. \tag{2.9}$$

Here, $N_0/2$ is the normal-state DOS per spin and ω_{D} is the Debye frequency, so $\hbar\omega_{\mathrm{D}}$ bounds the phonon mediated pairing shell. At each temperature, Eq. (2.9) implicitly determines $\Delta(T)$. Its zero-temperature solution defines $\Delta_0 \equiv \Delta(0)$, while T_{c} is the temperature at which the nonzero solution vanishes. Comparing these limits gives the weak-coupling relation

$$\Delta_0 \approx 1.764\, k_{\mathrm{B}} T_{\mathrm{c}}\,. \tag{2.10}$$

Thus, Δ_0 or T_{c} sets the material scale, while the self-consistency equation fixes the normalized temperature dependence $\Delta(T)/\Delta_0$. [35, 36] Representative bulk aluminum values are

$$\Delta_0 \approx 180\,\mu\mathrm{eV}\,, \qquad T_{\mathrm{c}} \approx 1.2\,\mathrm{K}\,. \tag{2.11}$$

Thin-film disorder, thickness, and fabrication can shift both values. Unless stated otherwise, the following figures use these representative scales. [19, 36]

Rather than solving Eq. (2.9) numerically at every temperature, we use the convenient interpolation

$$\frac{\Delta(T)}{\Delta_0} \approx \tanh\left(1.74\sqrt{\frac{T_\mathrm{c}}{T}-1}\right). \quad (2.12)$$

The gap vanishes continuously at T_c. At low temperature, it sets the approximate quasiparticle onset at $|eV| = \Delta$ for an NIS junction and $|eV| = 2\Delta$ for an identical SIS junction. [36, 42]

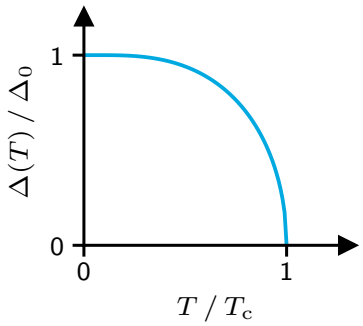


Figure 2.1. BCS gap as a function of temperature (Eq. 2.12).

Density of States

Pairing changes the relation between the normal-state energy ξ and the quasiparticle energy E while preserving the total spectral weight. The superconducting DOS therefore follows from the Jacobian of the homogeneous BdG dispersion,

$$\frac{N_\mathrm{S}(E)}{N_0} = \left|\frac{\mathrm{d}\xi}{\mathrm{d}E}\right| = \frac{E}{\sqrt{E^2-\Delta^2}} \qquad (E \geq \Delta). \quad (2.13)$$

No real ξ exists for $0 \leq E < \Delta$, so the DOS vanishes inside the gap. Particle–hole symmetry extends the result to negative energies through $N_\mathrm{S}(-E) = N_\mathrm{S}(E)$. The Jacobian diverges at the gap edges. [35, 36]

Nonideal spectra are commonly described by Dynes broadening,

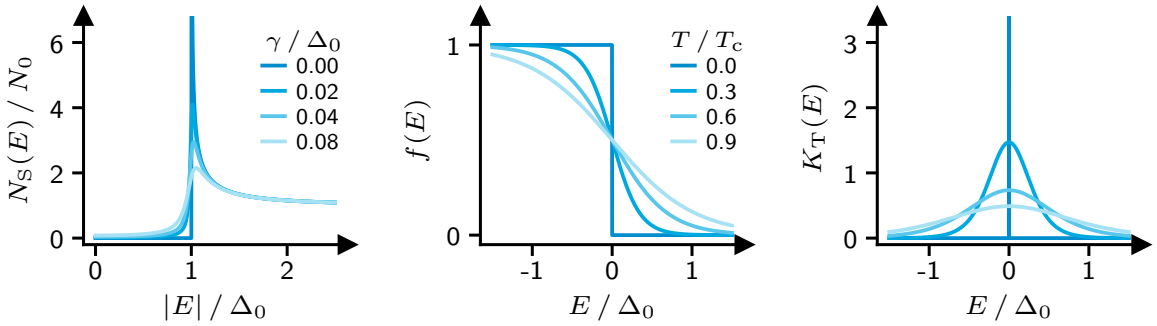
$$\frac{N_\mathrm{S}(E)}{N_0} = \left|\Re\left(\frac{E+\mathring{\imath}\gamma}{\sqrt{(E+\mathring{\imath}\gamma)^2-\Delta^2}}\right)\right|. \quad (2.14)$$

The parameter γ describes finite subgap spectral weight and smeared quasiparticle peaks but does not identify a unique microscopic mechanism. [43, 44]

Occupation and Thermal Broadening

The DOS specifies the available states, while the Fermi function determines their equilibrium occupation,

$$f(E) = \frac{1}{1+\exp(E/k_\mathrm{B}T)}. \quad (2.15)$$



(a) Superconducting DOS for varying γ/Δ_0.

(b) Fermi function for varying temperature T/T_c.

(c) Thermal kernel for varying temperature T/T_c.

Figure 2.2. Calculated quasiparticle DOS, Fermi function, and thermal kernel. Dynes broadening modifies the spectrum, while temperature broadens the occupation edge and the thermal kernel.

The derivative of this function defines the thermal broadening kernel

$$K_T(E) := -\frac{\partial f(E)}{\partial E}\,. \tag{2.16}$$

In the zero-temperature limit, $f(E) \to \theta(-E)$ and $K_T(E) \to \delta(E)$. Dynes broadening modifies the DOS itself, whereas temperature broadens the energy window sampled in transport. Figure 2.2 separates these effects.

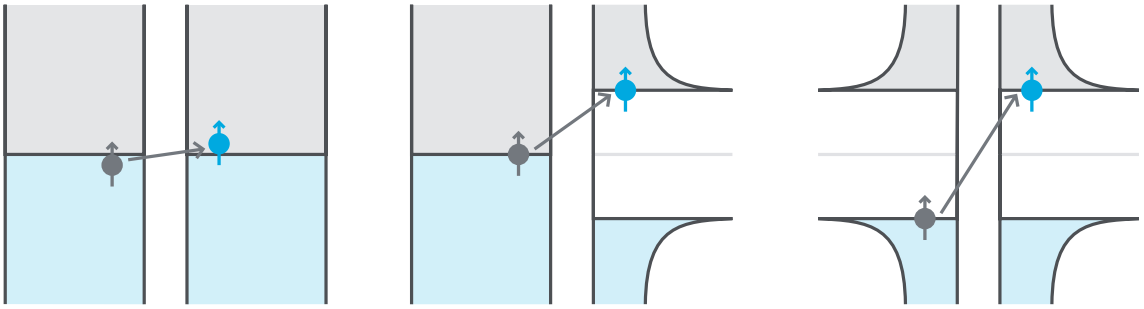
The DOS and occupation function provide the spectral and statistical inputs for the tunnel current model that follows.

2.1.2 Quasiparticle Tunneling

We consider two electrodes separated by a thin tunnel barrier. Although the barrier classically forbids transfer, an electronic wavefunction decays rather than vanishes inside it, leaving a small overlap with states in the opposite electrode. A quasiparticle can therefore tunnel from an occupied state in one electrode into an available state in the other, with the applied bias controlling their relative energies. In the low-transmission limit, we describe this incoherent single-particle process by a momentum averaged tunneling matrix element. [18, 19, 45]

In the expressions below, $N_i(E)$ is the DOS normalized to the constant normal-state value of electrode i. Thus, a normal electrode has $N_i(E) = 1$. This normalization, the junction geometry, and tunneling matrix element are incorporated into the normal-state conductance G_N.

We define the bias by $eV = \mu_1 - \mu_2$. An energy E relative to the chemical potential of electrode 1 therefore corresponds to $E + eV$ relative to that of electrode 2.



(a) NIN junction (b) NIS junction (c) SIS junction

Figure 2.3. Energy schematics of quasiparticle tunneling in NIN, NIS, and SIS junctions. The horizontal direction follows position from electrode 1 across the insulating barrier to electrode 2, while the vertical direction represents energy. Horizontal reference lines mark the electrode chemical potentials. The vertical span of each arrow indicates the bias energy required for the depicted quasiparticle transfer. The curves show the DOS, while blue and gray denote filled and empty states.

Under these conditions, the microscopic current flowing from electrode 1 to electrode 2 is proportional to the number of occupied states at energy E in electrode 1 and the number of empty states at energy $E + eV$ in electrode 2,

$$I_{1\to 2}(V) = \frac{G_\mathrm{N}}{e} \int_{-\infty}^{\infty} \left(N_1(E) f_1(E)\right) \cdot \left(N_2(E+eV)\left(1 - f_2(E+eV)\right)\right) \mathrm{d}E\,, \tag{2.17}$$

where V is the externally applied bias voltage. The reverse process $2 \to 1$ is described by an analogous expression. Subtracting the two contributions yields the net tunnel current

$$I(V) = \frac{G_\mathrm{N}}{e} \int_{-\infty}^{\infty} N_1(E)\, N_2(E+eV)\, \left(f_1(E) - f_2(E+eV)\right) \mathrm{d}E\,, \tag{2.18}$$

where N_1 and N_2 are the normalized DOSs defined above [18, 19].

For numerical evaluation, it is useful to express the current through a convolution[1]

$$I(V) = G_\mathrm{N}/e\, \left[(N_1 f_1) \otimes (N_2(1-f_2)) - (N_1(1-f_1)) \otimes (N_2 f_2)\right](eV)\,. \tag{2.19}$$

Depending on the electronic state of the electrodes, Eq. (2.18) describes the three configurations in Fig. 2.3. Here, N denotes a normal electrode, I the insulating barrier, and S a superconducting electrode [19, 45].

[1] Throughout this thesis, we define convolution as $(f \otimes g)(x) = \int_{-\infty}^{\infty} f(E) g(E+x)\, \mathrm{d}E$.

NIN Tunnel Junction

For a fully normal tunnel junction, both electrodes provide a flat density of states near the Fermi level. The current reduces to

$$I_{\mathrm{NIN}}(V) = \frac{G_{\mathrm{N}}}{e} \int_{-\infty}^{\infty} (f(E) - f(E + eV))\,\mathrm{d}E\,, \tag{2.20}$$

If both electrodes share the same temperature, the difference of Fermi functions integrates to the applied energy shift eV. The result is Ohm's law,

$$I_{\mathrm{NIN}}(V) = G_{\mathrm{N}} V\,, \tag{2.21}$$

showing that an NIN junction is the Ohmic reference for the tunneling limit.

NIS Tunnel Junction

In an NIS junction, the flat normal-state DOS leaves the current sensitive to the superconducting quasiparticle spectrum,

$$I_{\mathrm{NIS}}(V) = \frac{G_{\mathrm{N}}}{e} \int_{-\infty}^{\infty} N_{\mathrm{S}}(E)\,(f(E) - f(E + eV))\,\mathrm{d}E\,. \tag{2.22}$$

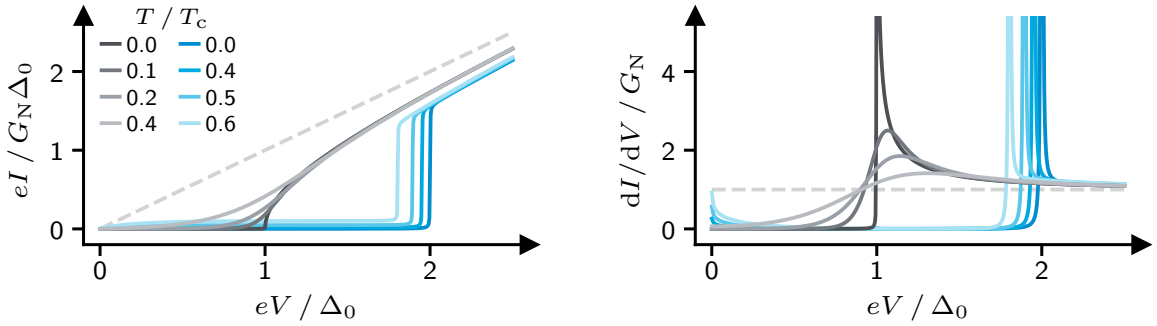
In the NIS case, the *dI–dV* characteristic probes the superconducting DOS through its convolution with the thermal broadening kernel from Eq. (2.16). Temperature broadens the measured *dI–dV* response without introducing spectral broadening into the equilibrium DOS[2].

Differentiating Eq. (2.22) with respect to V acts only on $f(E + eV)$ because both $N_{\mathrm{S}}(E)$ and $f(E)$ are independent of bias. The chain rule produces the thermal broadening kernel at $E + eV$. Using the particle–hole symmetry of $N_{\mathrm{S}}(E)$ and the even symmetry of $K_{\mathrm{T}}(E)$ gives

$$\tfrac{\mathrm{d}}{\mathrm{d}V} I_{\mathrm{NIS}}(V) = G_{\mathrm{N}}\,(N_{\mathrm{S}} \otimes K_{\mathrm{T}})\,(eV) \overset{T \to 0}{=} G_{\mathrm{N}}\,N_{\mathrm{S}}(eV)\,. \tag{2.23}$$

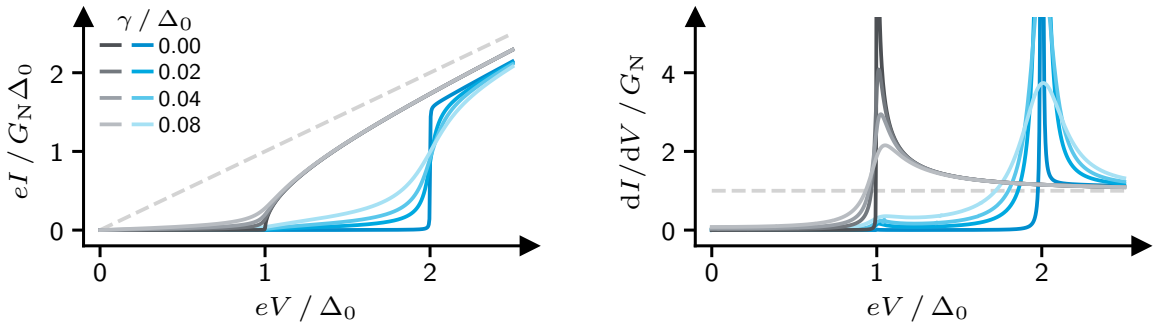
These considerations underlie scanning tunneling microscopy, where a normal metal tip above a superconducting surface realizes an NIS junction in the tunneling limit. At sufficiently low electronic temperatures, *I–V* and *dI–dV* characteristics provide a local probe of the superconducting quasiparticle DOS.

[2] The full width at half maximum of $K_{\mathrm{T}}(E)$ is approximately $3.5\,k_{\mathrm{B}}T$. At 30 mK this corresponds to about 10 µeV, whereas at 300 mK it is about 90 µeV. The relevant temperature is the electronic temperature, which can exceed the cryostat base temperature due to imperfect filtering, finite electron–phonon coupling, or microwave leakage.



(a) *I–V* characteristics for varying T/T_c.

(b) *dI–dV* characteristics for varying T/T_c.



(c) *I–V* characteristics for varying γ/Δ_0.

(d) *dI–dV* characteristics for varying γ/Δ_0.

Figure 2.4. Quasiparticle tunneling in NIN, NIS, and SIS junctions. Temperature and Dynes broadening of NIS (gray) and SIS (blue) *I–V* and *dI–dV* curves. The gray dashed line represents the NIN reference.

SIS Tunnel Junction

In a junction where both electrodes are superconducting, the tunnel current depends on the interaction of two gapped densities of states at E and $E + eV$,

$$I_{\mathrm{SIS}}(V) = \frac{G_{\mathrm{N}}}{e} \int_{-\infty}^{\infty} N_{\mathrm{S}}(E)\, N_{\mathrm{S}}(E+eV)\, \left(f(E) - f(E+eV)\right) \mathrm{d}E\,. \tag{2.24}$$

Unlike the NIS case, the SIS differential conductance is not reduced here to a convolution of one DOS with the thermal kernel. We evaluate it numerically from the full current. For identical electrodes, the dominant quasiparticle threshold occurs near $|eV| = 2\Delta$, producing a pronounced feature in both *I–V* and *dI–dV*.

More generally, this threshold sits at $|eV| = \Delta_1 + \Delta_2$ for electrodes with different gaps, since only then does a filled state in one electrode align with an empty state above the gap in the other. It reduces to the familiar 2Δ threshold for the identical electrodes considered throughout this thesis. We refer to this feature as the quasiparticle peak.

Figure 2.4 summarizes the effects of temperature and Dynes broadening in NIS and SIS junctions.

In NIS junctions, temperature broadens the occupation factor and therefore rounds the quasiparticle peak and smooths the gap edge, while Dynes broadening directly smears the superconducting DOS.

In SIS junctions, temperature produces thermally activated subgap current and weakens the threshold feature as T approaches T_c. For unequal gaps, quasiparticles already excited above the larger gap can additionally tunnel into the empty states just above the smaller gap without the bias supplying the full gap sum, which produces a separate and much weaker peak at $|eV| = |\Delta_1 - \Delta_2|$. Since it requires a thermal quasiparticle population, this feature vanishes as $T \to 0$. For the identical electrodes considered throughout this thesis it sits at $eV = 0$, merging into the thermally activated subgap current described above rather than appearing as a distinguishable peak. Dynes broadening introduces subgap spectral weight in both electrodes and smears the threshold.

Fits to measured *I–V* or *dI–dV* curves can provide estimates of Δ_0, T, and γ. However, correlations among these parameters, voltage noise, and spatial gap variations can produce similar line shapes. Reliable extraction therefore requires independent calibration or comparison across multiple spectra.

2.1.3 Photon Assisted Tunneling

The static NIN, NIS, and SIS characteristics derived above each define a reference current $I_0(V)$. Photon assisted tunneling is therefore not specific to superconducting junctions, although distinct sidebands are visible only when $I_0(V)$ contains nonlinear spectral features. In the ideal NIN case, the redistribution leaves the linear *I–V* characteristic unchanged. Under microwave irradiation, a quasiparticle may absorb or emit $|n|$ photons while tunneling, changing its energy from E_0 to

$$E_n = E_0 + nh\nu \qquad (n \in \mathbb{Z}), \tag{2.25}$$

where $n > 0$ corresponds to absorption and $n < 0$ to emission. These energy exchanges shift static tunnel features by integer multiples of $h\nu/e$. Figure 2.5 illustrates this process for an SIS junction, whose sharp quasiparticle threshold makes the shifts particularly clear. To determine the sideband weights, we represent the harmonic voltage by a time dependent phase factor in the tunneling amplitude. Its Fourier expansion then leads to the Tien-Gordon Eq. (2.32) [27].

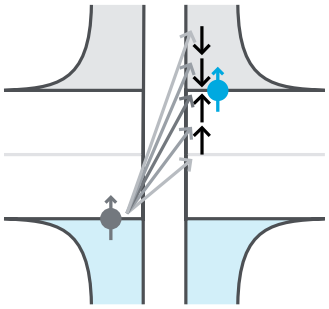


Figure 2.5. Energy schematic of photon assisted tunneling in an SIS junction. Exchange of $nh\nu$ shifts the threshold. Black and gray arrows show photon exchange and quasiparticle transfer.

Harmonic Drive and Sideband Formation

We model the effective voltage across the junction as

$$V(t) = V_0 + A\cos(2\pi\nu t)\,, \tag{2.26}$$

where V_0 is the static bias, A is the effective ac voltage amplitude at the junction, and ν is the drive frequency. We treat the coherent tone as a classical drive, neglect junction backaction, and work in the tunneling limit. The electrodes are assumed to remain locally thermal.

Gauge freedom allows the voltage dependence to be represented either through time dependent electrode energies or through a phase in the charge transfer amplitude. We use the phase factor representation. For a transferred charge q, the accumulated phase and corresponding coupling factor are

$$\phi_q(t) = \frac{q}{\hbar}\int_0^t V(t')\,\mathrm{d}t'\,, \qquad U_q(t) \equiv \exp\!\left[-\mathring{\imath}\phi_q(t)\right]\,. \tag{2.27}$$

Here, $q = e$ applies to quasiparticle tunneling.

For the harmonic voltage in Eq. (2.26), the phase factor separates into static and oscillatory contributions,

$$U_q(t) = \exp\!\left[-\mathring{\imath}\phi_{q,0}(t)\right]\exp\!\left[-\mathring{\imath}\alpha_{\nu,q}\sin(2\pi\nu t)\right]\,, \tag{2.28}$$

$$\phi_{q,0}(t) = \frac{qV_0 t}{\hbar}\,, \qquad \alpha_{\nu,q} = \frac{qA}{h\nu}\,. \tag{2.29}$$

The oscillatory phase is expanded with the Jacobi-Anger identity,

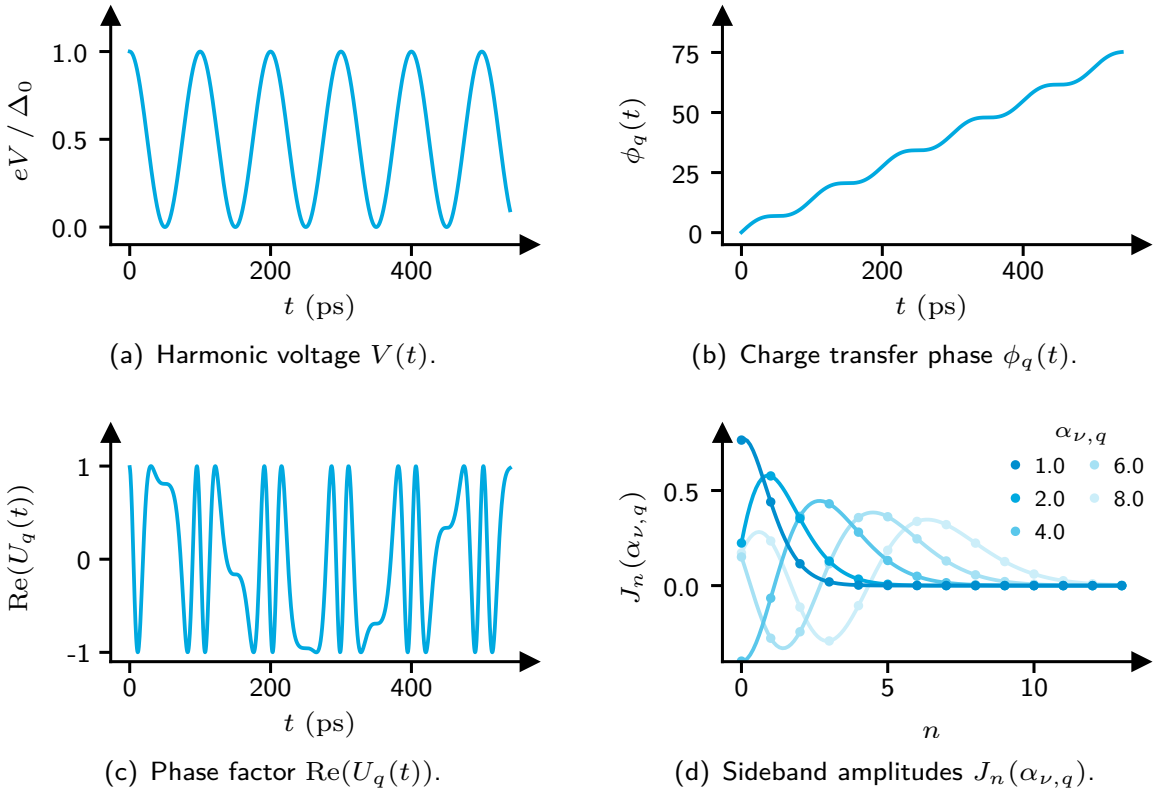
$$\exp\!\left[\pm\mathring{\imath}\alpha_{\nu,q}\sin(2\pi\nu t)\right] = \sum_{n=-\infty}^{\infty} J_n(\alpha_{\nu,q})\,\exp\!\left(\pm\mathring{\imath}n\,2\pi\nu t\right)\,. \tag{2.30}$$

This expansion produces the energy shifted processes introduced above, with amplitudes set by the Bessel coefficients $J_n(\alpha_{\nu,q})$ [46]. Figure 2.6 summarizes the drive and resulting sideband amplitudes.

The sideband picture assumes that the drive redistributes existing quasiparticle tunneling processes without substantially changing the electrode properties. At sufficiently high photon energies, radiation can break Cooper pairs and generate additional quasiparticles, so the equilibrium DOS and local thermal assumptions may fail. At low temperature, the single-photon pair breaking threshold is

$$h\nu_{\mathrm{pb}} = 2\Delta_0\,. \tag{2.31}$$

Using the representative gap in Eq. (2.11) gives $\nu_{\mathrm{pb}} \approx 87\,\mathrm{GHz}$ [35, 36]. The installed microwave paths transmitted frequencies up to approximately $20\,\mathrm{GHz}$, as established from microwave induced changes in measured *dI–dV* spectra [47]. This experimental upper frequency is below the single-photon pair breaking



(a) Harmonic voltage $V(t)$.

(b) Charge transfer phase $\phi_q(t)$.

(c) Phase factor $\mathrm{Re}(U_q(t))$.

(d) Sideband amplitudes $J_n(\alpha_{\nu,q})$.

Figure 2.6. Visualization of (a) harmonic voltage, (b) charge transfer phase, (c) phase factor, and (d) sideband amplitudes. The parameters are $q = e$ and $eV_0 = eA = 0.5\,\Delta_0$.

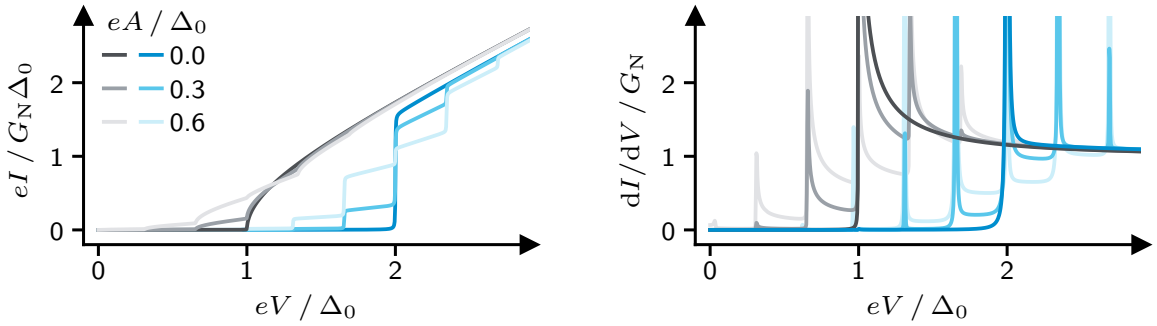
threshold. Direct single-photon pair breaking is therefore excluded within this estimate, although heating, multiphoton absorption, and nonequilibrium quasiparticle generation may still occur.

Tien-Gordon Quasiparticle Tunneling

We now specialize to quasiparticle tunneling with $q = e$. The Fourier components of $U_e(t)$ describe energy transfers $nh\nu$. For a stationary incoherent tunnel current, averaging over one drive period removes cross terms between different components and leaves the probabilities J_n^2. The resulting Tien-Gordon current is

$$I(V_0) = \sum_{n=-\infty}^{\infty} J_n^2\left(\frac{eA}{h\nu}\right) I_0\left(V_0 - \frac{nh\nu}{e}\right) , \tag{2.32}$$

where $I_0(V)$ is the static current derived above. Each term contributes a replica of the static characteristic shifted by $nh\nu/e$ and weighted by the sideband probability $J_n^2(\alpha_{\nu,e})$ [27].



(a) *I–V* characteristics.

(b) *dI–dV* characteristics.

Figure 2.7. Photon assisted NIS (gray) and SIS (blue) spectra. Sidebands are separated by $h\nu/e$ and weighted by $J_n^2(eA/h\nu)$.

Photon assisted tunneling redistributes an existing quasiparticle current rather than introducing a new transport mechanism. Differences between driven NIS and SIS junctions therefore follow from their static characteristics, as shown in Fig. 2.7. The treatment excludes junction backaction, drive induced heating, and coherent subgap transport.

2.2 Josephson Dynamics

The microscopic section introduced the complex pair potential $\widetilde{\Delta}(\vec{r})$ in Eq. (2.5), whose magnitude characterizes the local pairing strength and whose phase describes condensate coherence. When two superconductors are separated by a sufficiently thin weak link, superconducting pair correlations extend into the link from both sides and retain a finite overlap. The two condensates are therefore no longer independent.

Since their magnitudes remain fixed by the bulk electrodes, their coherent coupling is governed by the phase difference

$$\phi = \phi_1 - \phi_2 \,, \tag{2.33}$$

which becomes the central variable of the macroscopic description and determines the supercurrent through the weak link. Figure 2.8 depicts the resulting coherent Cooper pair transfer across an SIS junction.

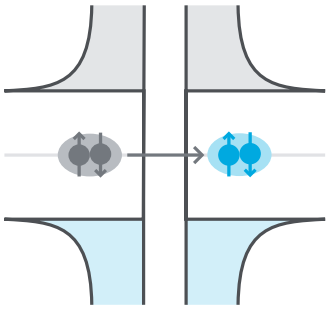


Figure 2.8. Schematic Cooper pair tunneling in an SIS junction.

A Josephson junction denotes such a weak link between two superconductors. It can be formed by an insulating barrier, a metallic constriction, or a short normal region. In the low-transmission limit considered here, the overlap and resulting Josephson coupling are weak, but remain sufficient for coherent Cooper pair transfer. Finite-transmission weak links require the mesoscopic description of Sec. 2.3. [22, 48]

We first introduce the Josephson relations, the Ambegaokar-Baratoff critical current, and the coupling energy. We then connect the ideal voltage biased SIS spectrum to current bias using a generalized resistively shunted junction (RSJ) model that retains the nonlinear quasiparticle branch. The resistively and capacitively shunted junction (RCSJ) model changes the modeling goal by replacing this spectrum with a linear shunt and adding capacitance to describe phase dynamics. Finally, we discuss the distinct voltage and current biased manifestations of microwave phase locking in the Shapiro effect.

2.2.1 Josephson Effect

The phase difference ϕ defined in Eq. (2.33) is the collective variable that describes the coherent coupling between the two superconductors. Its static and time-dependent behavior are expressed by the Josephson relations, which connect condensate coherence to the measurable current and voltage [22].

AC Josephson Relation

The ac Josephson relation connects the voltage across the weak link directly to the temporal evolution of the phase difference,

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{2e}{\hbar}\, V(t)\,. \tag{2.34}$$

It follows from the condensate charge $2e$ and is independent of the detailed structure and transmission of the junction. The ac Josephson relation therefore applies to all junction regimes considered in this chapter. A voltage drives the phase difference rather than fixing the current through the weak link [22].

For a constant voltage $V(t) = V_0$, the phase evolves linearly as

$$\phi(t) = \phi_0 + 2\pi\nu_0 t\,, \qquad\qquad \nu_0 = \frac{2e}{h}\, V_0\,, \tag{2.35}$$

where ϕ_0 is the phase difference at $t = 0$ and ν_0 is the Josephson frequency. Thus, a dc voltage produces a phase that advances periodically modulo 2π. An oscillating current follows only after this phase evolution is combined with the current–phase relation introduced below.

DC Josephson Relation

In the absence of a voltage, Eq. (2.34) gives a stationary phase difference. The coherent coupling between the condensates can then support a time independent supercurrent. To leading order in the weak coupling, its dependence on phase is described by the sinusoidal current–phase relation (CPR)

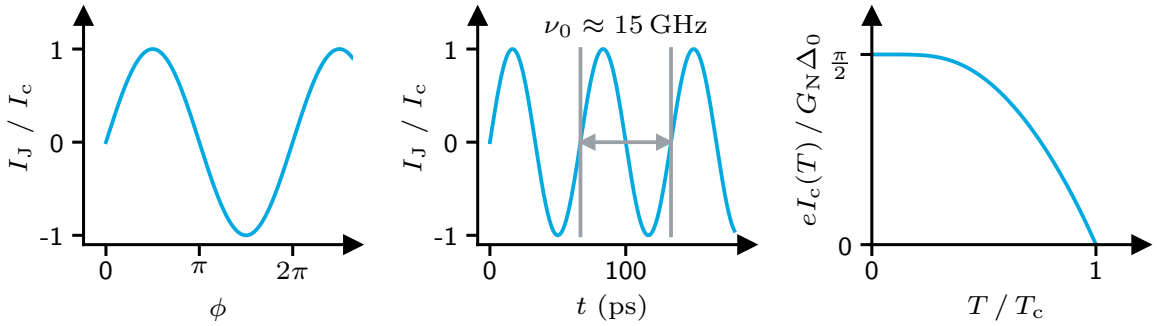
$$I_{\mathrm{J}}(\phi) = I_{\mathrm{c}} \sin\phi\,, \tag{2.36}$$

where I_{c} is the maximum supercurrent supported by the junction. Figure 2.9(a) shows this sinusoidal phase dependence. A finite phase difference can therefore drive a dissipationless current without a voltage drop. This is the dc Josephson effect [22].

Under a constant finite voltage, inserting the phase evolution from Eq. (2.35) into the CPR gives

$$I_{\mathrm{J}}(t) = I_{\mathrm{c}} \sin(\phi_0 + 2\pi\nu_0 t)\ . \tag{2.37}$$

The supercurrent then oscillates at the Josephson frequency ν_0, as shown in Fig. 2.9(b). The dc and ac Josephson effects are therefore two bias conditions of the same phase dynamics. Under microwave irradiation, synchronization of this intrinsic oscillation with the external drive produces voltage plateaus. This effect is known as the inverse ac Josephson effect or as Shapiro steps and is discussed below.



(a) Current–phase relation of a weak-coupling Josephson junction (Eq. 2.36).

(b) Josephson current for a constant voltage $V_0 = 31\,\mu\mathrm{V}$ over time (Eq. 2.37).

(c) Critical current versus temperature (Eq. 2.38).

Figure 2.9. Phase and temperature dependence of the Josephson current in the weak-coupling limit. (a) The sinusoidal current–phase relation reaches its maximum magnitude at I_c. (b) A constant voltage drives an oscillating current at the Josephson frequency. (c) The Ambegaokar-Baratoff critical current decreases with increasing temperature and vanishes as the superconducting gap closes at T_c.

Ambegaokar-Baratoff Relation

The Josephson relations determine the phase dependence and temporal evolution of the supercurrent, but do not fix its magnitude. For an SIS junction formed by identical BCS superconductors, the Ambegaokar-Baratoff (AB) theory derives this magnitude from coherent pair tunneling through the barrier. The resulting critical current is

$$I_c(T) = \frac{\pi}{2}\,\frac{G_N \Delta(T)}{e}\,\tanh\!\left(\frac{\Delta(T)}{2k_B T}\right), \tag{2.38}$$

where G_N is the normal-state conductance of the junction. The temperature dependence combines the reduction of the superconducting gap from Eq. (2.12) with the thermal occupation factor, as shown in Fig. 2.9(c). At zero temperature, Eq. (2.38) reduces to $I_c = \pi G_N \Delta_0/(2e)$ [49].

Together with the sinusoidal current–phase relation in Eq. (2.36), the AB result determines the complete equilibrium supercurrent for the junction considered here.

Josephson Coupling Energy

The strength of the coherent phase coupling is quantified by the Josephson energy

$$E_{\mathrm{J}} = \frac{\hbar I_{\mathrm{c}}}{2e}\,, \tag{2.39}$$

which sets the energetic stiffness of the phase and the robustness of coherent Cooper pair transfer. Large E_{J} favors a well-defined phase difference, while competing fluctuations weaken this phase locking. The influence of thermal fluctuations is considered phenomenologically in the RCSJ model introduced below. The charge dominated regime, in which the charging energy competes with or exceeds E_{J}, is discussed in Sec. 2.4 [36, 50].

DC Josephson Spectroscopy

We first consider dc spectroscopy of an ideal voltage biased SIS Josephson junction. At finite voltage, the phase evolves at the Josephson frequency according to Eq. (2.35). The supercurrent consequently oscillates and averages to zero over one period, so the measured dc current follows the SIS quasiparticle characteristic derived in Eq. (2.24). At $V = 0$, the phase remains at its initial value ϕ_0 and the junction can support the finite current $I_{\mathrm{J}} = I_{\mathrm{c}} \sin\phi_0$. In this idealized limit, the voltage fixes only the phase velocity, while the stationary value ϕ_0 is set by the initial condition or the external circuit. Figures 2.10(a) and (b) show the resulting voltage biased spectrum and its dependence on the initial phase.

Under current bias, the voltage is no longer fixed externally and instead becomes the dynamical response of the junction. We describe this response with a generalized RSJ model. Coherent Cooper pair transfer and dissipative quasiparticle tunneling occur in parallel and experience the same time dependent voltage, as shown in Fig. 2.11. In contrast to the conventional linear shunt V/R, the generalized model retains the microscopic nonlinear quasiparticle characteristic $I_{\mathrm{qp}}(V)$.

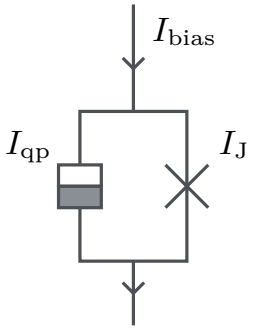
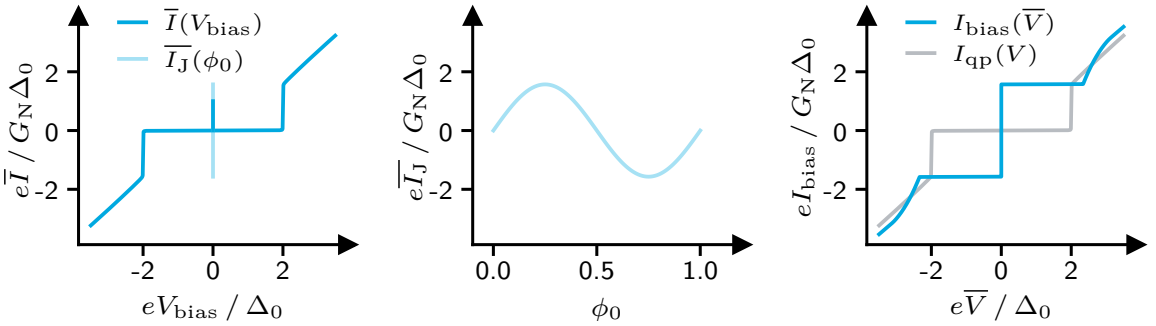


Figure 2.11. Generalized RSJ circuit with parallel Josephson and quasiparticle currents.

The applied current therefore divides according to

$$I_{\mathrm{bias}} = I_{\mathrm{c}} \sin\phi(t) + I_{\mathrm{qp}}(V(t))\ . \tag{2.40}$$

We denote the inverse of the BCS quasiparticle characteristic by $V_{\mathrm{qp}}(I)$. This construction assumes that the quasiparticle branch follows the instantaneous voltage and that its *I*–*V* characteristic can be inverted on the branch considered. Together with the ac Josephson relation, Eq. (2.40) determines the time evolution of the phase and voltage. For $|I_{\mathrm{bias}}| \leq I_{\mathrm{c}}$, a stationary phase can carry the applied current and the voltage remains zero. Above this range, the phase runs



(a) Time averaged current under voltage bias with quasiparticle and zero-voltage Josephson branches.

(b) Time averaged Josephson current at zero voltage as a function of the stationary initial phase ϕ_0.

(c) Time averaged voltage from generalized RSJ dynamics under current bias.

Figure 2.10. Josephson spectroscopy under voltage and current bias. Overbars denote time averages over the Josephson phase dynamics. (a) At finite voltage, the oscillating supercurrent averages to zero and $\overline{I}$ follows the quasiparticle characteristic. At zero voltage, the stationary phase supports a finite supercurrent. (b) The resulting $\overline{I_{\mathrm{J}}}$ depends on the initial phase ϕ_0. (c) Under current bias, a stationary phase gives $\overline{V} = 0$, while a running phase produces a finite $\overline{V}$.

through successive 2π periods. Its velocity changes during each rotation because the current available for quasiparticle tunneling depends on ϕ.

Consequently, the phase does not sample all values for equal durations. Although the unweighted phase average of $\sin\phi$ vanishes over one rotation, its time average need not vanish under current bias. This differs from ideal voltage bias, where the phase advances uniformly and the supercurrent averages to zero. The measured dc voltage is the time average over the nonuniform phase motion. For one complete phase rotation, it is

$$\overline{V} = 2\pi \left(\int_0^{2\pi} \frac{\mathrm{d}\phi}{V_{\mathrm{qp}}(I_{\mathrm{bias}} - I_{\mathrm{c}} \sin\phi)} \right)^{-1} . \tag{2.41}$$

The current balance and Eq. (2.41) thus convert the voltage biased quasiparticle spectrum into the current biased response shown in Fig. 2.10(c). This generalized RSJ description retains the nonlinear superconducting spectrum instead of replacing the dissipative branch by a linear resistance. It remains deterministic and neglects capacitive phase inertia and thermal fluctuations, which are considered phenomenologically with the RCSJ model below.

2.2.2 RCSJ Model

The generalized RSJ model above retains the nonlinear quasiparticle characteristic to describe dc spectroscopy. We now change the modeling goal from

reconstructing this spectrum to describing the classical phase dynamics. For this purpose, the RCSJ model approximates the dissipative branch by a linear shunt resistance R and adds a capacitance C in parallel with the Josephson element. The resistance accounts for damping, while the capacitance stores electrostatic energy and introduces phase inertia.

We assume the Josephson energy far exceeds the charging energy, such that quantum fluctuations of the phase can be neglected. Thermal fluctuations may nevertheless perturb the classical dynamics and are discussed phenomenologically below without introducing an explicit fluctuating current source.

The three parallel branches are shown in Fig. 2.12. Combining their current balance with the ac Josephson relation produces a second-order equation for ϕ. It is mathematically equivalent to the motion of a damped particle in a tilted periodic potential. In this phase-particle analogy, the capacitance sets the effective mass, the linear shunt provides viscous damping, and the Josephson coupling generates the periodic potential. The analogy connects stationary and running phase states to switching, retrapping, phase diffusion, and microwave phase locking [23, 24].

We first derive the RCSJ phase equation and then use the phase-particle picture to organize these dynamical regimes.

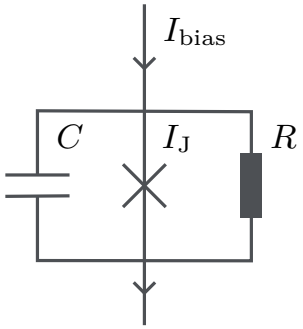


Figure 2.12. Circuit diagram of the RCSJ model. Josephson, resistive, and capacitive current paths are connected in parallel.

Current Bias

The phase dynamics follow by summing the currents through the three parallel elements,

$$I_{\mathrm{bias}} = I_{\mathrm{c}} \sin(\phi) + \frac{V}{R} + C\frac{\mathrm{d}V}{\mathrm{d}t} \,. \tag{2.42}$$

This current balance defines the standard RCSJ model [23, 24].

Using the ac Josephson voltage–phase relation, the current balance becomes

$$I_{\mathrm{bias}} = \underbrace{I_{\mathrm{c}} \sin\phi}_{\text{Josephson}} + \underbrace{\frac{\hbar}{2eR}\frac{\mathrm{d}\phi}{\mathrm{d}t}}_{\text{damping}} + \underbrace{\frac{\hbar C}{2e}\frac{\mathrm{d}^2\phi}{\mathrm{d}t^2}}_{\text{inertia}} \,. \tag{2.43}$$

The three terms represent the Josephson, dissipative, and capacitive contributions, respectively. In the mechanical analogy developed below, the latter two provide viscous damping and inertia. Together with the bias current, the Josephson contribution gives rise to the tilted-washboard potential, discussed below [23, 24].

Equation (2.43) is deterministic. The resistance introduces damping, but it does not add a fluctuating force. Thermal current fluctuations from the electro-

magnetic environment can perturb the phase dynamics, enabling thermally activated switching and phase diffusion. These effects are considered phenomenologically below without introducing a stochastic extension of the RCSJ equation.

Dimensionless RCSJ Model

A dimensionless form of Eq. (2.43) separates inertia, damping, and the nonlinear Josephson term.

The natural time scale of the junction is set by the Josephson plasma frequency,

$$\omega_{\mathrm{p}} = \sqrt{\frac{2eI_{\mathrm{c}}}{\hbar C}}\,, \tag{2.44}$$

which determines the small-oscillation angular frequency of the phase in a potential minimum. The corresponding ordinary frequency is $\nu_{\mathrm{p}} = \omega_{\mathrm{p}}/(2\pi)$. It is distinct from the AC Josephson frequency, which reflects voltage driven phase evolution, and instead characterizes the intrinsic phase resonance in the absence of bias [24, 36].

Introducing the rescaled time $t' = \omega_{\mathrm{p}} t$ and the normalized current $i = I_{\mathrm{bias}}/I_{\mathrm{c}}$ gives the compact dimensionless form

$$i = \sin\phi + \frac{1}{Q}\frac{\mathrm{d}\phi}{\mathrm{d}t'} + \frac{\mathrm{d}^2\phi}{\mathrm{d}t'^2}\,, \tag{2.45}$$

where the quality factor

$$Q = \sqrt{\frac{2eI_{\mathrm{c}}R^2C}{\hbar}} \tag{2.46}$$

quantifies the damping of the phase dynamics [24].

The single parameter Q distinguishes underdamped ($Q \gg 1$) from overdamped ($Q \ll 1$) junctions and sets the stage for the dynamical behavior discussed next.

Tilted Washboard Potential

The normalized RCSJ equation (2.45) has the form of Newton's equation for a unit mass phase particle subject to viscous damping. The phase difference ϕ acts as the particle coordinate, while $1/Q$ sets the normalized damping. The bias current i and the Josephson term $\sin\phi$ define the force acting on the phase particle. Identifying this force with the negative gradient of a dimensionless potential and integrating over ϕ gives

$$-\tfrac{\mathrm{d}}{\mathrm{d}\phi}u = i - \sin\phi\,, \qquad u(\phi) = -\cos\phi - i\phi\,. \tag{2.47}$$

The Josephson coupling produces the periodic cosine landscape, while the normalized bias current i produces its linear tilt. The corresponding dimensional

phase potential is $U(\phi) = E_J u(\phi)$. In the untilted case ($i = 0$), adjacent minima are separated by a barrier $\Delta U = 2E_J$. Figure 2.13 shows how this tilted washboard potential evolves with increasing bias current [23, 36].

For $i = 0$, the periodic potential has equivalent stable minima. A finite subcritical bias, $0 < i < 1$, tilts the potential and turns them into metastable minima separated by finite barriers. While the phase remains localized in one well, its average velocity and therefore the junction voltage are zero. Small oscillations about a minimum are governed by the plasma-frequency scale introduced in Eq. (2.44).

As the bias approaches $i = 1$, the barrier in the direction of the tilt decreases and vanishes at the deterministic critical current. For $i > 1$, no local minima remain and the phase runs continuously, corresponding to a finite-voltage state. Thermal and environmental fluctuations can induce escape already below this threshold and thereby cause premature switching. When the bias is subsequently reduced below $i = 1$, local minima reappear. The return from the running state to a localized minimum is called retrapping.

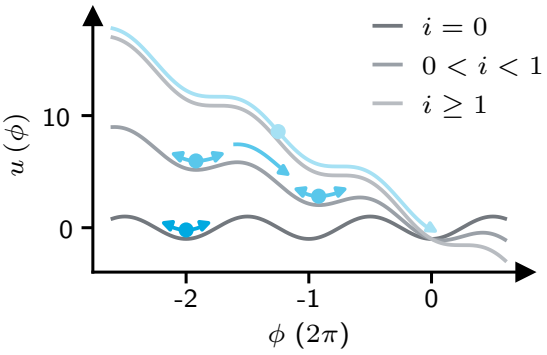


Figure 2.13. Tilted washboard potential of the RCSJ model (Eq. 2.47). Increasing the normalized bias current i tilts the potential and lowers the barrier in the direction of phase motion. Thermal fluctuations can induce 2π phase slips between neighboring minima before the barrier disappears at the critical tilt $i = 1$.

Not every escape leads to a sustained running state. If the phase is captured in an adjacent minimum, it has undergone a phase slip of approximately 2π. Through the Josephson voltage–phase relation, each slip produces a voltage pulse with time integral $h/(2e)$. Forward and backward slips balance at zero bias, whereas a directional imbalance produces a finite average phase velocity. Repeated slips interrupted by retrapping constitute phase diffusion.

Switching Dynamics and Phase Diffusion

The intrinsic critical current I_c is set by the Josephson coupling, while the switching current I_{sw} is the observed current at which the junction leaves the localized zero-voltage state and establishes a sustained running state. In the deterministic noise-free limit, the phase remains trapped until the barrier vanishes at $I_{bias} = I_c$. Thermal activation and environmental current noise can instead drive escape at a lower bias. The resulting I_{sw} varies between sweeps and depends on temperature, noise, and the bias ramp rate [36, 51].

Escape does not necessarily produce a sustained running state. The phase may be captured in a neighboring minimum after one or several slips. Under

finite bias, an imbalance between forward and backward slips generates a small average phase velocity and hence a finite voltage below I_c. This phase diffusion gives the supercurrent branch a finite slope and broadens the transition into the running state. It is not contained in deterministic Eq. (2.43) and is considered here phenomenologically [36, 51, 52].

Retrapping Dynamics

Once a sustained running state is established, the junction develops a finite average voltage. On a downward current sweep, metastable minima reappear below I_c, but the moving phase is not necessarily captured immediately. The bias at which it is eventually recaptured in a minimum defines the retrapping current I_{rt}.

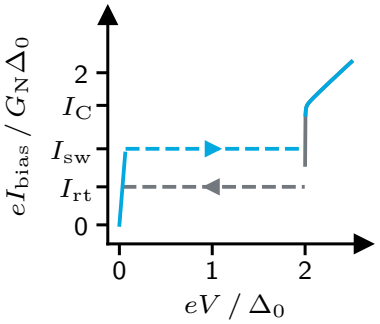


Figure 2.14. Phenomenological *I–V* characteristic of a Josephson junction with prescribed switching current I_{sw} and retrapping current I_{rt}. The finite slope represents phase diffusion, while the separation between I_{sw} and I_{rt} represents hysteretic switching and retrapping.

The relation between I_{sw} and I_{rt} is controlled primarily by the damping. In the overdamped limit, the phase loses energy rapidly, so it retraps as soon as stable minima reappear and switching and retrapping occur at similar currents. In the underdamped limit, the phase retains enough kinetic energy to remain in the running state well below I_c, so it retraps only at a much lower bias, producing pronounced hysteresis in the *I–V* characteristic. In the strongly underdamped limit, the retrapping current approaches zero relative to I_c [23, 24, 36].

Figure 2.14 shows a representative phenomenological *I–V* trace with a finite phase diffusion slope, switching at I_{sw}, and retrapping at I_{rt}. These currents are prescribed observables in the plotted characteristic rather than outputs of the deterministic equation. Thermal fluctuations, frequency dependent environmental damping, and heating can shift and broaden both transitions. For the atomic aluminum contact considered here, we use the overdamped limit.

Macroscopic Quantum Dynamics

The phase particle analogy provides the starting point for a quantum mechanical description of the junction. Quantizing the conservative junction degree of freedom while treating the dissipative environment separately promotes the phase difference and junction charge to conjugate observables. The capacitive energy supplies the kinetic term of the resulting Hamiltonian, while the tilted Josephson potential of Eq. (2.47) supplies its potential term. This Hamiltonian supports

discrete energy levels associated with quantized phase oscillations within each metastable minimum. Their observation demonstrated energy quantization of the collective phase difference across a current biased Josephson junction [53].

At sufficiently low temperature, the phase can escape a metastable minimum by tunneling through the potential barrier rather than by thermal activation. In an underdamped junction, this escape can initiate the running state. The process is known as macroscopic quantum tunneling because the tunneling coordinate is the collective phase difference across the junction. Its experimental observation, together with the quantized phase particle levels, established that an electrical circuit can exhibit quantum behavior in a collective macroscopic degree of freedom [54]. These quantum effects lie outside the classical RCSJ treatment used throughout this section.

2.2.3 Shapiro Effect

Under microwave irradiation, the Josephson oscillation can synchronize with the applied drive. The observable manifestation of this locking depends fundamentally on the bias condition. We first consider an ideal voltage biased junction, for which the junction voltage is prescribed and the coherent Josephson contribution appears only at discrete locking voltages. Current biased phase dynamics constitute a separate physical problem and are treated below [28, 51].

Driven Phase and Ideal Locking

To obtain the driven phase evolution, we integrate the ac Josephson relation of Eq. (2.34) using the harmonic voltage of Eq. (2.26), which gives

$$\phi(t) = \phi_0 + 2\pi\nu_0 t + \alpha_{\nu,2e}\sin(2\pi\nu t)\,, \quad \nu_0 = \frac{2eV_0}{h}\,, \quad \alpha_{\nu,2e} = \frac{2eA}{h\nu}\,. \tag{2.48}$$

The phase therefore contains uniform Josephson evolution and a periodic modulation set by the Cooper pair charge $2e$.

Inserting Eq. (2.48) into the sinusoidal CPR (Eq. 2.36) and using the Jacobi-Anger expansion (Eq. 2.30) produces

$$I_{\mathrm{J}}(t) = I_{\mathrm{c}} \sum_{n=-\infty}^{\infty} J_n(\alpha_{\nu,2e})\sin(\phi_0 + 2\pi(\nu_0 + n\nu)t)\ . \tag{2.49}$$

A harmonic becomes stationary whenever ν_0 is an integer multiple of ν. The corresponding locking voltages and full current intervals are

$$\begin{aligned} V_n &= nh\nu/2e\,, \\ \Delta I_n &= 2I_{\mathrm{c}}\,|J_n(\alpha_{\nu,2e})|\ . \end{aligned} \tag{2.50}$$

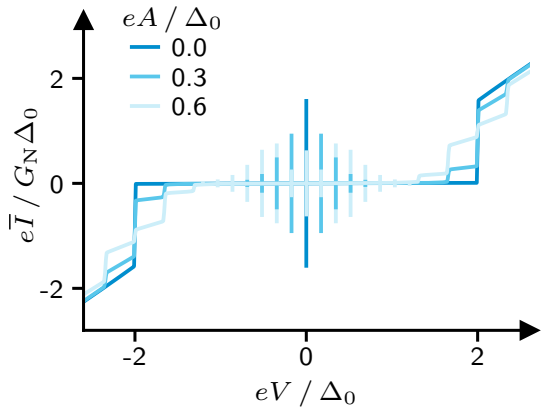


Figure 2.15. Under voltage bias, SIS curves show PAT. Vertical bars mark coherent phase locking.

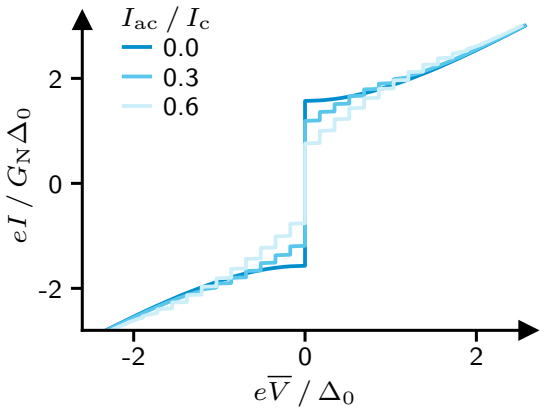


Figure 2.16. Overdamped current biased RCSJ response with $C = 0$ and linear R. The voltage $\overline{V}$ is time averaged.

The full interval follows by allowing the stationary phase offset to vary between the two extrema of the sinusoidal CPR. The integer locking voltages are independent of the critical current and damping. The ideal width instead depends on I_c and on the local microwave voltage amplitude [28].

The Josephson intervals and the PAT background in Fig. 2.15 arise from different physics. The unsquared Bessel function in ΔI_n reflects coherent phase evolution and Cooper pair transfer. Quasiparticle PAT is incoherent. Averaging over one drive period removes cross terms between photon channels and produces the squared Bessel weights in Eq. (2.32). The two contributions share only the prescribed junction voltage in this construction, with quasiparticles coupling through charge e and Cooper pairs through charge $2e$.

Under voltage bias, Eq. (2.32) gives the continuous time averaged quasiparticle background. At each V_n, the stationary Josephson contribution adds a current interval of width ΔI_n centered on $I_{\mathrm{PAT}}(V_n)$. Figure 2.15 visualizes this superposition.

Current Biased RCSJ Dynamics

Under current bias, the microwave drive enters the RCSJ current balance. Using the scales introduced in Sec. 2.2.2, we set $t' = \omega_p t$, $i = I_{dc}/I_c$, $a' = I_{ac}/I_c$, and $\nu' = \nu/\nu_p$. The driven RCSJ equation is

$$i + a' \sin(\nu' t') = \sin\phi + \frac{1}{Q}\frac{\mathrm{d}\phi}{\mathrm{d}t'} + \frac{\mathrm{d}^2\phi}{\mathrm{d}t'^2}\,, \tag{2.51}$$

The voltage is not prescribed. It follows from the phase velocity through the ac Josephson relation and is obtained by averaging the driven trajectory. The corresponding washboard potential from Eq. (2.47) is time dependent,

$$u(\phi, t') = -\cos\phi - \left(i + a' \sin(\nu' t')\right) \phi \,. \tag{2.52}$$

The ac current periodically raises and lowers the barriers of the tilted potential. In an underdamped junction, the capacitive term gives the phase particle sufficient inertia to continue moving after the instantaneous tilt has decreased. Locked trajectories can then compete with trapped and running states. Their stability depends on the initial condition and sweep history, and switching or retrapping can interrupt a locked branch. Underdamped Shapiro steps can therefore be less robust and protocol dependent, which makes them substantially more difficult to calculate as a unique stationary *I*–*V* characteristic.

We consequently use the strongly overdamped case for the illustrative current biased response. On the time scale of the driven motion, the capacitive contribution is then negligible compared with the damping. For Fig. 2.16, we set $C = 0$ in the dimensional RCSJ current balance of Eq. (2.43), while retaining the linear resistance. The phase therefore follows first order driven dynamics within the same RCSJ framework. Frequency locking occurs when it advances by $2\pi n$ during one microwave period, producing plateaus in the time averaged voltage at V_n. The figure shows the resulting characteristic for different drive amplitudes. It is a dynamical current biased solution, not a reinterpretation of the voltage biased spectrum above.

2.3 Mesoscopic Superconducting Transport

The microscopic section treated quasiparticle tunneling in the low-transmission limit, while the macroscopic section described phase dynamics of weakly coupled Josephson junctions. Atomic contacts require a mesoscopic description because their conductance is carried by a small number of phase-coherent channels with finite transmission. In this regime, the normal-state transmission eigenvalues, the mesoscopic pincode $\{\tau_i\}$, determine both the equilibrium spectrum and nonlinear subgap transport.

Finite transmission makes electron and hole degrees of freedom inseparable. Subgap current is no longer described by independent quasiparticle tunneling or by a lumped Josephson element, but by coherent Andreev reflection at superconducting interfaces. We first define the mesoscopic transport regimes and introduce atomic contacts. We then quantify a single NS interface with the Blonder-Tinkham-Klapwijk (BTK) model, extend the scattering picture to Andreev bound states in short SNS contacts, and finally treat finite-bias multiple Andreev reflection (MAR) and its charge resolved response.

2.3.1 Mesoscopic Regimes and Atomic Contacts

The transport description of a finite conductor is selected by comparing its length L with the elastic mean free path ℓ_{qp} and the phase-coherence length ℓ_ϕ. Elastic scattering randomizes momentum, while inelastic and dephasing processes limit phase coherence. A conductor is diffusive when $\ell_{\mathrm{qp}} \ll L$ and ballistic when $L \ll \ell_{\mathrm{qp}}$. In either case, transport remains phase coherent across the conductor when $L \lesssim \ell_\phi$. Figure 2.17 summarizes these regimes. [37, 55]

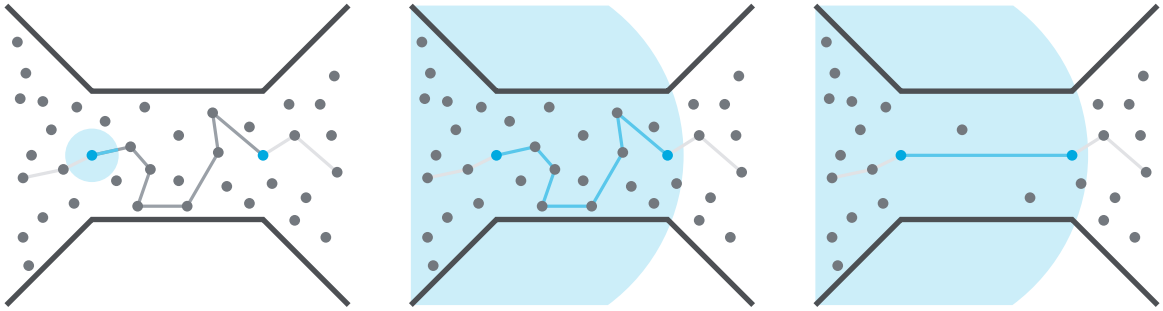
The atomic-scale constrictions considered in this thesis are much shorter than both ℓ_{qp} and ℓ_ϕ. They therefore lie in the ballistic and phase-coherent regime. Diffusive mesoscopic transport serves only as a reference, while the following discussion develops the transmission-channel description required for atomic contacts.

Ballistic Transport and Scattering Formalism

Ballistic transport is formulated as an elastic scattering problem. A finite scattering region connects ideal leads whose propagating modes carry the current. We denote the incoming amplitudes in the left and right leads by A_1 and A_2, and the outgoing amplitudes by B_1 and B_2, as shown in Fig. 2.18.

Linear scattering then gives

$$\begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = \underline{\underline{S}} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} . \tag{2.53}$$



(a) Incoherent diffusive (b) Phase-coherent diffusive (c) Ballistic

Figure 2.17. Sketch of the transport regime hierarchy selected by conductor length L, elastic mean free path ℓ_{qp}, and phase coherence length ℓ_ϕ. (a) Incoherent diffusive transport follows a local conductivity when $L \gg \ell_{\mathrm{qp}}, \ell_\phi$. (b) Elastic scattering remains frequent for $\ell_{\mathrm{qp}} \ll L \lesssim \ell_\phi$, but coherent path interference can modify the average diffusive response. (c) For $L \ll \ell_{\mathrm{qp}}$ and $L \lesssim \ell_\phi$, the constriction is ballistic and is described by transmission channels.

In the multi-mode case, $A_{1,2}$ and $B_{1,2}$ become vectors collecting all propagating modes in the corresponding lead. The scattering matrix decomposes into reflection blocks $\underline{\underline{r}}$ and $\underline{\underline{r}}'$ and transmission blocks $\underline{\underline{t}}$ and $\underline{\underline{t}}'$,

$$\underline{\underline{S}} = \begin{pmatrix} \underline{\underline{r}} & \underline{\underline{t}}' \\ \underline{\underline{t}} & \underline{\underline{r}}' \end{pmatrix} . \tag{2.54}$$

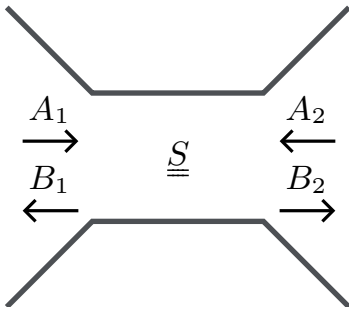


Figure 2.18. Two-terminal scatterer between ideal leads. The scattering matrix $\underline{\underline{S}}$ relates the incoming and outgoing mode amplitudes $\vec{A}$ and $\vec{B}$.

For purely elastic normal-state scattering, $\underline{\underline{S}}$ is unitary because probability current is conserved.

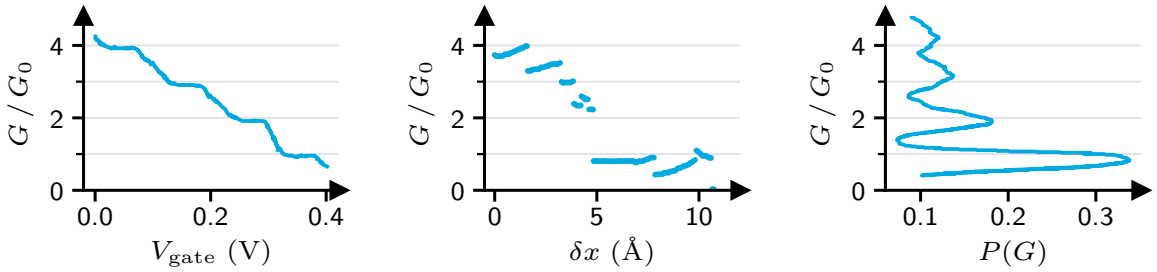
The Hermitian matrix $\underline{\underline{t}}^\dagger \underline{\underline{t}}$ can be diagonalized by a unitary transformation. Its eigenvectors define independent transmission eigenchannels, and its eigenvalues

$$\{\tau_i\} = \mathrm{eig}\left(\underline{\underline{t}}^\dagger \, \underline{\underline{t}}\right) , \quad \tau_i \in [0,1] , \tag{2.55}$$

give their transmission probabilities.

In linear response, the normal-state conductance is then given by the Landauer formula,

$$G_{\mathrm{N}} = G_0 \sum_{i=1}^{N} \tau_i , \quad G_0 = \frac{2e^2}{h} , \tag{2.56}$$



(a) QPC conductance, showing quantized plateaus. [58]

(b) Aluminum atomic contact opening trace. [16]

(c) Aluminum atomic contact conductance histogram. [60]

Figure 2.19. Normal-state conductance characterizes channel opening in a quantum point contact and the mechanical evolution of an atomic contact, but does not determine the individual transmission probabilities. (a) A split-gate quantum point contact shows quantized conductance plateaus as transverse modes open. (b) An aluminum opening trace records conductance changes as the atomic configuration evolves during elongation. (c) A conductance histogram identifies preferred conductance values over repeated breaking cycles.

where τ_i are the transmission eigenvalues and G_0 is the conductance quantum[3] [56, 57].

The normal-state conductance determines only the total transmission $\sum_i \tau_i$. Different channel sets can therefore produce the same G_N. The set $\{\tau_i\}$ is referred to throughout this thesis as the mesoscopic pincode. In a superconducting contact, different pincodes produce distinct multiple Andreev reflection characteristics, which allows the individual transmissions to be extracted from measured *I–V* and *dI–dV* curves. [16, 55]

Transmission Channels in Atomic Contacts

A split-gate quantum point contact (QPC) in a high-mobility two-dimensional electron gas provides a controllable model system for the channel description. Its ballistic constriction is tuned electrostatically, and the conductance increases in quantized steps as transverse modes open. Each fully open spin-degenerate mode defines a conduction channel and contributes one conductance quantum G_0. The resulting staircase $G(V_\mathrm{gate})$ is shown in Fig. 2.19(a) [58, 59].

In a semiconductor QPC, the channels can be opened electrostatically. In a metallic atomic contact, their number, orbital character, and transmission probabilities are instead determined by the atomic geometry and electronic structure of the complete scattering region.

[3] Throughout this thesis the conductance quantum is defined as $G_0 \equiv 2e^2/h = 77.48\,\mu\mathrm{S}$. This value includes spin degeneracy and corresponds to the conductance of a fully transmitting normal-state channel. The corresponding resistance quantum is $R_0 = h/(2e^2) = 12.9\,\mathrm{k\Omega}$.

Metallic atomic contacts are modeled as short, ballistic, and phase-coherent scatterers whose narrowest cross section contains one or a few atoms. During mechanical elongation, atomic rearrangements produce conductance changes such as those in the opening trace in Fig. 2.19(b). Repeating this process yields the conductance histogram in Fig. 2.19(c), whose peaks mark preferred conductance values. Opening traces and histograms characterize the mechanical evolution and statistical occurrence of atomic configurations, but they do not determine the individual transmission probabilities [16, 17, 60].

The eigenchannels are scattering states of the complete contact rather than isolated atomic orbitals. Their character nevertheless reflects the valence orbitals near the narrowest cross section and their hybridization with the electrodes. For aluminum, experiments and calculations commonly identify three dominant channels in a single-atom contact. These can be associated predominantly with one longitudinal sp_z-like contribution and two transverse p-like contributions. Their precise orbital composition and transmission probabilities depend on the contact geometry, and additional weakly transmitting channels may contribute [16, 61].

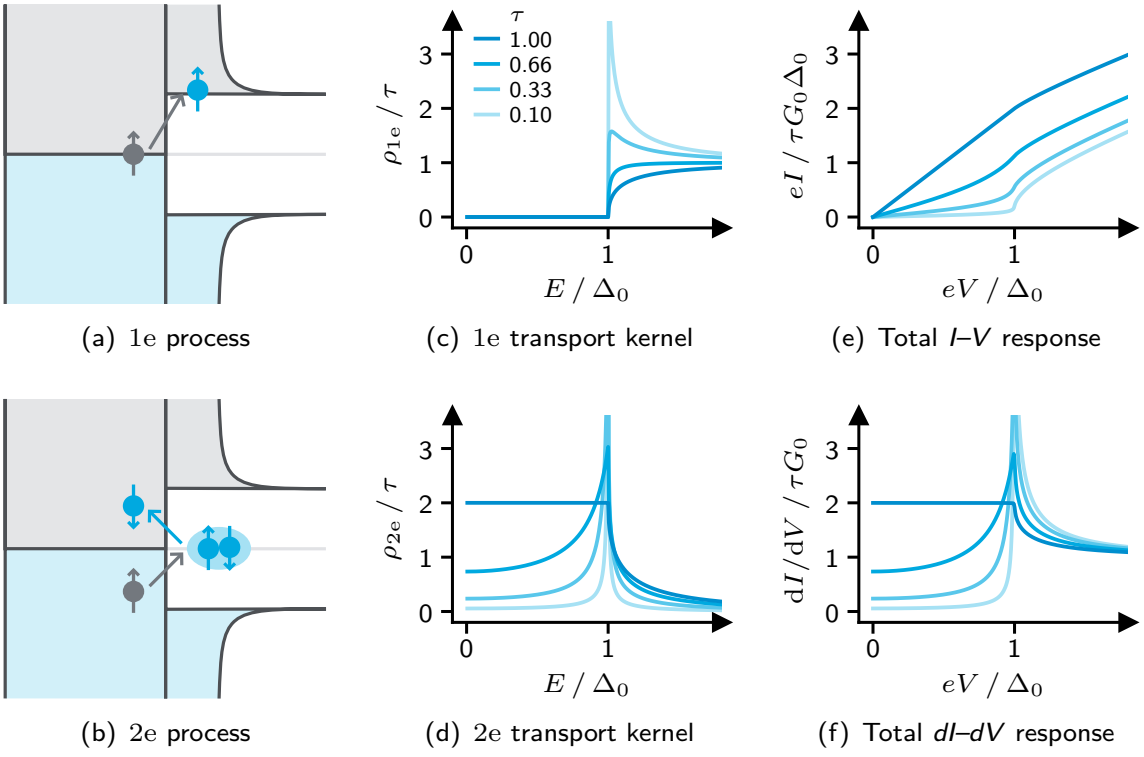
Normal-state measurements determine only the sum $G_{\mathrm{N}} = G_0 \sum_i \tau_i$. In the superconducting state, the nonlinear multiple Andreev reflection response depends separately on the individual τ_i, which allows the mesoscopic pincode to be extracted from measured *I–V* and *dI–dV* curves. The resulting transmission eigenvalues provide the input for the finite-transmission superconducting models developed below [16, 61].

In summary, atomic contacts lie in the ballistic and phase-coherent limit, where the transmission eigenvalues $\{\tau_i\}$ provide the natural transport variables. The normal-state conductance determines only the total transmission, while channel resolved superconducting spectroscopy supplies the additional information needed to determine the individual transmissions. We therefore turn to Andreev reflection, which makes the superconducting response of each channel depend nonlinearly on its transmission.

2.3.2 Andreev Reflection

At a normal–superconductor (NS) interface, no propagating quasiparticle states are available in the superconducting continuum for $|E| < \Delta$. An electron incident from the normal side therefore cannot enter the superconductor as a single quasiparticle at these energies. Instead, it can be reflected into the normal electrode as a hole while a Cooper pair with charge $2e$ enters the superconducting condensate. This coherent electron–hole conversion is Andreev reflection [62].

Figure 2.20(a) shows single-particle transmission into the superconducting quasiparticle continuum above the gap, while Fig. 2.20(b) shows subgap Andreev reflection. These panels are energy-space schematics rather than real-space trajectories. At a nonideal interface, normal reflection competes with Andreev



(a) 1e process (c) 1e transport kernel (e) Total *I*–*V* response

(b) 2e process (d) 2e transport kernel (f) Total *dI*–*dV* response

Figure 2.20. BTK energy schematics, transport kernels, and calculated characteristics for several normal-state transmissions τ. The single-particle and Andreev contributions determine the total *I*–*V* and *dI*–*dV* response. Increasing τ enhances the Andreev contribution, which increases the subgap conductance and reduces the quasiparticle peak.

reflection, and their relative probabilities depend on the normal-state transmission.

Andreev reflection converts quasiparticle current in the normal electrode into Cooper pair current in the superconductor. In a junction with two superconducting electrodes, repeated coherent conversion produces discrete Andreev bound states (ABS) in equilibrium and multiple Andreev reflection (MAR) under finite bias. In atomic contacts, both processes are resolved by channels with transmission probabilities τ_i. The Blonder-Tinkham-Klapwijk (BTK) model introduced next quantifies normal reflection, quasiparticle transmission, and Andreev reflection at a single NS interface [63, 64].

Blonder-Tinkham-Klapwijk Model

The Blonder-Tinkham-Klapwijk (BTK) model provides a quantitative scattering description of an NS interface with normal-state transmission probability τ. An electron incident from the normal electrode can undergo Andreev reflection with probability $A(E)$, normal reflection with probability $B(E)$, or quasiparticle

transmission into the superconductor with probability $C(E)$. Conservation of probability current requires

$$A(E) + B(E) + C(E) = 1\,. \tag{2.57}$$

This relation follows from probability-current conservation in the BTK scattering problem [63].

The BTK scattering amplitudes are expressed in terms of the electron and hole components of the Bogoliubov quasiparticles introduced in Sec. 2.1.1. Solving the homogeneous BdG eigenproblem in Eq. (2.8) gives the BCS coherence factors below. To incorporate finite quasiparticle lifetimes phenomenologically, we follow Plecenník *et al.* and evaluate them with the replacement $E \to |E| + \mathring{\imath}\gamma$ [44],

$$\begin{aligned} \xi(E) &= \sqrt{E^2 - \Delta^2}\,, \\ u_0(E) &= \sqrt{\tfrac{1}{2}\left(1 + \xi(E)/E\right)}\,, \\ v_0(E) &= \sqrt{\tfrac{1}{2}\left(1 - \xi(E)/E\right)}\,. \end{aligned} \tag{2.58}$$

Using the normalization factor d, the Andreev and normal reflection amplitudes a and b are

$$\begin{aligned} &A(E) = aa^*\,, \quad a = u_0 v_0/d\,, \quad d = \left(u_0^2 - (1-\tau)v_0^2\right)\big/\,\tau\,, \\ &B(E) = bb^*\,, \quad b = -\left(u_0^2 - v_0^2\right)\left((1-\tau) + \mathring{\imath}\sqrt{\tau(1-\tau)}\right)\Big/\,\tau d\,, \end{aligned} \tag{2.59}$$

Here u_0 and v_0 are evaluated at E. The quasiparticle transmission probability then follows from Eq. (2.57) as $C(E) = 1 - A(E) - B(E)$ [44, 63].

The NS current can be decomposed into single-particle and Andreev contributions through the energy resolved BTK transport kernels

$$\begin{aligned} \rho_{1\mathrm{e}}(E) &= 1 - A(E) - B(E) = C(E)\,, \\ \rho_{2\mathrm{e}}(E) &= 2\,A(E)\,, \end{aligned} \tag{2.60}$$

The first kernel describes quasiparticle transmission. The factor of two in the second kernel accounts for the charge $2e$ transferred by one Andreev reflection event. For ideal subgap transport at $\gamma = 0$, $C(E) = 0$ and the single-particle kernel vanishes. Finite lifetime broadening can introduce subgap quasiparticle weight. Although the kernels resemble densities of states, they enter the current integral and should not be interpreted as densities of states.

The current through an NS interface then follows from these kernels,

$$\begin{aligned} I_{1e}(V) &= \frac{G_0}{e}\int_{-\infty}^{\infty} \rho_{1e}(E)\,(f(E) - f(E+eV))\,\mathrm{d}E\,, \\ I_{2e}(V) &= \frac{G_0}{e}\int_{-\infty}^{\infty} \rho_{2e}(E)\,(f(E) - f(E+eV))\,\mathrm{d}E\,, \\ I_{\mathrm{NS}}(V) &= I_{1e}(V) + I_{2e}(V)\,. \end{aligned} \tag{2.61}$$

In the tunneling limit, $\tau \ll 1$, Eq. (2.61) reduces to the conventional quasiparticle tunneling expression. As τ approaches unity, Andreev reflection dominates the subgap response and charge is transferred predominantly in units of $2e$ [63].

Figure 2.20 summarizes this evolution across the transmission range. Panels (c) and (d) show the transport kernels $\rho_{1e}(E)$ and $\rho_{2e}(E)$ from Eq. (2.60). At low transmission, the single-particle contribution dominates and produces a quasiparticle peak near $|E| = \Delta$. Increasing τ transfers spectral weight to the Andreev contribution, which enhances the subgap conductance in *I–V* and *dI–dV*.

The BTK model describes scattering at one NS interface. When a normal region connects two superconducting electrodes, successive Andreev reflections at the two interfaces form coherent closed electron–hole trajectories. Their phase dependent quantization produces Andreev bound states whose spectra are set by the channel transmission probabilities τ_i.

2.3.3 Andreev Bound States

Andreev reflection becomes phase-coherent and spectrally quantized when two superconductors are connected by a mesoscopic weak link. In an SNS geometry, a subgap quasiparticle ($|E| < \Delta$) cannot escape into the superconducting continua and instead undergoes successive Andreev reflections at both interfaces, alternating between electron- and hole-like character as shown in Fig. 2.21. One cycle acquires the phase difference

$$\phi = \phi_1 - \phi_2\,. \tag{2.62}$$

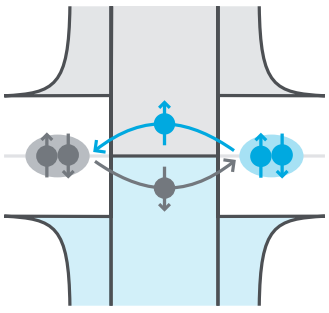


Figure 2.21. Andreev bound-state formation in an SNS weak link.

The resulting constructive-interference condition quantizes the motion into discrete subgap eigenstates localized around the junction, known as Andreev bound states (ABS) [62, 64, 65].

These ABS form the microscopic origin of the dc Josephson effect in short mesoscopic junctions. Their phase dependent energies $E_n(\phi)$ determine the equilibrium supercurrent through their occupation [64, 66].

Andreev Approximation and Short-Junction Limit

A transparent route from BdG to ABS is to treat Andreev reflection as an energy dependent boundary condition at the superconducting leads, while representing the weak link by its normal-state scattering properties.

In the Andreev approximation, $|E|, \Delta \ll E_{\mathrm{F}}$, only quasiparticles in a narrow shell around the Fermi surface contribute. For an isotropic parabolic band, the dispersion can be linearized in the radial direction,

$$\varepsilon(\vec{k}) - \mu \approx \frac{\hbar^2}{m}\, \vec{k}_{\mathrm{F}} \cdot \left(\vec{k} - \vec{k}_{\mathrm{F}}\right) , \tag{2.63}$$

so electron-like and hole-like components have nearly equal momentum magnitudes. For an excitation at energy E, the corresponding wave numbers satisfy

$$\delta k = \frac{k_{\mathrm{e}} - k_{\mathrm{h}}}{2} \approx \frac{k_{\mathrm{F}}}{2}\, \frac{E}{E_{\mathrm{F}}} , \tag{2.64}$$

implying that the Andreev-reflected hole retraces the incoming trajectory up to corrections of order E/E_{F} [41, 62].

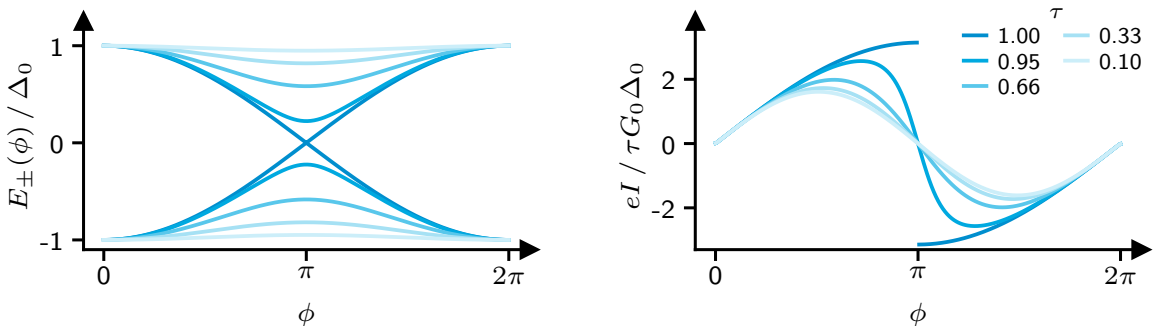
In the Andreev limit of a clean, specular, and highly transparent NS interface with negligible normal reflection, subgap Andreev reflection occurs with approximately unit probability. The corresponding electron-to-hole reflection amplitude can then be written as a pure phase factor,

$$r_{\mathrm{A}}(E) = e^{-\mathring{\imath}\,\phi_{\mathrm{A}}(E)}\, e^{-\mathring{\imath}\,\phi} , \qquad \phi_{\mathrm{A}}(E) \equiv \arccos(E/\Delta) . \tag{2.65}$$

Here $\phi_{\mathrm{A}}(E)$ is the energy dependent Andreev phase, while ϕ denotes the condensate phase of the electrode. For nonideal interfaces, finite backscattering reintroduces normal reflection and reduces the Andreev amplitude below unity [62, 64].

The second assumption is the short-junction limit, $L \ll \xi$, where ξ is the relevant superconducting length in the electrodes. The aluminum films can be diffusive even though the atomic constriction is a short coherent scatterer. Representative clean and dirty aluminum length estimates are collected in Appendix B. These superconducting lengths remain much larger than the atomic scattering region [16, 67].

In this limit, the propagation phase acquired across the weak link varies negligibly over the superconducting gap. Its normal-state scattering matrix can therefore be treated as energy independent in the relevant energy window. Each channel is characterized by one transmission eigenvalue τ_i, and its ABS spectrum



(a) ABS Energy Spectra

(b) ABS Current–Phase Relation

Figure 2.22. Andreev bound-state spectrum and equilibrium current–phase relation of a short single-channel contact for several transmissions τ. (a) Increasing τ strengthens the phase dispersion and reduces the minimum positive ABS energy at $\phi = \pi$, approaching a crossing for perfect transmission. (b) The corresponding current–phase relation evolves from an approximately sinusoidal tunnel response to a strongly skewed ballistic response with a cusp at $\phi = \pi$.

depends only on τ_i, Δ, and the phase difference ϕ. These assumptions lead to the universal short-junction spectrum introduced next.

Energy Spectrum and Current–Phase Relation

A complete Andreev cycle through the junction converts an electron into a hole at one interface and back into an electron at the other. A bound state forms when this closed electron–hole trajectory reproduces itself, i.e. when the total phase accumulated in one round trip is an integer multiple of 2π.

For a single channel of transmission τ, the constructive-interference condition reduces to the compact form

$$\sin^2\phi_{\mathrm{A}}(E) = \tau\,\sin^2(\phi/2)\;, \tag{2.66}$$

which directly yields the familiar Andreev bound-state spectrum

$$E_\pm(\phi) = \pm\Delta\sqrt{1-\tau\sin^2(\phi/2)}\,. \tag{2.67}$$

This universal spectrum applies to a short single-channel junction with energy-independent normal scattering [64, 66]. Figure 2.22(a) shows these spectra for different transmissions. In the tunneling regime, the levels remain close to the gap edges and depend weakly on phase. Increasing τ strengthens the phase dispersion and lowers the positive-energy branch near $\phi = \pi$.

The phase dependence of the ABS spectrum implies that these states carry a nondissipative supercurrent. In equilibrium, the current follows from the derivative of the junction free energy with respect to phase.

Equivalently, one may express the contribution of each ABS branch through its occupation, which yields the standard spin-degenerate relation

$$I(\phi) = -2\pi \frac{G_0}{e} \frac{\partial E(\phi)}{\partial \phi} \tanh\left(\frac{E(\phi)}{2k_{\mathrm{B}}T}\right) . \tag{2.68}$$

Here $E(\phi) > 0$ denotes the positive-energy ABS spectrum, and the hyperbolic tangent accounts for thermal occupation [64, 66].

Inserting Eq. (2.67) yields

$$I(\phi) = \frac{\pi}{2} \frac{G_0}{e} \Delta \frac{\tau \sin\phi}{\sqrt{1 - \tau \sin^2(\phi/2)}} \tanh\left(\frac{\Delta\sqrt{1 - \tau \sin^2(\phi/2)}}{2k_{\mathrm{B}}T}\right) . \tag{2.69}$$

In the zero-temperature limit, $\tanh(E/2k_{\mathrm{B}}T) \to 1$ and the current is set solely by the phase dispersion of the occupied negative-energy branch [66]. Figure 2.22(b) shows that increasing τ drives the current–phase relation away from a sinusoidal form and shifts its maximum towards $\phi = \pi$.

In the tunneling limit ($\tau \ll 1$), the current–phase relation becomes sinusoidal and reduces to the standard Josephson form discussed in Sec. 2.2. In the same limit, the critical current is consistent with the Ambegaokar-Baratoff result when expressed in terms of the normal-state resistance.

In the opposite ballistic limit ($\tau \to 1$), the bound-state spectrum becomes particularly simple,

$$E_{\pm}(\phi) = \pm\Delta \, |\cos(\phi/2)| . \tag{2.70}$$

The corresponding current at $T = 0$ follows from Eq. (2.69),

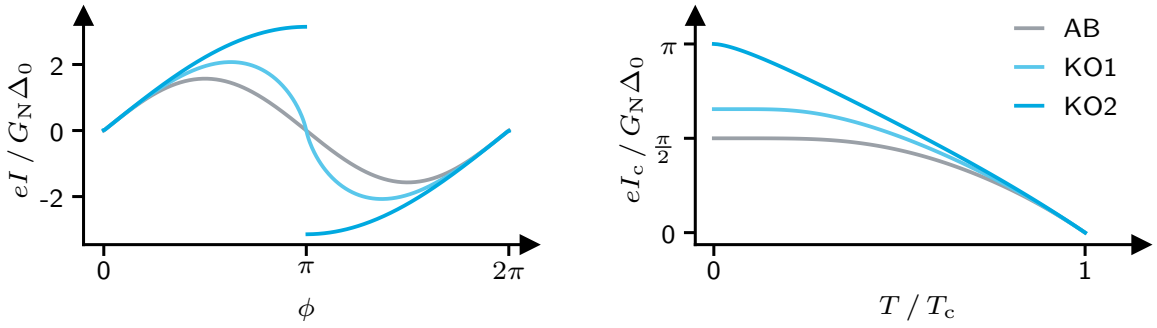
$$I(\phi) = \tfrac{\pi}{2} \tfrac{G_0}{e} \Delta \, \sin(\phi/2) \, \mathrm{sgn}(\cos(\phi/2)) . \tag{2.71}$$

At perfect transmission, the two equilibrium energies in Eq. (2.70) meet in cusps at $\phi = \pi$. The underlying parity resolved dispersions, proportional to $\pm\Delta\cos(\phi/2)$, cross at this phase. Any finite normal reflection opens a gap between these dispersions. The minimum positive ABS energy is

$$E_{\min} \equiv E_{+}(\pi) = \Delta\sqrt{1 - \tau} . \tag{2.72}$$

Deviations from perfect transmission therefore remove the zero-energy crossing, while finite temperature rounds the equilibrium current through the occupation factor.

In summary, the transmission τ controls the phase dispersion, minimum energy, and equilibrium current–phase relation of a short single-channel contact. For a multichannel contact, the pincode $\{\tau_i\}$ determines the complete set of ABS and their combined equilibrium supercurrent.



(a) Current–Phase Relation $I(\phi)$ (b) Critical Current $I_{\mathrm{c}}(T)$

Figure 2.23. Equilibrium current–phase relation and critical current of three canonical short-junction reference limits, Ambegaokar-Baratoff (AB) and Kulik-Omel'yanchuk (KO1 & KO2). (a) The AB limit corresponds to a sinusoidal CPR, whereas KO1 for a diffusive contact and KO2 for a clean ballistic contact are progressively more skewed towards $\phi = \pi$. (b) The corresponding critical currents follow from maximizing each current–phase relation with respect to ϕ using a BCS-like gap $\Delta(T)$.

Multi-Channel Contacts

Real atomic contacts generally support a small number of discrete conduction channels. In the short-junction limit, each channel i is characterized by a transmission eigenvalue τ_i and supports a pair of Andreev branches $E_{i,\pm}(\tau_i, \phi)$ given by Eq. (2.67). The equilibrium supercurrent follows by summing the single-channel contribution from Eq. (2.69) over all channels,

$$I(\phi, T) = \sum_i I(\tau_i, \phi, T)\,. \tag{2.73}$$

For a conductor with many modes, the discrete channel sum may instead be represented by a continuous density of transmission eigenvalues $\rho(\tau)$,

$$I(\phi, T) \simeq \int_0^1 \rho(\tau)\, I(\tau, \phi, T)\, \mathrm{d}\tau\,. \tag{2.74}$$

Here $\rho(\tau)\,\mathrm{d}\tau$ gives the number of channels with transmissions in the interval $[\tau, \tau + \mathrm{d}\tau]$. It is therefore a channel density rather than a normalized probability distribution. Its conductance normalization is $G_{\mathrm{N}} = G_0 \int_0^1 \rho(\tau)\,\tau\,\mathrm{d}\tau$ [64, 66].

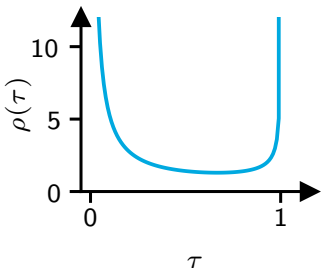


Figure 2.24. DMPK transmission density for a short diffusive wire.

In the metallic diffusive limit, the transmission eigenvalues of a quasi-one-dimensional wire follow the bimodal Dorokhov-Mello-Pereyra-Kumar (DMPK) distribution [55, 68],

$$\rho(\tau) = \frac{G_{\mathrm{N}}}{G_0} \frac{1}{2\tau\sqrt{1-\tau}}\,, \quad 0 < \tau < 1\,, \tag{2.75}$$

as shown in Fig. 2.24. Its bimodal form represents a large population of almost-closed channels together with almost-open channels that contribute strongly to transport.

Equation (2.75) gives the first Kulik-Omel'yanchuk (KO1) relation for a dirty diffusive contact. At $T = 0$, the current is

$$I_{\mathrm{KO1}}(\phi) = \pi \frac{G_{\mathrm{N}}}{e} \Delta \, \cos(\phi/2) \, \operatorname{arctanh}(\sin(\phi/2)) \,. \tag{2.76}$$

Although $\operatorname{arctanh}[\sin(\phi/2)]$ diverges as $\phi \to \pi$, the factor $\cos(\phi/2)$ vanishes more rapidly, such that $I_{\mathrm{KO1}}(\phi) \to 0$ [69, 70].

The KO1 limit applies when the weak link itself is diffusive. It does not follow merely from diffusive transport in the superconducting electrodes.

In the complementary clean ballistic limit, all modes are perfectly transmitting, $\tau_i = 1$. Summing the ballistic single-channel result gives the second Kulik-Omel'yanchuk relation (KO2),

$$I_{\mathrm{KO2}}(\phi, T) = \pi \, \frac{G_{\mathrm{N}}}{e} \, \Delta(T) \, \sin\!\left(\frac{\phi}{2}\right) \, \tanh\!\left(\frac{\Delta(T)\cos(\phi/2)}{2k_{\mathrm{B}}T}\right) . \tag{2.77}$$

This clean ballistic result is the KO2 reference limit [69, 70]. Channels with finite normal reflection, $\tau_i < 1$, are instead described by the general sum in Eq. (2.73). KO1 and KO2 therefore provide many-channel reference limits rather than generic models of an individual atomic contact, whose discrete channel set $\{\tau_i\}$ should normally be retained.

Figure 2.23 compares the current–phase relations and critical currents for the Ambegaokar-Baratoff, KO1, and KO2 limits. In each case, $I_{\mathrm{c}}(T)$ is found by maximizing $I(\phi, T)$ over phase with a BCS-like gap $\Delta(T)$. The channel set or its continuum density determines the observable current–phase relation and critical current, connecting microscopic transmission eigenvalues to equilibrium measurements.

Fractional Shapiro Steps

The transmission dependent ABS spectrum also determines the harmonic content of the equilibrium current–phase relation:

$$I(\phi) = \sum_{p=1}^{\infty} I_p \sin(p\phi) \,. \tag{2.78}$$

In the tunnel limit, the first harmonic I_1 dominates and the current is approximately sinusoidal. Highly transmitting channels produce the skewed current–phase relations shown in Figs. 2.22(b) and 2.23(a), which contain appreciable higher harmonics $I_{p>1}$.

Under microwave irradiation, these harmonics introduce phase locking conditions beyond the integer Shapiro steps discussed in Sec. 2.2.3. We use the driven phase of Eq. (2.48), with Josephson frequency $\nu_0 = 2eV/h$ and drive amplitude $\alpha_{\nu,2e} = 2eA/(h\nu)$. Inserting this phase into Eq. (2.78) and applying the Jacobi-Anger expansion gives

$$I(t) = \sum_{p\geq 1} \sum_{n\in\mathbb{Z}} I_p J_n(p\alpha_{\nu,2e}) \sin[p\phi_0 + 2\pi\,(p\nu_0 + n\nu)\,t] \,. \tag{2.79}$$

The p-th CPR harmonic experiences the enlarged modulation $p\alpha_{\nu,2e}$, and its n-photon locking contribution is weighted by $I_p J_n(p\alpha_{\nu,2e})$ rather than by a squared Bessel function. A dc contribution appears when $p\nu_0 + n\nu = 0$. Relabeling the sideband index gives the rational locking voltages

$$V_{n/p} = \frac{n}{p}\frac{h\nu}{2e} \,, \qquad n \in \mathbb{Z}, \quad p \in \mathbb{N}^+ \,. \tag{2.80}$$

An appreciable second harmonic produces half-integer steps for odd n. Their visibility is expected to increase when highly transmitting channels contribute strongly to the supercurrent [29, 31, 52].

In an ideal overdamped junction whose response at $V_{n/p}$ is dominated by one CPR harmonic, the full locking interval in dc current is approximately

$$\Delta I_{n/p} \simeq 2\,|I_p J_n(p\alpha_{\nu,2e})| \,. \tag{2.81}$$

The factor of two is the full interval obtained by varying the phase offset ϕ_0. For $p = 1$, Eq. (2.81) reduces to the familiar integer-step modulation $\Delta I_n \simeq 2|I_1 J_n(\alpha_{\nu,2e})|$. The oscillations and zeros of $J_n(p\alpha_{\nu,2e})$ predict suppression and revival of a fractional plateau as the microwave amplitude is varied [52].

Equation (2.81) is a single-dominant-harmonic estimate, not a general addition rule. Several pairs (p, n) can correspond to the same reduced rational voltage, and their phase dependent contributions can interfere. Realistic plateau widths require the driven RSJ or RCSJ phase dynamics introduced in Sec. 2.2.2 and also depend on damping, capacitance, temperature, noise, calibration of A, and

nonequilibrium ABS occupation. Conventional higher harmonics of a 2π-periodic current–phase relation already produce fractional steps. Their observation alone therefore does not establish a 4π-periodic Josephson effect.

In summary, repeated Andreev reflection quantizes the subgap spectrum of a short junction into transmission dependent ABS. Their phase dispersion and minimum gap determine the equilibrium current–phase relation, while the channel set fixes the total supercurrent. Highly transmitting channels generate higher CPR harmonics, which become observable through fractional Shapiro steps under microwave irradiation. The ABS description therefore connects microscopic channel transmissions to equilibrium supercurrent and measurements of phase locking.

A finite dc voltage changes this regime by driving the superconducting phase continuously in time. Repeated Andreev reflection then produces nonequilibrium charge transport rather than stationary bound-state supercurrent, leading to multiple Andreev reflection.

2.3.4 Multiple Andreev Reflection

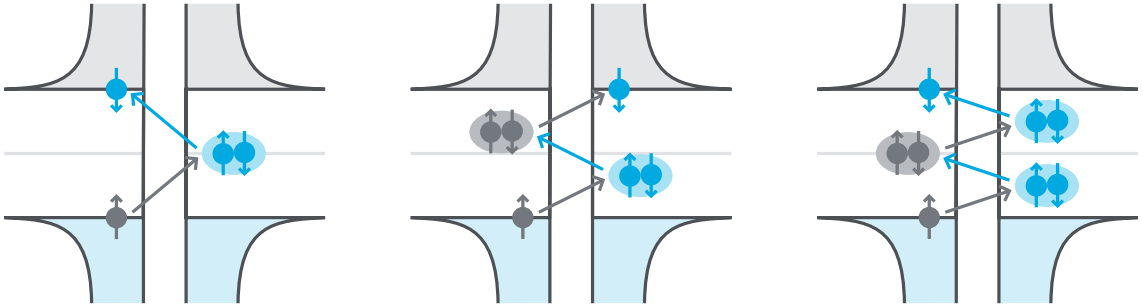
Multiple Andreev reflection (MAR) is the microscopic mechanism of subgap transport in voltage biased SNS weak links and short superconducting point contacts. Figure 2.25 shows the corresponding energy ladder for several MAR orders. For $|eV| < 2\Delta$, a quasiparticle cannot directly enter the continuum of the opposite electrode. Successive Andreev reflections instead alternate its electron- and hole-like character, while each traversal changes its energy by eV. In the trajectory picture of Klapwijk, Blonder, and Tinkham (KBT), a process that reaches the opposite continuum after m energy increments has the threshold

$$eV_m = \frac{2\Delta}{m}\,, \qquad m \in \mathbb{N}^+\,. \tag{2.82}$$

We refer to these voltages as MAR thresholds. Here m labels the MAR order. In the energy-ladder picture, it counts the energy increments required to connect the continua. In the charge resolved description, the corresponding process transfers the effective charge $q_m = me$. The MAR thresholds generate the subharmonic-gap structure observed as slope changes in *I–V* and peaks or kinks in *dI–dV* [71, 72].

The semiclassical OTBK model provides an important historical realization of this ladder picture and explains the origin of the subharmonic gap structure [72]. It neglects phase coherence and therefore does not provide the quantitative description required for short atomic contacts. For these contacts, we use the channel resolved Hamiltonian approach of Cuevas *et al.*, which treats coherent transport across the full transmission range.

At high transmission, the dynamics can be viewed through the ABS spectrum. A dc bias drives the phase through the avoided crossing near $\phi = \pi$, where



(a) Second-order MAR (b) Third-order MAR (c) Fourth-order MAR

Figure 2.25. Energy-ladder picture of multiple Andreev reflection for processes of order $m = 2$, 3, and 4. Successive electron–hole conversions at the two superconducting interfaces raise the quasiparticle energy by eV per traversal until it reaches the continuum. Lower voltage therefore requires a larger number of traversals to bridge the gap 2Δ.

nonadiabatic Landau-Zener transitions can change the ABS branch occupation. This two-level picture is useful for a nearly ballistic channel, while the coherent transport model below treats the complete MAR current at arbitrary transmission [20].

Coherent Channel-Resolved Transport

We now replace the trajectory picture by the coherent single-channel model used for quantitative dc MAR. A short single-channel contact is modeled as two bulk BCS reservoirs coupled by a local hopping amplitude [21]. This is the finite-voltage continuation of the ABS model: at $V = 0$ it produces the equilibrium spectrum discussed in Sec. 2.3.3, while a running phase generates coherent MAR.

The junction Hamiltonian is written as

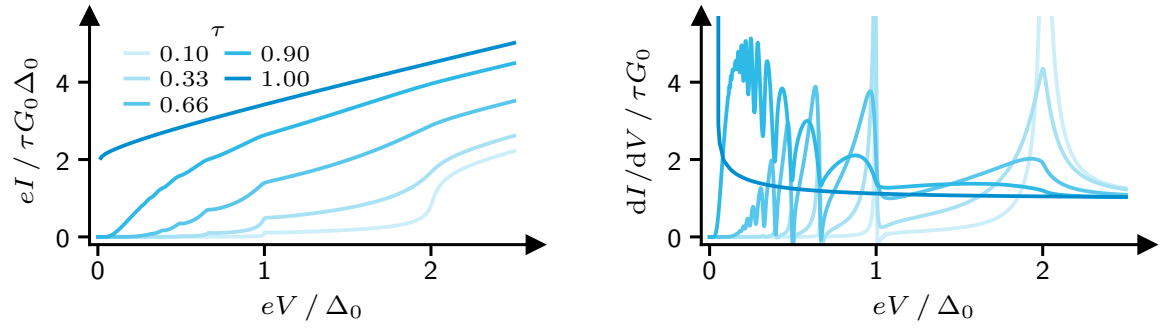
$$\hat{H} = \hat{H}_L + \hat{H}_R + \hat{H}_T(t)\,, \tag{2.83}$$

$$\hat{H}_T(t) = \sum_{\sigma} \left(t\, e^{+\mathring{\imath}\phi(t)/2}\, \hat{c}^{\dagger}_{L\sigma} \hat{c}_{R\sigma} + t\, e^{-\mathring{\imath}\phi(t)/2}\, \hat{c}^{\dagger}_{R\sigma} \hat{c}_{L\sigma} \right) . \tag{2.84}$$

This local-hopping Hamiltonian is the starting point of the coherent channel resolved MAR calculation [21].

The phase factors in $\hat{H}_T(t)$ incorporate the dc voltage through the $q = 2e$ phase of Eq. (2.27). The resulting time-periodic problem is solved with Keldysh Green functions in Nambu space and a Floquet representation in energy. For one channel, the calculation returns $I(V, \tau, T, \Delta, \gamma)$ from the transmission τ, gap Δ, temperature T, broadening γ, and voltage V.

Figure 2.26(a) shows that increasing τ enhances the subgap current because successive traversals are less suppressed. The MAR thresholds remain fixed by



(a) MAR *I–V* characteristic (b) MAR *dI–dV* characteristic

Figure 2.26. Coherent MAR transport through a short single-channel contact at selected transmissions τ. (a) The subgap current increases strongly with τ, and a finite excess current remains above the gap. (b) The corresponding differential conductance displays subharmonic-gap features at the MAR thresholds of Eq. (2.82), which broaden and overlap as the transmission increases.

Eq. (2.82), but the structures overlap and the current approaches a smooth high-transmission response. Figure 2.26(b) shows the same evolution in *dI–dV*, where thresholds appear as peaks or kinks.

At voltages far above the gap, the superconducting current approaches the normal-state asymptote with a finite offset. This offset defines the excess current,

$$I_{\mathrm{exc}}(\tau) = \lim_{V\to\infty} \left[I(V,\tau) - \tau G_0 V \right] . \quad (2.85)$$

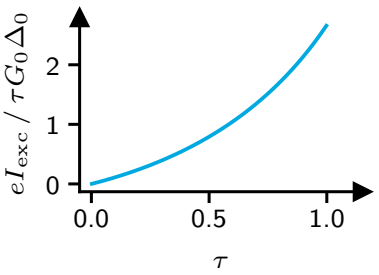


Figure 2.27. Normalized excess current versus transmission.

It results from Andreev processes and therefore depends nonlinearly on the channel transmission. For a short single-channel contact between identical BCS superconductors at zero temperature, the excess current has the analytical form

$$I_{\mathrm{exc}}(\tau) = \frac{G_0\Delta_0}{e}\,\frac{\tau^2}{1-\tau}\left[1 - \frac{\tau^2}{2(2-\tau)\sqrt{1-\tau}}\ln\left(\frac{1+\sqrt{1-\tau}}{1-\sqrt{1-\tau}}\right)\right] . \quad (2.86)$$

Figure 2.27 shows that the normalized excess current approaches $8/3$ in the ballistic limit $\tau \to 1$ [73].

Multichannel Atomic Contacts

The single-channel result becomes experimentally useful when it is combined with the mesoscopic pincode $\{\tau_i\}$ of an atomic contact. Independent channels contribute additively to the coherent dc current,

$$I(V, \{\tau_i\}, T, \Delta, \gamma) = \sum_i I(V, \tau_i, T, \Delta, \gamma)\,. \tag{2.87}$$

Each transmission produces a characteristic *I–V* and *dI–dV* line shape because MAR probabilities depend differently on τ_i. Their sum therefore remains sensitive to the mesoscopic pincode. This transmission sensitivity makes nonlinear SIS spectra a channel resolved probe of atomic contacts [16, 21].

The normal-state conductance determines only the total transmission, so different mesoscopic pincodes can have the same G_N. Their nonlinear SIS spectra can nevertheless differ and can be fitted to extract the pincode within the coherent short-contact model. Channels with the same transmission produce identical spectral contributions and cannot be distinguished individually. The fitting procedure is described in Sec. 3.3.2.

Charge-Resolved Multiple Andreev Reflection

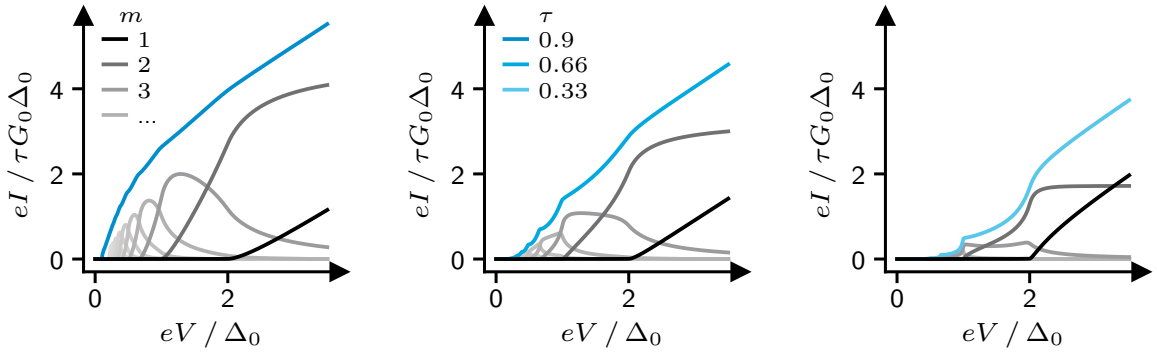
To connect dc MAR to charge transfer, we use full counting statistics as a process resolved extension of the coherent MAR calculation [74 - 76]. Here we use it only to decompose the dc current into contributions associated with transfer of an effective charge $q = me$,

$$I(V) = \sum_{m=1}^{\infty} I_m(V)\,. \tag{2.88}$$

The index m is the MAR order introduced in Eq. (2.82), and $I_m(V)$ denotes the contribution associated with transfer of the effective charge $q_m = me$. Several MAR orders can contribute at the same bias, so m characterizes an individual transport process rather than the total current [76].

Figure 2.28 shows this redistribution for three transmissions. Near the gap edge, low-order processes carry most of the current. Lower bias requires larger transfer orders, although these processes are strongly suppressed in the tunneling regime. As τ increases, repeated traversal becomes more probable and several charge-transfer orders acquire appreciable weight.

Figure 2.29 shows the same charge resolved quantities in a form that emphasizes the MAR thresholds. Panels (a) and (c) display $I_m(V)$ and the corresponding differential conductance on the usual voltage axis, while panels (b) and (d) replot them against inverse voltage. On this axis, the threshold positions of Eq. (2.82) fall at integer values, making the alignment between effective charge order and subharmonic-gap structure visible.



(a) $\tau = 0.90$ (b) $\tau = 0.66$ (c) $\tau = 0.33$

Figure 2.28. Charge resolved MAR current for three single-channel transmissions. Each panel decomposes the total current into contributions $I_m(V)$ associated with effective charge transfer $q = me$. Lower bias requires access to larger transfer orders, while increasing τ makes repeated traversal more probable and allows several orders to carry appreciable current.

The resolved functions $I_m(V)$ provide a possible bridge from dc MAR to photon assisted MAR because each component is associated with the effective charge $q = me$. However, dc full counting statistics does not establish that these components can be dressed independently under microwave irradiation.

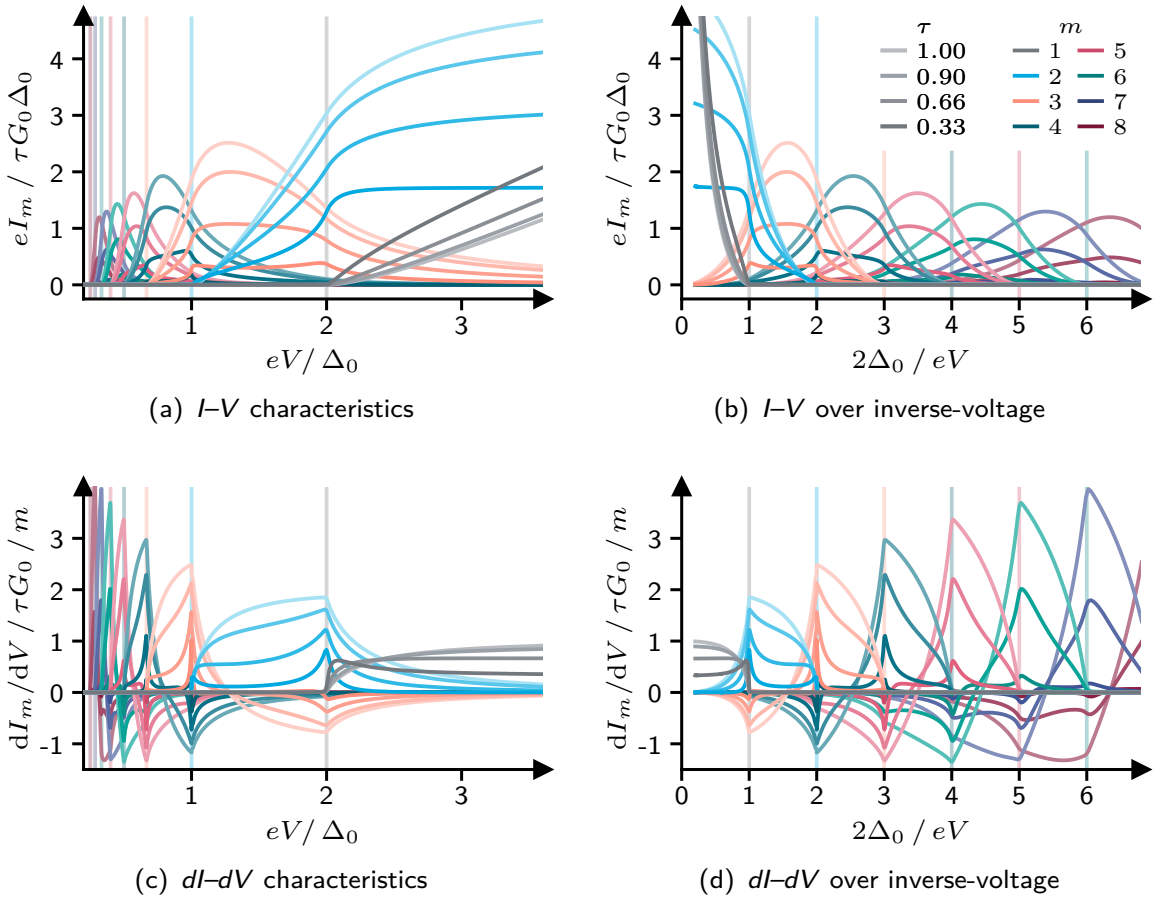
Photon Assisted Multiple Andreev Reflection

Irradiated MAR is more complex than dc MAR because a voltage biased superconducting contact already contains the Josephson frequency $\nu_0 = 2eV/h$. A microwave drive at frequency ν adds a second time scale. A useful phenomenological description is the unified charge dependent Tien-Gordon expression. Soerensen *et al.* introduced this viewpoint for microwave modified subgap structure, and Carrad *et al.* used the charge dependent sideband spacing to identify high-order MAR processes [34, 77]. If the dc current is decomposed as $I(V) = \sum_m I_m(V)$ and each component is assigned the effective charge $q_m = me$, the expression reads

$$I_{\mathrm{PAMAR}}(V, A) = \sum_{m\in\mathbb{N}^+} \sum_{n\in\mathbb{Z}} J_n^2\left(\frac{meA}{h\nu}\right) I_m\left(V + \frac{nh\nu}{me}\right), \tag{2.89}$$

where A is the effective microwave voltage amplitude at the junction and n labels the photon sideband. The corresponding replicas of the MAR thresholds appear approximately at

$$eV_{m,n} \simeq \frac{2\Delta + nh\nu}{m}, \qquad n \in \mathbb{Z}. \tag{2.90}$$



(a) *I–V* characteristics

(b) *I–V* over inverse-voltage

(c) *dI–dV* characteristics

(d) *dI–dV* over inverse-voltage

Figure 2.29. Charge resolved MAR current and differential conductance for selected single-channel transmissions. (a,c) Voltage representation. (b,d) Inverse-voltage representation, where the thresholds $eV_m = 2\Delta/m$ align at integer positions. Curve color identifies the MAR order m, and color intensity identifies the transmission τ.

Photon assisted MAR can identify the effective charge order associated with a subgap feature, complementing the dc charge resolution in Fig. 2.29 [34, 77].

Equation (2.89) does not determine how the measured or calculated dc current should be separated into the components $I_m(V)$. The FCS currents of Eq. (2.88) are a natural candidate because they resolve transfer events by charge. Their independent Tien-Gordon dressing is nevertheless an additional assumption, not a consequence of the dc FCS theory. In Sec. 4.2.2, we test this candidate and find that it does not reproduce the measured sideband pattern. We therefore use a phenomenological threshold based decomposition for the practical analysis.

The microscopic reference for irradiated coherent MAR is the two-frequency Keldysh-Floquet theory developed by Cuevas *et al.* for superconducting point contacts of arbitrary transmission [29]. It treats the Josephson oscillation and the external microwave drive within the same calculation and therefore includes coherent interference between driven MAR paths. The unified Tien-Gordon ex-

pression remains an interpretable approximation for sideband positions and practical modeling, but it does not replace the double-Floquet theory.

In summary, the transmission eigenvalues provide a common description of mesoscopic superconducting transport. Andreev reflection determines the interface process, ABS describe the equilibrium spectrum and supercurrent, and MAR governs finite-bias subgap transport. Channel resolved models thereby connect the mesoscopic pincode to measured I–V and dI–dV characteristics.

We now turn from phase coherent channel transport to the stochastic regime, where voltage fluctuations and energy exchange with the electromagnetic environment determine the charge-transfer dynamics.

2.4 Environment and Charge Dynamics

The electromagnetic environment couples voltage fluctuations to the charge dynamics of a junction. When these fluctuations destroy phase memory between successive tunneling events, transport becomes stochastic. The junction voltage then contains a fluctuating contribution $\delta V(t)$ around its deterministic bias value. This contribution produces a random phase accumulation and allows the environment to absorb or supply energy during a tunneling event. We therefore describe transport by transition probabilities and charge state occupations rather than by deterministic phase evolution.

The crossover from phase coherent to stochastic transport follows from the conjugacy of the superconducting phase difference ϕ and the junction charge operator Q,

$$[\phi, Q] = 2e\mathring{\imath}\,, \tag{2.91}$$

so charge localization is accompanied by fluctuations of the condensate phase difference. More generally, a tunneling process that transfers charge q acquires the fluctuating phase

$$\delta\phi_q(t) = \frac{q}{\hbar}\int_0^t \delta V(t')\,\mathrm{d}t'\,. \tag{2.92}$$

Here $\delta V(t)$ is the fluctuating part of the voltage across the junction relative to its mean bias value. For Cooper pair transfer, $q = 2e$, and $\delta\phi_{2e}$ is the fluctuating contribution to the condensate phase difference obtained from the ac Josephson relation in Eq. (2.34). For single-electron tunneling, $\delta\phi_e$ instead multiplies the single-electron tunneling amplitude. It represents the environmental phase acquired by transferring charge e and is not an independent superconducting condensate phase [25, 26].

Strong phase fluctuations suppress phase memory. They broaden coherent Josephson and Andreev processes. When independent tunneling events dominate, their rates are governed by energy exchange with the environment. This crossover depends on the frequency dependent impedance, junction parameters, temperature, and transferred charge.

In this incoherent limit, *P(E)* Theory gives the probability density for energy exchange between a discrete tunneling event and the environment. We first construct this environmental kernel, then apply it to dynamical Coulomb blockade of single-electron tunneling, incoherent Cooper pair tunneling and its photon assisted counterpart, and superconducting single-electron transistors.

2.4.1 *P(E)* Theory

The circuit impedance determines the voltage fluctuations seen by the junction. These fluctuations produce the phase correlator and the energy exchange probability that dresses the tunneling rate. The label q denotes the net charge transferred in one tunneling event, for example $q = e$ for quasiparticle tunneling and $q = 2e$ for Cooper pair transfer. Because the phase acquired from a voltage fluctuation is proportional to q, the environmental coupling in the phase correlator scales as q^2. We use the environmental tunneling formulation of Falci *et al.* and the conventions summarized by Ingold and Nazarov [25, 26].

The relevant impedance is the effective impedance $Z(\omega)$ seen by the junction rather than the impedance of an isolated circuit element. For a junction capacitance C in parallel with an environmental impedance $Z_{\mathrm{env}}(\omega)$, it is

$$Z(\omega) = \left[\mathring{\imath}\omega C + Z_{\mathrm{env}}^{-1}(\omega)\right]^{-1} . \tag{2.93}$$

The junction capacitance suppresses the high-frequency response and supplies a physical cutoff. For a more complex circuit, $Z(\omega)$ denotes the corresponding network impedance evaluated at the junction terminals [25, 26].

The effective impedance sets the spectrum of voltage fluctuations. Since voltage fluctuations integrate into the tunneling phase, the impedance enters through the charge dependent phase correlation function

$$J_q(t) = \frac{2q^2}{h}\int_0^\infty \underbrace{\frac{\mathrm{Re}\,Z(\omega)}{\omega}}_{\text{coupling}} \left[\underbrace{\coth\!\left(\frac{\hbar\omega}{2k_{\mathrm{B}}T}\right)(\cos(\omega t)-1)}_{\text{diffusion}} - \underbrace{\mathring{\imath}\sin(\omega t)}_{\text{winding}}\right] \mathrm{d}\omega\,, \tag{2.94}$$

This equilibrium correlator follows from the fluctuation spectrum of a linear impedance [25, 26].

Equation (2.94) can be read from left to right through the three annotated factors. The coupling factor gives the dissipative environmental spectrum seen by the junction. The division by frequency appears because phase is the time integral of voltage, making low-frequency noise especially effective. The diffusion factor is the real part of the correlator and describes loss of phase memory. The winding factor is the imaginary part and tracks the deterministic phase shift generated by the environment.

We obtain the energy exchange probability by Fourier transforming this phase correlator,

$$P_q(E) = \frac{1}{h}\int_{-\infty}^{\infty} \exp\left(J_q(t) + \mathring{\imath}Et/\hbar\right)\,\mathrm{d}t\,. \tag{2.95}$$

A narrow peak near zero energy describes nearly elastic tunneling. A broad or structured distribution produces suppressed onsets, shifted thresholds, and satellites in measured *I–V* and *dI–dV* [25, 26].

For a passive linear environment in thermal equilibrium, this probability density is nonnegative, normalized, and obeys detailed balance,

$$P_q(E) \geq 0\,, \quad \int_{-\infty}^{\infty} P_q(E)\,\mathrm{d}E = 1\,, \quad P_q(-E) = \exp\!\left(-\frac{E}{k_{\mathrm{B}}T}\right) P_q(E)\,. \tag{2.96}$$

Detailed balance fixes the relative probabilities for environmental emission and absorption. A coherently driven or nonequilibrium environment does not generally satisfy this relation and requires an explicitly time dependent or nonequilibrium kernel [26].

We obtain the direct connection to transport by separating the environmental and electronic parts of the tunneling process. The general forward rate is

$$\Gamma_q^{\rightarrow}(V) = \int_{-\infty}^{\infty} P_q(E) F_q^{\rightarrow}(E,V)\,\mathrm{d}E\,. \tag{2.97}$$

The electronic kernel $F_q^{\rightarrow}(E,V)$ is the bare tunneling rate for a process in which the environment absorbs the fixed energy E. It contains the electrode DOS, Fermi occupations, tunneling matrix elements, bias dependence, and the initial and final electronic states of the process. Its form distinguishes quasiparticle tunneling from incoherent Cooper pair tunneling, while $P_q(E)$ describes the environment. The reverse rate has the same structure with the corresponding reverse electronic kernel. The net current is

$$I(V) = q\left[\Gamma_q^{\rightarrow}(V) - \Gamma_q^{\leftarrow}(V)\right]\,. \tag{2.98}$$

This factorization separates the electronic tunneling kernel from environmental energy exchange [25, 26].

Thus $P_q(E)$ redistributes the available energy, while the electrode DOS and occupations remain in $F_q(E,V)$. Environmental dressing changes transport characteristics without redefining the electronic DOS, which is the key distinction from phenomenological broadening.

Elastic Limit

The elastic limit is the weak-coupling limit $\alpha_q \to 0$ of the Ohmic family. When the impedance is much smaller than the relevant resistance quantum over the frequencies involved in tunneling, voltage and phase fluctuations are weak and the environment exchanges little energy with the transferred charge. The energy exchange probability approaches a delta function,

$$P_{\mathrm{low}}(E) \approx \delta(E)\,, \qquad \mathrm{Re}\,Z(\omega) \ll \frac{h}{q^2}\,. \tag{2.99}$$

In the limit $P_q(E) = \delta(E)$, Eq. (2.97) reduces to

$$\Gamma_q^{\rightarrow}(V) = F_q^{\rightarrow}(0, V)\,. \tag{2.100}$$

The environment exchanges no energy with the tunneling charge. For single-electron tunneling, Eq. (2.100) recovers the elastic microscopic tunnel rate introduced in Sec. 2.1.2. A finite but small α_q broadens the delta peak and introduces weak inelastic corrections [26].

Finite Ohmic Coupling

As the Ohmic coupling increases, phase fluctuations broaden the elastic peak into a continuous energy exchange spectrum. A simple zero-temperature cutoff model gives the normalized form

$$P_{\mathrm{Ohmic}}(E) = \frac{1}{E_{\mathrm{cut}}\Gamma(\alpha_q)} \left(\frac{E}{E_{\mathrm{cut}}}\right)^{\alpha_q - 1} \exp\!\left(-\frac{E}{E_{\mathrm{cut}}}\right), \quad \alpha_q = \frac{2q^2 R}{h}, \quad E > 0\,. \tag{2.101}$$

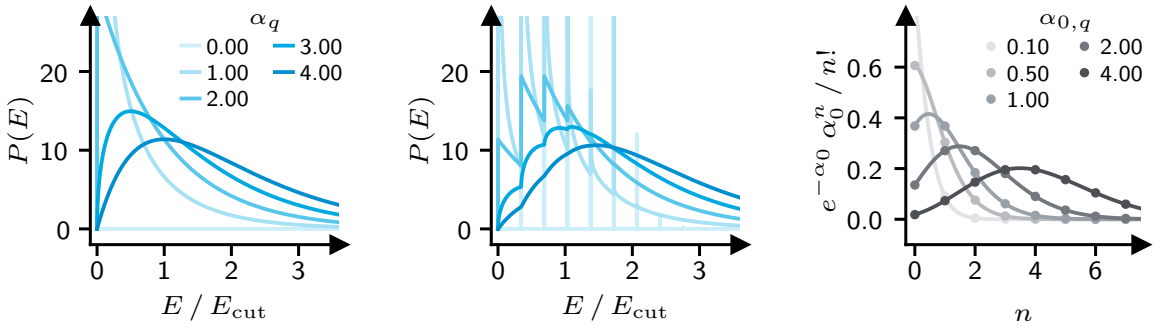
The cutoff energy is $E_{\mathrm{cut}} = \hbar\omega_{\mathrm{cut}}$. The Euler Gamma function $\Gamma(\alpha_q)$ normalizes the distribution. The power law describes the universal Ohmic low-energy behavior, while the exponential factor regularizes the high-energy tail of an environment with finite bandwidth. Equation (2.101) is therefore a convenient cutoff model and not the exact kernel of every Ohmic circuit [26].

At zero temperature this form has support only for $E > 0$, because the environment can absorb energy but cannot emit energy from its ground state. At finite temperature, thermal modes add a negative-energy tail to $P(E)$, corresponding to energy emitted by the environment into the tunneling process. The full finite-temperature kernel is usually evaluated from Eq. (2.95). The low-energy suppression of $P(E)$ underlies dynamical Coulomb blockade and also sets the scale for incoherent Cooper pair tunneling.

Resonant Environment

If the impedance contains discrete resonances from inductive elements or cavity modes, the real part of the impedance acquires sharp features at the corresponding frequencies. These features imprint themselves onto $P(E)$, which develops additional weight at the energies of the environmental modes.

The simplest analytic case is a single harmonic mode at frequency ω_0 added to an Ohmic background. A tunneling event linearly displaces this oscillator. For



(a) Ohmic environment (b) Resonant environment (c) Mode weights

Figure 2.30. Energy exchange probability for Ohmic and resonant electromagnetic environments. (a) The Ohmic kernel evolves from nearly elastic tunneling at small α_q to a broad distribution with suppressed low-energy weight as the environmental resistance increases. (b) Coupling to a harmonic mode redistributes this kernel into replicas separated by the mode energy $\hbar\omega_0$. (c) The replica amplitudes follow the Poisson weights, so increasing the mode coupling $\alpha_{0,q}$ transfers weight from the elastic contribution to higher-order mode excitations. Unless indicated by the legend, $\alpha_q = \alpha_{0,q} = 1$.

a mode initially in its ground state, the number of excited oscillator quanta is Poisson distributed, which gives

$$P_{\mathrm{resonant}}(E) = e^{-\alpha_{0,q}} \sum_{n=0}^{\infty} \frac{\alpha_{0,q}^n}{n!}\, P_{\mathrm{Ohmic}}(E - n\hbar\omega_0)\,, \tag{2.102}$$

where $\alpha_{0,q}$ quantifies the coupling between charge q and the resonant mode. The shifted kernels describe tunneling events accompanied by excitation of n mode quanta. The Poisson weights are not a generic property of any peak in $Z(\omega)$, but a consequence of a linearly coupled harmonic mode. Damping, thermal occupation, nonlinearities, or strong driving modify the sideband weights and line shapes. In the measured current, this structured $P_q(E)$ dresses the electronic kernel in Eq. (2.97) and produces satellites whose voltage spacing depends on the transferred charge [26].

Figure 2.30 summarizes these three environmental limits. Weak coupling concentrates the probability near zero exchanged energy. Ohmic coupling broadens this contribution and suppresses low-energy tunneling, while a resonant mode produces discrete inelastic contributions. After convolution with the electronic kernel, these structures appear in measured transport as zero-bias suppression, broadened thresholds, and resonant satellites.

The transferred charge selects the environmental kernel because $J_q(t) \propto q^2$. Dynamical Coulomb blockade uses $P_e(E)$ for single-electron tunneling, whereas incoherent Cooper pair tunneling uses the more strongly coupled $P_{2e}(E)$. In a superconducting single-electron transistor, a master equation combines these charge specific rates with the stationary charge state occupations.

2.4.2 Dynamical Coulomb Blockade

Dynamical Coulomb blockade (DCB) applies $P_e(E)$ to the elastic electronic kernel from the microscopic tunnel description. Figure 2.31 shows the energy balance for an SIS tunneling event. The electron creates quasiparticle excitations in the electrodes. At zero temperature, any energy not carried by the final electronic states must be absorbed by the environment. At finite temperature, thermally occupied environmental modes can also emit energy into the tunneling process. If $P_e(E)$ is small at the required exchange energy, this channel is suppressed [25, 26].

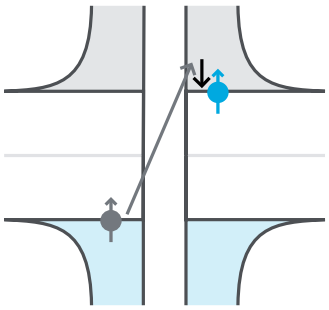


Figure 2.31. Environmental energy absorption in SIS junctions.

A single electron crossing a junction with capacitance C changes its electrostatic energy by an amount of order $E_\mathrm{C} = e^2/2C$. The environment carries this energy away through its voltage fluctuations. Small low-energy weight in $P_e(E)$ therefore appears as a zero-bias suppression in *dI–dV* [25, 26].

We write the environmental dressing most cleanly for directional tunneling branches. We obtain the total current after dressing both directions,

$$I(V) = I_{1\to2}(V) - I_{2\to1}(V)\,, \tag{2.103}$$

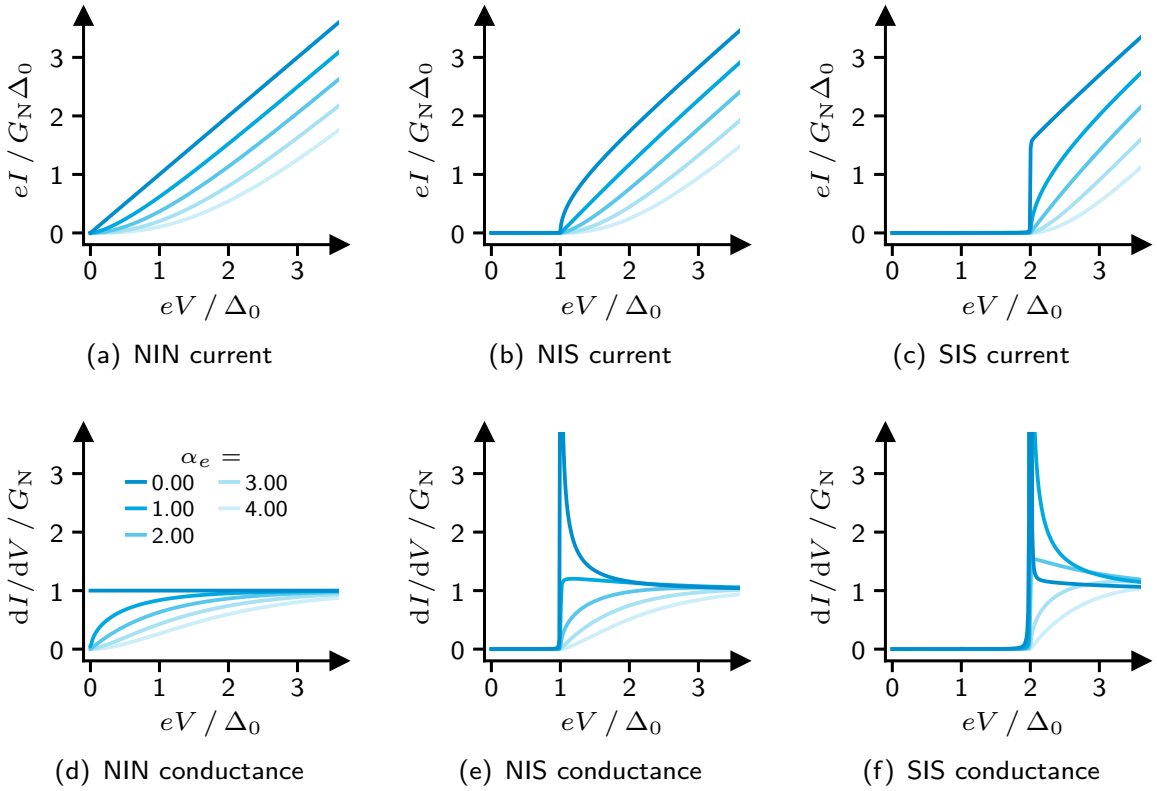
$$I_{1\to2}(V) = \int_{-\infty}^{\infty} P_e(\varepsilon)\, I^{(0)}_{1\to2}\left(V - \frac{\varepsilon}{e}\right) \mathrm{d}\varepsilon\,, \tag{2.104}$$

$$I_{2\to1}(V) = \int_{-\infty}^{\infty} P_e(\varepsilon)\, I^{(0)}_{2\to1}\left(-V - \frac{\varepsilon}{e}\right) \mathrm{d}\varepsilon\,. \tag{2.105}$$

Here $I^{(0)}_{i\to j}$ is the corresponding elastic branch without environmental energy exchange. It contains the electrode density of states, occupation factors, and tunneling matrix element for that direction, as in Eq. (2.17). This branch-wise form makes the voltage symmetry transparent. For a passive symmetric junction in equilibrium, the negative-bias current is the sign-reversed positive-bias current even though the zero-temperature kernel has support only for positive exchanged energy. The environment absorbs energy according to the favored tunneling direction at either voltage polarity [25, 26].

Figure 2.32 compares *I–V* and *dI–dV* curves for NIN, NIS, and SIS junctions in an Ohmic environment. The NIN panels isolate the zero-bias anomaly, while NIS and SIS show how the same dressing reshapes superconducting thresholds.

The same $P_e(E)$ therefore yields a zero-bias anomaly in NIN junctions and dressed DOS thresholds in superconducting junctions, which are the main DCB observables used below.



(a) NIN current (b) NIS current (c) SIS current

(d) NIN conductance (e) NIS conductance (f) SIS conductance

Figure 2.32. DCB dressing of tunnel *I–V* and *dI–dV* curves for NIN, NIS, and SIS junctions. The calculation uses the Ohmic kernel from Eq. (2.101) with cutoff energy $E_{\mathrm{cut}} = \Delta_0$. Increasing the environmental coupling deepens the NIN zero-bias conductance dip, broadens the NIS threshold, and redistributes spectral weight near the SIS pair breaking onset.

Photon Assisted Dynamical Coulomb Blockade

The photon assisted tunneling mechanism and its Bessel weights are derived in Sec. 2.1.3. For DCB, we apply the same time averaged dressing to the complete undriven current $I_{\mathrm{DCB},0}(V)$, which already contains environmental energy exchange. The resulting photon assisted DCB current is

$$I_{\mathrm{PADCB}}(V_0) = \sum_{n=-\infty}^{\infty} J_n^2(\alpha_{\nu,e})\, I_{\mathrm{DCB},0}\left(V_0 - \frac{nh\nu}{e}\right), \quad \alpha_{\nu,e} = \frac{eA}{h\nu}. \tag{2.106}$$

This current replica formulation applies to coherently driven tunnel junctions in the stationary DCB approximation [25, 78 - 80]. Here A is the effective ac voltage amplitude after environmental filtering and voltage division. The replicas are separated by $h\nu/e$.

Equation (2.106) assumes weak tunneling, a coherent classical drive, and an environment whose response can be absorbed into the stationary DCB characteristic and effective amplitude A. It neglects drive induced changes of the

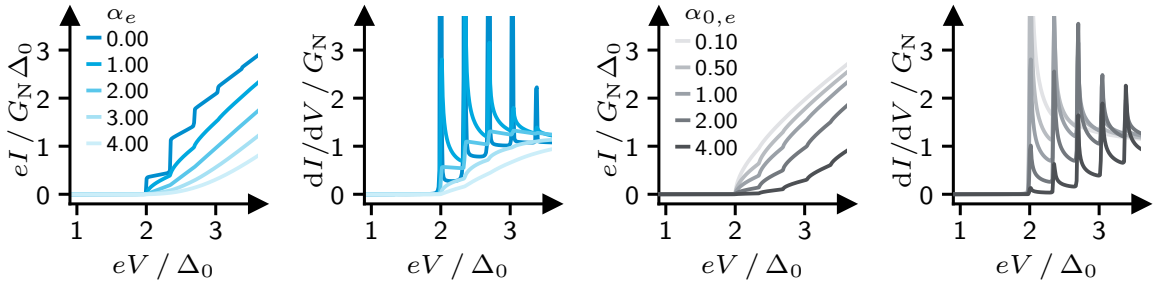


Figure 2.33. Resonant DCB in an SIS junction. (a) and (b) Increasing α_e at fixed $\alpha_{0,e} = 1$ suppresses and broadens the quasiparticle features. (c) and (d) Increasing $\alpha_{0,e}$ at fixed $\alpha_e = 1$ transfers weight to Poisson weighted satellites spaced by $\hbar\omega_0/e$.

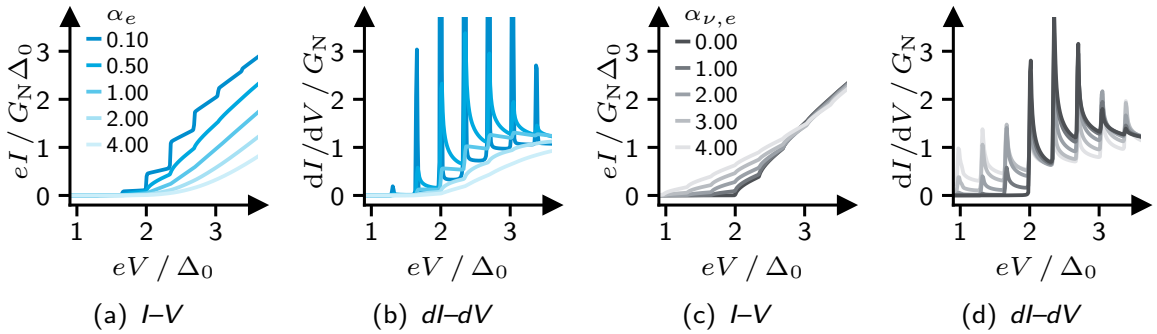


Figure 2.34. Photon assisted DCB in an SIS junction. (a) and (b) Increasing α_e at fixed $\alpha_{\nu,e} = 1$ strengthens environmental suppression. (c) and (d) Increasing $\alpha_{\nu,e}$ at fixed $\alpha_e = 1$ creates Bessel weighted replicas spaced by $h\nu/e$.

DOS, heating, nonequilibrium electrode occupations, and nonlinear environmental backaction.

Houle *et al.* found that current and photon assisted noise required different effective ac amplitudes. The stationary replica model therefore misses part of the driven junction and environment dynamics and is limited as a quantitative microwave calibration [80].

Figures 2.33 and 2.34 compare passive resonant and photon assisted DCB in an SIS junction. Both generate satellites separated by $\hbar\omega_0 = h\nu$, but passive modes produce Poisson weighted inelastic channels and coherent drives produce Bessel weighted current replicas. At zero temperature, passive mode satellites lie on the higher-$|V|$ side of the quasiparticle threshold because the mode can only absorb energy. A coherent drive permits absorption and stimulated emission, creating replicas on both sides. Thermal mode occupation weakens this distinction through detailed balance.

DCB and ICPT are two charge specific consequences of the same environmental interaction. Small low-energy weight in $P_e(E)$ suppresses stochastic $q = e$

tunneling, while finite-energy weight in $P_{2e}(E)$ enables the inelastic $q = 2e$ transfer discussed next.

Dynamical Coulomb Blockade of Multiple Andreev Reflection

The tunnel junction treatment above assigns one elementary charge to each stochastic event. MAR instead consists of correlated electron and hole traversals through a finite-transmission superconducting contact. Its charge resolved description in Sec. 2.3.4 associates a transfer of effective charge $q = me$ with each current component $I_m(V)$. This suggests an enhanced environmental sensitivity at low voltage because the phase correlator in Eq. (2.94) scales with q^2. However, DCB of MAR is not generally obtained by independently convolving every $I_m(V)$ with a kernel $P_{me}(E)$. The traversals within one MAR process remain coherent, different charge transfer orders overlap at a given voltage, and the environment can also renormalize the elastic MAR probabilities.

Levy Yeyati, Cuevas, and Martín-Rodero calculate the leading environmental correction for a superconducting quantum point contact coupled to a weak macroscopic impedance [81]. In the low-transmission limit, the relative blockade of an mth-order process is enhanced approximately as m^2, consistent with the environmental coupling to a transferred charge me. At higher transmission, Pauli correlations reduce this enhancement and elastic renormalization becomes important. The correction can then change sign in parts of the subgap region, so environmental coupling enhances rather than suppresses the MAR current. This antiblockade shows that the effective charge alone does not determine the environmental correction.

The experimental work by Cron *et al.* established the two ingredients separately. Shot noise measurements in superconducting atomic contacts resolved the enhanced effective charge associated with MAR, while normal-state measurements on the same class of contacts showed that DCB decreases as a channel approaches perfect transmission [82, 83]. The latter experiment used a characterized series impedance and provided the platform proposed by Levy Yeyati *et al.* for testing their superconducting prediction. It did not, however, measure DCB of MAR. Cron's thesis explicitly identifies the blockade of MAR processes as an open problem.

The expected observables are changes in subgap current and in the amplitude and line shape of the MAR structures in *dI–dV*. Their interpretation requires comparison with a coherent MAR reference at the same channel transmissions. A suppression near a MAR threshold is not by itself evidence for DCB because temperature, lifetime broadening, voltage noise, and imperfect knowledge of the channel transmissions can produce similar changes. Conversely, environmental enhancement between thresholds would be missed by a model that treats DCB only as a positive charging energy cost.

In summary, DCB of MAR couples environmental fluctuations to coherent multiparticle transfer through a finite-transmission contact. The low-transmission effective charge picture explains the enhanced blockade, but it is incomplete once several MAR orders overlap and elastic renormalization becomes important. A quantitative interpretation must therefore retain the channel transmissions, coherent MAR dynamics, and electromagnetic environment together. The resulting transmission dependent blockade and possible antiblockade distinguish this problem from a simple charging energy shift or an independent dressing of each charge order. A complete quantitative theory that treats arbitrary channel transmissions, overlapping MAR orders, and environmental backaction on equal footing remains an open problem.

2.4.3 Incoherent Cooper Pair Tunneling

In a voltage biased SIS junction, environmental phase fluctuations can suppress coherent Josephson transport while retaining discrete Cooper pair transfer with $q = 2e$. The Josephson coupling energy then enters perturbatively at second order, and the environment supplies or absorbs the energy associated with each pair transfer event.

The standard forward and backward incoherent Cooper pair tunneling (ICPT) rates follow directly from the charge-$2e$ kernel introduced in Sec. 2.4.1,

$$\Gamma^{\rightarrow/\leftarrow}_{\mathrm{ICPT}}(V) = \frac{\pi E_{\mathrm{J}}^2}{2\hbar} P_{2e}(\pm 2eV)\,. \tag{2.107}$$

The upper sign describes forward pair transfer, while the lower sign describes backward transfer [25, 26].

For an equilibrium environment, we therefore write the current as

$$I_{\mathrm{ICPT}}(V) = \frac{\pi e E_{\mathrm{J}}^2}{\hbar}\left[1 - \exp(-2eV/k_{\mathrm{B}}T)\right] P_{2e}(2eV)\,. \tag{2.108}$$

This form follows from the rate difference and the equilibrium detailed balance relation in Eq. (2.96) [26].

The voltage axis therefore samples the environmental kernel through $E = 2eV$. For an Ohmic environment, its low-energy behavior determines the low-bias ICPT line shape. A resonant mode produces satellites whenever $2eV = n\hbar\omega_0$, so neighboring features are separated by $\hbar\omega_0/(2e)$ in voltage.

This perturbative description assumes weak E_{J}, statistically independent pair transfer events, and a linear equilibrium environment. It therefore predicts environmental sidebands for charge $2e$, but not coherent Shapiro locking.

Photon Assisted Incoherent Cooper Pair Tunneling

A coherent microwave drive adds photon assisted replicas of the stationary ICPT characteristic. Using the PAT result from Sec. 2.1.3, the effective charge $2e$ gives

$$I_{\mathrm{PAICPT}}(V_0) = \sum_{n=-\infty}^{\infty} J_n^2(\alpha_{\nu,2e})\, I_{\mathrm{ICPT}}^{(0)}\left(V_0 - \frac{nh\nu}{2e}\right), \quad \alpha_{\nu,2e} = \frac{2eA}{h\nu}. \tag{2.109}$$

The squared Bessel factor is the probability for an incoherent pair transfer event to exchange $|n|$ drive photons, producing replicas separated by $h\nu/(2e)$ [25, 30].

This differs from the coherent Josephson response in Sec. 2.2.3, where Bessel coefficients enter linearly at the amplitude level and determine frequency locked Shapiro steps. Linear Bessel coefficients and squared Bessel weights describe the coherent and incoherent limits, respectively. In the intermediate phase diffusion regime, broadened Shapiro features and stochastic sidebands can coexist, so neither limit is complete. Their common spacing reflects the transferred charge $2e$, while the weights distinguish probabilistic tunneling from phase locking. Photon assisted ICPT with squared Bessel weights has been observed in Josephson scanning tunneling microscopy [30].

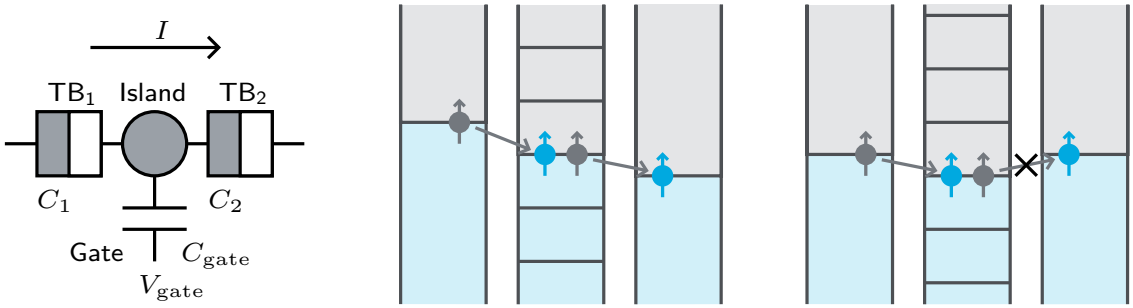
In summary, $P_{2e}(E)$ governs stochastic Cooper pair transfer and fixes the voltage spacing of its environmental and photon assisted satellites. Squared Bessel weights characterize the fully incoherent microwave limit and distinguish it from phase locked Josephson transport.

2.4.4 Superconducting Single-Electron Transistor

A superconducting single-electron transistor (SSET) combines charge quantization with quasiparticle and Cooper pair transport. We first introduce the normal conducting tunnel junction device and its charge state master equation, then discuss tunable and hybrid extensions [26, 50].

Normal Conducting SET

A normal conducting SET consists of a metallic island connected to source and drain by two tunnel junctions and coupled capacitively to a gate. The barriers permit charge quantization, while the gate shifts the charge state energies, as shown in Fig. 2.35(a). Replacing the normal metals with superconductors gives an SSET.



(a) SET circuit schematic (b) Sequential tunneling (c) Coulomb blockade

Figure 2.35. Gate control of sequential transport in a normal conducting SET. The two tunnel junctions define a metallic island whose addition energy determines whether both steps of a transport cycle are allowed. Blue and gray indicate occupied and empty states, while arrows indicate single-electron transfer.

Adding one electron to an isolated metallic island requires electrostatic energy. For junction capacitances C_1 and C_2 and gate capacitance C_{gate}, the charging energy is

$$E_{\mathrm{C}} = \frac{e^2}{2C_{\Sigma}}\,, \qquad C_{\Sigma} = C_1 + C_2 + C_{\mathrm{gate}}\,. \tag{2.110}$$

This is the orthodox charging scale for the lumped capacitance island model [26, 50].

At finite bias, the junction capacitances determine how strongly the applied voltage changes the island energy for each tunneling step. A transport onset occurs when the initial and final free energies become degenerate. The gate and bias dependence of this condition produces lines in a gate–bias map, with slopes fixed by capacitance ratios. The transition rates and stationary occupations determine the intensity along these lines.

Charge quantization requires weak thermal activation and weak quantum charge fluctuations. Thus $E_{\mathrm{C}} \gg k_{\mathrm{B}}T$, and in the tunnel limit each normal-state junction resistance must be large compared with $R_{\mathrm{K}} = h/e^2$. The island can then be described by an integer excess electron number n. A gate voltage induces the offset charge $n_{\mathrm{gate}} = C_{\mathrm{gate}}V_{\mathrm{gate}}/e$ and changes the electrostatic energy according to

$$E_n(V_{\mathrm{gate}}) = E_{\mathrm{C}}\left(n - n_{\mathrm{gate}}\right)^2\,. \tag{2.111}$$

The expression omits a charge independent constant and assumes that other induced charges are absorbed into n_{gate} [26, 50].

For a normal tunnel junction with resistance R_i, the sequential tunneling rate for a transition whose final free energy exceeds the initial free energy by ΔE is

$$\Gamma_i(\Delta E) = \frac{\Delta E}{e^2 R_i \left[\exp\left(\Delta E/k_{\mathrm{B}}T\right) - 1\right]}\,. \tag{2.112}$$

Negative ΔE describes an energetically favorable transition, while positive ΔE is thermally activated. We evaluate separate rates for both directions across each junction.

Neighboring charge states become degenerate at half-integer gate charge. Between these points, one state has the lowest electrostatic energy and charge transfer costs energy. This gate controlled cost is the origin of static Coulomb blockade [26, 50].

In normal-state sequential transport, an electron enters through one junction and leaves through the other, restoring the initial island charge. A dc current flows when both steps are energetically favorable, as illustrated in Fig. 2.35(b). The gate opens or closes this cycle by shifting the charge state energies.

If either step is inaccessible, the sequential current is blocked, as in Fig. 2.35(c). Conductance is largest near charge degeneracy and periodic in gate induced charge. Far from degeneracy, weaker cotunneling can bypass the intermediate charge state [26, 50].

Superconducting SET

We now consider superconductivity. In a conventional SSET, the island and both leads are superconducting, while the two junctions remain in the tunnel limit. The electrostatic charge states remain those of the normal SET, but the superconducting gap shifts the threshold energies of the transitions. Quasiparticle tunneling changes the island charge by one elementary charge and requires occupied and empty quasiparticle states. Cooper pair tunneling changes the island charge by two elementary charges and connects charge states of the same parity. The measured current can therefore contain competing single-electron and pair transfer cycles. A stationary current flows only when energetically allowed transitions at both junctions form a complete transport cycle.

The superconducting island distinguishes even and odd electron numbers. An odd charge state contains an unpaired quasiparticle and carries an additional free energy cost governed by the gap. At low temperature, this parity effect favors even charge states and can change the gate periodicity from e-periodic to $2e$-periodic. Nonequilibrium quasiparticles weaken this distinction and can restore single-electron periodicity. Here we retain parity as an energy cost for odd charge states rather than develop a separate quasiparticle population model [50].

The Josephson-quasiparticle (JQP) cycle combines quasiparticle and Cooper pair transport. In the first step, a Cooper pair tunnels coherently onto the island through one junction while a quasiparticle simultaneously leaves through the other junction. The island charge therefore increases by one elementary charge rather than two. This correlated step avoids the electrostatic cost of an isolated Cooper pair addition. A second quasiparticle subsequently leaves the island and restores the initial charge state. The complete cycle transfers charge through

both junctions and produces a localized feature in bias and gate voltage [84-86].

Describing the JQP cycle with a population master equation replaces the coherent Josephson dynamics by an effective transition rate. This approximation is valid when coherence between the coupled charge states decays much faster than the Josephson evolution, $\Gamma_{\mathrm{deph}} \gg E_{\mathrm{J}}/\hbar$. Otherwise, the off-diagonal density matrix elements must be retained in a generalized master equation.

Orthodox theory describes the island by classical charge state probabilities and independent junction transitions. This population only description assumes well-defined charge states, Markovian events, and negligible off-diagonal coherences. Each junction and direction contributes a rate set by the corresponding free energy difference and microscopic junction properties. In an SSET, these rates include the superconducting DOS and incoherent Cooper pair transfer [26, 50].

Let $\Gamma_{n\to m}$ denote the total rate for all junction processes that change the island from charge state n to charge state m. In the stationary state, probability flow into every charge state balances probability flow out of it,

$$\sum_m \left(\Gamma_{m\to n} p_m - \Gamma_{n\to m} p_n\right) = 0\,, \qquad \sum_n p_n = 1\,. \tag{2.113}$$

This is the orthodox stationary population master equation [26, 50].

Solving this linear system gives the occupation probability p_n of each charge state. The current at junction i follows by summing all signed charge transfer events through that junction,

$$I_i = \sum_n p_n \sum_{m\neq n} q_{i,n\to m}\,\Gamma_{i,n\to m}\,. \tag{2.114}$$

Here $q_{i,n\to m}$ is the signed charge transferred through junction i, while $\Gamma_{i,n\to m}$ is its rate. Charge conservation gives the same stationary current through both junctions. The master equation is a population balance method that can use orthodox single-electron rates or additional superconducting rates [26, 50].

The rates specify the allowed transport steps, while the master equation supplies their initial state occupations. Together they determine *I–V*, *dI–dV*, and gate dependent transport maps.

The orthodox description is valid when both junctions remain in the tunnel limit, island charge states are well defined, and successive tunneling events are independent. It does not describe transport through a highly transmitting contact or regimes with strong environmental backaction, large quantum charge fluctuations, or coherent multiparticle transfer.

Tunable and Hybrid SSETs

Replacing one tunnel junction by a mechanically tunable contact extends the SSET beyond the tunnel limit. Increasing transmission enables Andreev reflection and MAR and enhances quantum charge fluctuations. Charge state probabilities can remain useful over part of this crossover if the rates include multiparticle transfer through the contact. The microscopic MAR mechanism has been described in Sec. 2.3.4, while its environmental correction is discussed in Sec. 2.4.2.

Tunable SSETs place a few-channel superconducting contact in a charge sensitive circuit and therefore provide access to the crossover between island Coulomb blockade, coherent MAR, and environmental backaction on multiparticle transfer. This device concept motivates part of the experimental approach and fabrication developed in this thesis. It also provides an instructive framework for combining charge resolved MAR rates with stochastic island occupations and for identifying where the separation into independent contact events and well-defined island charge states can fail.

Thomas Lorenz and Susanne Sprenger study all-superconducting SETs in which an oxide tunnel barrier and a tunable superconducting break junction define the island. Their measurements span the weak-coupling to atomic contact regimes. Coulomb blockade suppresses low-bias transport, while increasing contact transmission activates higher-order MAR. They extend orthodox theory by adding MAR rates to the charge state transitions and thereby describe the principal gate dependent transport features qualitatively. The approach also reveals where the tunnel limit picture becomes incomplete, including reduced effective charging energy toward strong coupling and transport features that cannot be assigned uniquely. [84 - 86]

Sobral Rey *et al.* use a hybrid SSN SET to separate the roles of the two junctions more clearly. In this device, a superconducting island is connected to a superconducting lead by the tunable break junction and to a normal metal lead by an oxide tunnel barrier. Multiparticle superconducting transport is concentrated at the break junction, while the tunnel barrier mainly provides single quasiparticle transfer. This asymmetry reduces the number of competing cycles and makes JQP, Andreev, and MAR contributions easier to identify. Their simulations retain the stationary master equation for the island occupations and use charge resolved, time averaged MAR rates obtained from full counting statistics. [87, 88]

The strong-coupling limit illustrates why the charge state picture remains useful but cannot fully describe transport. A low-impedance superconducting contact can suppress charge quantization by pinning the island potential, while Coulomb blockade signatures can return after the contact switches to a resistive state. The observed transport then reflects competition between charging energy, superconducting multiparticle transport, and contact transmission rather than a simple disappearance of Coulomb blockade. [87, 88]

In summary, the charge state master equation combines device specific quasiparticle, JQP, Andreev, and MAR rates when successive events are separable and charge state coherence is negligible. Coherent mixing requires a generalized master equation with off-diagonal density matrix elements. At stronger coupling, sharply defined island charge states and the charge state description itself break down.

3 Experimental Methods

This chapter describes the experimental methods used to obtain reproducible measurements. The experiment comprises not only the superconducting nanostructure, but also the complete measurement chain around it: sample fabrication, cryogenic mounting, filtered electrical wiring, room-temperature instrumentation, digital acquisition, and numerical evaluation. Each part determines the quality of the measured I–V curves and derived dI–dV spectra, and constrains the interpretation of extracted model parameters.

I first describe the fabrication of mechanically controllable break junction samples in Sec. 3.1. The process follows established group experience, but I adapt the design and fabrication protocol to improve sample stability, contacting, and reproducibility. The discussion focuses on the steps that determine the mechanical and electrical properties of the final devices. Practical fabrication notes are provided in Appendix C.

I then follow the measurement chain from the sample environment to the room-temperature electronics in Sec. 3.2. The discussion covers mechanical actuation, cold wiring, filtering, microwave coupling, voltage and current reconstruction, and grounding. It emphasizes the choices that affect electronic temperature, low-frequency noise, microwave irradiation, and the reliability of the measured I–V curves. Further details of the cryogenic environment are given in Appendix D.

Finally, Sec. 3.3 connects the physical setup to the data products analyzed in later chapters. I describe the Python based instrument control, structured measurement files, evaluation pipeline, offset correction, and numerical interfaces for tunnel regime BCS fits and finite transmission MAR analysis. Together, the fabrication, measurement environment, and digital infrastructure define the experimental basis for the results chapters.

3.1 Sample Fabrication

Single-electron transistors (SETs) consist of two junctions that define a small island capacitively coupled to a dc gate electrode. In the present approach, I realize one of these junctions as a mechanically controllable break junction (MCBJ) and the second as an oxide tunnel barrier. I use aluminum as the device material and integrate an ac gate to couple microwave fields to the device. Figure 3.1 shows such a sample.

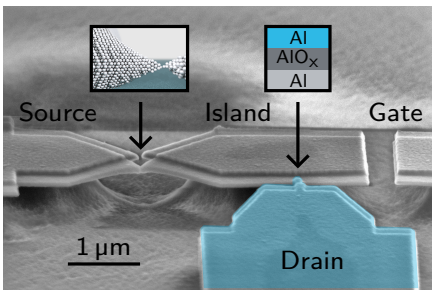


Figure 3.1. Scanning electron micrograph of the central sample region. The colored region highlights the top layer that forms one electrode of the oxide tunnel barrier. The schematic insets illustrate the atomic contact and the cross section of the tunnel barrier.

This section describes work carried out together with Patrick Raif as part of his project practicum[1] [89].

The following subsections present the sample preparation process from complementary perspectives. First, I introduce the conceptual framework. Next, I discuss the main design choices and their rationale. The fabrication procedure, including practical considerations and typical challenges encountered in the laboratory, is given in Appendix C.

3.1.1 Device Concept

In this subsection, I introduce the general concept of the sample fabrication process and its connection to the underlying device physics.

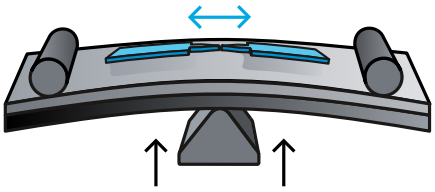
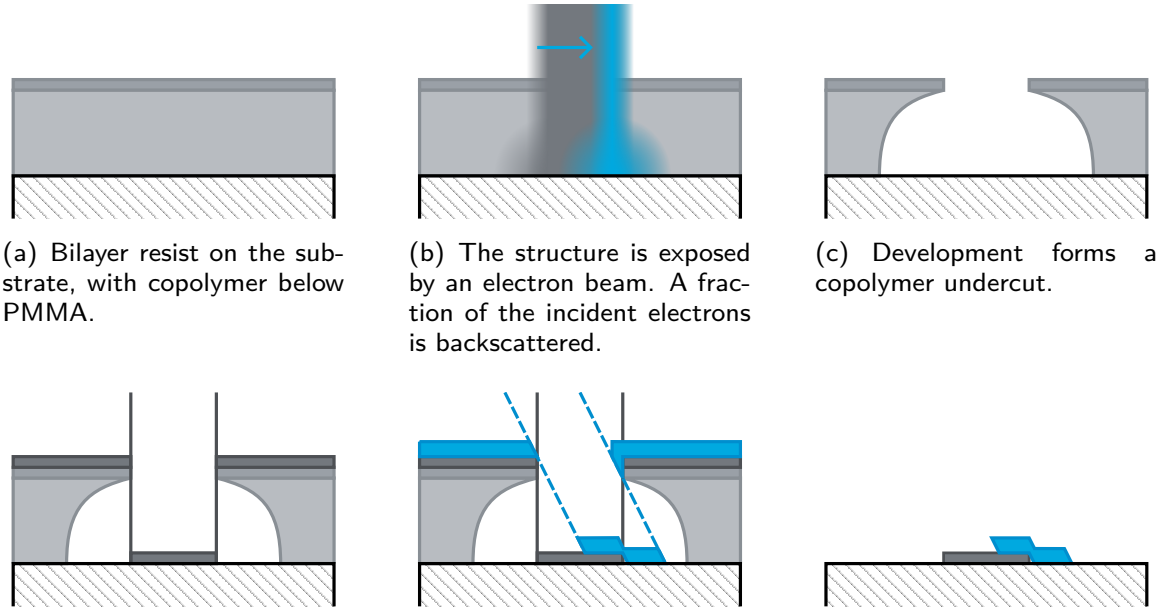


Figure 3.2. MCBJ bending principle. Adapted from Ref. [90].

The sample design must satisfy several requirements simultaneously. Mechanically, it must provide a freestanding bridge on a bendable substrate so that an atomic contact can be formed in a controlled manner. Lithographically, it must enable suspended shadow mask structures for double angle evaporation. Electrically, it must define an island between two weak links and provide gate coupling. Finally, the design should allow efficient coupling of microwave fields.

I realize the MCBJ as a freestanding bridge on a bendable substrate, as shown in Fig. 3.2. A stamp from below or counter supports from above bend the substrate and elongate the bridge, allowing the junction to enter the atomic contact and tunnel regimes. Subsection 3.2.1 describes the mechanical setup in detail. A polyimide (PI) layer insulates the substrate from the structures

[1] An optional component of the master's curriculum completed within four weeks.



(a) Bilayer resist on the substrate, with copolymer below PMMA.

(b) The structure is exposed by an electron beam. A fraction of the incident electrons is backscattered.

(c) Development forms a copolymer undercut.

(d) First metal deposition normal to the sample plane, followed by in situ oxidation.

(e) Second metal deposition at an oblique angle. The thicker second layer ensures electrical continuity.

(f) The resist undercut facilitates lift-off after deposition.

Figure 3.3. Schematic cross section of the fabrication process used to realize an oxide tunnel barrier by electron beam lithography and double angle evaporation.

produced in the subsequent steps. More importantly, the PI layer serves as a sacrificial layer. During later processing, I partially etch it off, allowing the structure to become freestanding. Next, I apply a bilayer of positive electron sensitive resist. I then use electron beam lithography (EBL) to expose the desired pattern with high precision. After development, I remove the exposed resist, leaving behind the lithographically defined structure.

In addition to the MCBJ, the SET requires a second weak link, which I realize as an oxide barrier. A convenient method for fabricating this barrier is shadow evaporation [91, 92]. In this technique, two evaporation steps are carried out at different incident angles, producing a lateral offset between the deposited structures. Between the two depositions, a controlled oxidation step is performed without breaking vacuum. This yields overlapping regions with a metal–oxide–metal cross section, as illustrated in Fig. 3.3.

I use aluminum (Al) in both evaporation steps to obtain a superconducting single-electron transistor (SSET) comprising an Al break junction (BJ) and an Al–AlO_x–Al oxide barrier, as described by Thomas Lorenz [85] and Susanne Sprenger [86]. The second electrode material can also be changed. Laura Sobral Rey, for example, realized Al–AlO_x–Cu structures [88].

In the final fabrication step, I remove approximately 500 nm of the PI layer by oxygen plasma etching. Because the etch proceeds essentially isotropically, it

creates a pronounced undercut, as illustrated in Fig. 3.1. This step releases the bridge and renders the break junction freestanding.

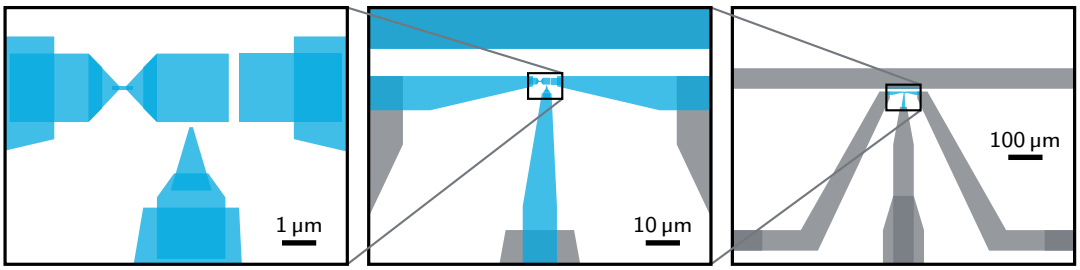
3.1.2 Design Choices

This section discusses the fabrication process from the perspective of the underlying design decisions. While there is no single unique route to a working sample, individual choices can have important practical consequences. The following paragraphs summarize the reasoning behind the most relevant steps.

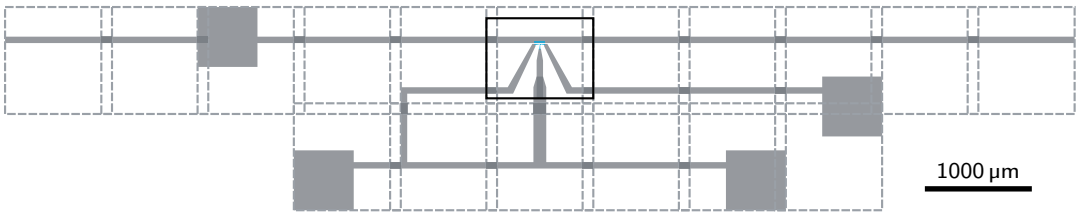
Bronze has long been used as a substrate material for MCBJ samples. A key reason is that it bends with comparatively little force while retaining excellent elastic behavior. This is crucial for MCBJs, where the junction configuration must be adjusted repeatedly and reproducibly. Other materials, such as copper, bend less uniformly and tend to deform mainly in the center beneath the stamp. This bending behavior leads to larger angular changes and poorer elastic reproducibility. In addition, copper often shows nonmonotonic bending behavior, which makes fine adjustment in the atomic contact regime impractical. Kapton, by contrast, bends homogeneously along its length but does not allow sufficiently strong bending for the present MCBJ geometry. Copper–beryllium would be mechanically suitable, but its toxicity makes it undesirable for routine use. Finally, the substrate must be nonmagnetic and contain as few magnetic impurities as possible. Considering all these requirements, bronze has proven to be the most suitable substrate material for this type of MCBJ [90].

I spin-coat a polyamic acid (PA) layer onto the entire bronze wafer and convert it into polyimide (PI) during a subsequent hard-bake step. I then apply two resist layers. The thin top layer is optimized for high-resolution lithography and defines sharp feature edges. The thicker bottom layer serves two purposes. It lifts the top layer away from the substrate and promotes undercut formation during development. Two effects contribute to this undercut. First, electrons spread as they pass through the resist stack, so the exposed volume widens with depth. Backscattered electrons from the substrate can also reenter the resist from below and increase the effective exposure in the lower resist region. As a result, the bottom layer develops more rapidly and forms an undercut. Second, prolonged rinsing in isopropyl alcohol (IPA) after development in diluted methyl isobutyl ketone (MIBK) further enhances the undercut.

I carry out EBL using a Zeiss `Crossbeam` system that is modified for significantly faster beam blanking. The beam blanker operates synchronously with the beam deflection, allowing arbitrary structures to be exposed precisely within the selected field of view. For small high-resolution structures, I typically use a field size of 100 µm, whereas I write coarser and larger features in a 1000 µm field. I expose structures exceeding these dimensions over multiple adjacent fields by moving the sample stage. To ensure smooth transitions between neighboring fields, I include a deliberate overlap for stitching.



(a) Central region, successively magnified, showing the break junction, the island, the oxide barrier finger, the nearby dc gate, and the adjacent ac gate. Blue structures (■) are written in a 100 µm writing field, whereas gray structures (■) are written in 1000 µm writing fields.



(b) Overview of the full pattern. Individual writing fields are indicated by dashed squares. Their overlap avoids stitching problems. The outer leads are arranged to simplify electrical contacting, and a shorting structure is included to protect the sample against electrostatic discharge.

Figure 3.4. Electron beam lithography design of the sample at different length scales.

At the beginning of my doctoral work, only one of the two previously used EBL software packages, `Elphy Plus` by Raith, is functional and properly supported. I therefore have to reimplement the sample design from scratch because my predecessors use a different software environment. The dimensions of the break junction and the gate follow the established geometry and are consistent with earlier implementations in Refs. [85, 86, 88].

The tip of the bottom electrode connects to the island during the second evaporation at an oblique angle. The angle and height of the mask define the lateral offset of this deposition. I encounter difficulties with the mechanical stability of the evaporation mask. During evaporation, the mask sags, and the shadow electrode does not connect electrically to the island. I modify the finger geometry to reduce the freestanding length of the evaporation mask. A new resist further improves the stability of the mask, and the finger then reliably reaches the island during the second evaporation. The complete electron beam lithography design is shown in Fig. 3.4.

I arrange the larger leads and contact pads to simplify electrical contacting. In addition, I place an ac gate or stripline at a distance of 8 µm from the junction to maximize microwave coupling while keeping the electron beam writing time within reasonable limits. I also include a shorting structure to protect the sensitive tunnel barrier against electrostatic discharge. Finally, I make the central vertical lead wider to improve its visibility, which helps to align the break junction precisely with the stamp during mounting.

The evaporator used in this work has a comparatively long source–sample distance. As a result, the aluminum deposition is highly directional. For the same reason, however, the effective deposition cross section is reduced. The system offers two methods for aluminum evaporation: thermal evaporation and electron beam evaporation. In thermal evaporation, a large current heats a small crucible until the aluminum evaporates. The crucible used for this method is fragile and must be loaded and completely emptied during each run to avoid crack formation.

This procedure is laborious, prone to error, and offers no clear advantage over electron beam evaporation. In electron beam evaporation, a focused electron beam locally heats the aluminum charge, resulting in a more efficient and reliable deposition process. For this reason, and in contrast to Refs. [85, 86, 88], I use electron beam evaporation throughout this work.

I perform the oxidation step between the two evaporations in the load lock of the evaporator at an oxygen pressure of approximately 3.0(2) mbar for 3 min [88]. In the past, this step was found to be insufficiently reproducible. I therefore monitor it carefully in the present process [86, 93].

I carry out reactive ion etching (RIE) using the `PlasmaPro 80` ICP RIE system from Oxford Instruments. I establish the break junction etching process on this newly introduced system. Although this tool is not primarily designed for highly isotropic etching, I achieve acceptable uniformity by combining high inductively coupled plasma (ICP) radio-frequency (RF) power, low RF power at the sample table, and high chamber pressure. The ICP source generates a high-density plasma by coupling energy into the chamber through an electromagnetic field, whereas the RF power applied to the table controls the ion energy at the sample surface. A heated sample table increases the chemical etching component and further improves etch isotropy. I monitor the etched depth in real time by laser interferometry, which allows precise process control.

The final step is electrical contacting of the sample. Wedge bonding is unsuitable because the bonding wire can press through the pad and penetrate the soft PI layer, thereby causing a short to ground. Instead, I attach thin copper wires to the pads using conductive silver paste. This approach provides reliable electrical contact without damaging the underlying structure.

Overall, by modifying the finger geometry and resist process, I stabilize the suspended evaporation mask and enable the shadow electrode to connect reliably to the island. Together with the isotropic etching, the protected contact layout, and the nondestructive electrical contacting, I produce freestanding MCBJ devices that combine a tunable atomic contact, an oxide tunnel barrier, and integrated gate coupling. The resulting geometry provides the mechanical stability and electrical access required for the measurements described in the following sections.

3.2 Experimental Setup

The experiments in this thesis require atomic scale control of superconducting contacts, a millikelvin temperature environment, low-noise nonlinear transport measurements, and microwave excitation. The mechanically controllable break junction (MCBJ) provides the contact control, while the dilution refrigerator establishes the low-temperature environment and thermal anchors for the sample and wiring. The bias circuit, filtering, microwave coupling, instrumentation, and grounding determine the experimentally accessible voltages, currents, and drive conditions.

These parts cannot be treated as independent accessories. A low base temperature is only useful if the electrical lines are filtered and thermalized well enough that the electronic temperature follows the cold stage. A nominal bias voltage is meaningful only if the voltage division between reference resistors, filters, leads, and the sample is measured rather than assumed. A microwave source power is likewise not the same as a calibrated local microwave voltage at the junction. The setup therefore defines both the experimental control parameters and the limits of their interpretation.

The platform is inherited from earlier MCBJ experiments in the group, but I modify it substantially during this work for superconducting devices, microwave irradiation, local thermometry, and more robust repeated cooldowns. I replace mechanical parts, redistribute gearbox ratios, remove fragile wiring, revise filter and reference resistor layouts, and select the grounding configuration empirically from noise measurements. These practical details determine whether the data are reproducible. They also establish which assumptions are justified when later chapters interpret I–V curves, derived dI–dV spectra, and microwave induced features.

I first describe the MCBJ mechanics and the conversion from motor motion to the relative contact control coordinate in Sec. 3.2.1. I then define the cold electrical measurement environment, including voltage and current reconstruction, filtering, and microwave coupling in Sec. 3.2.2. Finally, I describe the instrumentation and grounding, including the practical cryostat ground reference and the isolation choices used for low-noise measurements in Sec. 3.2.3. The operation and performance of the dilution refrigerator are documented separately in Appendix D.

3.2.1 Mechanically Controllable Break Junction

The in situ control coordinate of the experiment is given by the mechanically controllable break junction (MCBJ). Its task is to convert macroscopic motor motion at room temperature into a reproducible, much smaller elongation of the metallic constriction at millikelvin temperature. The sample is mounted in a three-point bending geometry, as shown in Fig. 3.5. The three contact points are located at the bottom of the rendering. The two counter supports are anchored in the lower plate, while the central pushing element lies on the vertical symmetry axis and projects partway through this plate. The fixed counter supports hold the substrate, while the central pushing element bends it from the opposite side. In this implementation, the counter supports remain fixed and the central element is displaced. The suspended metallic bridge on the elastic substrate then elongates by only a small fraction of the macroscopic bending motion. This mechanical reduction allows atomic scale contacts to be formed, opened, and held over long measurement times.

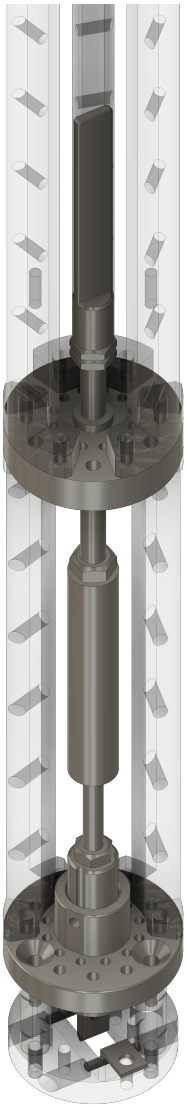


Figure 3.5. Cryogenic MCBJ mechanics with the differential screw, pushing element, counter supports, and sample holder.

The cryogenic mechanics are based on the LD400-compatible design developed by Ansgar Fischer and Martin Prestel [94] and built by the scientific workshop.[2] At the sample stage, a differential screw converts the rotary motion of the drive train into linear motion of the central pushing element. The screw combines an M4 and an M5 thread, so one full revolution advances the bending mechanism only by the difference between the two thread pitches. The resulting macroscopic feed is 0.1 mm per revolution.

I treat the motor position operationally rather than as a universal measure of junction elongation. The motor encoder reports integer values up to about 1.8×10^9, which are not useful for describing contact tuning. The laboratory control software therefore maps the motion to a relative coordinate from 0 - 18. This range corresponds to 30 revolutions of the differential screw and thus to 3 mm of macroscopic screw feed. For an individual opening trace, the motor coordinate can be converted into an approximate relative electrode displacement by

[2] Scientific workshop, 07062016.

fitting the exponential conductance decay in the tunnel regime, as demonstrated in Sec. 4.2.1. I treat this conversion as sample dependent because the effective mechanical reduction can vary with the realized bridge, mounting geometry, and thickness of the polyimide layer. The operational coordinate is therefore used for reproducible opening and closing, while a displacement scale is calibrated from the tunnel data whenever it is required.

Cryogenic Drive Train

The cryogenic drive train must transmit controlled rotary motion from room temperature to the sample space without creating excessive heat load or mechanical stress. The established design meets the thermal requirement by dividing the train into several gearbox stages that are thermalized at the corresponding cryostat stages. Hollow stainless steel tubes connect the gearboxes.[3] The gearboxes transmit the motion and interrupt the thermal path along the shaft.

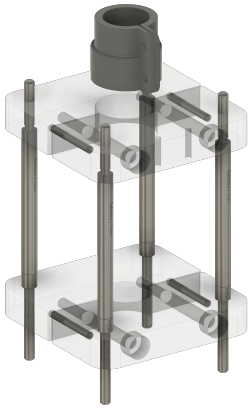


Figure 3.6. Redesigned room-temperature mounts for the motor and vacuum feedthrough.

The existing drive train uses the same 1:3.1 reduction at each of its four commercial spur gearheads.[4] Repeated failures during this work show that this distribution loads the upper cryogenic gearboxes too strongly when a lower stage becomes stiff or blocked. The resulting torque could damage the small gears inside the gearboxes or disengage the coupling to the following shaft. I redistribute the reduction by replacing the gearhead close to the motor with a 1:9.7 reduction and the first two cryogenic gearboxes with 1:5.4 reductions. I leave the remaining 1:3.1 stages unchanged. This configuration reduces the torque applied by the first cryogenic gearbox. It requires more rotations at the 40 K and 4 K stages and therefore increases the heat dissipated there. This additional heat load is acceptable because these stages provide much greater cooling power than the millikelvin stages.

Room-Temperature Drive and Alignment

The room-temperature drive must hold the motor and rotary vacuum feedthrough on a common axis so that the cryogenic shaft is not loaded by lateral motion. The previous mount uses a massive plastic disk with a central bore and a narrow radial

[3] Faulhaber, `401.4659.04 "Ritzel" 2.0x4.35x4.7 Z12 Weld`, welded to stainless steel tubes with 4 mm outer diameter and 0.1 mm wall thickness.

[4] Faulhaber, `22/2K`, 0.1 N m.

slit. Tightening the clamp mainly deforms the plastic body instead of reducing the bore diameter, and the thread cut directly into the plastic is damaged by the required clamping force. The mount therefore does not hold the motor or feedthrough reproducibly. Coaxial alignment of the two components is not possible, which imposes substantial lateral load on the entire drive train.

I replace it with the two-piece mount shown in Fig. 3.6, designed by Oliver Bruttel and me.[5] Separate clamps hold the motor and vacuum feedthrough, while rods connect the two plates and maintain their alignment. A 3D-printed clamp holds the separately ordered motor and gearhead in place instead of glue, allowing either component to be replaced independently. Under excessive load, the printed clamp also provides a sacrificial mechanical weak point that can fail before the motor is damaged. The revised mount is stiffer, can be assembled reproducibly, and improves the maintainability of the drive.

Friction and misalignment at the vacuum feedthrough and slip clutch produce a second source of excessive motor load. I therefore lubricate the rotary feedthrough[6] regularly with vacuum grease[7] to keep its friction low. In the original slip clutch, repeated compression deforms the clamping threads cut into the plastic body and displaces the clutch from the rotation axis. The resulting misalignment contributes to premature motor failure. I replace the plastic threads with brass inserts. This improves the centering and keeps the coupling stable under repeated clamping.

Assembly and Commissioning

Reliable operation also depends on assembly and preparation. I fix the rods inside the cryostat while rotating them slowly so that small angular or lateral offsets can relax before locking the drive train in place. This reduces the risk of residual misalignment that can block motion after cooldown. I strip each gearbox of its original lubricant, clean it overnight in an ultrasonic acetone bath, and rinse it with IPA. Immediately before cooldown, I heat the gearboxes with a hot air gun to remove residual solvent and adsorbed moisture. This preparation prevents lubricant, solvent, or water residues from freezing and blocking the gears at cryogenic temperature.

The drive train cannot be inspected directly at low temperature, so I use the heat generated at the cryostat stages as an operational diagnostic. Mechanical motion dissipates heat at the stages through which it is transmitted. In some tests, motion of the cold mechanism increases the sample stage temperature to about 1 K, illustrating the magnitude of this dissipation. A temperature increase at successive stages therefore indicates that the corresponding parts of the drive train are moving. If this response stops at one stage or changes abruptly relative

[5] Scientific workshop, 13042026.
[6] Pfeiffer Vacuum, DD 016 A, rotary feedthrough, DN 16 ISO-KF.
[7] Syensqo Specialty Polymers, Fomblin YL VAC 25/6.

to earlier operation, the blockage or failure is likely near the following gearbox or coupling. The gas handling system must be monitored during such tests because additional heat at the still can increase the return pressure and may cause overpressure if the motion is too aggressive or repeated too quickly.

I reach the final configuration through iterative commissioning. Each change requires a full cryogenic test, and a failed test usually requires another warm-up, repair or replacement, pump-down, and cooldown. After I revise the gearbox distribution, motor mount, slip clutch, feedthrough lubrication, and assembly procedure, the mechanics operate reproducibly during cooldown and at base temperature. The samples can then be broken and tuned without further mechanical failures. The MCBJ position therefore provides a reproducible conductance control coordinate, while the absolute atomic displacement remains inferred from the measured transport response.

3.2.2 Electrical Measurement Environment

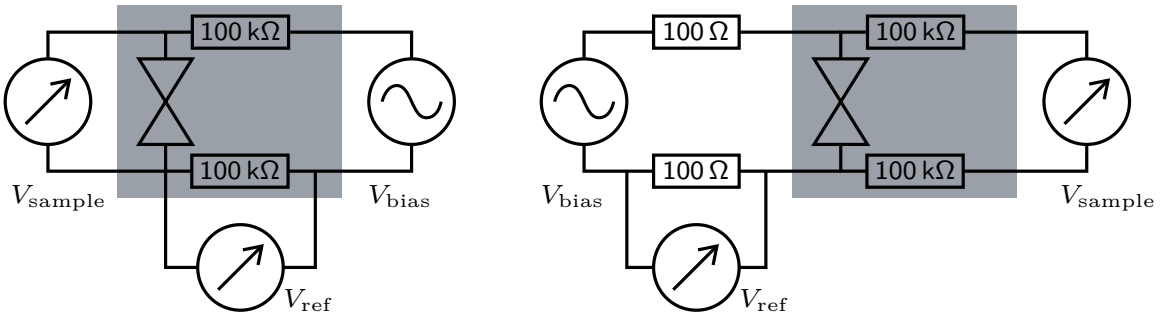
The electrical setup must bias the junction and determine its nonlinear *I*–*V* characteristic while the device resistance changes by orders of magnitude. It must also maintain a low electronic temperature and provide controlled microwave excitation. I first define how the sample voltage and current are obtained from the circuit. I then describe the cryogenic wiring, filtering, microwave coupling, thermometry, and room-temperature instrumentation that establish the measurement conditions.

Measurement Schematic

Figure 3.7 shows the two bias and readout configurations used in this work. In both configurations, the transport circuit is operated in a quasi-dc mode. A programmed voltage V_{bias} is applied to the bias line, while the synchronized voltage drops V_{sample} and V_{ref} are recorded across the sample region and a known reference resistor. The sample current is inferred from V_{ref}. The physical *I*–*V* characteristic is therefore reconstructed from measured voltage drops rather than from the programmed source voltage.

Both configurations use the same cryostat wiring. I select between them by reconfiguring the room-temperature connections and reusing the available internal lines, without warming up the cryostat or modifying the sample wiring. This allows the same junction to be measured during one cooldown as it is tuned between the high-impedance tunnel regime and the low-impedance contact regime.

I use the standard configuration in Fig. 3.7(a) for tunnel junctions and few-channel contacts. Two cold reference resistors are placed symmetrically on either side of the sample. This keeps the sample potential close to the cryostat ground



(a) Standard high-impedance wiring.

(b) Modified low-impedance wiring.

Figure 3.7. Bias and readout configurations used to reconstruct sample voltage and current. The programmed quasi-dc bias V_{bias} is applied through the bias line, while V_{sample} and V_{ref} are recorded as synchronized voltage drops. Opposing triangles denote the sample, and gray shading (■) marks components thermalized inside the cryostat. (a) The standard high-impedance configuration places two cold reference resistors symmetrically around the sample, keeping the current readout in the cold filtered environment. (b) For low sample impedances, two additional warm 100 Ω resistors provide a lower-resistance bias and current readout path. The current is inferred from the voltage drop across one of these resistors.

during a bias sweep and reduces common-mode offsets in the voltage readout. The voltage V_{ref} is measured across one of these resistors. Its independently measured 4-point resistance at base temperature is

$$R_{\mathrm{ref}} = 101.473(1)\,\mathrm{k\Omega}\,. \tag{3.1}$$

The second cold reference resistor contributes to the total series resistance of the bias path, but it is not part of the measured reference voltage drop. The sample current is obtained directly from

$$I_{\mathrm{sample}} = V_{\mathrm{ref}} \,/\, R_{\mathrm{ref}}\,. \tag{3.2}$$

The programmed voltage V_{bias} is not equal to V_{sample} because the cold reference resistors, filters, leads, contacts, and device form a bias divider. In a linearized resistive approximation, the voltage and current would follow

$$V_{\mathrm{sample}} \simeq V_{\mathrm{bias}}\, R_{\mathrm{sample}} \,/\, R_{\mathrm{total}}\,, \tag{3.3}$$

$$I_{\mathrm{sample}} \simeq V_{\mathrm{bias}} \,/\, R_{\mathrm{total}}\,. \tag{3.4}$$

These expressions illustrate the voltage division, but the analysis uses the simultaneously measured values of V_{sample} and V_{ref}. This is essential for nonlinear superconducting transport because the differential resistance changes with bias.

A useful measure of the instantaneous bias condition is

$$\eta = R_{\mathrm{series}} \,/\, R_{\mathrm{sample}} \,, \tag{3.5}$$

where R_{sample} is the instantaneous device resistance and R_{series} is the remaining resistance in the bias path,

$$R_{\mathrm{series}} \simeq 2\, R_{\mathrm{ref}} + R_{\mathrm{filters}} + R_{\mathrm{leads}} + \dots \,. \tag{3.6}$$

In the limit $\eta \ll 1$, most of the applied voltage drops across the sample and the circuit approaches voltage bias. In the opposite limit, $\eta \gg 1$, the series resistance dominates and the circuit approaches current bias. The circuit moves between these limits as the MCBJ is tuned. A closed junction can have a resistance of order 100 Ω, a single atom contact has a resistance of order 12.9 kΩ, and the tunnel regime is characterized by resistances in the range 100 - 500 kΩ. Because the standard filtered circuit contains about 200 kΩ from the two cold reference resistors, its effective bias condition changes with the junction state.

For low sample resistances, the standard configuration directs most of the programmed bias across the cold reference resistors. The accessible sample voltage range can then become too small to leave the zero-voltage supercurrent branch or reach spectroscopic features such as the superconducting gap. This regime requires a lower series resistance rather than a different reconstruction principle.

I therefore use the modified configuration in Fig. 3.7(b) for low-impedance measurements. Two additional warm resistors with nominal resistances of 100 Ω provide the lower-resistance bias path. The voltage drop across one of these resistors provides the current readout. The Johnson–Nyquist voltage noise density of a resistor is [95, 96]

$$S_V^{1/2} = \sqrt{4 k_{\mathrm{B}} T R} \,. \tag{3.7}$$

Although the additional resistor is warmer, its resistance is about three orders of magnitude smaller than that of the standard reference resistor. Its voltage noise density therefore remains of the same order. The smaller resistance also produces a correspondingly smaller reference voltage for a given current. For a fixed readout voltage range, this increases the measurable current range by approximately three orders of magnitude. The tradeoff is reduced current sensitivity. I use this configuration when the higher sample voltage and current ranges are required.

Both configurations therefore produce the same physical observables, the measured sample voltage and the current inferred from a reference voltage drop. Switching the room-temperature connections sets the available sample voltage range, the source impedance, and the associated noise environment while preserving the junction state and the internal cryostat wiring. The two configurations thus provide access to opposing impedance regimes of the same sample within one cooldown.

Cryogenic Measurement Lines

The cryostat wiring has to satisfy two requirements at the same time. It must define a low-noise electrical environment for high-impedance tunnel devices and few-channel atomic contacts, and it must not carry excessive heat from room temperature to the millikelvin stage. The first requirement motivates symmetric wiring, cold reference resistors, shielding, and filtering of external high-frequency noise. The second requirement motivates low-thermal-conductance cables and careful thermal anchoring. The wiring concept follows earlier MCBJ and single-electron-transistor setups in the group [85, 94], but I change several parts for this experiment.

Figure 3.8 shows the custom wiring used near the sample. The cryostat contains separate cable systems for low-frequency signals and microwave excitation. The low-frequency system carries the sample voltage and current signals, applies the bias and dc gate voltage, reads the thermometers, and supplies the heater power. The ac gate and antenna lines are high-frequency coaxial cables, described below. The dc gate uses two physical lines that are shorted before the thin copper wires connect to the sample. This arrangement provides diagnostic and backup access while the sample sees a single gate terminal. From room temperature to the base temperature region, the low-frequency lines use commercial Bluefors twisted-pair DC wiring, in which the pairs are combined in round protective bundles, in contrast to earlier implementations [47, 94]. At the base temperature stage, they are continued with double-twisted coaxial cable[8] with brass inner and outer conductors.

The base temperature cables are soldered to pin headers. From there, 90 µm insulated copper wires connect to the sample pads through conductive silver paste. This short copper section gives a low contact resistance and provides a local thermal anchor close to the device. The voltage and current leads define a four-terminal measurement with respect to the cryostat wiring. They do not form an ideal Kelvin measurement of the atomic contact alone because the local copper wires, silver paste contacts, and sample pads remain inside the measured sample region. For the high-impedance devices considered here, this residual series resistance is negligible compared with the device resistance.

Low-Frequency Filtering

The dc wiring carries the slowly varying bias and gate voltages as well as the voltage signals used to determine sample current and voltage. Its task is not only to pass low-frequency signals. It must also suppress high-frequency radiation that would heat the junction or create photon assisted features unrelated to the

[8] GVL Cryoengineering, GVLZ 169, connected with two-pole LEMO, FGG.00.302.CLAD35 plugs and EGG.00.302.CLL sockets. Copper tubes with 2 mm outer diameter and 0.45 mm wall thickness are used as bend protection.

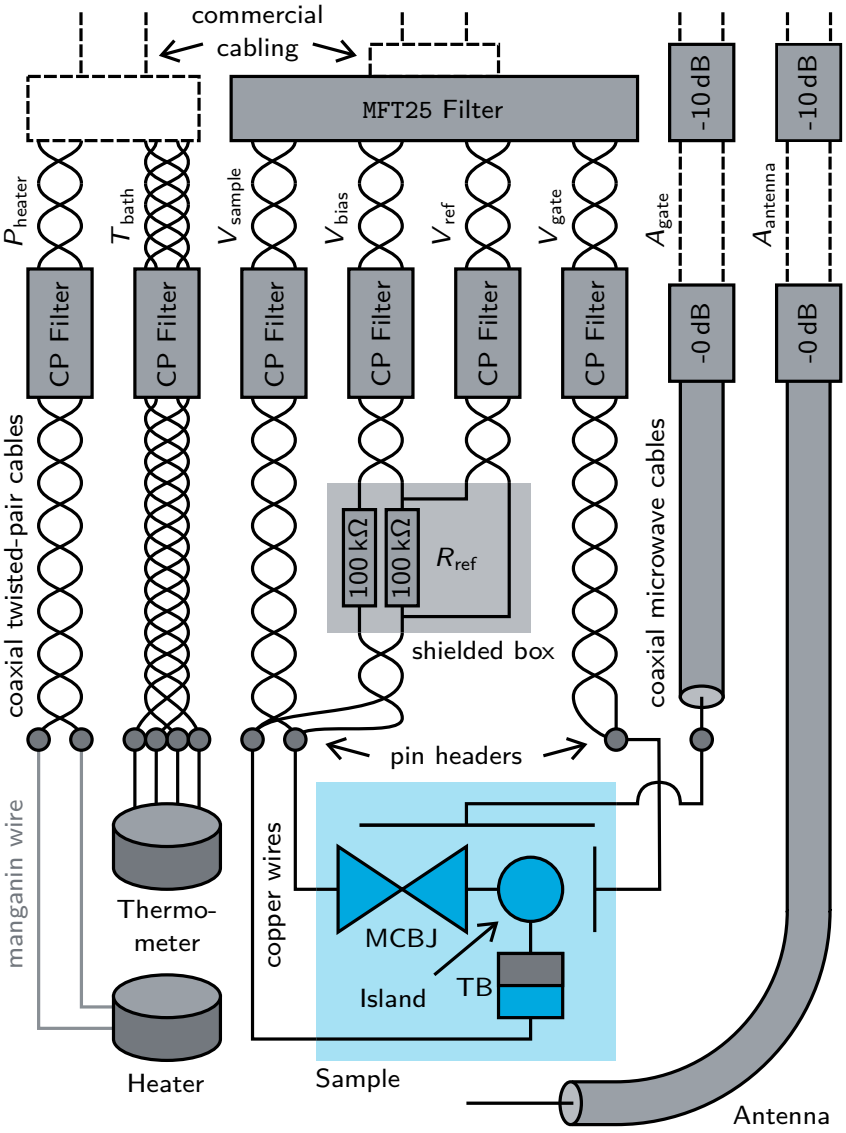


Figure 3.8. Custom cryostat cabling at the sample stage. All components shown are mounted at the base temperature stage, except for the microwave attenuators at the 4 K stage. The island is coupled to the MCBJ (left), the tunnel barrier (bottom), the dc gate (right), and the ac gate (top). Thin copper wires connect the sample pads to pin headers, which are connected either to low-frequency coaxial twisted pair cables or to high-frequency coaxial cables. The cold reference resistors R_{ref} are mounted in an additionally shielded box. Copper powder (CP) filters and the commercial MFT25 filter suppress high-frequency leakage and thermalize the lines.

intended microwave drive. I therefore design the filter stack as both an electrical low-noise environment and a thermalization stage.

The low-frequency lines are filtered and thermalized in several stages. Four copper powder filters built in our group are mounted at the base temperature stage, one each for the voltage, bias, current, and dc gate lines. They follow the filter concept described in Ref. [97]. In this filter type, a copper signal wire is embedded in compacted copper powder inside a gold plated copper housing. The filter has low resistance at dc and low frequency but becomes lossy at high frequency. The relevant mechanism is the skin effect. High-frequency currents are confined closer to the surface of the signal wire, while the surrounding copper powder provides a dissipative electromagnetic environment. The filter therefore attenuates signals over a broad frequency range and along its full length rather than above one well-defined RC cutoff frequency.

The copper powder filters act as a broadband protection stage against high-frequency leakage. At the same time, the large metal–metal contact area between wire, powder, and housing provides an efficient thermal anchor for the electronic degrees of freedom in the line. Their role in the present stack is to add high-frequency loss without adding a large dc resistance to the measurement circuit.

I use a commercial microwave filter–thermalizer[9] (`MFT25`) in series with the copper powder filters. According to the manufacturer, the `MFT25` embeds the signal conductors in a highly conductive silver matrix that acts as a heat exchanger, and the compact package contains 25 filter lines in a gold plated non-magnetic copper box. The manufacturer specifies strong microwave attenuation above 130 MHz together with low capacitance and good thermal contact to the millikelvin stage [98]. In the present setup, the `MFT25` provides the compact commercial filter–thermalizer stage, while the copper powder filters add additional broadband high-frequency loss.

The `MFT25` also improves the practical reliability of the dc wiring. Earlier versions of the setup use fragile steel capillary wiring, similar to the configuration described in Ref. [94]. These lines could break during assembly or thermal cycling, and inherited copper powder filters fail intermittently. The `MFT25` replaces most of this capillary filter chain on the transport and dc gate lines while adding a well thermalized commercial filtering stage. The result is a mechanically more robust wiring system that is better suited to maintaining a low electronic temperature at the sample.

The reference resistors are mounted in the shielded base temperature region close to the sample. In an earlier configuration, they are located at the still stage. Moving them to the base temperature stage reduces their Johnson–Nyquist voltage noise by lowering the resistor temperature while keeping the current readout in the same cold filtered environment as the sample [95, 96]. I also make the design mechanically more robust. An earlier through-hole metal oxide resistor develops intermittent contacts, while the revised design uses an

[9] Basel Precision Instruments, `MFT25-1000hm SN0219`. The unit is connected through a Glenair, `25-way MWDM Micro-D` panel mount receptacle.

SMD resistor[10] on a small chip. This provides better thermal contact and a less fragile electrical connection. The reference resistor region is additionally shielded, as indicated in Fig. 3.8, to reduce capacitive pickup from neighboring lines. Before cooldown, I check the wiring by measuring the resistance matrix between the relevant ports and by verifying the ground connections of the lines.

I therefore optimize the filtering stack for low electronic temperature and robust wiring. The MFT25 and the revised reference resistor design reduce electronic heating and measurement noise. This is essential for resolving sharp superconducting features in the *dI–dV* spectra.

Microwave Cabling

In addition to the dc bias and gate lines, the setup contains two microwave paths for excitation. I refer to them as the ac gate line and the antenna line. The ac gate line terminates on the on-chip ac gate, which is capacitively coupled to the sample. The antenna line is a stripped coaxial cable placed close to the sample. These names describe the hardware geometry, not a calibrated lumped circuit. At GHz frequencies, both paths can launch local electromagnetic fields that couple to the device region.

Both microwave paths are routed as coaxial lines. The commercial microwave cabling inside the cryostat uses superconducting cables to keep the transmission high. At each temperature stage, the microwave line passes through a connector that also provides a thermalization point. The cryostat is initially delivered with attenuators installed at these positions. I replace them with pass-through connectors[11] except at the 4 K flange. The remaining -10 dB attenuators at 4 K provide a defined thermalization point while placing the dissipated power at a stage with comparatively large cooling power. For future measurements, moving the attenuation closer to the mixing chamber would likely be preferable because the electronic temperature at the device is more important than maximizing the available cooling power.

Patrick Raif and I install the base temperature part of the microwave cabling[12] [47]. Frequencies up to about 20 GHz are transmitted reliably in the installed configuration, as inferred from microwave induced changes in the *dI–dV* spectra. The network analyzer specifies its output as a power $P_{\mathrm{out,dBm}}$. For the amplitude sweeps discussed later, I convert this setting to the nominal source amplitude A_{out} using the matched load convention

$$P_{\mathrm{out,W}} = 10^{P_{\mathrm{out,dBm}}/10-3}, \qquad A_{\mathrm{out}} = \sqrt{2Z_0 P_{\mathrm{out}}}, \qquad Z_0 = 50\,\Omega. \tag{3.8}$$

[10] Scientific workshop, 2481023. Vishay/Draloric, TNPU1206100KBZEN00.
[11] Amphenol SV Microwave, SF1597-6003.
[12] Elspec, DA50047-821, 50 Ω, CuAg/PTFE/Cu, 1.19 mm. Rosenberger, 02S141-270L5.

Here A_{out} is the nominal peak voltage amplitude[13]. The maximum applied setting of 10 dBm therefore corresponds to 10 mW and 1 V. Equation (3.8) assumes a matched 50 Ω load and defines only a source side coordinate. It does not account for attenuation, reflections, standing waves, or antenna coupling and therefore does not give the microwave amplitude at the junction [99]. The ac gate and antenna paths have different relative transmissions as functions of frequency, providing access to different effective power windows for different microwave tones.

Patrick Raif and I show that both paths act in practice like antennas for the device [47]. Part of the microwave power is launched near the pin headers. From the perspective of the sample, the two paths therefore behave more like two different antennas than like two sharply separated drive mechanisms. The coupling is not impedance matched, and the break junction architecture is not designed as a calibrated microwave transmission structure. Reflections, standing waves, and antenna coupling consequently produce an unknown transfer function between the source setting and the local voltage at the junction. This transfer function depends on frequency, coupling path, and device configuration and cannot be determined once as a fixed setup calibration. A matched calibration would require a substantially different architecture, closer to the dedicated microwave design used by Bretheau *et al.* [100].

I therefore use the applied source power only as a reproducible control coordinate. Whenever the local amplitude is required quantitatively, I determine an effective microwave voltage from an internal reference in the transport data, in particular from photon assisted tunneling as described in Sec. 2.1.3. I perform this calibration separately for the corresponding device, frequency, and coupling path in the results sections. It incorporates the complete unknown microwave coupling chain into a data derived effective amplitude rather than assuming a universal conversion from source power to junction voltage.

[13] The corresponding root-mean-square voltage is $V_{\mathrm{out,rms}} = A_{\mathrm{out}}/\sqrt{2} = \sqrt{Z_0 P_{\mathrm{out}}}$.

Thermometry and Heating

The sample space is equipped with dedicated thermometer and heater wiring to monitor and control the local sample holder temperature. These lines use a dedicated Bluefors twisted-pair DC bundle and a separate breakout box rather than the bundle used for transport and gate signals.[14] They are filtered by copper powder filters. This separation keeps heater and thermometer signals away from the sensitive transport circuit, where thermometer readout and heater operation could otherwise interfere with low-current measurements. The thermometer arrangement follows the earlier MCBJ setup used by Martin Prestel [94]. In the present setup, I screw a ruthenium oxide resistor thermometer[15] to the underside of the lower plate shown at the bottom of Fig. 3.5, as close as possible to the two counter supports.

The heater is a local resistive heater made from thin insulated manganin wire with a resistance of about 100 Ω. I wrap it around one of the posts that connect the sample holder to the mechanical break junction assembly. This geometry heats the sample holder locally while remaining mechanically simple and robust. I use the heater for controlled temperature dependent measurements.

The resistance thermometer measures its own temperature at its mounting position on the sample holder. Because it is thermally anchored to the holder, I use its calibrated reading as the local bath temperature T_{bath}. Finite thermal resistance along the pushing rod, stamp, and counter rods can produce temperature gradients between the heater, thermometer, sample holder, and junction. The thermometer therefore does not directly determine the effective electronic temperature T_{fit} inferred from junction transport. The junction electrons can be warmer than the thermometer because of imperfect line thermalization, residual high-frequency radiation, dissipated bias power, or microwave induced heating. This distinction is particularly important under microwave irradiation. A change in the thermometer reading can indicate heating of the sample holder, but it can also include a direct microwave induced readout artifact.

The thermometer and heater lines are filtered and thermalized for the same reasons as the transport lines. They connect room temperature electronics to the millikelvin stage and can therefore act as thermal and electrical noise channels. Their voltage noise requirements are less restrictive than those of the transport lines, but they still have to be well anchored to avoid parasitic heating.

I calibrate the heating protocol empirically by recording the steady sample holder temperature T_{bath} for a series of applied heater powers P_{heater}. Here, T_{bath} denotes the temperature obtained from the factory calibrated resistance curve of the ruthenium oxide thermometer. It is therefore the local thermometer reading used by the control software and by the PID loop. The heater power denotes the electrical power dissipated in the manganin heater. Figure 3.9 shows the resulting relation for the temperature sweep used in this thesis.

[14] The heater connection uses a LEMO, `FGG.0S.302.ZLAT`, plug.

[15] Lake Shore Cryotronics Inc., `RX102B-CB`, serial number `U06030`, curve name `MCBJ Sample ORI`.

Because the thermal conductance between the sample holder and the mixing chamber increases with temperature, equally spaced power setpoints do not produce equally spaced temperatures. I therefore select the heater setpoints from the empirical relation

$$P_{\mathrm{heater}} \approx 0.11\,\mu\mathrm{W\,K^{-2.5}} \cdot T_{\mathrm{bath}}^{2.5}\,. \tag{3.9}$$

This choice produces an approximately uniform temperature axis for repeated *I*–*V* sweeps. Equation (3.9) is a calibration of the applied setpoints for this assembly, not a microscopic model of its thermal conductance. It can change after remounting the sample holder or modifying the thermal links and must then be measured again.

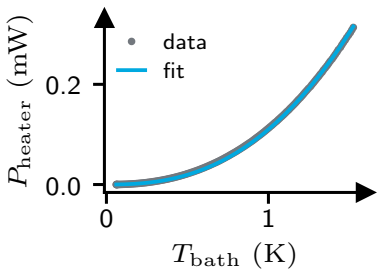


Figure 3.9. Applied heater power P_{heater} as a function of the sample holder temperature T_{bath}. The line shows the empirical power law used to choose the heater setpoints.

The calibrated thermometer reading estimates the local bath temperature, while controlled heating and temperature dependent *I*–*V* curves provide consistency checks on the electronic response. Together, the cold reference resistors, thermalized wiring, and microwave lines define the current readout, limit noise and heating, and provide reproducible irradiation without an absolute calibration.

3.2.3 Instrumentation and Grounding

Figure 3.10 summarizes the room-temperature instruments and their grounding connections. This part of the setup has two tasks. It has to provide synchronized voltage readout and bias control, and it has to avoid additional ground paths that bypass the filtered cryostat wiring. I treat the cryostat (Z) as the local ground reference because its large metallic body provides a stable electrical potential. The working rule is to connect each instrument to this reference only through its intended signal path or through a deliberately isolated control link. I avoid uncontrolled parallel connections between building ground, computer interfaces, instrument shields, and the cryostat body.

Instrument control is part of the grounding task because control cables can introduce additional shield and ground paths. I connect the control computer (H) to the laboratory network and interface GPIB and RS-232 instruments through USB adapters[16]. USB connections can introduce additional shield and ground paths, so I isolate them either by USB-over-LAN links[17] or by galvanic USB isolators[18]. I include the cryostat control unit (C)[19] in this isolated control chain. I use LAN connected instruments where available because their network interfaces

[16] National Instruments, `GPIB-USB-HS`, used for GPIB access.
[17] Icron, `USB 2.0 Ranger 2304`.
[18] CESYS, `C028149`.
[19] `Bluefors LD400`, serial number `BF2015-05`, connected by USB.

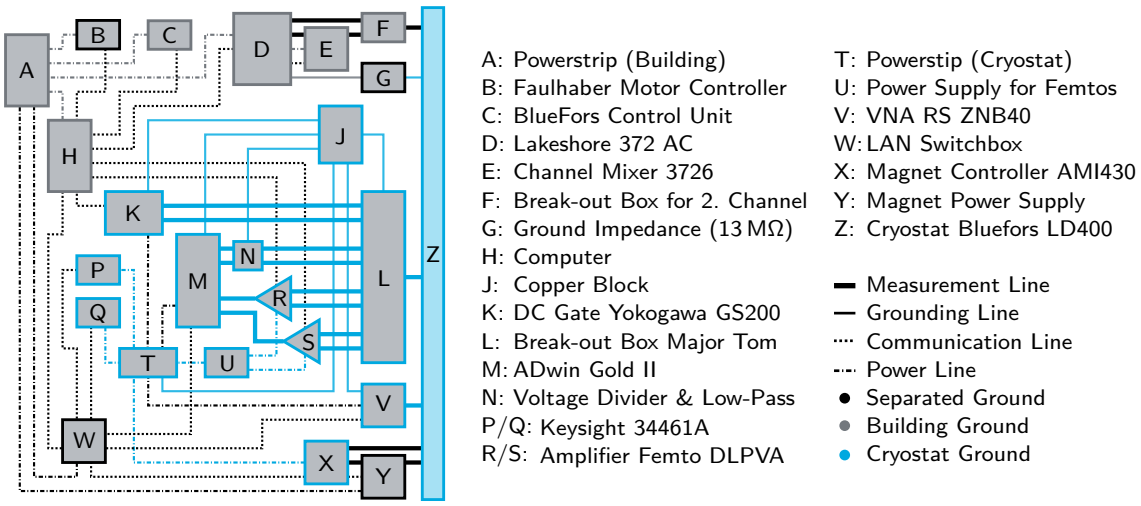


Figure 3.10. Grounding and instrument schematic. Building ground and cryostat ground are separated except for a 13 MΩ connection (G) through the temperature controller. The motor controller (B) and LAN switch (W) are powered by two-wire power supplies without a protective earth conductor. Operating the magnet power supply (Y) without protective earth minimizes noise in the I–V traces, but creates a potential touch voltage hazard and must not be reproduced.

provide galvanic isolation and therefore avoid direct ground paths through control cables to the measurement computer. I power the motor controller (B) and LAN switch (W)[20] with two-wire power supplies without a protective earth conductor, as indicated in Fig. 3.10.

The voltage signals are first amplified by voltage preamplifiers (R/S), supplied by their shared power supply (U).[21] These amplifiers provide selectable gains and a low-pass bandwidth of about 100 kHz. Their ground referenced output voltages are digitized by the analog inputs of the measurement card (M).[22] Although the card can sample at rates up to 100 kHz, the quasi-dc sweeps used here do not require this bandwidth. I therefore reduce the data stream on the card to 137 Hz.

The acquisition strategy is to store synchronized voltage traces without additional processing and to postpone filtering, smoothing, and gridding to offline evaluation. The measurement card writes the digitized traces directly to the measurement file, preserving the raw data for later correction and reanalysis.

The bias voltage is generated by an analog output of the same measurement card. Before entering the cryostat wiring, it passes through a warm symmetric voltage divider and a low-pass filter (N).[23] The divider reduces the full output range of the card to the voltage window relevant for the sample and improves the

[20] D-Link, `DGS-108`.
[21] Femto, `DLPVA`, preamplifier. Scientific workshop, `WW-98000627`, power supply. Femto, `Luci10`, USB control interface.
[22] Jäger Computergesteuerte Messtechnik GmbH, `ADwin Gold II`, connected by LAN.
[23] Thorlabs Inc., `EF120`.

effective bias resolution. I choose its resistances low enough that their thermal noise and their contribution to the total source impedance remain limited.

I keep the divider at room temperature, in contrast to the cold shunt used in Ref. [94]. This preserves access to filter, wiring, and grounding diagnostics during cooldown and low-temperature operation, following the practical motivation of Ref. [85]. A shunt mounted inside the cryostat would have fixed part of the bias environment locally, but it would also have made several grounding and continuity checks inaccessible during a cooldown.

I test several combinations of available voltage sources, amplifiers, and multimeters before settling on the configuration shown in Fig. 3.10. The practical optimization criterion is the low-frequency noise in the sample voltage and reference voltage channels under the same filtering and grounding conditions used for transport measurements. The relevant observable is not only electrical continuity but also the stability and cleanliness of the measured I–V traces.

Using one measurement card for both voltage channels ensures that the sample voltage and reference voltage readings are acquired as synchronized pairs. During a typical measurement, I apply the bias symmetrically around ground so that the common mode voltage of the sample remains close to zero. I ramp the voltage slowly down and up, with full periods of 5 min or longer.

The remaining instruments provide gate control, microwave excitation, temperature control, mechanical actuation, magnetic field control, and diagnostics. The dc gate voltage is applied with respect to ground by a high-precision voltage source (K).[24] All dc lines are routed through a breakout box (L)[25] into the commercial cryostat cabling. The microwave excitation paths are driven by the two ports of a vector network analyzer (V)[26], which are connected to the ac gate and antenna lines.

Temperature monitoring and local heating are handled by an ac resistance bridge (D) and channel mixer (E).[27] One channel continuously reads the sample thermometer, while the second channel and mixer read the auxiliary thermometers through the separate thermometry cable tree. The bridge also controls the local heater. I replace the previous bridge with this two-channel unit so that the sample thermometer remains available continuously during auxiliary readout. This requires an additional breakout box (F)[28] to separate the dedicated sample thermometer line from the auxiliary thermometry lines. The continuous reading enables proportional–integral–derivative (PID) control of the local bath temperature at a chosen setpoint.

Mechanical actuation of the break junction is provided by an encoded motor and driver (B).[29] A one-axis 7 T magnet system, comprising a controller (X)

[24] Yokogawa, GS200, connected by GPIB.
[25] Scientific workshop, WW-2280090, “Major Tom”.
[26] Rohde & Schwarz, ZNB40, connected by LAN.
[27] Lake Shore Cryotronics Inc., 372AC and 3726, connected by USB.
[28] Scientific workshop, WW-2480259, thermometry breakout box.
[29] Faulhaber, Motor 2232R024S, Encoder R IE2-1024 4613, Driver MCDC3002S RS, Gear Head 22/2S, connected by RS-232.

and a separate high-current power supply (Y), is used when superconductivity has to be suppressed.[30] Additional diagnostic access is provided by a precision source and digital multimeters (P/Q).[31] I connect these diagnostic instruments only when their information is needed because each additional instrument could introduce another pickup or ground path.

For the lowest-noise I–V traces, I disconnect devices that are not required during the actual sweep. In particular, I usually disconnect both the magnet controller and its high-current power supply when not in use. The power supply can couple low-frequency pickup or ground offsets into the cryostat environment even when the magnetic field is constant. I remove the communication cables of the preamplifiers whenever remote gain control is not needed. This reduces the number of possible ground loop and pickup paths connected to the sensitive voltage readout circuit.

I develop the final grounding configuration empirically for this cryostat, instrument rack, and wiring. It preserves a well-defined cryostat reference, minimizes pickup in the voltage channels, and documents the wiring and isolation choices that produce stable I–V traces.

Together, acquisition synchronization, bias range and resolution, and low-frequency pickup determine the noise floor of the reconstructed I–V traces.

[30] American Magnetics Inc., `AMI430`, connected by LAN.
[31] Keysight, `B2962A`, connected by LAN. Keysight, `34461A`, connected by LAN.

3.3 Digital Infrastructure

The transport measurements in this thesis depend on more than the electrical circuit described above. Each I–V trace also has to be associated unambiguously with the state of the surrounding experiment: amplifier gains, reference resistance, bias waveform, temperature, microwave settings, magnetic field, gate voltage, and the mechanical position of the break junction. Without this context, a recorded voltage trace would not be a reproducible physical measurement, but only a time series from the acquisition card. A central design goal is therefore to avoid relying on undocumented manual instrument settings. Before a measurement, the scripts initialize the controllable devices, apply the relevant settings digitally, and record their state. The measurement protocol thus defines the conditions explicitly. A new measurement does not silently inherit an undocumented setting from an earlier measurement or manual intervention.

The digital infrastructure is therefore treated as part of the experimental apparatus. Its role is to connect instrument control, synchronized acquisition, metadata logging, and data evaluation into one traceable path from raw voltages to calibrated I–V curves, derived dI–dV spectra, and fitted model parameters. This section first describes the measurement architecture used to acquire and store the data. It then describes the evaluation layer that converts raw measurement records into the common data products used by the different analysis schemes.

3.3.1 Digital Measurement Architecture

The measurements in this work are controlled by a layered Python infrastructure. The first layer, the Python package `p5control`, provides the general abstraction for instrument access, server based control, and structured data storage. The second layer, `p5control-bluefors`, adapts this framework to the Bluefors break junction setup by defining the relevant device drivers, the calculated diagnostic channels and spectroscopy protocols. Together, these layers connect the physical instruments described in Sec. 3.2.3 to the raw voltage traces and metadata used in the subsequent data evaluation.

`p5control`: Instrument Abstraction and Data Acquisition

The experiment requires a control layer that can handle many slow instruments during quasi-dc spectroscopy without turning each measurement script into a collection of device specific commands. The task of this layer is to keep instrument control, data acquisition, and metadata logging in one reproducible structure.

The general framework for this task is the Python package `p5control`, which is developed by Florian Kayatz during his student assistant work under my su-

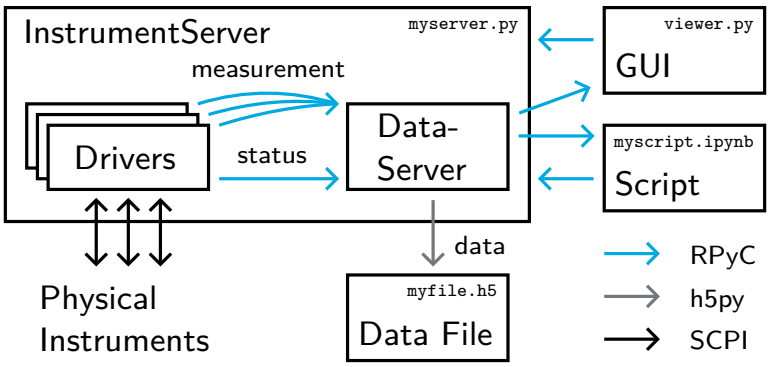


Figure 3.11. Architecture of the `p5control` measurement framework. Device specific drivers connect physical instruments to an `InstrumentServer`. A separate `DataServer` writes status and measurement data to the measurement file. Scripts and graphical user interfaces communicate with both servers.

pervision [101]. The central idea of the framework is to separate device specific communication from measurement scripts and graphical user interfaces. Physical instruments are represented by drivers. These drivers expose the relevant control parameters and readout values in a common form, while the details of LAN, GPIB, RS-232, USB, or device specific command syntax remain hidden inside the driver implementation.

Figure 3.11 shows the resulting architecture. The `InstrumentServer` and `DataServer` separate instrument control from file writing, while measurement scripts and graphical user interfaces communicate with both servers through RPyC [102]. This makes the same instrument layer available for interactive control widgets and scripted measurement protocols, and it keeps the file writing logic independent of individual measurement scripts.

During operation, `p5control` distinguishes between status data and measurement data. Status data are recorded continuously. They document the slowly changing state of the setup, for example temperatures, motor position, magnetic field, microwave settings, amplifier gains, or output states. Measurement data are written when a measurement is started. They contain the time traces acquired for a specific sweep or protocol. This distinction is important because the physical context of an I–V trace can change on a slower timescale than the trace itself.

The measurement file uses HDF5, a hierarchical data format accessed here through `h5py`, as sketched in Fig. 3.12 [103, 104]. It contains both the raw traces used for later evaluation and the surrounding instrument state needed to interpret them. It therefore acts as both a measurement record and a digital laboratory notebook.

This framework coordinates slow quasi-dc measurement sequences and stores the resulting data streams in a common time indexed structure. Individual sweeps last several minutes, so strict timing guarantees and specialized synchronization mechanisms would add substantial complexity without improving the acquisition.

```
measurement/mpath0/dev0
                    dev1
          mpath1/dev0
                    dev1
      status/dev0
             dev1
```

(a) `myfile.h5`

t	x1	x2
0.00	0.12	3.45
0.01	1.23	4.56
0.02	2.34	5.67
0.03	3.45	6.78
...	...	...

(b) `.../mpath0/dev0`

t	a	b
0.0	0.5	0.1
1.0	1.0	0.1
2.0	1.0	0.2
3.0	0.0	0.2
...	...	...

(c) `status/dev0`

Figure 3.12. Structure of the measurement file used by `p5control`. The `measurement` branch stores sweep specific datasets, while the `status` branch stores continuously logged setup quantities. Both are written as time indexed tables.

Each recorded trace can therefore be related to the continuously logged status data that describe the measurement conditions. This provides the traceability required for later reconstruction and comparison of the *I*–*V* curves.

`p5control-bluefors`: Setup Specific Control Layer

The generic `p5control` layer provides the server structure, data storage, and driver abstraction, but it does not define the measurement logic of the Bluefors break junction setup. This setup requires a spectroscopy oriented control layer that knows which instruments belong to one transport measurement and which quantities have to be stored with each *I*–*V* trace. I implement this setup specific layer in the Python package `p5control-bluefors`, including device specific drivers, graphical control widgets, calculated transport channels, and the measurement protocols used for the data in this thesis [105].

This layer represents each controlled object using the same practical model, summarized in Fig. 3.13: parameters, values, and state. Parameters are quantities set by the user or by a measurement script, for example bias amplitude, amplifier gain, microwave source settings, heater power, magnetic field target, or motor displacement. Values are quantities read back from instruments or calculated from measured channels. State records whether a device is actively sweeping, irradiating, heating, ramping, moving, outputting, or reporting a diagnostic condition such as amplifier overload. This model keeps slow control variables and safety relevant state changes in the same status log as the measured *I*–*V* traces.

The primary measurement record consists of the synchronized voltage traces recorded by the acquisition card and the instrument state stored around the sweep. Together with the selected Femto gains and the reference resistance, these raw voltage traces define the later reconstruction of V_{sample} and I_{sample}. In `p5control-bluefors`, the same conversion is also implemented in the `Calc` driver as a diagnostic layer for live monitoring. This layer is useful during setup work and interactive checks, but it is usually inactive during actual measure-

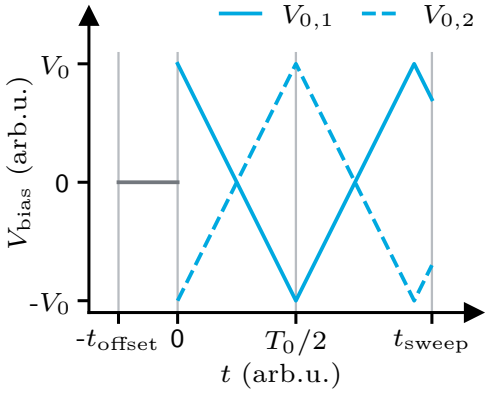
driver	parameters	values	state
ADwin	V_0, T_0, f_0	V_1, V_2, i_0	Output, Sweeping
Femtos	A_1, A_2		Overload (2)
VNA	A_{out}, ν	S_{11}, S_{21}	Irradiating
BlueFors	P_{heater}	T_{bath}, $(T,p)_{\mathrm{aux}}$	Heating
Yoko	V_{gate}	I_{leak}	Output
Magnet	$\mu_0 H$, $\frac{\partial}{\partial t}\mu_0 H$	$\mu_0 H$	Ramping
Motor	Δx, $\frac{\partial}{\partial t}\Delta x$	Δx	Moving
Calc	R_{ref}	$(I,V,G,R)_{\mathrm{sample}}$ $(V_1,V_2)_{\mathrm{offset}}$	Calculating

Figure 3.13. Setup specific driver model used by `p5control-bluefors`. Each driver exposes set parameters, read or calculated values, and state variables. Parameters and states are logged as status data, while measurement values are stored with the corresponding *I*–*V* trace. Gray entries mark diagnostic quantities.

ments. The *I*–*V* curves used for analysis are derived separately from the stored raw voltage traces, as described in Sec. 3.3.2. The remaining drivers provide the physical context of each trace: microwave amplitude and frequency, sample temperature, displacement of the break junction, magnetic field, and gate voltage. In this sense, `p5control-bluefors` connects the electrical reconstruction described in Sec. 3.2.2 to the HDF5 data structure shown in Fig. 3.12.

This connection is enforced by the measurement protocol shown in Fig. 3.14. A single *I*–*V* trace is more than a voltage ramp. The relevant gains, reference resistance, bias waveform, timing parameters, and instrument states have to be fixed before the sweep, and the synchronized voltage traces have to be stored together with these settings. The script `single_iv.py` therefore defines one complete trace. It sets the sweep parameters, waits during an offset interval, records the sweep, and writes the acquired data and metadata to the measurement file.

Parameter dependent measurements are built by repeating this elementary *I*–*V* trace. The script `iv_script.py` steps through an external parameter axis, for example microwave amplitude, frequency, sample temperature, break junction displacement, magnetic field, or gate voltage. At each point, it starts one `single_iv.py` measurement and stores the result in the corresponding measurement path. This structure makes the resulting series reproducible because each *I*–*V* trace is linked to the raw voltage traces and to the status of the surrounding setup at the time of acquisition.



(a) Sequence of one *I*–*V* measurement

`single_iv.py`

(V_0, T_0):	sweep (amplitude, period)
$(t_{\mathrm{offset}}, t_{\mathrm{sweep}})$:	time of (offset, sweep)
(A_1, A_2):	channel gains
f_0:	sampling rate

`iv_script.py`

(A_{out}, ν):	microwave irradiation
T_{bath}:	bath temperature
Δx:	sample displacement
$\mu_0 H$:	magnetic field
V_{gate}:	gate voltage

(b) Settings of measurement sequences

Figure 3.14. Protocol for quasi-dc *I*–*V* spectroscopy. (a) One *I*–*V* trace consists of an offset interval followed by a controlled bias sweep. (b) `single_iv.py` defines the sweep settings, gains, and sampling rate. `iv_script.py` repeats this trace over an external parameter axis.

3.3.2 Data Analysis with `superconductivity`

The raw measurement files produced by the measurement stack contain synchronized instrument outputs and setup metadata, but they are not yet the physical data products used in the analysis. This separation is intentional. The measurement files store the minimally processed useful record, while offset correction, time domain filtering, gridding, and smoothing are applied later with explicit parameters. I first implement many of the basic evaluation ideas in the setup specific repository `p5control-bluefors-evaluation` [106]. For this thesis, I then develop the broader Python package `superconductivity` [107] as the evaluation layer between acquisition and interpretation. In addition to trace evaluation, `superconductivity` contains numerical models for BCS and MAR spectra, fitting routines, a graphical user interface, and a visualization pipeline. The graphical interface and visualization tools are useful for daily work but are not discussed in detail here. This section focuses on the reproducible data path from raw voltage traces to calibrated *I*–*V* curves, derived *dI*–*dV* spectra, uniformly sampled grids, and fitted model parameters.

Data Evaluation

The evaluation converts each recorded measurement path into a calibrated *I*–*V* curve in six steps. It selects one sweep branch, converts the recorded channels to sample voltage and current, corrects their offsets, filters the resulting time trace, resamples it, and finally bins it onto a regular voltage or current grid. This order is important because the low-pass filter operates on the ordered time

trace. Binning removes that time coordinate and is therefore performed only after filtering.

The acquisition card records the synchronized channels V_1, V_2, and i_0 on a common time base. The channels V_1 and V_2 contain the amplified room temperature readout voltages. The trigger channel i_0 identifies the offset interval and the upward and downward sweep branches. I use this channel to select one monotonic branch without inferring its boundaries from the noisy voltage ramp. Unless stated otherwise, the curves shown in this thesis use the downward branch.

I next convert the selected branch to sample voltage and current. With voltage gains A_1 and A_2, reference resistance R_{ref}, and channel offsets $V_{1,\mathrm{off}}$ and $V_{2,\mathrm{off}}$, these quantities are

$$\begin{aligned} V(t) &= \frac{V_1(t) - V_{1,\mathrm{off}}}{A_1} , \\ I(t) &= \frac{V_2(t) - V_{2,\mathrm{off}}}{A_2 R_{\mathrm{ref}}} . \end{aligned} \tag{3.10}$$

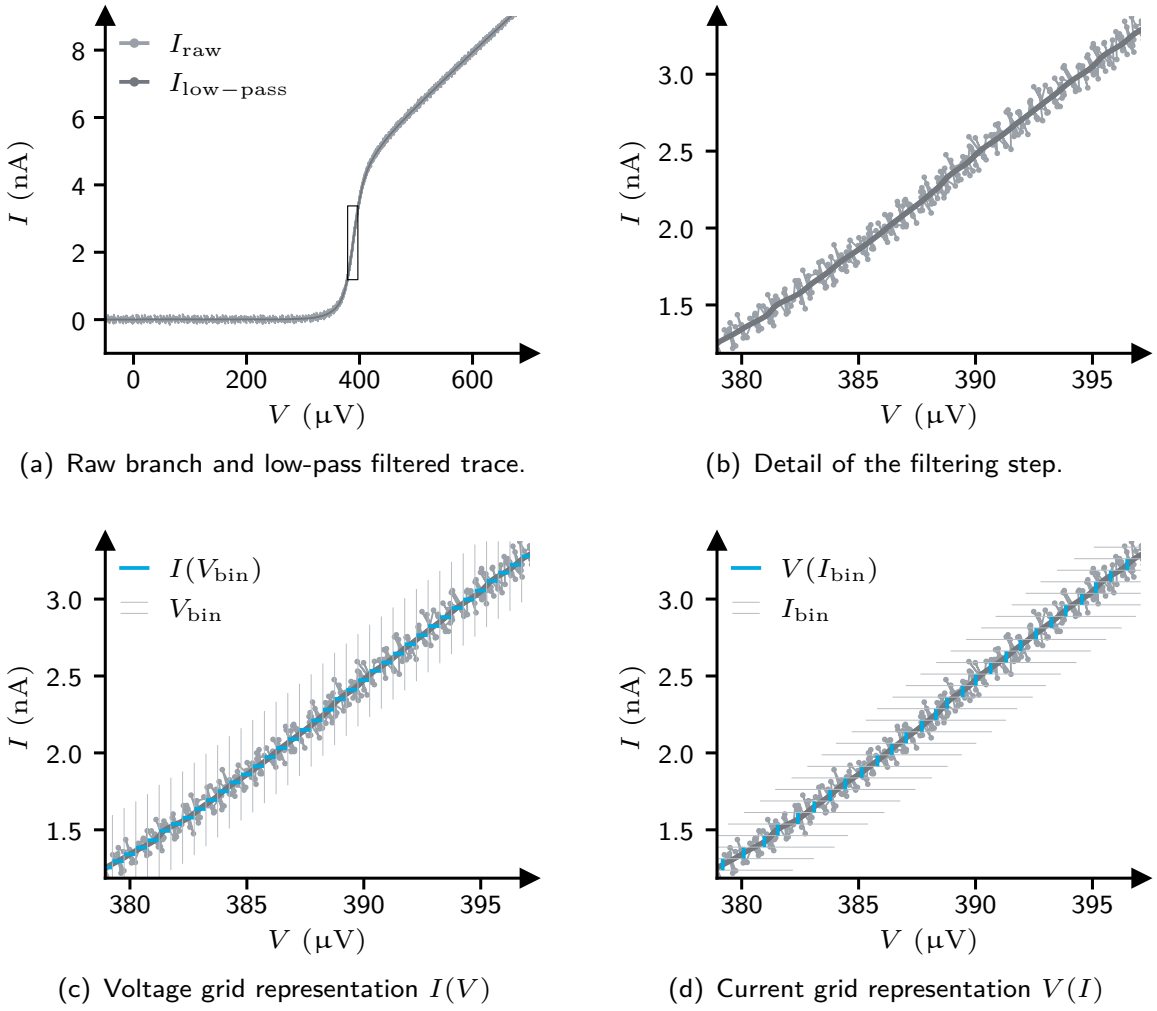
The offsets are obtained from the interval recorded before the sweep and are subtracted from every trace. This correction compensates for slow thermal drifts in the amplifier outputs. When the data contain sufficient symmetric structure, the finer symmetry based correction described below can also be applied. Both corrections are performed while the data remain an ordered pair of time traces, $V(t)$ and $I(t)$.

The corrected branch is then low-pass filtered to suppress high-frequency pickup that is not resolved by the quasi-dc sweep. I filter only the selected branch, which prevents discontinuities at the boundaries between the offset interval and the sweep branches from affecting the result. The standard analysis preset uses a fourth-order Butterworth filter with a 43.7 Hz cutoff. The filtered voltage and current traces are subsequently linearly interpolated onto a uniform time grid at 137 Hz. This numerical upsampling does not add independent measurements. It provides a denser representation of the filtered trace and regularizes its subsequent mapping onto the voltage or current grid.

Figure 3.15 shows the transition from the time trace to the regular data products. After filtering and resampling, the data are assigned to equally spaced voltage or current bins. For a voltage based representation, I average the current values whose measured voltages fall into the bin around V_k,

$$I(V_k) = \langle I(t_j) \rangle_{V(t_j) \in [V_k - \Delta V/2, V_k + \Delta V/2]} . \tag{3.11}$$

Exchanging the axes gives the current grid representation $V(I_k)$. This representation is useful for branches with steep current changes, where a voltage grid would place many measurements within a narrow voltage interval. The regular curve can finally be smoothed with a median filter followed by Gaussian smoothing. This smoothing is applied after binning, so numerical derivatives are calculated from a regular data product rather than from the irregular time trace.



(a) Raw branch and low-pass filtered trace.

(b) Detail of the filtering step.

(c) Voltage grid representation $I(V)$

(d) Current grid representation $V(I)$

Figure 3.15. Evaluation pipeline for one selected sweep branch. After offset correction, the ordered time trace is low-pass filtered and resampled. It is then binned onto a regular voltage or current grid, which removes the time coordinate.

All filtering, resampling, gridding, and smoothing parameters remain part of the evaluation record.

The result is a calibrated and regularly sampled *I*–*V* curve that remains linked to its selected sweep branch and acquisition metadata. This curve is used for plotting, numerical differentiation, and model comparison. The data can be represented as *I*–*V* curves or as derived *dI*–*dV* spectra and compared with the appropriate theoretical model. Each model dependent analysis therefore starts from the same reproducible data product with defined units, offsets, grids, and processing parameters.

Offset Analysis

Beyond the coarse subtraction of amplifier drift, the evaluation pipeline provides an optional symmetry based correction that aligns the residual voltage and current origins. It is not applied by default, and its use and fitted offsets remain part of the evaluation record.

The key advantage of the method is that differentiation separates the two offset coordinates. For an intrinsic point symmetric quasi-dc *I*–*V* curve, the differential conductance is even in voltage. A constant current offset drops out when calculating *dI*–*dV*, so the symmetry search depends on the voltage offset but not on the current offset. The remaining task is the horizontal alignment of the conductance trace. If the correct voltage origin is chosen, the derivative overlaps with its reflection. The implementation determines this alignment by minimizing a mirror symmetry error.

In practice, the analysis scans a list of candidate voltage offsets. For a candidate δV, the trace is shifted by subtracting δV, then passed through the same filtering, resampling, binning, and smoothing pipeline as the final data. The shifted trace defines a sampled curve $I_{\delta V}(V_k)$ on a symmetric voltage grid. The score is the mean absolute mirror mismatch of its differential conductance,

$$\epsilon_V(\delta V) = \left\langle \left| \frac{\mathrm{d}I_{\delta V}}{\mathrm{d}V}(V_k) - \frac{\mathrm{d}I_{\delta V}}{\mathrm{d}V}(-V_k) \right| \right\rangle_k . \tag{3.12}$$

The voltage offset assigned to the trace is the candidate with the smallest finite value of ϵ_V. I then exchange the axes and apply the same search to *dV*–*dI*. In this representation, a constant voltage offset drops out under differentiation, while the symmetry alignment determines the current offset. The two searches therefore determine δV and δI independently. Subtracting both values corrects the complete origin of the measured trace in the *I*–*V* plane.

The procedure is a diagnostic correction, not a physical model fit. It requires corresponding conductance features at positive and negative bias and a distinct error minimum. Traces with rectification, strong hysteresis, insufficient branch overlap, or no distinct minimum are not corrected because symmetry cannot determine a unique voltage origin.

BCS Simulation and Fitting

The BCS simulation predicts tunnel regime quasiparticle spectra from a set of physical parameters. The microscopic tunnel expressions are introduced in Sec. 2.1.2. Here I describe how they are evaluated numerically and used in fits.

The core model evaluates the SIS tunnel current from Eq. (2.18). The two electrodes are represented by Dynes broadened BCS densities of states and finite-temperature Fermi functions. The model parameters are the normal conductance

G_{N}, electronic temperature T, zero-temperature superconducting gap Δ_0, and Dynes broadening γ.

For repeated evaluations, the production model uses the convolution form of the tunnel current, Eq. (2.19). Instead of recomputing the full energy integral for every bias point, the occupied and empty spectral functions are convolved on an energy grid and interpolated back to the requested voltage axis. This makes parameter sweeps and fits substantially faster while keeping the same physical model.

Photon assisted tunneling is included by applying the Tien-Gordon sideband sum of Eq. (2.32) to the BCS current. This extension adds the irradiation amplitude A and frequency ν as model parameters. After evaluating the static or photon assisted current $I_0(V)$, the model can account for unresolved voltage fluctuations by averaging over a Gaussian distribution of instantaneous bias offsets δV,

$$I_{\sigma_V}(V) = \int_{-\infty}^{\infty} \frac{\mathrm{d}(\delta V)}{\sqrt{2\pi}\sigma_V} \exp\left[-\frac{(\delta V)^2}{2\sigma_V^2}\right] I_0(V+\delta V)\,, \qquad \delta V \sim \mathcal{N}(0,\sigma_V^2)\,. \tag{3.13}$$

Here σ_V is the standard deviation of the voltage offset distribution, not its full width at half maximum. The corresponding full width is $2\sqrt{2\ln 2}\,\sigma_V$. The implementation evaluates I_0 on a voltage grid extended by six standard deviations on either side, performs the normalized Gaussian convolution, and interpolates the broadened current back to the requested bias points. Numerical derivatives used for the simulated *dI–dV* and *dV–dI* curves are obtained from this broadened *I–V* curve.

The parameter σ_V is an effective instrumental broadening rather than a microscopic property of the junction. It represents bias fluctuations that are not resolved during the acquisition of one data point and can also absorb broadening caused by averaging over a changing bias within the measurement bandwidth. It differs from the electronic temperature T, which changes the Fermi occupations, and from the Dynes parameter γ, which broadens the quasiparticle density of states and introduces subgap spectral weight. These mechanisms can still be correlated in a fit. The fitting interface therefore allows σ_V to be optimized together with the other model parameters or fixed to an independently determined value or a value obtained from a reference trace, depending on the purpose and constraints of the analysis.

I design the fitting interface so that each model parameter can either be optimized or kept fixed. This is important for data sets containing many *I–V* spectra, where global parameters such as Δ_0, γ, G_{N}, ν, or σ_V may be determined independently or from a reference trace and then held constant while only T or A is fitted for each spectrum. Fixing well-constrained parameters reduces correlations between fit parameters and makes the extracted trends easier to interpret.

Repeated fits require many evaluations of the same model over complete data sets. I therefore use the JAX backend for vectorized evaluation with just-in-time compilation [108]. Together with the convolution kernel, this reduces typical fitting tasks to minutes on a standard laptop. I retain a direct NumPy implementation as a transparent numerical reference.

The resulting fits should be interpreted as tunnel regime quasiparticle fits. They return effective parameters within the BCS-Dynes description, but they do not describe finite transmission MAR, coherent supercurrent dynamics, Coulomb blockade, or environmental phase fluctuations.

MAR Simulation

The MAR simulation predicts finite transmission spectra of atomic contacts from the channel transmissions and the remaining model parameters. The physical model is described in Sec. 2.3.4. Here I describe only the numerical layer used for evaluation. The core calculations are based on three compiled Fortran backends from Juan Carlos Cuevas: a symmetric Hamiltonian approach solver, an asymmetric Hamiltonian approach solver, and a full counting statistics (FCS) solver. The two Hamiltonian approach solvers evaluate the dc MAR current described in Ref. [21], while the FCS backend gives the charge resolved decomposition of Ref. [76]. I do not modify these numerical kernels.

My contribution is the shared Python layer around all three backends. This layer handles units, model dispatch, temperature dependent gaps, deterministic HDF5 caching, odd symmetry reconstruction, segmented backend calls, and parallel evaluation. The user facing parameter is the zero-temperature gap, while the backend is called with the temperature dependent gap. This is important when temperature is scanned or fitted, because a change in T affects both the Fermi occupations and the MAR voltage scale.

The reason for this additional layer is the high computational cost. In my evaluation environment, an uncached single-channel calculation on a sufficiently resolved voltage grid typically takes a few tens of seconds with the symmetric Hamiltonian approach backend, about one to two minutes with the asymmetric backend, and about ten minutes with the FCS backend. Dense parameter scans would therefore be impractical if every voltage grid and parameter point were evaluated from scratch. The wrapper reduces this cost without changing the physical model. Symmetric electrodes are sent to the faster symmetric backend, asymmetric electrodes to the asymmetric backend, and charge resolved calculations to the FCS backend only when needed. Cached curves are indexed by normalized model parameters and voltage axes, so numerically equivalent floating point inputs map to the same entry instead of creating duplicate curves. Since the dc I–V curve is point symmetric, only positive voltages are evaluated and cached, while the negative branch is reconstructed.

MAR Fitting

To determine the mesoscopic pincode $\{\tau_i\}$ of an atomic contact, I fit the measured finite transmission *I–V* characteristic with a sum of independent single-channel MAR currents. The adjustable parameters are the channel transmissions $\{\tau_i\}$, the zero-temperature gap Δ_0, and the effective voltage noise width σ_V.

I fix the electronic temperature and Dynes broadening to $T = 0$ and $\gamma = 0$. The effective voltage noise width σ_V already accounts phenomenologically for the rounding relevant to these fits. Allowing additional broadening through T or γ would introduce strongly correlated parameters that cannot be distinguished reliably at the available voltage and current resolution. The fitted transmissions should therefore be interpreted within this restricted zero-temperature model rather than as an independent determination of electronic temperature or lifetime broadening.

Each adjustable parameter can be varied within prescribed bounds on a predefined grid or fixed to a selected value. Because the transmission grid includes $\tau_i = 0$, a fit allowing N channels also includes every solution with fewer than N contributing channels. Increasing the maximum number of allowed channels therefore does not require all of them to contribute to the fitted current.

I first evaluate and cache a database of single-channel currents on a discrete grid in τ and Δ_0. At every point in this grid, I apply the same Gaussian voltage convolution used for the tunnel regime model and store the resulting curves for discrete values of σ_V. I then obtain a multichannel prediction by summing the stored currents at the selected channel transmissions. I interpolate the measured current on the voltage grid of this database. I discard nonfinite samples and mask the central voltage interval containing the switching or supercurrent feature because it is not described by the dc MAR model. Additional nonnegative point weights and voltage limits can be used to restrict the fit to informative parts of the trace.

The fit minimizes a weighted current space residual on this discrete parameter grid. If the normal-state conductance G_{N} can be determined reliably from the measured high-bias response, I restrict it to a trustworthy interval. The objective function is

$$\chi^2(\{\tau_i\}, \Delta_0, \sigma_V) = \sum_{k \in \mathcal{K}} w_k \cdot \left(I_{\mathrm{exp}}(V_k) - I_{\mathrm{MAR}}(V_k, \{\tau_i\}, \Delta_0, \sigma_V)\right)^2 , \quad (3.14)$$

where $\mathcal{K}$ denotes the retained voltage samples. The weights can be chosen to balance different parts of the spectrum. For example, $w_k \propto 1/|V_k|$ reduces the influence of the larger absolute currents at high bias while retaining sensitivity to the subgap MAR structure. In this case, the zero-voltage point is assigned zero weight, while any finite central interval not described by the dc model is excluded through $\mathcal{K}$.

An exhaustive global search is not computationally feasible beyond three channels, even on a sparse transmission grid, because the number of possible pincodes

grows combinatorially with the number of channels. The search therefore starts from initial parameter guesses and tests all valid states reached by moving one free parameter by one grid point. It accepts the neighboring state with the largest reduction in χ^2 and repeats this procedure until no tested move improves the fit. In addition to single-channel steps, paired exchange moves increase one transmission and decrease another. These moves allow the channel decomposition to change even when a narrow constraint on G_{N} prohibits an isolated transmission step. Independent random restarts within the parameter bounds reduce the sensitivity to the initial pincode. For a sequence of contact configurations, the result of one spectrum can be used as the starting point for the next spectrum.

This procedure is a local discrete optimization and does not guarantee the global minimum. The restarts improve robustness but do not replace a complete search of the pincode space. Allowing additional channels can introduce nearly degenerate solutions containing very weak transmissions, so the number of physically resolved channels remains limited by the data quality and transmission grid resolution. The parameter resolution is set by the database axes, and the reported values are grid points without interpolation or confidence intervals. The extracted transmissions are therefore effective channel transmissions within the coherent short-contact MAR model. The procedure does not model the dc supercurrent branch itself, environmental phase fluctuations, Coulomb blockade, or nonequilibrium heating beyond the effective parameters included in the grid.

4 Experimental Results

This chapter presents the experimental results on superconducting charge transport in nanostructures with progressively increasing transmission. The measurements span the tunnel regime, in which quasiparticle transport provides a direct probe of the superconducting density of states, and atomic-scale contacts, in which transport is governed by a small number of channels with finite transmission. Microwave irradiation introduces an additional control parameter and allows us to examine photon assisted, nonlinear, and nonequilibrium transport across these regimes.

Throughout the chapter, I characterize the devices using measured I–V curves and the numerically derived dI–dV and dV–dI responses. I compare these observables with minimal transport models appropriate to each regime. The models provide reference responses, support the extraction of experimentally relevant parameters, and identify conditions under which the underlying assumptions cease to describe the measurements. Particular attention is given to microwave calibration, spectral resolution, reproducibility, and the distinction between effects intrinsic to the device and those associated with its electromagnetic environment.

4.1 Tunnel Barrier

SIS tunnel spectroscopy provides indirect access to the quasiparticle density of states. Under microwave irradiation, photon assisted tunneling produces features displaced from the original spectral structure by integer multiples of the photon energy. Tunnel junctions therefore provide a controlled platform for testing the SIS transport model and the limits of the Tien-Gordon description.

In this section, I investigate static and microwave driven quasiparticle transport in two similar aluminum tunnel junctions, denoted TB1 and TB2. I measure the devices with different versions of the setup, and their fabrication and local circuit environments are not identical. Their comparison therefore tests which features follow the common tunnel junction physics and which depend on the measurement conditions or electromagnetic environment.

I first use the higher resolution measurements of TB1 to establish the static junction parameters, calibrate the effective temperature and microwave amplitude, and compare the irradiated response with the Tien-Gordon model. I then turn to TB2 and characterize the bias asymmetric response that appears under microwave irradiation. This progression establishes the conventional reference behavior before examining the regime in which the minimal model is no longer sufficient.

4.1.1 High-Resolution Characterization

This section starts with the simplest microwave driven superconducting junction investigated in this thesis. The goal is to understand one device as completely as possible before moving to junctions where additional mechanisms complicate the interpretation. TB1 is an SIS tunnel barrier (TB) without an observable supercurrent. Its static spectra are therefore expected to be described by quasiparticle tunneling between two BCS electrodes, and its microwave response should follow photon assisted tunneling as introduced in Sec. 2.1.

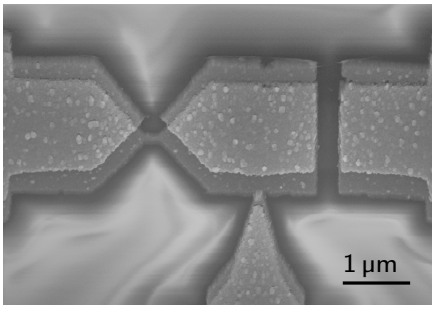


Figure 4.1. SEM image of TB1. The 70 nm lower Al layer is oxidized in situ for 2:15 min at 4 mbar. The upper Al layer is 92 nm thick. The barrier overlap is about $(50\times100)\,\mathrm{nm}^2$.

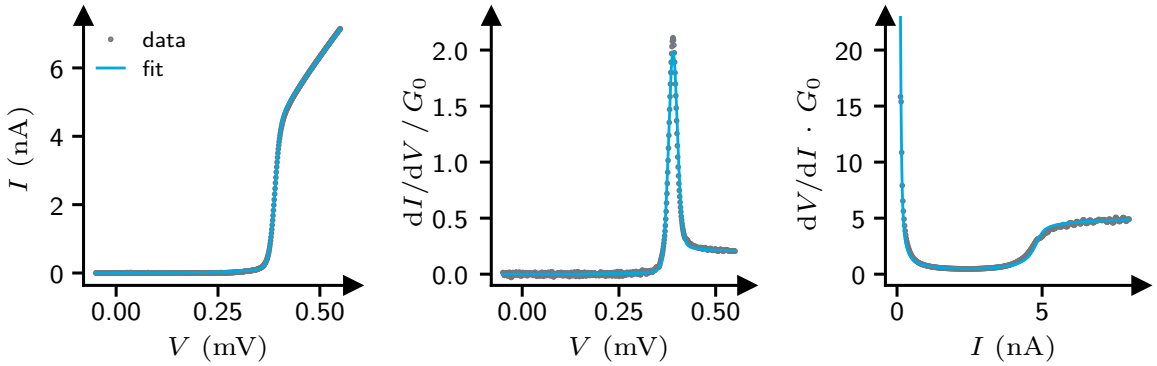
I use this device as a reference system for three connected questions. First, I determine how well the unirradiated *I*–*V* curve is described by the TB parameters. Second, I use controlled heating to calibrate the temperature axis and to introduce the transport landscape visualization used throughout the chapter. Third, I drive the same junction with microwaves and use the remapped amplitude axis to test how far photon assisted sidebands can be resolved. The high-amplitude data show sidebands up to very high photon order and therefore provide a direct benchmark for the spectral resolution and microwave coupling of the setup.

The sample TB1 is fabricated as described in Sec. 3.1. Figure 4.1 shows an SEM image of the junction. Its caption gives the layer thicknesses, barrier overlap area, and in situ oxidation conditions that define the sample geometry and barrier fabrication. Irradiation experiments are performed with the antenna in the setup described in Sec. 3.2. All measurements discussed here are taken with the BJ in its virgin state as fabricated.

A typical *I*–*V* curve is shown in Fig. 4.2. The quasiparticle onset appears at approximately $2\Delta/e$, as expected for an SIS junction (Sec. 2.1.2). I do not observe a supercurrent, even though both electrodes are superconducting. A finite Josephson coupling may nevertheless remain below the experimental resolution. Using the Ambegaokar-Baratoff estimate (Eqs. 2.38 and 2.39) with parameters from the first column of Table 4.1 gives

$$I_\mathrm{c} \approx 4.5\,\mathrm{nA}\,, \qquad E_\mathrm{J} \approx 9.3\,\mu\mathrm{eV}\,, \qquad E_\mathrm{T} \approx 18.5\,\mu\mathrm{eV}\,. \tag{4.1}$$

The estimated critical current is large enough to be observable in principle, but the thermal energy E_T is of the same order and already about twice as large as E_J. This relation places the junction in a thermally fluctuating regime where the Josephson coupling cannot maintain a stable phase difference. This provides a natural explanation for the absence of a resolved zero-voltage branch.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.2. Representative static transport trace of TB1 at $T_{\mathrm{bath}} = 33\,\mathrm{mK}$. The panels show the measured *I–V* curve and the numerically derived *dI–dV* and *dV–dI* responses. The *I–V* curve is fitted with the SIS BCS model, with the extracted parameters listed for M1 in Table 4.1.

The capacitance of TB1 is not measured independently. Using the same geometric parallel-plate approximation as Lorenz *et al.* gives an estimated range of $C_{\mathrm{TB}} \approx 0.2\text{-}0.4\,\mathrm{fF}$ [84, 85]. This range is consistent with related MCBJ SET estimates by Susanne Sprenger, who found junction capacitances of about $0.1\,\mathrm{fF}$ [86], and Laura Sobral Rey, who found total island capacitances of about $1.2\,\mathrm{fF}$ [87, 88].

The corresponding charging energy for $C_{\mathrm{TB}} \sim 0.2\text{-}0.4\,\mathrm{fF}$ is $E_{\mathrm{C}} = e^2/2C_{\mathrm{TB}} \sim 200\text{-}400\,\mu\mathrm{eV}$. Even allowing for stray capacitance that increases the effective capacitance to several fF, this scale remains comparable to or larger than the Josephson energy. I therefore interpret the missing zero-voltage branch as consistent with a junction whose dc phase relation is strongly affected by thermal fluctuations, charge fluctuations, voltage noise, and the unengineered microwave environment. The electrodes remain superconducting, but the phase difference across the barrier is not stiff enough to produce a resolvable zero-voltage branch [36, 49, 50].

Because no dc Josephson branch is resolved, I attribute the expected microwave response primarily to quasiparticle photon assisted tunneling rather than to Shapiro steps from coherent Cooper pair transport. The sharp superconducting gap edge provides a well-defined spectral reference, and microwave irradiation should generate replicas displaced from it by integer multiples of the single-photon energy.

The lack of a supercurrent does not prevent me from characterizing the junction. The quasiparticle branch gives access to the superconducting gap Δ_0, the normal-state conductance G_{N}, the fitted electronic temperature T_{fit}, the Dynes broadening γ, and the effective voltage broadening σ_V. Assuming symmetric electrodes and negligible magnetic field, I obtain the values listed in Table 4.1 for the measurement blocks used in this section.

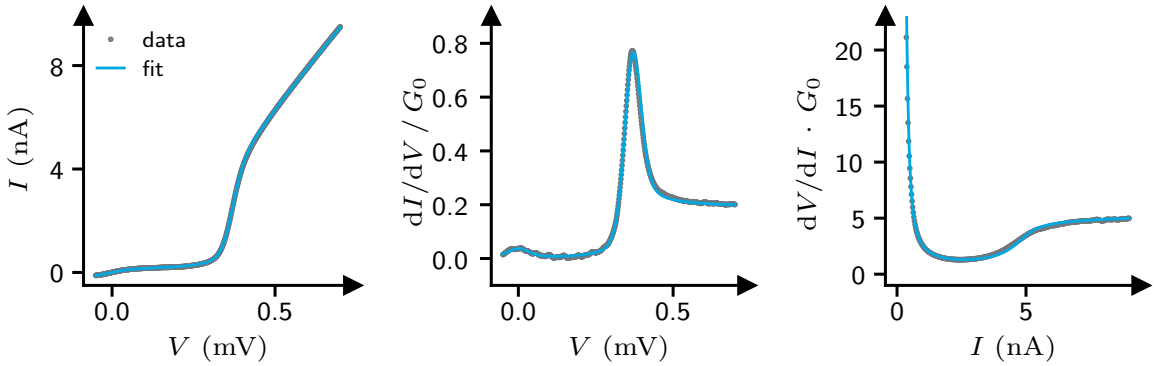
Table 4.1. Static reference parameters and microwave calibration values by measurement block. Each column corresponds to the listed figure group and collects the parameters from the representative single *I–V* fit used for that measurement block. The small differences between columns reflect block-to-block drift and changing broadening or temperature conditions, while the values remain consistent with the same SIS tunnel junction.

Measurement	M1	M2	M3	M4
Figures	4.2	4.3 - 4.6	4.9 - 4.11	4.12 - 4.15
G_{N} (G_0)	0.1894995(4)	0.18723(4)	0.189190(7)	0.188933(3)
T_{fit} (mK)	215(13)	614.8(34)	243(30)	223(59)
Δ_0 (μeV)	194.800(45)	194.48(16)	195.26(16)	195.85(24)
γ (μeV)	1.1282(38)	5.53(12)	1.289(14)	2.079(43)
σ_V (μV)	10.315(10)	21.32(31)	15.756(45)	14.24(12)
ν (GHz)	–	–	18.3	13.6
A (μV)	–	–	121.793(40)	324.881(47)
A_{out} (mV)	–	–	150	150
T_{bath} (mK)	33(1)	697(1)	63(1)	187(1)

In principle, one static *I–V* fit would define the junction parameters for all later analysis. In practice, the extracted static parameters vary slightly between measurement blocks acquired up to several days apart. Small changes in electronic temperature, thermalization, residual voltage noise, or the electromagnetic environment can alter the measured line shape and shift correlated fit parameters. Because G_{N} and Δ_0 remain similar across the measurement blocks, I attribute the variation primarily to changing measurement conditions rather than to a modified junction. I therefore use a representative static reference fit for each measurement block (M1 - M4) listed in Table 4.1. Within each measurement block, I hold the corresponding static parameters fixed while I extract the intended calibration parameter, namely T_{fit} for the temperature sweep and the effective microwave amplitude A_{fit} for the microwave amplitude sweeps. Unless stated otherwise, the parenthetical uncertainties of fitted quantities correspond to one standard deviation obtained from the covariance matrix.

The fitted gaps cluster near $\Delta_0 \approx 195\,\mu\mathrm{eV}$, slightly larger than the literature value for bulk aluminum of about $180\,\mu\mathrm{eV}$ used in Sec. 2.1.1. This difference is typical for thin-film aluminum. The weak-coupling estimate from Δ_0 gives $T_{\mathrm{c}} \approx 1.274\,\mathrm{K}$, consistent with the temperature calibration in Sec. 4.1.1 [19, 36]. The normal-state conductance remains near $G_{\mathrm{N}} \simeq 0.19\,G_0$ in all reference fits, which places TB1 in the tunnel limit. BCS quasiparticle tunneling and Tien-Gordon photon assisted tunneling are therefore the appropriate minimal models for the measurement blocks in this section.

The fitted electronic temperature T_{fit} is difficult to estimate from an SIS junction far below the critical temperature because the spectrum changes only weakly in that temperature range. Although the static fit gives $T_{\mathrm{fit}} = 215(13)\,\mathrm{mK}$, the controlled temperature sweep in Sec. 4.1.1 shows a low-temperature fit plateau



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.3. Representative *I–V* fit from the temperature sweep at $T_{\mathrm{bath}} = 697\,\mathrm{mK}$. The SIS BCS fit is used to extract the static parameters, shown in Table 4.1.

near $T_{\mathrm{base}} = 282(2)\,\mathrm{mK}$. I therefore treat the single-trace value as a weakly constrained model parameter rather than as a direct temperature measurement.

The remaining two fit parameters describe phenomenological broadening. The Dynes parameter γ accounts for a finite quasiparticle lifetime and other mechanisms that broaden the singularity in the DOS itself, while σ_V describes effective voltage broadening from residual bias noise and averaging during the voltage sweep. I keep both parameters because they affect the fit differently. Increasing γ alone can round the curve at the gap edge, but it also increases the modeled subgap conductance. The additional voltage broadening allows the gap edge width to be reproduced without producing an overall rounding of the DOS.

The BCS fit captures the gap position and the overall line shape, but it does not reproduce the full height of the quasiparticle peaks. The *dI–dV* curve also shows small systematic deviations near the gap edge. I attribute these deviations primarily to effective voltage broadening and DOS broadening. As discussed in Sec. 2.1.2 and Fig. 2.4, temperature in an SIS spectrum mainly changes the gap, subgap conductance, and the 2Δ peak height in differential conductance near T_{c}. In the low-temperature reference fits, σ_V accounts for the residual gap edge width without requiring γ to absorb too much of the rounding.

The reference fits therefore identify TB1 as a well-behaved SIS tunnel junction across the measurement blocks. The least direct parameter is T_{fit}, so I next test its meaning with a controlled temperature sweep and use the same data to introduce the transport landscape visualization used below.

Temperature Calibration

The sample holder is heated locally while a thermometer next to the device records the bath temperature T_{bath}. For each temperature point, I fit the SIS

BCS model to the acquired *I*–*V* curve and extract the fitted temperature T_{fit}. I use T_{bath} for the thermometer reading and T_{fit} for the effective electronic transport temperature required by the tunnel model. For this sweep (M2), I use the static reference parameters in the temperature column of Table 4.1. Only T_{fit} varies point by point. The calibration therefore tests how the fitted SIS response relates the externally controlled bath temperature axis to an effective transport temperature axis.

Figure 4.4 compares T_{fit} with T_{bath}. The calibration curve has three regimes: a base plateau, an almost linear middle region, and saturation at high bath temperature. I describe all three regimes with a clipped affine calibration,

$$T_{\mathrm{fit}}(T_{\mathrm{bath}}) = \min\left[T_{\mathrm{c}},\, \max\left(T_{\mathrm{base}},\, \eta_{T_{\mathrm{bath}}}^{(T_{\mathrm{fit}})} T_{\mathrm{bath}} + T_{\mathrm{off}}\right)\right] . \tag{4.2}$$

Here T_{base} is the effective temperature floor, T_{c} is the upper saturation temperature, $\eta_{T_{\mathrm{bath}}}^{(T_{\mathrm{fit}})}$ is the slope relating the thermometer reading and fitted temperature in the intermediate regime, and T_{off} is the extrapolated temperature offset between the two axes. This procedure avoids manually excluding the plateau and saturation points. The full sweep is described with the same mapping, while the lower and upper bounds encode the weak temperature sensitivity far below and near T_{c}. This procedure gives

$$\begin{aligned} T_{\mathrm{c}} &= 1.256(26)\,\mathrm{K}\,, \\ T_{\mathrm{base}} &= 0.2819(17)\,\mathrm{K}\,, \\ T_{\mathrm{off}} &= -0.0929(36)\,\mathrm{K}\,, \\ \eta_{T_{\mathrm{bath}}}^{(T_{\mathrm{fit}})} &= 1.0023(40)\,. \end{aligned} \tag{4.3}$$

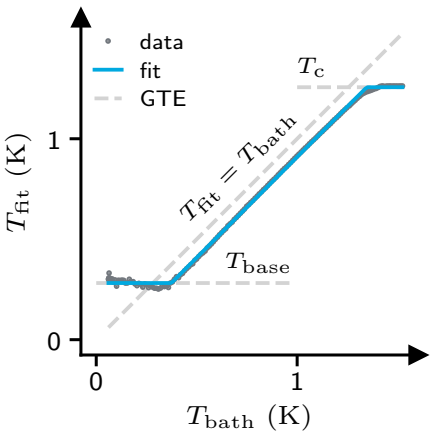


Figure 4.4. Measured TB1 temperature. Guides mark the low-temperature plateau and high-temperature saturation. Within these limits, the relation is linear.

The critical temperature inferred from the zero-temperature gap is about $1.274\,\mathrm{K}$, so the value obtained from the temperature sweep agrees within the uncertainty. The slope $\eta_{\mathrm{bath}}^{(\mathrm{fit})}$ is close to unity, as expected when the thermometer and electronic transport temperature follow each other in the intermediate region. Limited equilibration may contribute to the negative offset because I wait only a few seconds after each heater setpoint change before recording the next *I*–*V* curve. The present data do not isolate this contribution from other systematic temperature offsets.

The low-temperature plateau should not be read as a direct thermometer measurement. Far below T_{c}, an SIS spectrum changes only weakly with temperature, so the fit has limited sensitivity to small changes in T_{fit}. In addition, residual radiation, imperfect line thermalization, and bias dissipation can produce a real electronic temperature floor above the thermometer reading. I therefore interpret

T_{base} as the lowest effective transport temperature resolved by this calibration. The high-temperature plateau corresponds to the critical temperature T_{c}. At this temperature, superconductivity is completely suppressed, and the *I*–*V* curve becomes linear.

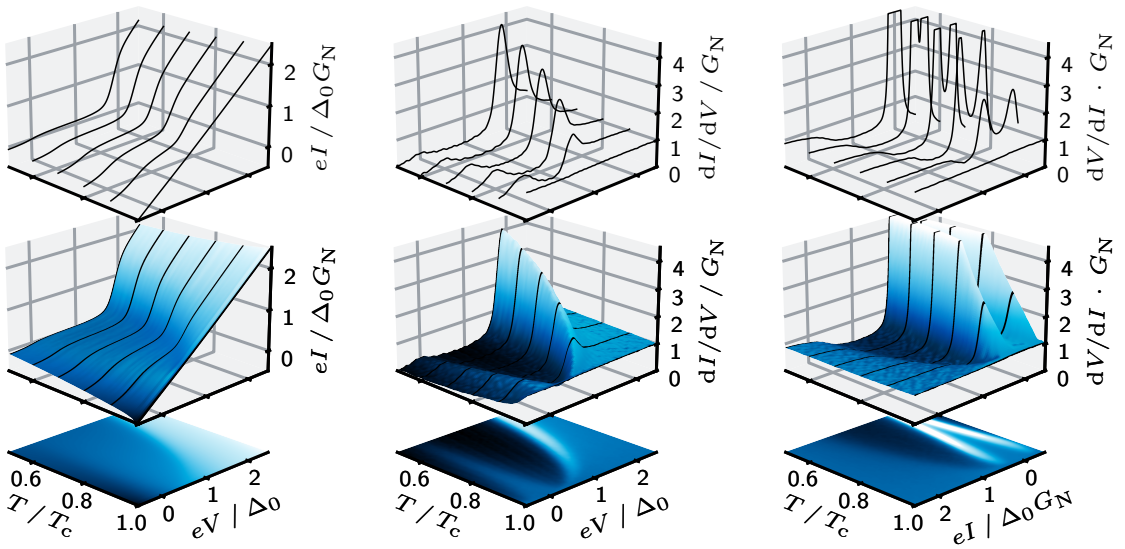
I remap the measured temperature sweep from the recorded T_{bath} coordinate to the corresponding effective T_{fit} coordinate. The transport data themselves are unchanged, but I place each trace at the pointwise fitted transport temperature shown in Fig. 4.4. The linear relation in Eq. (4.2) summarizes the fitted-temperature data, but the remapped landscape uses the fitted temperature values. Figure 4.5 shows the transport landscape after remapping the vertical axis to T_{fit}. The gap closes near T_{c}, the quasiparticle peaks at the gap edge shrink, and the zero-bias conductance increases near the high-temperature end of the sweep.

This figure also introduces the transport landscape format used throughout the experimental results chapter. A sweep consists of a series of transport curves recorded while one control parameter is increased. The three columns show *I*–*V*, *dI*–*dV*, and *dV*–*dI* representations of the same sweep. The top row shows representative curves, which preserve the detailed line shape. The middle row combines all curves into a surface. The bottom row shows the same data as a two-dimensional false-color map, with the sweep parameter and bias as coordinates and the transport response encoded by color. I refer to this last representation as a map, not as a separate measurement. Surface height and false-color map color use the same value range, so both views can be compared directly. I show both derivative representations because *dI*–*dV* is natural for voltage biased gap features, while *dV*–*dI* on a current axis remains useful when comparing to current biased views. The same nine-panel layout is reused below for sweeps in which microwave amplitude is the control parameter.

The corresponding SIS BCS simulation in Fig. 4.6 uses the same T_{fit} axis and the static reference parameters from the temperature column of Table 4.1. The simulation reproduces the robust trends of the measurement, namely the gap closing, the reduction of the quasiparticle peak height, and the increase in zero-bias conductance. This agreement shows that the same SIS model captures the controlled temperature response of TB1 within the limits of the fitted-temperature calibration.

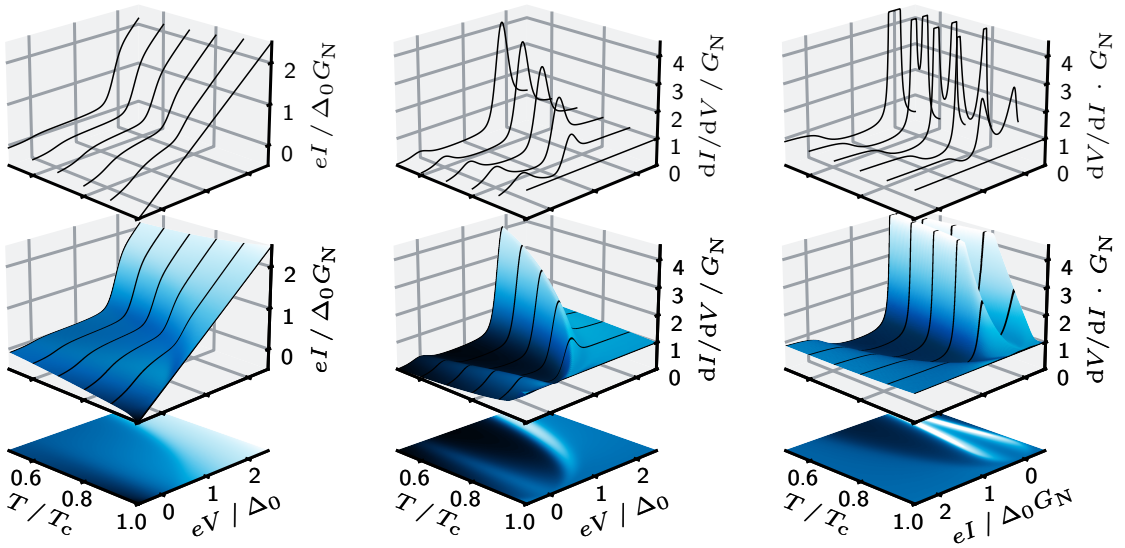
Microwave Amplitude Calibration

Next, I use TB1 to calibrate the microwave amplitude axis and to test the dynamic range of the PAT measurement. The two drive frequencies emphasize complementary aspects of the response. At 18.3 GHz, the larger single-photon energy produces a wider voltage separation between adjacent replicas, while the weaker microwave coupling limits the maximum reduced amplitude reached in the sweep. The resulting uncluttered maps make the sideband evolution and



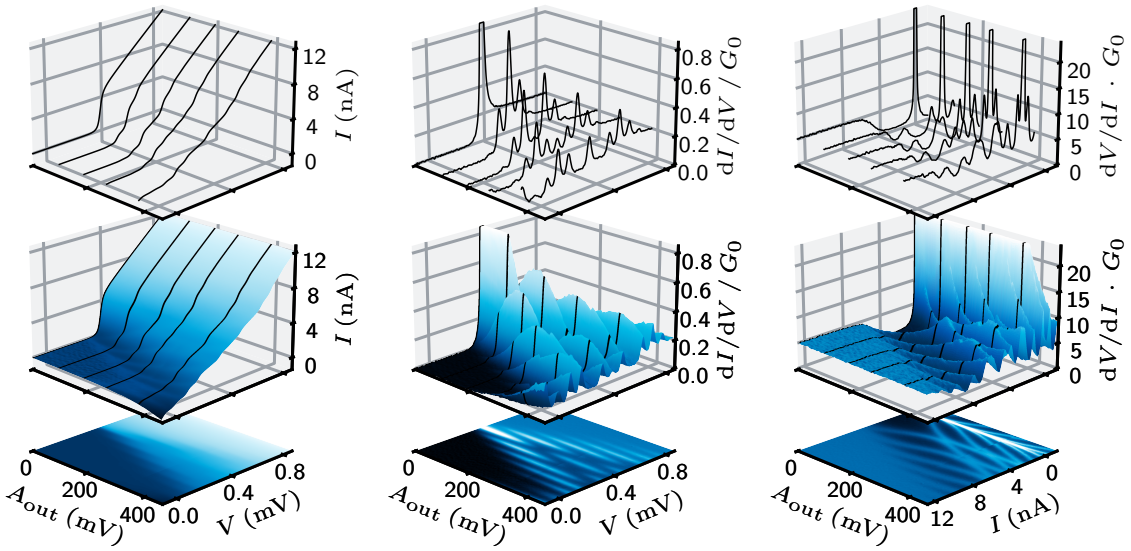
(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.5. Transport landscape of TB1 as a function of temperature on the remapped T_{fit} axis. The three columns show the measured *I–V*, *dI–dV*, and *dV–dI* responses, while the rows show representative traces, a surface view, and a false-color map of the same data.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.6. SIS BCS simulation of the temperature sweep. The panels show the simulated *I–V*, *dI–dV*, and *dV–dI* responses using the same temperature axis as Fig. 4.5. The static reference parameters are taken from the temperature column of Table 4.1.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.7. Raw microwave amplitude sweep of TB1 at 18.3 GHz. The vertical axis is the nominal VNA output amplitude for a 50 Ω load. Photon assisted replicas of the $2\Delta_0$ peaks appear at spacings set by the single-photon energy, but the amplitude axis is still a hardware setting rather than the effective drive at the junction.

amplitude calibration particularly clear. I therefore use this measurement first to introduce the analysis. At 13.6 GHz, the smaller photon energy and stronger coupling provide access to a much wider range of reduced amplitudes and photon orders. I use this second measurement to test how far the sidebands remain resolvable in the multiphoton regime. For both sweeps, I fit the effective junction amplitude point by point.

In this quasiparticle PAT regime the relevant charge is $q = e$, so I use the reduced drive amplitude $\alpha_e = eA/h\nu$ and the voltage equivalent of the single-photon energy, $V_\nu = h\nu/e$.

TB1 is the cleanest reference system for this purpose. The static data show sharp quasiparticle onsets, the normal-state conductance places the device in the tunnel limit, and I do not resolve a supercurrent. I therefore do not have to separate quasiparticle photon assisted tunneling from Shapiro steps or coherent Cooper pair transport. The expected microwave response is the quasiparticle PAT process introduced in Sec. 2.1.3. For both microwave amplitude sweeps (M3 and M4), I use the corresponding static reference parameters from the matching column of Table 4.1.

Figure 4.7 shows the raw 18.3 GHz amplitude sweep as a function of nominal VNA output. This hardware setting does not specify the ac voltage across the junction because the microwave path contains attenuation, impedance mismatch, antenna coupling, and standing waves. The effective transfer to the junction can

therefore depend strongly on frequency. It may also depend on the exact contact configuration through its local impedance and electromagnetic environment. All measurements in this section are taken on the same unchanged TB1 contact, so these data cannot determine or exclude such a contact dependence. The resulting calibration is consequently an empirical conversion for this contact and each measured frequency, not a transferable property of the microwave line. The transport landscape figure nevertheless already identifies the mechanism because the main SIS quasiparticle peaks are accompanied by replicas shifted by integer multiples of the single-photon energy.

I convert the nominal output setting into an effective junction amplitude by fitting the driven spectra with the Tien-Gordon model. The only fitted microwave parameter at each nominal output setting is the effective ac amplitude A at the junction. This strongly narrows the fit because it cannot improve the agreement by moving the gap edge, changing the normal-state conductance, or broadening the spectrum.

The single-trace fit in Fig. 4.9 is the most direct check that the calibration is quantitative rather than only qualitative. The same reduced amplitude controls all sideband weights at once, so agreement over the main quasiparticle peaks, satellite peaks, and spectral weight tests the full PAT redistribution.

Figure 4.8 shows the pointwise fitted amplitudes as a function of nominal VNA output. These fitted amplitudes are the values I use to remap the sweep to the reduced PAT coordinate. I fit them again with a linear relation only to compare the effective coupling between measurement blocks.

Over the indicated parameter range of the sweep, this comparison fit gives

$$A = \eta A_{\mathrm{out}} , \qquad \eta \approx 8.38(3) \times 10^{-4} . \quad (4.4)$$

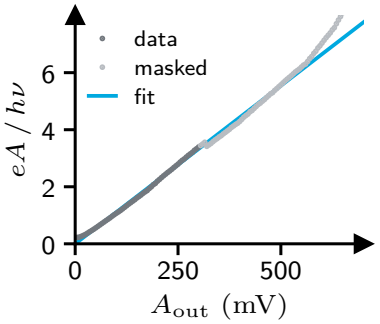
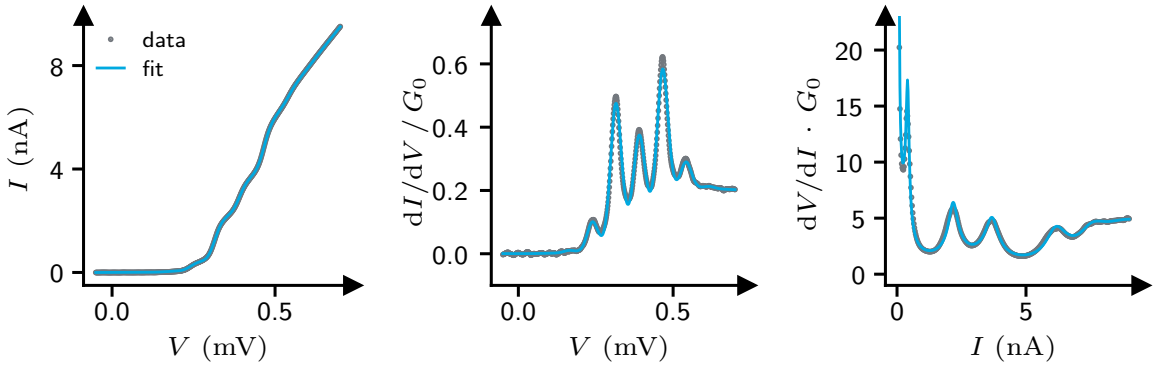


Figure 4.8. Microwave amplitude calibration of M3. The points give the fitted ac voltage used for remapping, while the slope summarizes the effective coupling.

The abrupt feature around $A_{\mathrm{out}} \approx 300\,\mathrm{mV}$ is not a change of junction physics. It is consistent with a change in the internal source amplification state of the network analyzer. At the largest source settings the fitted amplitudes also deviate from a linear relation, which I treat as a limit of using the VNA output as a microwave power coordinate.

Using the pointwise fitted amplitudes, I remap the measured sweep from nominal VNA output to the reduced PAT coordinate. Figures 4.10 and 4.11 compare the remapped 18.3 GHz amplitude sweep with the corresponding simulation. Their agreement validates the calibration procedure because I replace the hardware dependent source setting with the dimensionless drive coordinate that controls the expected PAT structure.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.9. Representative photon assisted tunneling fit at 18.3 GHz and $A_{\mathrm{out}} = 150\,\mathrm{mV}$. The *I–V* curve is measured at $T_{\mathrm{bath}} = 63(1)\,\mathrm{mK}$. This fit provides the static reference parameters and microwave coupling listed for this measurement block in Table 4.1.

High Dynamic Range

I then use the 13.6 GHz amplitude sweep (M4) as a dynamic-range test of the same remapped-axis picture. This frequency is favorable for high-order sidebands because the microwave coupling to TB1 is stronger and the photon energy is smaller than at 18.3 GHz. For a comparable nominal source range, both effects increase the reduced amplitude and push spectral weight into higher Bessel sidebands. A representative *I–V* curve is shown in Fig. 4.12.

I again use the fitted amplitude values in Fig. 4.13 directly for the remapped axis. A secondary linear fit to these values gives the comparison slope

$$\eta \approx 21.50(1) \times 10^{-4}\,. \tag{4.5}$$

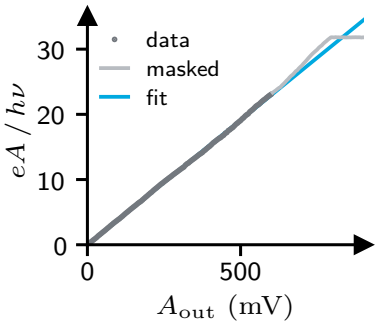
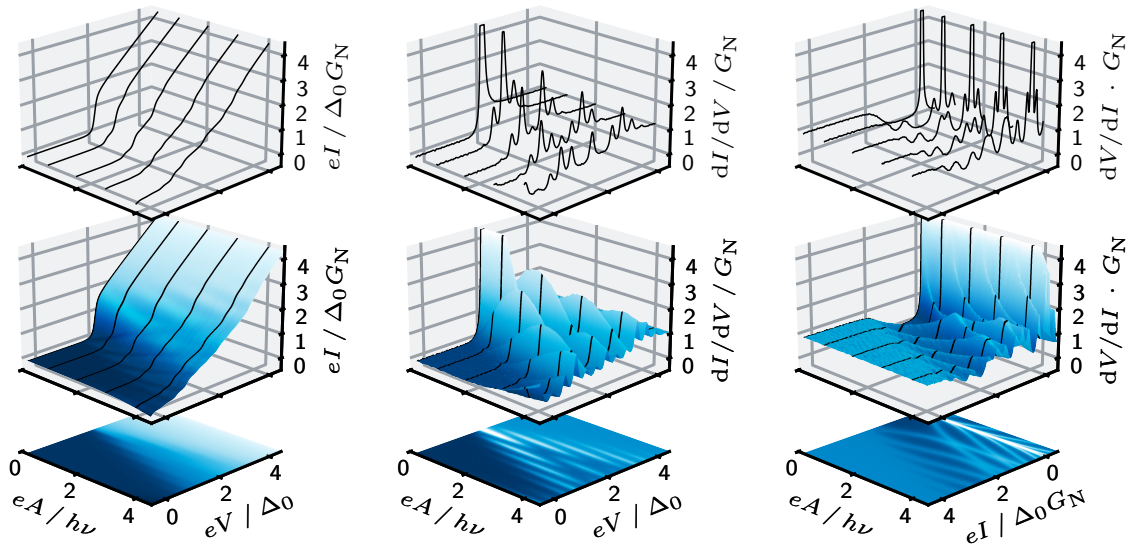


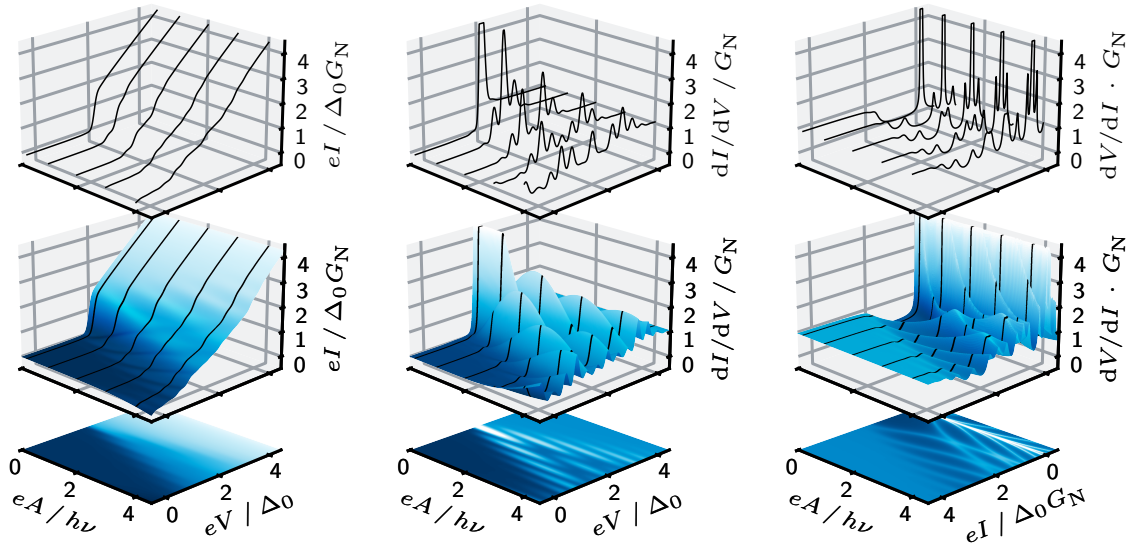
Figure 4.13. Microwave amplitude calibration at 13.6 GHz. The points give the fitted ac voltage used for remapping, while the slope summarizes the effective coupling for comparison.

At the highest source settings, Fig. 4.13 shows a plateau because the VNA reaches its maximum available output power. For this fixed contact, the coupling is about 2.6 times as large as the fitted coupling at 18.3 GHz. This comparison demonstrates a strong frequency dependence, but it does not establish a universal frequency response because the coupling may change when the contact configuration changes. The η values listed in Table 4.1 are therefore summary slopes from these secondary fits, not the calibration inputs used to remap the transport landscapes. Together with the lower photon energy, the larger slope makes the 13.6 GHz amplitude sweep the high-amplitude benchmark of the section.



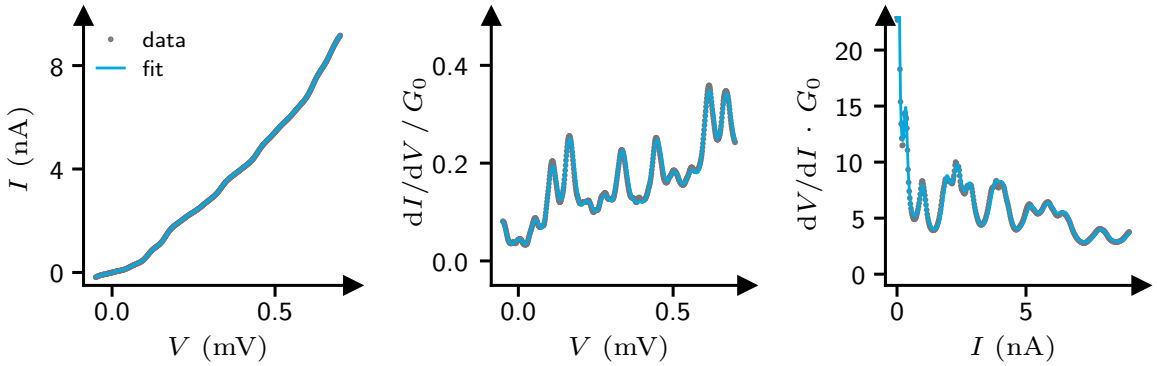
(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.10. Remapped amplitude sweep of TB1 at 18.3 GHz. The same data as in Fig. 4.7 are plotted on the reduced amplitude axis obtained from the pointwise amplitude fits shown in Fig. 4.8, so the sideband evolution can be compared directly with the Tien-Gordon model.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.11. Tien-Gordon simulation of the amplitude sweep at 18.3 GHz. The static reference parameters are taken from the matching column of Table 4.1, and the vertical axis is the same reduced drive amplitude as in Fig. 4.10. The simulation reproduces the sideband spacing and the redistribution of spectral weight with increasing drive.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

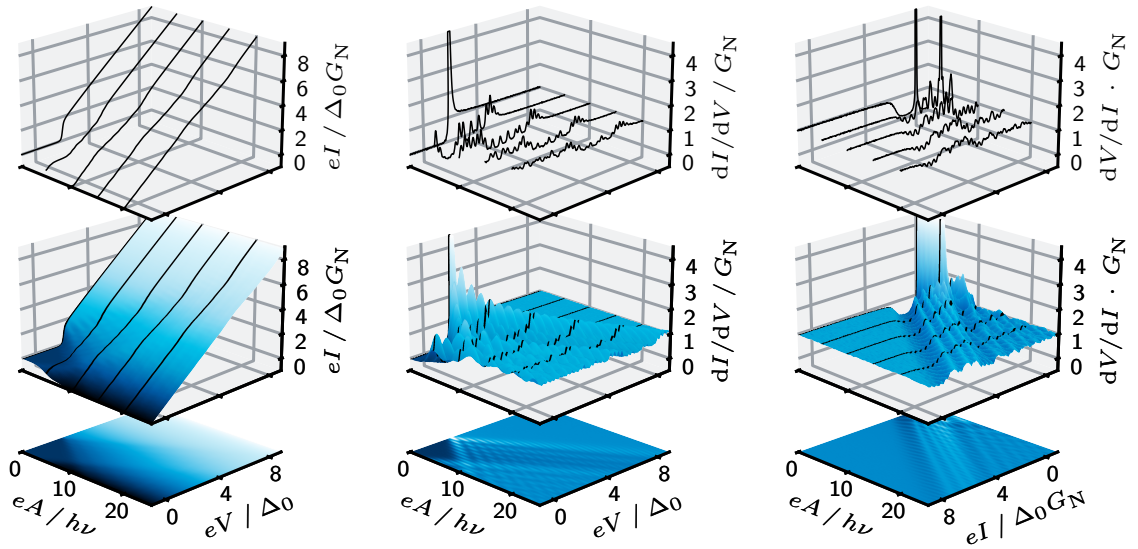
Figure 4.12. Representative photon assisted tunneling fit at $\nu = 13.6\,\mathrm{GHz}$ and $A_{\mathrm{out}} = 150\,\mathrm{mV}$. The *I–V* curve is measured at $T_{\mathrm{bath}} = 187(1)\,\mathrm{mK}$. This fit provides the static reference parameters for this measurement block and the microwave coupling used for the remapped 13.6 GHz amplitude sweep.

Figures 4.14 and 4.15 show the remapped 13.6 GHz amplitude sweep (M4) and the corresponding simulation. The 2Δ peak acts as a sharp energy marker, and the microwave drive creates many photon shifted copies of this marker. Individual replicas remain visually resolved to at least order 25.

This high-order sideband resolution is the strongest result of the section because it requires four things at once: strong microwave coupling at the junction, enough voltage resolution to separate closely spaced photon replicas, a stable contact geometry throughout the sweep, and enough spectral stability that the underlying SIS feature remains identifiable after its weight has been redistributed over many sidebands. Here, I call a replica visually resolved when it appears at the expected displacement, remains distinguishable from neighboring features, and can be followed continuously through the full microwave amplitude map. The experiment therefore resolves a well-organized Tien-Gordon response over a sideband ladder that is much wider than 2Δ.

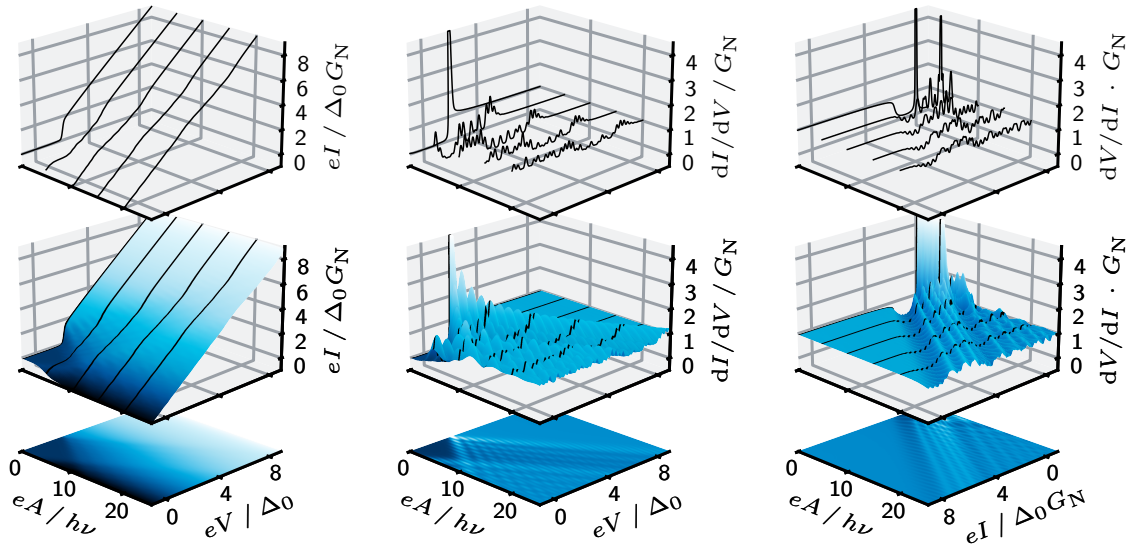
The remapped amplitude axis is essential for this claim. In nominal VNA units, the high-order structure would only show that the source setting is large. For each trace, I instead infer the effective ac voltage at the junction by fitting the complete *I–V* curve. On the resulting reduced-amplitude axis, the structure can be read as a PAT redistribution over many Bessel weighted channels. The high photon order therefore does not mean that every tunneling event absorbs exactly 25 photons. It means that photon shifted tunneling channels with order numbers up to at least 25 remain visually distinguishable in the measured quasiparticle spectrum.

This interpretation also sets the limits of the result. I do not use the photon order to infer a quasiparticle tunneling time or traversal time. The PAT order labels energy shifted contributions in a driven tunneling probability, not a directly



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.14. Remapped amplitude sweep of TB1 at 13.6 GHz. The vertical axis is the fitted reduced drive amplitude from Fig. 4.13 rather than the nominal VNA output. The microwave coupling is stronger than at 18.3 GHz, and the lower photon energy increases the reduced drive amplitude for the same ac voltage. The false-color map resolves individual photon assisted replicas to at least order 25. The parameters are listed in Table 4.1.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.15. Tien-Gordon simulation of the remapped 13.6 GHz amplitude sweep. The static reference parameters are taken from the matching column of Table 4.1. The calculated multiphoton sidebands provide the model reference for the high-order structures visible in Fig. 4.14, without assigning a deterministic photon count history to individual tunneling events.

timed sequence of absorption events. The calibration is also local to this device, frequency, and antenna configuration. Heating and nonequilibrium quasiparticle generation may still contribute at high drive, even though the dominant visible structure remains organized by the Tien-Gordon sideband pattern.

To my knowledge, no previous amplitude map of quasiparticle photon assisted tunneling in an SIS junction resolves and continuously tracks individual replicas to a higher order under this definition. The amplitude map reported by Kot *et al.* shows approximately 13 visually distinguishable replica orders [32]. The stable full amplitude map presented here follows the individual branches to at least order 25. This comparison concerns visually resolved quasiparticle replicas, not the reduced drive reached in a single spectrum, high-order Shapiro response, or photon assisted subgap transport in finite-transmission weak links.

The amplitude calibration data therefore serve two purposes. They establish a quantitative conversion from nominal microwave output to the reduced drive amplitude used on the y-axis, and they demonstrate that the setup resolves photon assisted structure deep into the multiphoton regime. With this reference established, the next step is to study a related tunnel barrier experiment where the plain BCS plus Tien-Gordon description is no longer sufficient.

4.1.2 Rectification and Asymmetry

The measurements in this section are an earlier attempt to test whether microwave driven quasiparticle transport can be reproduced in an Al TB. I measure the similar device TB2 with an earlier version of the setup, before performing the optimized measurements on TB1 discussed in Sec. 4.1.1. Although the spectra of TB2 are less sharply resolved, they show reproducible features that depart from the expected bias symmetric SIS response and motivate the analysis presented here.

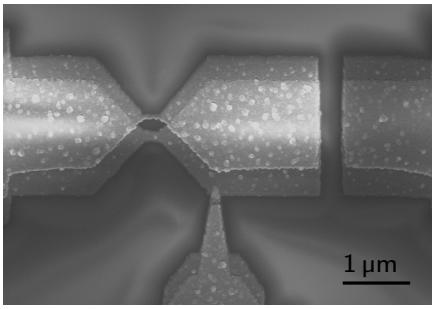


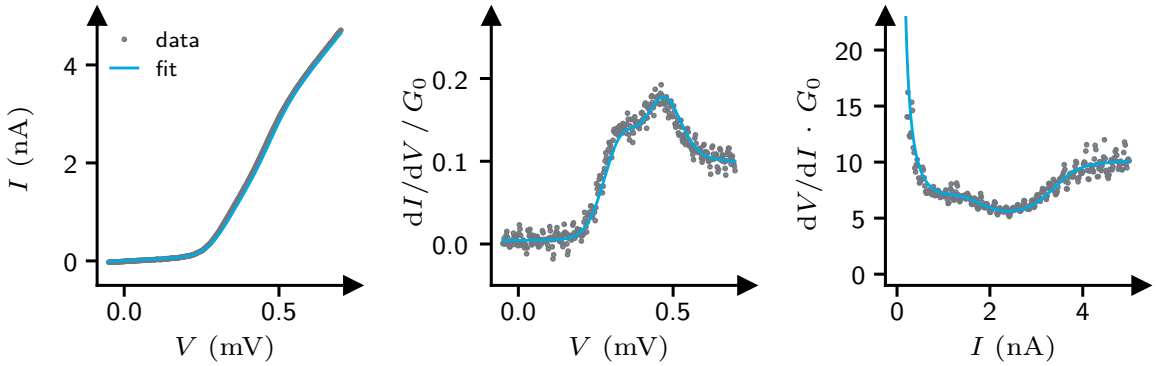
Figure 4.16. SEM image of TB2. The 60 nm lower Al layer is oxidized in situ at 3.4 mbar for 3 min. The angled top layer corresponds to 77 nm Al.

TB2 also provides a useful comparison because its fabrication differs from TB1 in several identifiable ways. The junction finger of TB2 is shifted to the left compared to TB1. This shift reduces the unsupported length of the evaporation mask after repeated mask collapse during evaporation. The aluminum layers are slightly thinner, and the oxidation pressure and duration differ moderately. Most importantly, the shadow electrode near the break junction remains connected because the evaporation mask has more undercut. I nevertheless expect the junction itself to remain in a similar tunnel regime.

Static Transport and Reference Parameters

The static transport response initially supports this expectation. TB2 shows a clear quasiparticle onset and no resolved supercurrent, and its *I*–*V* curve can be described by the same SIS model used for TB1. Under microwave irradiation, however, the transport becomes strongly bias asymmetric within a narrow frequency range. The purpose of this section is therefore to characterize this unexpected rectification, establish its dependence on microwave amplitude and frequency, and test whether it correlates with microwave coupling or sample heating. The available measurements limit possible explanations but do not identify a unique microscopic mechanism.

I use the representative *I*–*V* curve in Fig. 4.17 to determine the static junction parameters G_{N}, Δ_0, γ, and σ_V. The fit also returns the trace dependent fitted temperature T_{fit} and effective microwave amplitude A_{fit} for this curve. The corresponding values and measurement conditions are listed in Table 4.2. I do not apply the additional offset-correction algorithm discussed in Sec. 3.3.2 to these data. This algorithm assumes a symmetric *I*–*V* response, which is not the case for this measurement.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.17. Representative transport trace of TB2. The panels show the measured *I–V* curve and the numerically derived *dI–dV* and *dV–dI* responses. The *I–V* curve is fitted with the SIS BCS model, with the extracted parameters listed in Table 4.2.

Compared with TB1, TB2 has approximately half the normal-state conductance, while its fitted superconducting gap remains similar. The representative trace has a higher T_{fit} and shows stronger spectral rounding, reflected in the larger fitted values of γ and σ_V. I attribute the reduced spectral resolution primarily to the non-optimized measurement setup and its less effective filtering. The fit alone, however, cannot distinguish instrumental broadening from intrinsic broadening of the junction.

Table 4.2. Static parameters and microwave calibration values.

G_{N} (G_0)	0.092746(8)
T_{fit} (mK)	534(41)
Δ_0 (µeV)	203.1(10)
γ (µeV)	5.5(17)
σ_V (µV)	41.0(37)
ν (GHz)	7.78
A_{fit} (µV)	105.9(12)
A_{out} (mV)	200
T_{bath} (mK)	277(1)

I perform these measurements with the setup described by Patrick Raif and Martin Prestel, with several changes to the low-temperature wiring [47, 94]. Instead of the commercial thermalizer and filter, I use a sequence of copper-powder filters and stainless steel capillaries. The reference resistor is already located at low temperature, in contrast to the configuration used by Martin Prestel. Microwave irradiation is again applied through the antenna.

For the microwave amplitude sweeps, I hold the static junction parameters G_{N}, Δ_0, γ, and σ_V fixed at the values obtained from the representative fit. I hold ν at the nominal applied frequency. I then allow T_{fit} and A_{fit} to vary for each irradiated trace. This procedure follows the analysis of TB1 and provides a common reference for comparing the different drive frequencies. Unlike TB1, however, TB2 develops a pronounced bias asymmetry under microwave irradiation. The symmetric SIS plus Tien-Gordon model cannot reproduce this response, so the extracted T_{fit} and A_{fit} should be interpreted as effective comparison parameters rather than as a complete physical description of the asymmetric curves.

Motivating Asymmetric Response

The deviation from the symmetric SIS response becomes pronounced at selected frequencies and sufficiently large microwave amplitudes. Figure 4.18 shows a representative trace at $\nu = 7.97\,\mathrm{GHz}$. To compare the two bias polarities, I reflect the negative-bias branch about the origin and plot it together with the positive-bias branch. The two branches differ substantially in the quasiparticle onset region and remain separated at larger bias. The corresponding *dI–dV* and *dV–dI* curves show that the asymmetry is associated with a redistribution and displacement of the spectral features rather than with a constant current or voltage offset.

I record the same trace with increasing and decreasing bias. The two sweep directions overlap within the experimental resolution. The asymmetry is therefore reproducible and does not originate from sweep direction hysteresis on the measurement timescale of 3 min.

To quantify the effect locally, I define normalized bias asymmetry parameters for the three transport representations,

$$\chi_I(V) = \frac{I(V) + I(-V)}{|I(V) - I(-V)|}\,, \tag{4.6}$$

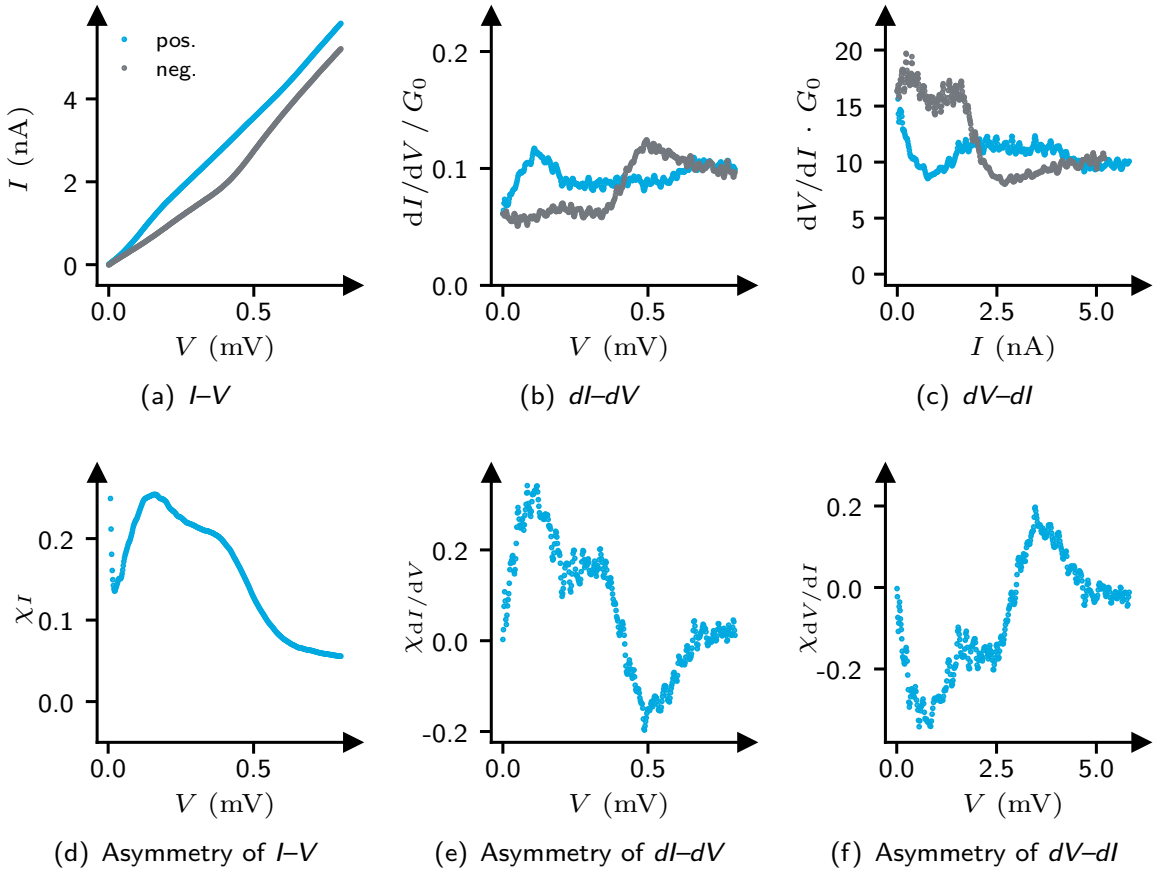
$$\chi_{\mathrm{d}I/\mathrm{d}V}(V) = \frac{\frac{\mathrm{d}I}{\mathrm{d}V}(V) - \frac{\mathrm{d}I}{\mathrm{d}V}(-V)}{\left|\frac{\mathrm{d}I}{\mathrm{d}V}(V) + \frac{\mathrm{d}I}{\mathrm{d}V}(-V)\right|}\,, \tag{4.7}$$

$$\chi_{\mathrm{d}V/\mathrm{d}I}(I) = \frac{\frac{\mathrm{d}V}{\mathrm{d}I}(I) - \frac{\mathrm{d}V}{\mathrm{d}I}(-I)}{\left|\frac{\mathrm{d}V}{\mathrm{d}I}(I) + \frac{\mathrm{d}V}{\mathrm{d}I}(-I)\right|}\,. \tag{4.8}$$

All three quantities vanish for an ideally bias symmetric transport response. Their signs indicate which bias branch has the larger response, while their magnitudes quantify the relative difference between the branches. Because the normalization can become small, particularly in the derivative representations, I use χ primarily to identify the bias regions in which the symmetry breaking occurs, rather than as a globally averaged measure.

For the representative trace in Fig. 4.18, the local current asymmetry reaches $\chi_I = 0.254$. The largest values occur in the voltage range where the positive- and negative-bias quasiparticle onsets evolve differently. This observation shows that the microwave response cannot be described by a single bias-symmetric SIS plus Tien-Gordon curve. The full amplitude sweep at this frequency is shown in Fig. E.2, in the Appendix.

This trace creates a calibration problem. For TB1, each irradiated curve can be fitted with the symmetric SIS plus Tien-Gordon model and remapped using its individual A_{fit}. When the rectification in TB2 becomes large, however, the measured response violates the bias symmetry assumed by this model. The resulting A_{fit} is only then a compromise between two differently evolving bias branches and no longer represents the physical microwave amplitude. A pointwise remap-



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

(d) Asymmetry of *I–V* (e) Asymmetry of *dI–dV* (f) Asymmetry of *dV–dI*

Figure 4.18. Representative asymmetric transport trace of TB2. Panels (a) - (c) compare the positive-bias branch with the negative-bias branch reflected about the origin for the measured *I–V* curve and the numerically derived *dI–dV* and *dV–dI* responses. Panels (d) - (f) show the corresponding local normalized asymmetry defined in Eq. (4.8). The trace is measured at 7.97 GHz, $A_{\mathrm{out}} = 720\,\mathrm{mV}$, and $T_{\mathrm{bath}} = 917(1)\,\mathrm{mK}$.

ping based on these values would therefore imprint the failure of the symmetric model onto the calibrated amplitude axis.

Effective Microwave Amplitude Calibration

I nevertheless fit every complete *I–V* curve with the symmetric SIS plus Tien-Gordon model and extract A_{fit} and T_{fit}. For asymmetric curves, these values are effective fit parameters because the model cannot reproduce both bias branches simultaneously. I therefore do not use the individual A_{fit} values as the remapped amplitude coordinate.

Instead, I determine one effective coupling coefficient for each frequency. For every nonzero source amplitude in a frequency sweep, I evaluate $A_{\mathrm{fit}}/A_{\mathrm{out}}$. All

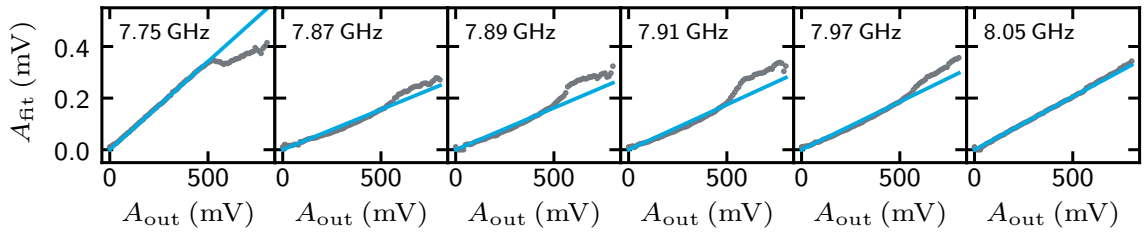


Figure 4.19. Effective microwave-amplitude calibration of TB2 at selected frequencies. The points show A_{fit} obtained from the complete signed *I–V* curves. The lines show the frequency dependent effective coupling used to define A_{cal} in Eq. (4.9). Deviations from the lines are largest where the bias asymmetry prevents a quantitative symmetric-model fit.

ratios are included without rejection based on fit quality or asymmetry. I define the frequency dependent coupling and its relation as

$$\eta(\nu) = \underset{A_{\mathrm{out}}>0}{\mathrm{median}}\left(\frac{A_{\mathrm{fit}}}{A_{\mathrm{out}}}\right), \qquad A_{\mathrm{cal}} = \eta(\nu)A_{\mathrm{out}}\,. \tag{4.9}$$

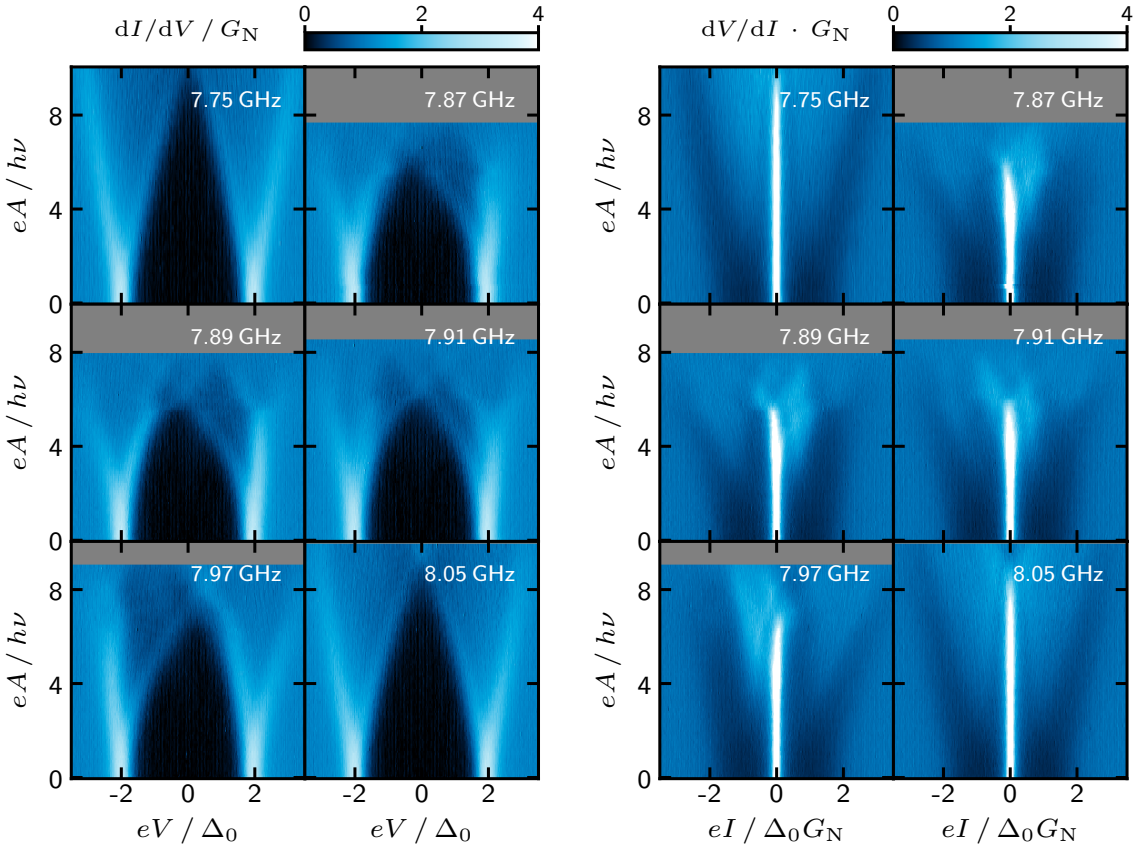
The median reduces the influence of scattered fits in the asymmetric region, while the symmetric parts of each sweep provide the amplitude scale. This procedure yields a common effective comparison axis but does not constitute an independent measurement of the ac voltage across the junction.

Frequencies with weaker coupling do not reach the full displayed A_{cal} range. With this effective amplitude coordinate established, I now compare how the measured transport evolves across the investigated frequency interval.

Frequency Dependent Transport Response

The representative trace establishes the presence of microwave induced rectification at one operating point. To determine whether the effect is isolated or evolves systematically, I measure microwave amplitude sweeps from 7.75 to 8.05 GHz in steps of 50 MHz. Figure 4.20 compares the resulting *dI–dV* and *dV–dI* landscapes at six selected frequencies. The corresponding branch resolved line cuts and the complete *I–V*, *dI–dV*, and *dV–dI* frequency comparisons are provided in Appendix E.

The *dI–dV* maps provide the clearest view of the spectral evolution. At the lowest and highest frequencies, the two quasiparticle onset ridges remain approximately symmetric. At intermediate frequencies, one ridge is suppressed and displaced relative to the other, and the direction of this deformation reverses across the investigated frequency interval. Near the reversal, an additional conductance feature appears between the two main ridges. The *I–V* and *dV–dI* representations show the same systematic amplitude and frequency dependence, although the narrow resistance features make the latter more sensitive to noise.



(a) *dI–dV* (b) *dV–dI*

Figure 4.20. Microwave amplitude evolution of the asymmetric transport response of TB2 at six selected frequencies. Panel (a) shows the *dI–dV* response as a function of voltage and calibrated amplitude, while panel (b) shows the *dV–dI* response as a function of current and calibrated amplitude. The quasiparticle features evolve approximately symmetrically at the lowest and highest frequencies but become strongly bias asymmetric at intermediate frequencies, with the direction of the deformation reversing across the investigated interval. Gray regions indicate amplitudes for which no measurement is available.

The calibrated landscapes establish that the symmetry breaking evolves systematically with both microwave amplitude and frequency. The symmetric calculation reproduces the overall photon assisted redistribution of spectral weight but not the unequal evolution of the two bias branches. I next quantify this difference over the complete data set.

Rectification Analysis

The local quantity χ identifies the bias regions in which the representative trace becomes asymmetric. To compare all measured curves without selecting a particular bias point, I assign each transport curve a signed integrated asymmetry $\mathcal{A}$,

$$\mathcal{A}_I = \frac{\int [I(V) + I(-V)]\,\mathrm{d}V}{\int |I(V) - I(-V)|\,\mathrm{d}V}\,, \tag{4.10}$$

$$\mathcal{A}_{\mathrm{d}I/\mathrm{d}V} = \frac{\int \left(\frac{\mathrm{d}I}{\mathrm{d}V}(V) - \frac{\mathrm{d}I}{\mathrm{d}V}(-V)\right)\mathrm{d}V}{\int \left|\frac{\mathrm{d}I}{\mathrm{d}V}(V) + \frac{\mathrm{d}I}{\mathrm{d}V}(-V)\right|\mathrm{d}V}\,, \tag{4.11}$$

$$\mathcal{A}_{\mathrm{d}V/\mathrm{d}I} = \frac{\int \left(\frac{\mathrm{d}V}{\mathrm{d}I}(I) - \frac{\mathrm{d}V}{\mathrm{d}I}(-I)\right)\mathrm{d}I}{\int \left|\frac{\mathrm{d}V}{\mathrm{d}I}(I) + \frac{\mathrm{d}V}{\mathrm{d}I}(-I)\right|\mathrm{d}I}\,. \tag{4.12}$$

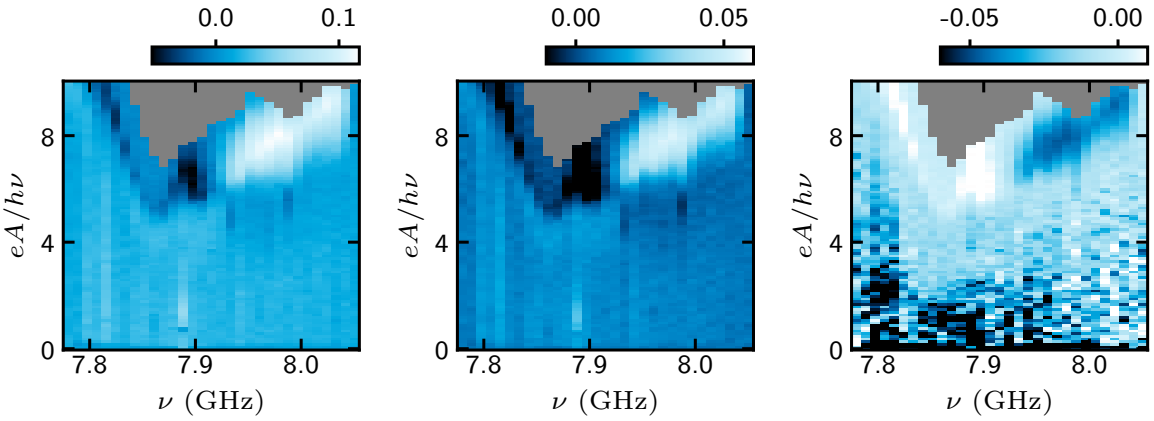
The integrations use the common symmetric bias interval available for the two reflected branches. In contrast to the local quantity χ, $\mathcal{A}$ summarizes the signed asymmetry of an entire curve. Positive and negative contributions can therefore cancel.

Figure 4.21 shows $\mathcal{A}$ as a function of frequency and calibrated amplitude. The magnitude of the *I–V* asymmetry generally increases with amplitude and changes sign across the investigated frequency interval. The *dI–dV* asymmetry follows the same broad frequency dependence, while the *dV–dI* asymmetry has the opposite sign. The derivative measures are noisy at low amplitude because their normalization becomes small, but the large-amplitude structure is consistent across the three transport representations.

To compare the frequency dependence on a single axis, I average each asymmetry measure over the same calibrated-amplitude interval. I evaluate the average by integration rather than by an arithmetic mean to account for the uneven spacing of the A_{cal} points,

$$\langle \mathcal{A}_X \rangle_{A_{\mathrm{cal}}} = \frac{1}{A_{\mathrm{cal,max}} - A_{\mathrm{cal,min}}} \int_{A_{\mathrm{cal,min}}}^{A_{\mathrm{cal,max}}} \mathcal{A}_X(A_{\mathrm{cal}})\,\mathrm{d}A_{\mathrm{cal}}\,, \tag{4.13}$$

where X denotes I, $\mathrm{d}I/\mathrm{d}V$, or $\mathrm{d}V/\mathrm{d}I$. Panels (a) - (c) of Fig. 4.23 show the resulting averages.



(a) Rectification $\mathcal{A}_I$ (b) Rectification $\mathcal{A}_{\mathrm{d}I/\mathrm{d}V}$ (c) Rectification $\mathcal{A}_{\mathrm{d}V/\mathrm{d}I}$

Figure 4.21. Frequency and amplitude dependent rectification of TB2. The panels show the signed integrated asymmetry $\mathcal{A}$ of the *I–V*, *dI–dV*, and *dV–dI* responses. Gray regions indicate amplitudes for which no measurement is available at the corresponding frequency.

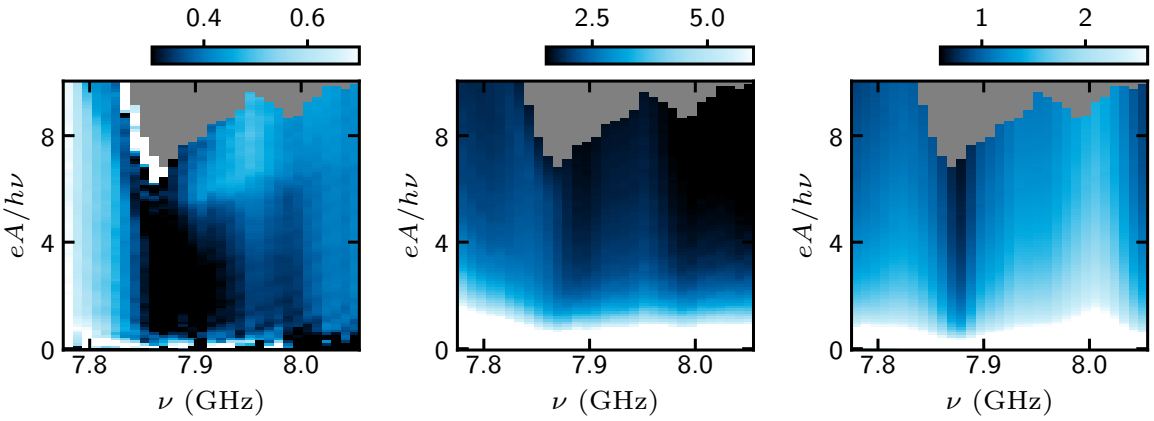
This frequency and amplitude dependent map is the central experimental characterization of the rectification effect. It shows a drive threshold, an increase in magnitude with amplitude, and a sign reversal within a narrow frequency interval, consistently across all three transport representations. The next section tests whether this frequency dependence traces back to microwave coupling or to sample heating.

Coupling and Heating Controls

I use separate diagnostic fits to test whether the frequency dependence of the asymmetry follows the microwave coupling or sample heating. These fits compare the effective amplitude, fitted temperature, and bath temperature against the available amplitude and temperature coordinates. Their definitions and the corresponding control plots are collected in Appendix E.

Figure 4.22 shows the pointwise amplitude and temperature ratios. Panels (d) - (i) of Fig. 4.23 summarize the slopes extracted from the diagnostic fits as functions of frequency. The clipped temperature fits use empirical lower and upper bounds. These bounds stabilize the comparison across frequencies but are not independent measurements of the electronic base temperature or superconducting critical temperature. The displayed coupling parameters are retrieved, as described in Appendix E.

Most diagnostic slopes neither correlate nor anticorrelate clearly with the amplitude averaged asymmetry. The bath heating slopes show the most similar frequency dependence, but this comparison alone does not establish heating as



(a) $A_{\mathrm{fit}}/A_{\mathrm{out}}$ (10^3) (b) $T_{\mathrm{fit}}/A_{\mathrm{out}}$ $(\mathrm{K\,V^{-1}})$ (c) $T_{\mathrm{bath}}/A_{\mathrm{out}}$ $(\mathrm{K\,V^{-1}})$

Figure 4.22. Frequency and amplitude dependent control quantities. The panels show the pointwise ratios $A_{\mathrm{fit}}/A_{\mathrm{out}}$, $T_{\mathrm{fit}}/A_{\mathrm{out}}$, and $T_{\mathrm{bath}}/A_{\mathrm{out}}$. Gray regions indicate amplitudes for which no measurement is available at the corresponding frequency.

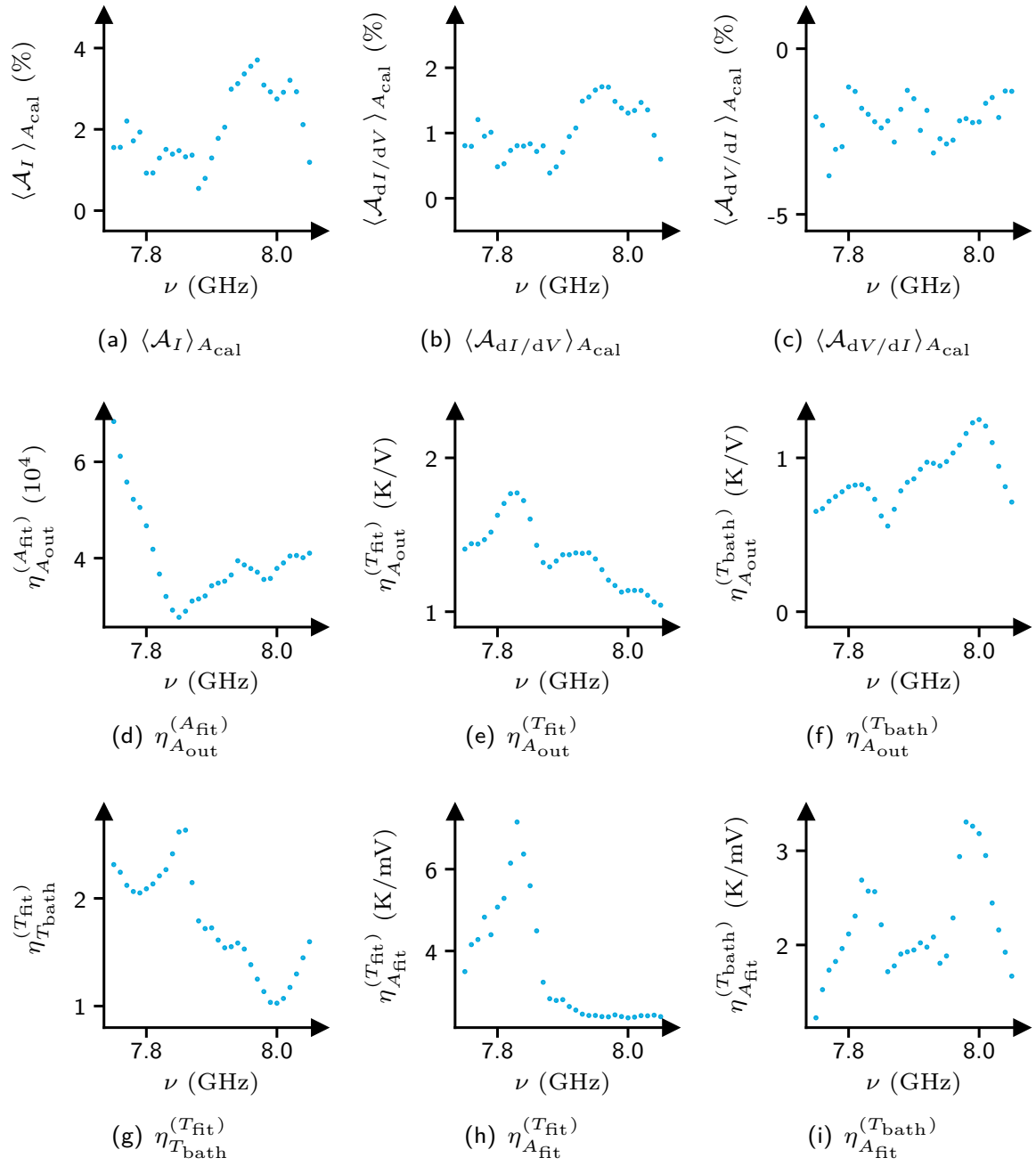
the cause of the rectification. Uniform heating changes the quasiparticle spectrum but cannot by itself break bias symmetry.

Interpretation and Model Limits

The measurements establish a reproducible microwave induced breaking of bias symmetry in TB2. At the representative operating point, increasing and decreasing bias sweeps overlap within the experimental resolution. The diode like response is therefore single valued and nonhysteretic on the measurement timescale. Its defining features are an onset at elevated drive, confinement to a narrow frequency interval, and a reversal of direction within that interval.

The static control measurements show conventional superconducting behavior. Magnetic field dependent maps contain no unexpected features, and the transport remains unchanged over a wide range of dc gate voltages. Together with the low-impedance connection through the intact break junction, these observations support treating the asymmetry as a microwave induced response rather than a property of a static flux sensitive or Coulomb blockaded state.

The narrow frequency interval points to resonant coupling through the microwave environment. In an earlier transmission experiment in the same cryostat, I observed a feature-rich transfer function and associated prominent features with different parts of the setup. The dominant resonance was consistent with the radius of the semicylindrical shielding cup used at the time [109]. The present cylindrical cup has a diameter of 38 mm, close to the free-space wavelength $\lambda_0 = c/\nu \approx 37.8\,\mathrm{mm}$ at 7.93 GHz. This correspondence supports an



(a) $\langle \mathcal{A}_I \rangle_{A_{\mathrm{cal}}}$ (b) $\langle \mathcal{A}_{\mathrm{d}I/\mathrm{d}V} \rangle_{A_{\mathrm{cal}}}$ (c) $\langle \mathcal{A}_{\mathrm{d}V/\mathrm{d}I} \rangle_{A_{\mathrm{cal}}}$

(d) $\eta^{(A_{\mathrm{fit}})}_{A_{\mathrm{out}}}$ (e) $\eta^{(T_{\mathrm{fit}})}_{A_{\mathrm{out}}}$ (f) $\eta^{(T_{\mathrm{bath}})}_{A_{\mathrm{out}}}$

(g) $\eta^{(T_{\mathrm{fit}})}_{T_{\mathrm{bath}}}$ (h) $\eta^{(T_{\mathrm{fit}})}_{A_{\mathrm{fit}}}$ (i) $\eta^{(T_{\mathrm{bath}})}_{A_{\mathrm{fit}}}$

Figure 4.23. Frequency dependence of the integrated asymmetry and diagnostic coupling coefficients. Panels (a) - (c) show the amplitude averaged asymmetry defined in Eq. (4.13). Panels (d) - (i) show the slopes extracted from the amplitude and temperature correlation fits. Similar frequency dependence indicates correlation only and does not establish a causal mechanism.

electromagnetic mode of the sample environment as the origin of the frequency selectivity.

Such a mode couples to the local circuit formed by the island, shadow electrode, TB counterelectrode, and BJ lead. The sample lies near the center of the cup, where a field node or another symmetry point can produce a strong spatial variation of the field. A displacement from this point, together with unequal lead impedances and electrical path lengths, can generate different microwave amplitudes and phases at the electrodes. Resonant enhancement then provides a natural origin for the drive threshold, while the phase evolution across the resonance can account for the reversal of the rectification direction.

The intact BJ provides a phase sensitive element in this circuit. A similar constriction in sample AC exhibits a conventional Josephson response in its pristine state, including a finite switching current (Sec. F.1). I therefore treat the BJ in TB2 as a Josephson weak link. Under irradiation, phase locking can generate an inverse ac Josephson response and Shapiro structure [28]. Near 7.8 GHz, the integer Shapiro spacing is $h\nu/(2e) \approx 16\,\mu\mathrm{V}$, below the fitted voltage broadening $\sigma_V \approx 41\,\mu\mathrm{V}$ of TB2. The absence of resolved steps is therefore consistent with the experimental resolution.

In this interpretation, the TB detects the influence of the driven weak link on photon assisted quasiparticle transport. A phase locked voltage contribution across the BJ changes the island potential and redistributes the PAT features of the combined device. This resembles Giaever's detection of the ac Josephson effect with a second tunnel junction, although the topology and coupling differ here [110]. Unequal relaxation paths in the island and counterelectrode may additionally produce different quasiparticle distributions, while heating can broaden these distributions and enhance the available spectral weight. The observed heating correlation is consistent with this supporting role.

The measurements thus support a working sequence in which a resonance redistributes the local microwave amplitude and phase, the BJ contributes a phase dependent island potential, and the TB converts this response into bias asymmetric quasiparticle transport. This sequence accounts qualitatively for the threshold, frequency selectivity, and sign reversal. Its microscopic details remain open because the experiment does not independently resolve the microwave fields, the voltage across the BJ, or the electrode distributions.

Significance and Outlook

The central result is a microwave controlled rectification effect whose magnitude and polarity are tuned jointly by dc bias, microwave amplitude, and frequency. The response reaches a local current asymmetry of 25 %, appears above a drive threshold, and reverses direction within a narrow frequency interval. It therefore represents a distinct transport regime beyond the bias symmetric Tien-Gordon model. To my knowledge, this combination of amplitude controlled onset and

frequency controlled polarity reversal has not previously been reported for an SIS tunnel junction.

The response also demonstrates that the electromagnetic environment can become an active part of quasiparticle transport. The measurements are consistent with resonant microwave coupling to a local circuit containing the intact Josephson weak link and the tunnel barrier. A deliberately designed resonator or impedance environment would provide the most direct test of this interpretation. Such an experiment could determine whether the observed rectification can be reproduced and controlled as a junction property rather than as a sample specific response.

If reproducible, the frequency sensitive sign reversal could provide a basis for a microwave controlled superconducting diode or a local probe of resonant microwave coupling. Establishing such functionality would require calibrated delivered power, zero-bias current or open-circuit voltage measurements, noise characterization, and measurements of nominally identical devices. The present experiment establishes the tunable rectification and identifies a testable physical route toward these applications.

4.2 Atomic Contacts

The tunnel junctions considered in the preceding section provide a fixed weak transmission reference. A mechanically controllable break junction extends this picture by allowing the same aluminum constriction to be tuned from a many-channel metallic connection through few-channel atomic configurations and finally into the tunnel regime. The normal-state conductance is a measure of the sum of the transmission probabilities. In the superconducting state, multiple Andreev reflection produces a nonlinear subgap spectrum that instead depends on the individual transmission eigenvalues. The central purpose of this section is therefore to use superconducting transport to follow how the mesoscopic channel composition changes as the junction is opened and to determine how this channel dependent transport responds to microwave irradiation.

I perform these measurements on sample AC ("atomic contact"), shown in Fig. 4.24. The device is fabricated as a mechanically controllable aluminum break junction. The cryogenic and electrical setup is described in Sec. 3.2 and Ref. [47]. The synchronized acquisition and evaluation procedure is described in Sec. 3.3.

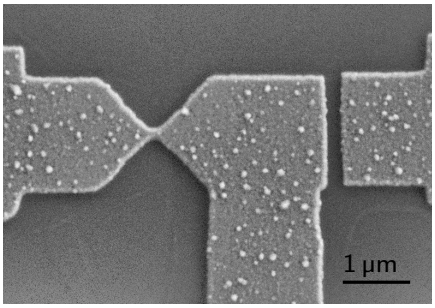


Figure 4.24. Scanning electron image of MCBJ sample AC with an 80 nm thick aluminum film.

Mechanical adjustment of the break junction produces several metastable contact configurations that remain sufficiently stable for detailed transport measurements. I denote these configurations as C0 - C3. The labels refer to distinct microscopic arrangements of the same junction. Each configuration has its own normal-state conductance and set of transmission eigenvalues. I select these configurations for measurements in different transport regimes.

Before discussing the higher-transmission contacts, I use C0 as a tunnel regime reference, documented in Appendix F.1. Its static spectrum is dominated by one weakly transmitting channel. Under irradiation, C0 recovers conventional Tien-Gordon photon assisted quasiparticle tunneling. A weak subgap contribution marks the onset of Andreev reflection.

In Sec. 4.2.1, I first follow the mesoscopic pincode along a complete opening trace. Channel resolved MAR fits reveal changes in the individual transmissions that remain hidden in the total normal-state conductance. The extracted channels evolve continuously along conductance plateaus and reorganize at abrupt jumps. I use the tunnel decay after rupture to establish an approximate relative displacement scale and compare the resulting channel evolution with plausible aluminum contact geometries. This comparison restricts the transition from a few-atom constriction to a tunnel junction, although the transport data do not

determine a unique microscopic atomic structure. Representative fits and the complete measured and reconstructed landscapes are provided in Appendix F.4.

Section 4.2.2 then examines microwave driven transport in the two few-channel contacts C1 and C2, which have contrasting pincodes. The measured response contains replicas associated with the quasiparticle threshold, the MAR structure, and a narrow zero-bias Cooper pair contribution. I calibrate the effective drive amplitude from the transport data and test how these components must be treated within a common description. Threshold weighted photon assisted MAR captures the dominant finite-bias trajectories, while photon assisted incoherent Cooper pair tunneling describes the zero-bias-derived replicas when that contribution can be separated. The comparison between C1 with moderate channel transmissions and C2 with a nearly ballistic leading channel tests both the useful range and the low-voltage limitations of this construction. Appendix F.5 collects the static and coupling parameters, charge order decompositions, model selection residuals, and complete transport landscapes.

Finally, Sec. 4.2.3 considers C3 after the junction is reclosed into a high-conductance state. Many channels contribute in this regime, so a unique channel-by-channel pincode cannot be extracted. The static characteristic also contains sharp MAR-like subgap features. Their conductance maxima are displaced from the nominal opening voltages, as expected when the line shape changes with transmission, and no separate maximum is resolved for the fourth order. None of the tested transmission distributions reproduces the complete line shape. I therefore use the measured static response as the input for a phenomenological replica model under irradiation. This procedure reproduces the principal amplitude dependent structures and exposes additional fractional trajectories, but it does not assign them a unique microscopic origin. C3 consequently marks the point at which the controlled few-channel description gives way to a more complex many-channel response. Its complete measured and phenomenological landscapes are shown in Appendix F.6.

Magnetic field is the remaining control parameter that I apply systematically to C0 - C3. The field dependent maps show that the superconducting transport features disappear at a common empirical scale near 28 mT despite the different contact regimes. These measurements are not required for the microwave models and are therefore collected in Appendix F.3. The three subsections below thus follow the junction across increasing model complexity, from channel resolved static transport through photon assisted few-channel transport to the phenomenological high-conductance regime.

I perform all measurements outside Sec. 4.2.1 together with Patrick Raif during his master's thesis work [47]. His thesis describes the corresponding state of the setup.

4.2.1 The Mesoscopic Pincode

What happens when a highly transmitting aluminum contact is opened very slowly in picometer scale steps? The constriction does not narrow uniformly. Its conductance evolves along tilted plateaus interrupted by abrupt jumps, suggesting elastic deformation of metastable structures followed by atomic rearrangements. The opening trace reveals when these changes occur, but not how the current is redistributed among the conducting channels. Superconducting transport provides this missing information because each channel produces a characteristic multiple Andreev reflection (MAR) structure. The channel resolved method introduced by Scheer *et al.* therefore turns the nonlinear subgap spectrum into a probe of the evolving contact [16]. The extracted mesoscopic pincode $\{\tau_i\}$ is the set of transmission eigenvalues obtained from the measured spectrum.

In this section, I follow the pincode along a complete opening trace rather than analyzing isolated contact states. The fitted spectra test whether the MAR model captures the measured spectral evolution, while the tunnel decay provides a relative displacement calibration. I then use the extracted mesoscopic pincode $\{\tau_i\}$ to distinguish continuous evolution within plateaus from abrupt reorganizations between them. Comparison with orbital based models for aluminum suggests plausible changes of the contact symmetry and atomic geometry, but it does not determine a unique microscopic structure.

Pincode Extraction and Fit Quality

I record superconducting *I*–*V* characteristics while opening a highly transmitting contact in small motor steps. I represent the measured current as a sum of independent MAR contributions and assign a best-fit pincode to every recorded contact state. Different sets of $\{\tau_i\}$ produce distinct MAR contributions, so the nonlinear subgap spectrum contains information that cannot be retrieved from the normal-state conductance G_N alone. The physical basis of this extraction is introduced in Secs. 2.3.1 and 2.3.4, and the numerical fitting procedure is described in Sec. 3.3.2. The model assumes a short coherent contact with energy independent transmissions. It therefore extracts the effective transmission eigenvalues that reproduce the measured dc MAR spectrum. The fit does not uniquely determine the underlying atomic orbitals or contact geometry. I infer plausible microscopic configurations later by comparing the number, magnitude, degeneracy, and evolution of the fitted channels with theoretical expectations for aluminum. Inelastic transport inside the contact and energy dependent scattering are not included in this interpretation.

In the final fits, I fix the temperature and Dynes broadening to $T = 0$ and $\gamma = 0$. The bath temperature remains below 50 mK, corresponding to 4.3 μeV. This energy is small compared with both the superconducting gap and the ob-

served spectral width, which makes the zero-temperature approximation appropriate for the present extraction. Setting $\gamma = 0$ is a minimal model assumption rather than evidence for an infinite quasiparticle lifetime. Finite electronic temperature, lifetime broadening, and voltage noise all smooth the MAR structure and cannot be separated reliably in this data set. I therefore do not fit these effects simultaneously. Instead, I convolve the calculated voltage dependence with a Gaussian of width σ_V, which represents the experimental voltage resolution together with unresolved thermal and lifetime broadening. I allow the zero-temperature gap Δ_0 and σ_V to vary independently for every spectrum in the same fitting step as the channel transmissions. Although the pincode changes substantially during opening, both nuisance parameters remain concentrated around

$$\Delta_0 = 188.5(5)\,\mu\mathrm{eV}\,, \qquad \sigma_V = 25(1)\,\mu\mathrm{V}\,. \tag{4.14}$$

Here, the central values are the means across the trace and the uncertainties are the corresponding standard deviations, not the statistical uncertainties of individual fits. The medians give the same central values within rounding. The clustering is physically reasonable because Δ_0 is set by the aluminum electrodes and σ_V mainly by the measurement chain, whereas the changing atomic configuration primarily affects $\{\tau_i\}$. No systematic drift of Δ_0 or σ_V is apparent along the trace.

I minimize the current-space residual with point weights proportional to the inverse absolute voltage $w_k \propto 1/|V_k|$. This weighting reduces the influence of larger absolute currents at high bias and retains sensitivity to the subgap MAR structure. The zero-voltage point carries zero weight, and I exclude the central interval containing the switching branch because the dc MAR model does not describe it.

Figure F.11 in the supplementary material, Sec. F.4, shows six representative contact states. For each state, the best-fit pincode reproduces the measured *I*–*V* characteristic and the corresponding differential conductance. This comparison covers states with different conductances and numbers of fitted channels and tests the MAR line shape over the full bias range. The derivative is calculated from the same measured *I*–*V* data and is therefore a complementary representation. The reported transmissions are best-fit values on a grid with spacing 0.001. This spacing is the numerical sampling resolution, not a confidence interval. Nearly equal channels cannot be distinguished, and sufficiently weak channels can fall below the sensitivity of the spectrum. The fitted channel count should consequently be understood as the number of contributions resolved within this model and data quality, not as an exact count of conduction channels.

The discrete optimization uses random restarts to reduce its sensitivity to the initial pincode, but it does not guarantee the global minimum. Independent restarts converge to the same dominant transmissions. Because the fit also per-

mits $\tau_i = 0$, solutions with a larger maximum channel count include solutions with fewer contributing channels and can contain nearly degenerate weak contributions. A full statistical uncertainty analysis is outside the scope of this section. Claims of channel degeneracy, splitting, and continuity below are therefore qualitative features of the fitted eigenvalue spectrum. They should not be read as confidence qualified identification or tracking of individual orbital channels.

Displacement Calibration

To calibrate the distance axis, I determine the scale from the conductance decay after the contact enters the tunnel regime. For a rectangular vacuum barrier, the normal-state conductance depends exponentially on the electrode separation x,

$$G_{\mathrm{N}}(x) \propto \exp(-2\kappa x)\,, \qquad \kappa = \sqrt{2m_{\mathrm{e}}U_{\mathrm{eff}}}\big/\,\hbar \approx 1.05\,\text{Å}^{-1}\,. \tag{4.15}$$

Here, m_{e} is the electron mass and U_{eff} is the effective barrier height. Because I separate the aluminum electrodes by breaking the constriction in situ, I assume a clean aluminum surface and use $U_{\mathrm{eff}} = 4.2\,\mathrm{eV}$ [111].

I fit $\ln(G_{\mathrm{N}}/G_0)$ as a linear function of the recorded motor coordinate in the tunnel regime. The slope of this fit, together with Eq. (4.15), converts the motor coordinate into a relative electrode displacement and gives approximately 2 pm per acquisition step. I define $x = 0$ at the beginning of the tunnel regime. Positions recorded while the electrodes are still in contact consequently have $x < 0$. I use this approximate calibration only as a relative displacement scale for visualization. Figure 4.25 shows the fitted data with the arbitrary motor coordinate converted to the calibrated displacement x.

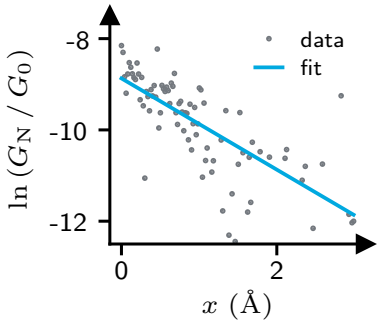


Figure 4.25. Tunnel decay fit used to calibrate displacement.

With this relative displacement scale established, Figs. F.9 and F.10, collected in the supplementary material in Sec. F.4, compare the complete measured and reconstructed transport landscapes. Figure F.9 displays the data supplied to the fitting workflow, while Fig. F.10 is calculated from the extracted pincodes and the mean values in Eq. (4.14), because both parameters describe position independent properties of the electrodes and measurement chain. The agreement between the landscapes provides a global consistency check complementary to the individual spectra in Fig. F.11. Because the reconstruction uses parameters fitted to the same data, it is not an independent validation. It nevertheless shows that the coherent MAR model follows the principal displacement dependent structures and their abrupt or continuous evolution. The *dI–dV* and *dV–dI* representations emphasize deviations that are less visible in the current itself.

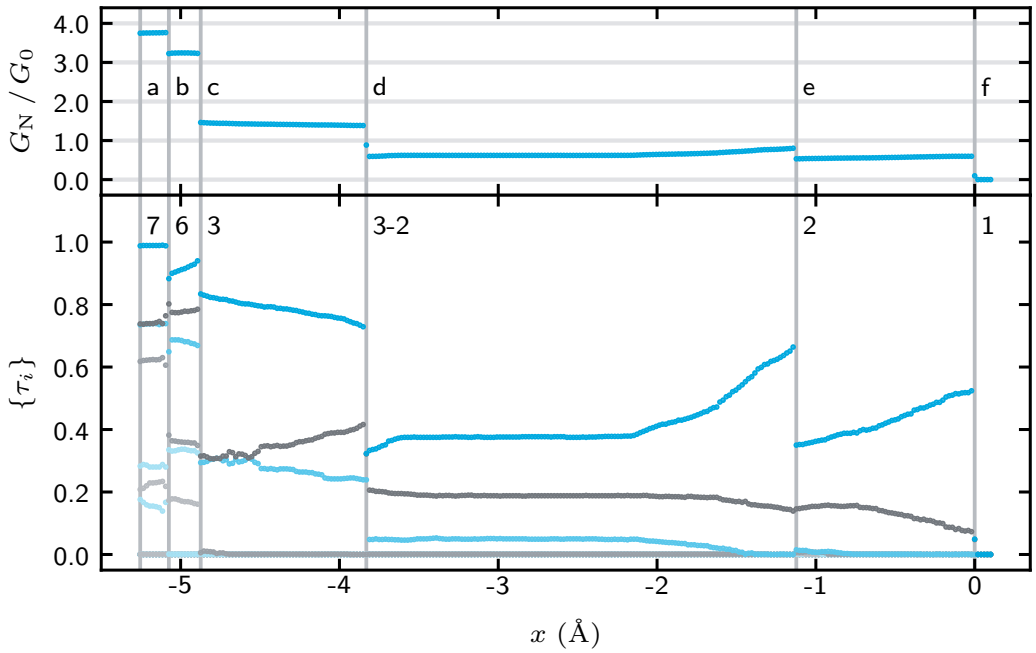
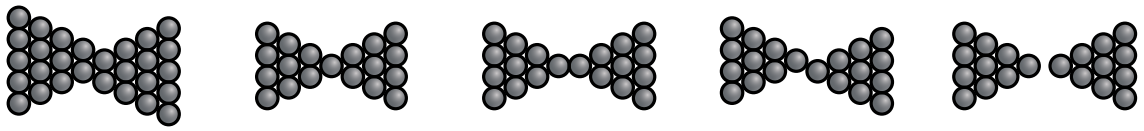


Figure 4.26. Evolution of the mesoscopic pincode while opening the atomic contact. The normal-state conductance G_N/G_0 and the extracted mesoscopic pincode $\{\tau_i\}$ distinguish six successive regions (a) - (f). The number of channels resolved within the MAR model is indicated in each region. Abrupt conductance changes coincide with a reorganization of the fitted pincode, while it evolves continuously within each region. Representative spectra and their MAR fits are shown in Fig. F.11.

Channel Evolution and Contact Geometry

Having established that the fitted pincodes reproduce the measured transport landscape, I now use their displacement dependence to interpret how the contact evolves. The minimal microscopic reference is the tight-binding model of an aluminum one-atom contact developed by Cuevas, Levy Yeyati, and Martín-Rodero [112]. It predicts one highly transmitting channel with mixed axial sp_z character and two weaker channels composed mainly of the transverse p_x and p_y orbitals. The transverse pair is degenerate in a rotationally symmetric contact and splits when the symmetry is broken [61]. Under elongation, the transverse transmissions decrease toward zero while the axial transmission approaches unity [113]. These signatures provide the reference for interpreting the number, degeneracy, and displacement dependence of the extracted channels, but they do not identify a unique atomic geometry.

Figure 4.26 condenses the complete extraction into the normal-state conductance and the individual channel transmissions. The six regions (a) - (f) mark successive metastable contact configurations separated by abrupt rearrangements. Figure 4.27 shows the schematic geometries used to discuss their possible microscopic origin.



(a) Few atoms (b) Monomer (c) Dimer (d) Off-center (e) Tunnel gap

Figure 4.27. Possible atomic configurations during contact opening. The schematics illustrate a few-atom constriction, symmetric monomer and dimer contacts, an off-center contact, and the tunnel regime. They provide microscopic interpretations of the channel evolution in Fig. 4.26, but the measured pincodes do not determine the geometry uniquely.

Configurations (a) and (b) contain up to seven contributing channels with substantial transmission. They are therefore consistent with a few-atom constriction, sketched in Fig. 4.27(a). The jump between them reorganizes the complete pincode. The channel count limits the complexity of the constriction, but not its precise atomic arrangement.

Configuration (c) is fitted with three resolved contributions and is the clearest candidate for a highly symmetric one-atom contact. Its dominant transmission and approximately degenerate pair of weaker transmissions agree qualitatively with the axial sp_z and transverse p_x, p_y contributions expected for aluminum [61]. The fitted pair separates as the plateau evolves, which is consistent with a gradual loss of rotational symmetry. Both the monomer and the axial dimer geometries in Figs. 4.27(b) and 4.27(c) are compatible with the initial state. The later separation therefore suggests that a dimer or one-atom contact becomes distorted or displaced from the contact axis.

Configurations (d) and (e) retain two principal channels even after the conductance falls below G_0. Within both plateaus, the dominant channel strengthens while the weaker contribution is suppressed. This behavior is consistent with the off-center geometry in Fig. 4.27(d), where reduced symmetry suppresses one transverse contribution.

The transition from (d) to (e) is unusually selective within the fitted eigenvalue spectrum. The dominant transmission changes abruptly, while the weaker transmission remains almost continuous and the third stays nearly closed. This pattern is consistent with a rearrangement that mainly modifies the axial transport path while retaining a weak transverse contribution of similar magnitude. The states related to the Al $3p_z$ orbital are especially sensitive to slight changes in position, and first-principles calculations show a strong geometry dependence of aluminum channel transmissions [73, 114]. Nevertheless, the fit determines the eigenvalues and not the eigenvectors, so it cannot verify whether the weaker contribution represents the same orbital channel across the jump. The predominantly axial rearrangement is a microscopic hypothesis.

Overall, the discontinuities between configurations are consistent with atomic rearrangements, while the continuous evolution within each plateau is consistent

with elastic deformation of a metastable structure [16]. This interpretation is supported by a related study of an aluminum atomic contact, in which superconducting transmission channel analysis combined with molecular dynamics and quantum transport calculations attributed bistable conductance switching to the reversible rearrangement of single atoms [115].

In the final configuration (f), a single weak channel enters the tunnel geometry shown in Fig. 4.27(e) and its conductance subsequently decays exponentially according to Eq. (4.15). The proposed sequence from a few-atom constriction through a symmetric monomer or dimer and an off-center contact to tunneling captures the channel evolution, but it is not a unique reconstruction of the atomic geometry.

Distribution of the Extracted Transmissions

The tight-binding results for randomly distorted one-atom aluminum contacts provide a reference for the transmission histogram. They place the dominant axial channel with mixed sp_z character approximately within $0.6 \lesssim \tau \lesssim 0.9$ and the two weaker transverse channels associated mainly with the p_x and p_y orbitals approximately within $0.1 \lesssim \tau \lesssim 0.3$ [112]. This channel pattern applies to one-atom contacts and should not be used to interpret the larger few-atom configurations. The separate deformation calculation does not provide a statistical distribution. It instead shows that the transverse transmissions can be of order 0.3 near a compact geometry and move below 0.2 as the contact is elongated, while the axial transmission moves toward unity [113]. The transmission ranges expected for the axial and transverse channels therefore depend on the geometry and its elongation.

Figure 4.28 is restricted to the active channels in configurations (c) - (e), which contain two or three resolved contributions and are the plausible one-atom states identified above. It excludes the larger few-atom configurations (a) and (b), for which the one-atom channel pattern is not applicable, as well as the tunnel configuration (f). Resolved by channel, τ_0 alone populates the high-transmission peak between approximately 0.7 and 0.9, while τ_1 and τ_2 remain below approximately 0.4 throughout. This directly separates a single dominant channel from two weaker ones, matching the dominant axial and weaker transverse channels ex-

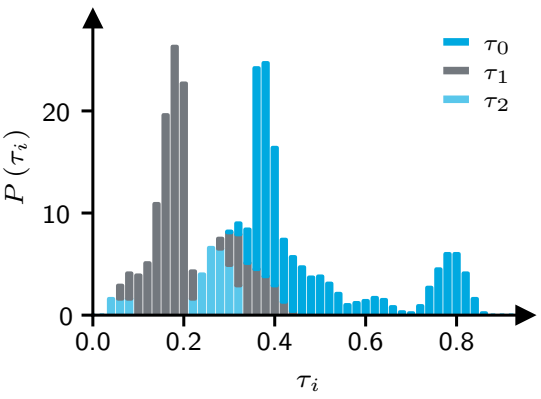
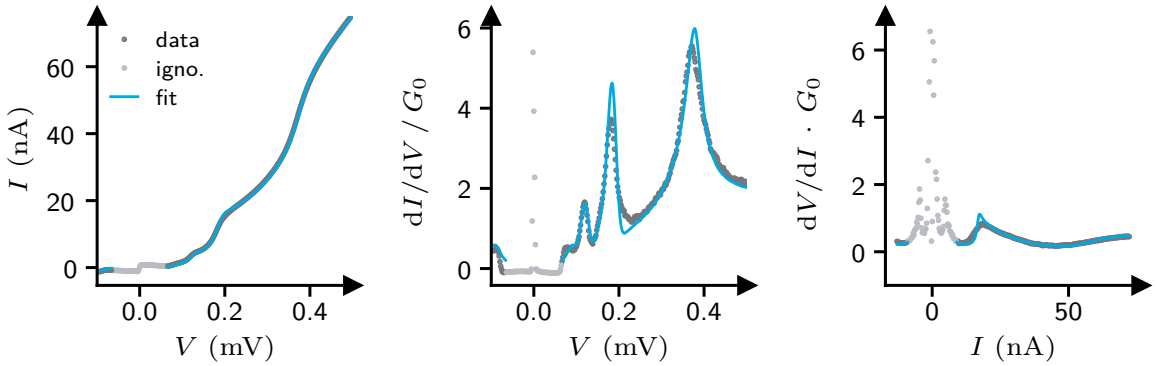


Figure 4.28. Distribution of fitted channel transmissions in regions (c) - (e) of Fig. 4.26. τ_0 is the leading, highest-transmission channel. τ_1 and τ_2 are the weaker subsequent channels.

pected for a one-atom contact. τ_0 is itself bimodal. Its high-transmission peak comes from the short configuration (c) plateau, and its broader weight between 0.3 and 0.6 comes from configurations (d) and (e), where the dominant channel is suppressed after the rearrangement and then partly recovers with elongation. Weight around $\tau \approx 0.3$ is plausible for compact or weakly elongated configurations, while the extension of τ_1 and τ_2 toward lower transmission is compatible with the suppression of transverse channels during stretching.

Two relatively sharp peaks near 0.2 and 0.4 are superimposed on the broad low-transmission distribution. Resolved by channel, the two peaks belong to different transmissions, τ_1 near 0.2 and τ_0 near 0.4. Both arise from the same long stretch of configuration (d), over which the fitted transmissions change very little. Consecutive spectra in this stretch repeatedly contribute nearly the same values to the histogram. That two channels peak together at one plateau reinforces that the sharp peaks reflect the sampled trajectory rather than intrinsically preferred channel transmissions. More generally, the spectra are correlated and configurations (c) - (e) contribute unequal numbers of samples, so the relative weights of the distributions are not statistical occurrence probabilities. Restricting the histogram to active channels in these regions nevertheless makes their transmission ranges meaningfully comparable with the one-atom orbital model. This comparison does not distinguish a monomer from an axial dimer or assign an orbital character to an individual fitted eigenvalue.

In summary, the coherent MAR model reproduces the measured spectral evolution along a complete opening trace with common values of Δ_0 and σ_V. This agreement is the robust result of the extraction. The extracted mesoscopic pincode $\{\tau_i\}$ evolves continuously on conductance plateaus and reorganizes at abrupt jumps, which is consistent with elastic deformation between atomic rearrangements. The number, approximate degeneracy, and separation of the fitted transmissions further suggest a progression from a few-atom constriction through a nearly symmetric three-channel monomer or dimer to an off-center contact and finally to the tunnel regime. The orbital assignments and the predominantly axial interpretation of the transition from (d) to (e) remain microscopic hypotheses because the fit does not determine channel eigenvectors or a unique atomic geometry. The restricted histogram provides an additional descriptive comparison with the low transverse and high axial transmission ranges expected for one-atom aluminum contacts, while its sharp peaks reflect repeated sampling along the opening trajectory. Channel resolved MAR spectroscopy therefore reveals changes in the mesoscopic pincode that remain hidden in the normal-state conductance alone and restrict, without uniquely reconstructing, the evolution of the contact.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.29. Static transport of C1 and multichannel MAR fit. The fit reproduces the finite-bias *I–V*, *dI–dV*, and *dV–dI* characteristics for the transmission eigenvalues in Eq. (4.16). Data near zero bias are excluded because the additional narrow peak is not contained in the dc MAR model.

4.2.2 Photon Assisted Transport

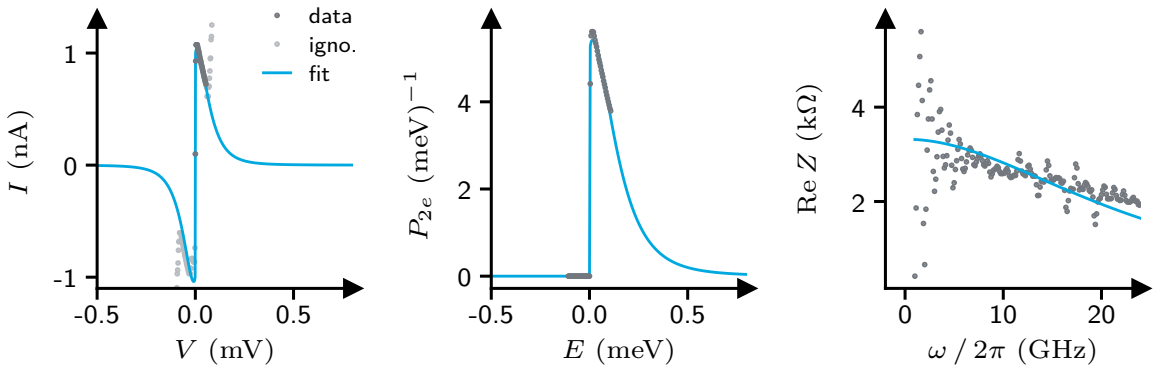
I first consider atomic configuration C1, whose static transport and multichannel MAR fit are shown in Fig. 4.29. In addition to the finite-bias MAR structure, the measurement contains a narrow zero-bias peak associated with a supercurrent contribution. This contribution is not included in the dc MAR model used to determine the transmission eigenvalues. I therefore exclude the data near zero bias from the fit and obtain

$$\{\tau_i\} = \{0.517, 0.399, 0.332, 0.305, 0.204\}\,, \qquad G_\mathrm{N} = 1.757\,G_0\,. \tag{4.16}$$

The largest transmission of this contact is only $\tau_1 = 0.517$. C1 therefore provides a comparatively low-transmission starting point for separating the photon assisted quasiparticle, MAR, and supercurrent contributions below. I use the same static MAR fitting procedure for all irradiated contacts. Their extracted pincodes are collected in Table F.1 in Appendix F.5, demonstrating that the undriven transport of the full contact set can be described within the same multichannel MAR framework. At the end of this section, I test the photon assisted model on C2. Its nearly ballistic leading channel, $\tau_1 = 0.996$, provides a deliberately contrasting validation case rather than a repetition of the C1 regime.

Incoherent Cooper Pair Transport (ICPT)

I now return to the narrow zero-bias feature that is excluded from the static MAR fit. Its location identifies it with a Cooper pair contribution, but the measurement does not show a stable zero-voltage supercurrent branch. This suggests that phase fluctuations induced by the electromagnetic environment



(a) Zero-bias *I*–*V* and ICPT fit. (b) Extracted $P_{2e}(E)$. (c) Effective dissipative $Z(\omega)$.

Figure 4.30. Effective environment analysis and forward model fit of the zero-bias feature of C1. (a) Measured *I*–*V* characteristic and fitted ICPT current. Data below $V = 0$ and in the quasiparticle branch are excluded. (b) Effective zero-temperature $P_{2e}(E)$ obtained from the measured current. (c) Effective dissipative impedance reconstructed from $P_{2e}(E)$, together with the parallel-RC form used in the subsequent forward fit. The resulting fit parameters are given in Eq. (4.18).

suppress coherent Josephson transport. I therefore test whether the feature is consistent with ICPT, for which individual charge-$2e$ transfer events exchange energy with the environment.

In the perturbative ICPT regime, the current samples the environmental probability density $P_{2e}(E)$ at $E = 2eV$ according to Eq. (2.108). I use the zero-temperature approximation, in which the environment can absorb energy released by a tunneling Cooper pair but cannot supply energy to it.

I first invert the measured current to visualize the environmental response before choosing a circuit model. Under the zero-temperature approximation, Eq. (2.108) gives an effective $P_{2e}(E)$ directly from the measured $I(V)$, as shown in Fig. 4.30(b). Equations (2.95) and (2.94) then yield the corresponding effective dissipative impedance in Fig. 4.30(c). This inversion is a diagnostic construction. The inferred frequency dependence motivates the simplest impedance ansatz that includes both low-frequency dissipation and a high-frequency cutoff. I represent the linear environment seen by the junction by an effective resistance R_{eff} in parallel with an effective capacitance C_{eff}. The resistance sets the low-frequency impedance, while the capacitance progressively shunts the impedance as the frequency increases. The resulting impedance is

$$Z(\omega, R_{\mathrm{eff}}, C_{\mathrm{eff}}) = \frac{R_{\mathrm{eff}}}{1 + i\omega R_{\mathrm{eff}} C_{\mathrm{eff}}} . \tag{4.17}$$

After selecting this circuit form, I reverse the calculation for the quantitative fit. For each trial, I calculate $P_{2e}(E)$ from the assumed impedance and substitute it into Eq. (2.108). This gives the forward model, which I fit directly to the

measured zero-bias *I*–*V* curve. Thus, the inversion motivates the parallel-RC model, while I obtain the fitted parameters from the forward calculation.

This forward fit gives

$$I_{\mathrm{c}} = 7.76(14)\,\mathrm{nA}\,, \qquad R_{\mathrm{eff}} = 3.32(5)\,\mathrm{k\Omega}\,, \qquad C_{\mathrm{eff}} = 2.02(10)\,\mathrm{fF}\,. \tag{4.18}$$

The static characteristic is therefore consistent with ICPT in an effective dissipative environment, but it does not by itself exclude a broadened coherent Josephson response. The photon assisted replicas provide an independent test below because incoherent transfer produces squared-Bessel weights, whereas coherent Shapiro response has a different amplitude dependence. The parameters extracted here define the undriven ICPT contribution used in that comparison. The remaining calibration step is to relate the nominal microwave output to the ac voltage across the junction.

Coupling Parameter

In contrast to the tunnel-junction calibration, the irradiated atomic contact curves cannot be fitted with a single complete PAT model, as in Sec. 4.1.1, because quasiparticle transport, MAR, and the zero-bias Cooper pair contribution coexist. I therefore use a self-referenced PAT approximation to estimate the amplitude coupling.

For all microwave amplitude sweeps on the atomic contacts, I record a series of *I*–*V* curves while stepping the source power uniformly in dBm. Conversion to the nominal voltage amplitude $A_{\mathrm{out}} \propto \sqrt{P_{\mathrm{out}}}$ therefore gives logarithmically spaced rather than equally spaced amplitude values. I acquire the unirradiated curve at $A_{\mathrm{out}} = 0$ separately. In the corresponding amplitude maps, bias and calibrated amplitude form the two coordinates, while current or differential response is encoded by color. The maps retain the measured amplitude coordinates after calibration.

I then use the measured unirradiated conductance as the undriven characteristic in the Tien-Gordon transformation from Eq. (2.32). This construction is self-contained because it does not require a separate microscopic fit of the static or irradiated curves. It preserves the measured line shape and asks which value of η best reproduces its photon assisted replicas over the complete amplitude sweep. I perform the comparison in *dI*–*dV*, which suppresses the approximately voltage independent excess current and emphasizes the replicated spectral features. The

map is dominated by quasiparticle PAT, so I use the transferred charge $q = e$. A trial coupling η therefore gives

$$\frac{\mathrm{d}I}{\mathrm{d}V}(\eta, A_{\mathrm{out}}, V) = \sum_{n=-\infty}^{\infty} J_n^2\left(\frac{q\eta A_{\mathrm{out}}}{h\nu}\right) \cdot \frac{\mathrm{d}I}{\mathrm{d}V}_{\mathrm{exp}}^{(A_{\mathrm{out}}=0)}\left(V - \frac{nh\nu}{q}\right), \tag{4.19}$$

$$\chi^2(\eta) = \left\langle \left(\frac{\mathrm{d}I}{\mathrm{d}V}(\eta, A_{\mathrm{out}}, V) - \frac{\mathrm{d}I}{\mathrm{d}V}_{\mathrm{exp}}(A_{\mathrm{out}}, V) \right)^2 \right\rangle_{(A_{\mathrm{out}}, V)}. \tag{4.20}$$

I evaluate the squared deviation at each sampled amplitude and voltage point and average it over the map to obtain the objective $\chi^2(\eta)$. Narrow MAR and zero-bias features are not expected to follow the charge-e transformation exactly, but they occupy a smaller part of the map than the quasiparticle replicas and therefore do not set the fitted coupling.

I minimize Eq. (4.20) with a gradient based optimizer. This continuous optimization is more efficient than a dense grid search because each objective evaluation transforms the complete measurement map and the minimum is narrow. The independently sampled objective landscape in Fig. 4.31 is not the fit procedure itself. It displays a wider parameter range and confirms that the optimizer converges to the clear global minimum. For C1, this gives

$$\eta = 0.006179(16) \qquad (\nu = 15.0\,\mathrm{GHz})\,. \tag{4.21}$$

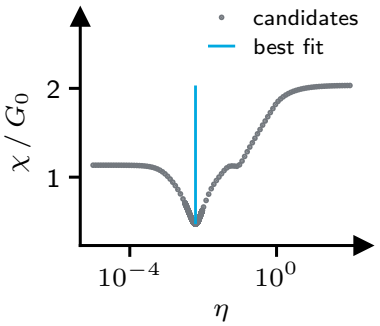


Figure 4.31. Coupling calibration for C1 at 15 GHz.

This calibration replaces the hardware dependent source setting with the effective amplitude at the contact. The coupling parameters obtained for all irradiated contacts are summarized in Table F.2 in Appendix F.5.

Photon Assisted Multiple Andreev Reflection (PAMAR)

The calibrated *dI–dV* map contains several families of photon assisted features. Replicas emerge from the quasiparticle threshold at $e|V| = 2\Delta_0$ and from the subgap MAR thresholds near $e|V| = 2\Delta_0/m$. The full counting statistics (FCS) decomposition introduced in Sec. 2.3.4 and Eq. (2.88) separates the dc MAR current into components $I_m(V)$ associated with the transfer of an effective charge $q = me$. I first test whether these components can be used directly in the unified charge dependent Tien-Gordon expression of Eq. (2.89).

This is a natural microscopic starting point because each FCS component already carries an effective charge assignment. Dressing each $I_m(V)$ with the corresponding Bessel argument and voltage spacing therefore requires no addi-

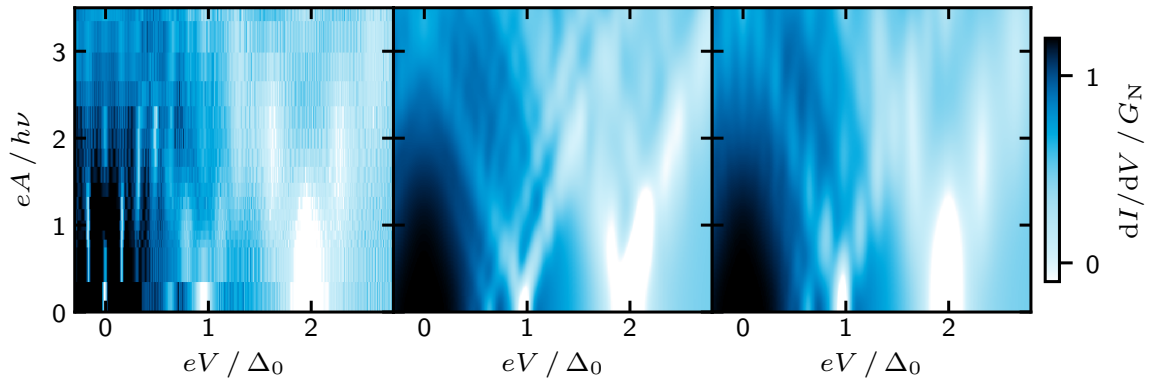


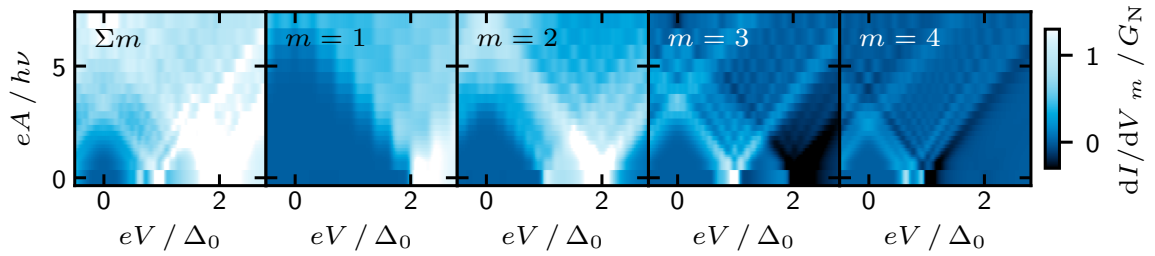
Figure 4.32. Test of the photon assisted MAR construction for C1 at 15.0 GHz. The left panel shows the measured *dI–dV* map on the logarithmically sampled amplitude grid described in Sec. 4.2.2. The center panel applies charge dependent Tien-Gordon dressing directly to the dc FCS components. This approach does not reproduce the measured sideband pattern because positive and neighboring negative conductance contributions are replicated with the same effective charge. The right panel instead dresses components localized around the MAR thresholds and recovers the principal measured sideband trajectories. All calculations use the calibrated drive amplitude.

tional assignment. The center panel of Fig. 4.32, however, shows that this direct construction does not reproduce the measured sideband pattern.

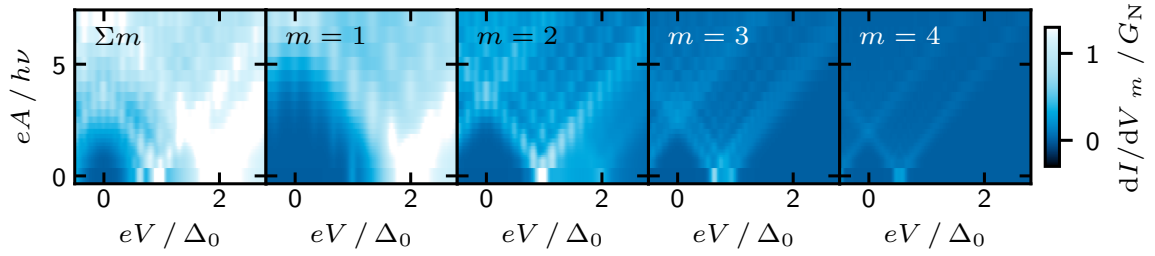
The difficulty is not the charge dependence of the Tien-Gordon transformation, but the choice of its input currents. The FCS contributions are charge resolved, but they are not localized exclusively around the corresponding MAR thresholds. Their differential conductance can contain both the dominant positive MAR feature and neighboring negative structure over the same voltage range, as discussed in Sec. 2.3.4. Direct photon dressing consequently replicates both contributions with the same effective charge, producing additional features that are absent or substantially weaker in the measurement. The component maps in Fig. 4.33(a) make this mixing visible.

I therefore retain the charge dependent Tien-Gordon transformation but replace the dc FCS currents with phenomenological currents localized around the MAR thresholds in Eq. (2.82). Carrad *et al.* used a related charge dependent sideband spacing to identify high-order MAR processes in individual traces [34]. Here I instead build explicit threshold localized input currents, so that the same decomposition predicts the full amplitude resolved response. I construct these currents from the simulated static MAR current calculated with the pincode in Eq. (4.16). The natural coordinate for the decomposition is $2\Delta_0/|eV|$, on which the thresholds occur at integer m. I define normalized Gaussian weights

$$w_m(V) = \frac{\tilde{w}_m(V)}{w_\mathcal{N}(V)}\,, \qquad \tilde{w}_m(V) = \exp\left(-\left(\frac{2\Delta_0/|eV| - m}{2}\right)^2\right)\,, \tag{4.22}$$



(a) FCS charge components



(b) Threshold weighted components

Figure 4.33. Charge resolved contributions to the photon assisted MAR model of C1. Each row shows the summed response and the components assigned to effective charge orders $m = 1$ to 4. (a) Direct dressing of the dc FCS components replicates positive and neighboring negative *dI–dV* contributions with the same effective charge. (b) Decomposing the simulated static MAR *dI–dV* around the thresholds before applying the charge dependent dressing suppresses this mixing and isolates the sidebands associated with each threshold. All panels use the calibrated drive amplitude at $15.0\,\mathrm{GHz}$.

I merge the contribution w_0 into w_1 because quasiparticle transport dominates above $e|V| = 2\Delta_0$. For practical reasons, I use a maximum order M that combines order M and all higher orders, typically $M = 10$. I choose the normalization factor $w_{\mathcal{N}}(V)$ so that the individual weights $w_m(V)$ sum to unity at every voltage. Figure 4.34 shows that the same smooth partition appears unequally spaced on the voltage axis but equally spaced on the inverse-voltage coordinate.

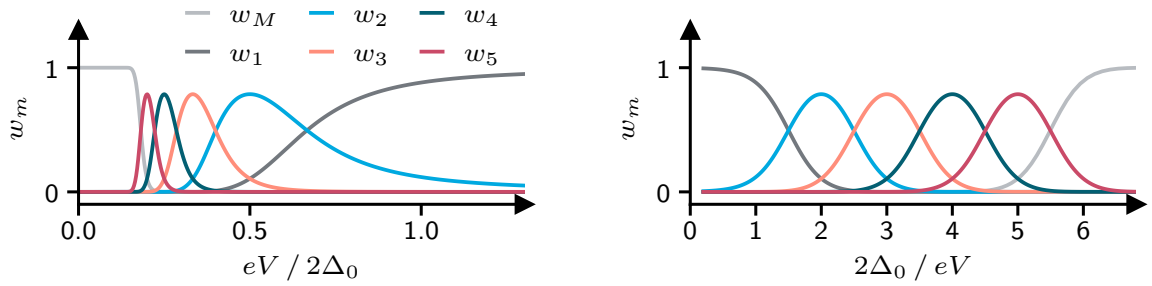
I apply the weights to the differential conductance of the simulated static MAR current and reconstruct an effective current for each threshold,

$$I_m(V) = \int w_m(V)\,\tfrac{\mathrm{d}I}{\mathrm{d}V}(V)\,\mathrm{d}V\,, \qquad I_m(0) \stackrel{!}{=} 0\,. \tag{4.23}$$

Partitioning *dI–dV* rather than multiplying the current directly avoids an artificial boundary term. In the latter construction,

$$\tfrac{\mathrm{d}}{\mathrm{d}V}(w_m\,I) = w_m\,\tfrac{\mathrm{d}I}{\mathrm{d}V} + I\,\tfrac{\mathrm{d}}{\mathrm{d}V}w_m\,, \tag{4.24}$$

so the derivative of the weight would generate artificial conductance structure that is not present in the simulated static MAR spectrum.



(a) Voltage representation (b) Inverse-voltage representation

Figure 4.34. Phenomenological weights used to decompose the simulated static MAR *dI–dV* by threshold. (a) On the voltage axis, each $w_m(V)$ selects the vicinity of $e|V| = 2\Delta_0/m$, while w_1 contains the quasiparticle region above $2\Delta_0$ and the terminal weight w_M collects all remaining higher orders near zero bias. (b) On the inverse-voltage axis, the explicitly resolved weights form equally spaced Gaussians centered at integer m. The weights are normalized to unity at every voltage before constructing the effective currents $I_m(V)$. $M = 6$ is the only free parameter, chosen for visual clarity.

I then use the effective currents from Eq. (4.23) in the unified Tien-Gordon expression. Figure 4.33(b) shows the resulting contributions by threshold. Their sum gives the right panel of Fig. 4.32 and reproduces the principal quasiparticle and MAR sideband trajectories without the mixing produced by direct FCS dressing.

This threshold weighted PAMAR construction is a phenomenological stationary replica model. It assigns the simulated static MAR structure near each threshold to its nominal effective charge order, but it does not replace a microscopic Floquet calculation of irradiated MAR. In particular, it neglects coherent interference between driven MAR paths, irradiation induced changes of the contact occupations, and nonequilibrium redistribution at high drive. Within these limits, it provides a transparent description of the principal finite-bias replicas. A detailed voltage cut decomposition into the individual charge orders is provided in Appendix Fig. F.12. The narrow replicas originating from the zero-bias feature require the separate Cooper pair contribution considered next.

Photon Assisted Incoherent Cooper Pair Tunneling (PAICPT)

Figure 4.32 shows experimental features that are not associated with PAMAR. Instead, they originate from the zero-bias feature identified above as ICPT. Their spacing alone does not distinguish ICPT from coherent Josephson transport. Both mechanisms involve charge $2e$ and therefore generate features at voltages separated by $h\nu/(2e)$. The distinction lies in how their spectral weight evolves with microwave amplitude.

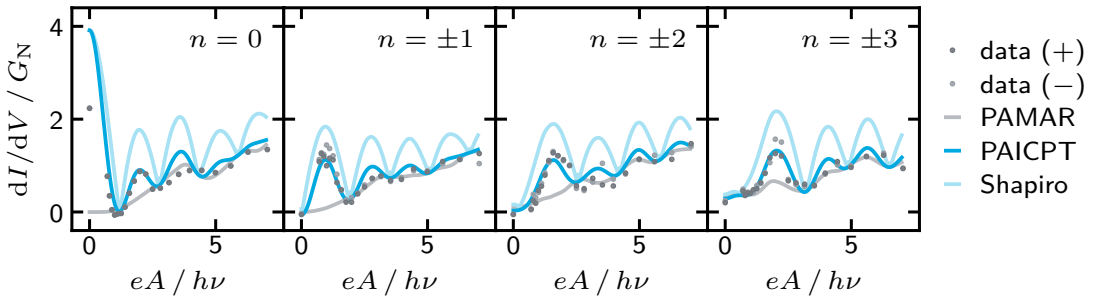
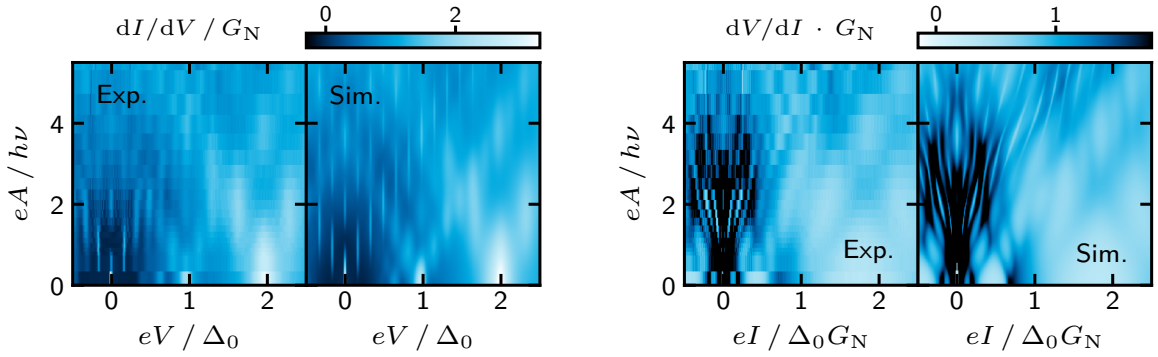


Figure 4.35. Amplitude dependence of the features originating from zero bias in C1 at 15.0 GHz. The four panels follow the central feature and the first three replica pairs, $n = 0, \pm1, \pm2, \pm3$. The measured positive- and negative-bias branches are compared with the squared-Bessel PAICPT calculation and an ideal coherent Shapiro response on a PAMAR background. The observed redistribution among the replica orders is described more consistently by PAICPT.

For PAICPT, the microwave drive dresses the static ICPT characteristic determined above with probabilities proportional to squared Bessel functions, as discussed in Sec. 2.4.3. An ideal coherent Shapiro response follows the phase-locking relation in Eq. (2.50), for which the Bessel functions enter at the amplitude level. I calculate both responses using the calibrated ac voltage at the contact. The PAICPT calculation additionally uses the static ICPT parameters in Eq. (4.18). The comparison therefore tests the amplitude dependence without refitting the individual replicas.

Figure 4.35 follows the central feature and the first three pairs of replicas along the microwave amplitude axis. The measured positive- and negative-bias branches show the same sequence of maxima and minima as the PAICPT calculation over most of the calibrated range. The ideal Shapiro response instead predicts a different redistribution between the central feature and successive replica orders. The finite PAMAR contribution provides a smooth background to these cuts but does not account for their oscillatory amplitude dependence.

Deviations remain at the largest amplitudes, where the measured line shapes broaden and the stationary PAICPT model becomes less accurate. Heating, nonequilibrium occupation, and drive dependent phase dynamics are not included in the calculation. The line cuts therefore do not establish ICPT from the microwave response alone. Together with the static $P(E)$ analysis, however, their squared-Bessel-like amplitude evolution supports the interpretation of the zero-bias feature as incoherent Cooper pair transport rather than an ideal coherent Shapiro response.



(a) *dI–dV* (b) *dV–dI*

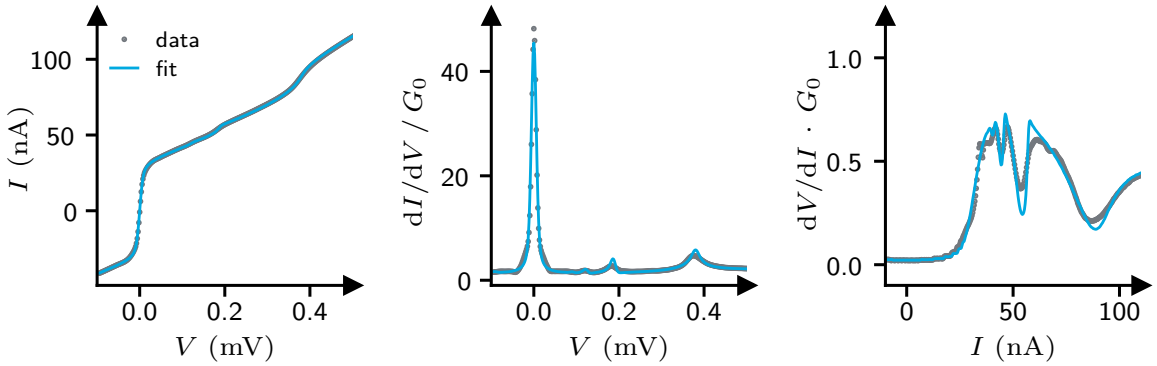
Figure 4.36. Comparison of the measured and simulated microwave response of C1 at 15.0 GHz. (a) The combined PAMAR and PAICPT model reproduces the quasiparticle and MAR sideband trajectories in *dI–dV*, together with the narrow replicas originating from the zero-bias feature. (b) The corresponding *dV–dI* maps emphasize the same agreement in the strongly nonlinear subgap region. Experiment and simulation use the calibrated drive amplitude and the independently determined static parameters.

Comparison of Experiment and Simulation

I finally combine the threshold weighted PAMAR and PAICPT contributions. The calculation uses the static pincode, the effective ICPT environment, and the microwave coupling determined above. I make no additional adjustment for the comparison in Fig. 4.36.

The agreement is exceptionally close considering the number and overlap of the observed structures. The simulation reproduces not only their positions but also the redistribution of spectral weight across the microwave amplitude range in both derivative representations. The remaining differences are concentrated at the largest drive amplitudes and in the detailed line shapes of the sharpest subgap features, consistent with the limits identified above.

Appendix Fig. F.13 provides the stricter model selection test. It compares all four combinations of the two MAR constructions and the two Cooper pair responses, together with their residual maps. The combination of threshold weighted PAMAR and PAICPT gives the best agreement. It reduces the structured residuals both along the MAR trajectories and around the replicas originating from zero bias. The complete measured and simulated transport landscapes are shown in Appendix Figs. F.14 and F.15. The full comparison therefore confirms that both ingredients are required to capture the dominant microwave response of C1 within a common description constrained by independently determined parameters.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.37. Static transport of C2 and multichannel MAR fit. The nearly ballistic leading channel produces a stronger excess current and sharper subgap structure than in C1. The simultaneous comparison of *I–V*, *dI–dV*, and *dV–dI* establishes the undriven reference used for the photon assisted calculation.

Validation with C2

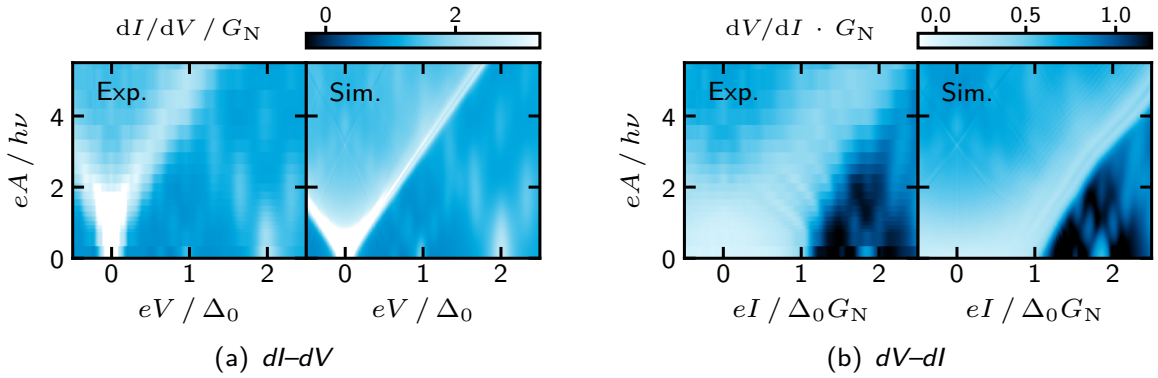
I next test whether the threshold weighted PAMAR construction remains useful for a contact with a nearly ballistic leading channel. The static MAR fit gives

$$\{\tau_i\} = \{0.997, 0.440, 0.184, 0.184, 0.182\}\,, \qquad G_\mathrm{N} = 1.987\,G_0\,. \tag{4.25}$$

This pincode is deliberately different from that of C1. Its leading channel produces a strong MAR contribution that dominates the measured response around zero voltage. Therefore, a separate static supercurrent contribution cannot be isolated and fitted reliably. For this validation, I use only the threshold weighted PAMAR model constructed from the simulated static MAR current and do not add a PAICPT contribution.

Figure 4.38 shows that PAMAR alone already reproduces the overall microwave response remarkably well. The simulation captures the dominant quasiparticle and MAR trajectories, their charge dependent spacing, and the broad redistribution of spectral weight over the calibrated amplitude range. This agreement is obtained despite the nearly ballistic leading channel and without using the ICPT analysis developed for C1.

The PAMAR simulation provides a reference that separates the dominant MAR response from weaker experimental structure. The tilted lines parallel to the principal simulated features in Fig. 4.38 arise from the numerical calculation and do not have physical significance. The measured *dI–dV* map additionally contains faint vertical lines, with corresponding lines perpendicular to the principal features in *dV–dI*. Their absence from the PAMAR reference identifies them as a separate contribution, which I interpret as PAICPT replicas originating from ICPT near zero bias.



(a) *dI–dV* (b) *dV–dI*

Figure 4.38. PAMAR validation for the high-transmission C2 at 15.0 GHz. (a) Measured and simulated *dI–dV* maps. (b) The corresponding *dV–dI* maps emphasize the strong subgap response generated by the nearly ballistic leading channel. Threshold weighted PAMAR reproduces the overall trajectories and amplitude evolution without an additional PAICPT contribution. Tilted lines parallel to the main simulated features are numerical artifacts. Vertical structures in the measured *dI–dV* map and structures perpendicular to the main measured *dV–dI* ridges indicate a contribution missing from the PAMAR-only calculation.

The full measured and simulated landscapes in Appendix Figs. F.16 and F.17 support this interpretation. PAMAR reproduces the dominant zero-bias response, while the different redistribution of part of the low-voltage spectral weight reveals the weaker contribution as the drive amplitude increases. In the static *I–V* characteristic, ICPT overlaps directly with the strong MAR response of the nearly ballistic channel, so its independent input cannot be extracted for a quantitative PAICPT simulation. The comparison nevertheless resolves the physical picture. The zero-bias response of C2 contains a dominant MAR contribution together with a weaker ICPT contribution.

C2 therefore validates the principal PAMAR picture and reveals a weaker ICPT contribution within the MAR dominated zero-bias response. Together, C1 and C2 test opposite pincode regimes, from a moderate leading transmission to an almost ballistic channel, and establish the complementary roles of PAMAR and PAICPT. The same static MAR, microwave calibration, and photon assisted modeling procedures describe the overall response of the full atomic contact set summarized in Appendix Tables F.1 and F.2. Within the tabulated range, PAMAR captures the dominant finite-bias structure, while PAICPT accounts for the contribution originating from zero bias when it can be separated from the MAR background.

4.2.3 Subharmonic Transport at High Conductance

I measure C3 after reclosing the mechanically controllable break junction. The resulting contact reaches a conductance of approximately 29(1) G_0, indicating transport through many contributing channels rather than a small set of individually resolvable channels. This high-conductance regime differs substantially from the few-channel contacts considered in the preceding section. Its static and irradiated transport exhibits pronounced subgap structure and additional replicas beyond the standard PAMAR and PAICPT descriptions introduced above.

Series Resistance Correction

The low-voltage superconducting branch exhibits a small linear slope. I assume that this slope is dominated by a residual series resistance between the contact and the effective four-point measurement terminals. Under this assumption, the linear fit shown in Fig. 4.39 gives the effective series resistance R_s, which I use to define the corrected voltage V' as

$$R_\mathrm{s} = 12.61(7)\,\Omega\,, \quad V' = V - IR_\mathrm{s}\,. \quad (4.26)$$

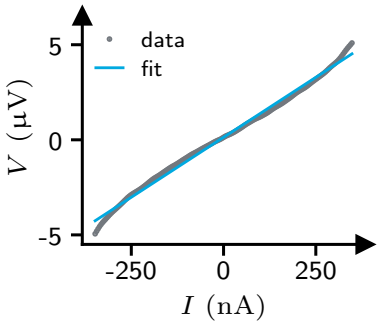


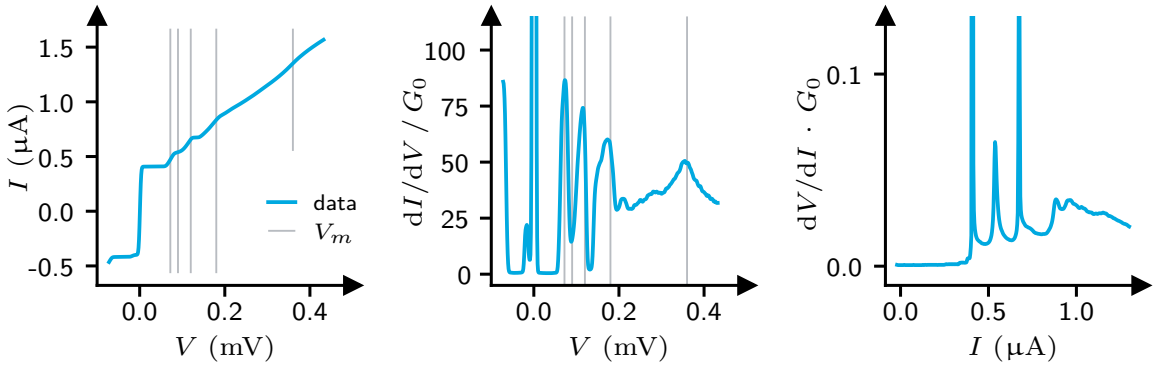
Figure 4.39. Linear fit of the low-voltage superconducting branch.

The extracted resistance probably originates from the copper wire and silver paste contacts to the sample. I do not expect it to include a substantial contribution from the on-chip aluminum leads. Their large cross section gives them a high critical current, so they most likely remain superconducting throughout this measurement. This procedure provides an approximate series correction rather than an independent measurement of the lead resistance. In particular, any intrinsic linear voltage, e.g. from phase diffusion, or environmental fluctuations, would also be included in the fitted value.

Static *I*–*V* Description

The unirradiated *I*–*V* characteristic combines a zero-voltage branch up to approximately 404 nA, pronounced subgap structure, and several abrupt voltage changes. After correcting the series resistance and voltage offset, I compare the *I*–*V* characteristic with MAR calculations. I consider fits using individual transmission channels, randomly generated pincodes, grouped channels, and a continuous transmission distribution. The large conductance of approximately 29 G_0 makes the channel-by-channel pincode inaccessible.

The expected MAR thresholds do not coincide exactly with the measured *dI*–*dV* maxima, as shown in Fig. 4.40. Most orders are slightly displaced, and no



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure 4.40. Series resistance corrected static transport of C3. The panels show the *I–V*, *dI–dV*, and *dV–dI* characteristics. Gray lines mark the MAR opening voltages $V_m = 2\Delta_0/(me)$. Differential conductance maxima need not coincide with these thresholds because their positions and line shapes depend on transmission. No separate maximum is resolved for fourth-order MAR. The tested multichannel fits do not reproduce the complete sharp, step-like characteristic.

distinct feature appears at the fourth-order threshold. The gray lines indicate the opening voltages of the MAR processes, not the positions at which their differential conductance must be maximal. Calculations for a single channel show that the amplitude, line shape, and position of each maximum depend strongly on the transmission. Only in the low-transmission limit do sharp maxima remain close to the thresholds. With increasing transmission, the maxima broaden and move to lower voltage, between neighboring nominal thresholds, while minima can remain near the subharmonic voltages [116]. Broad, split, and nonideally scaled MAR maxima have also been observed in high-transparency diffusive junctions [117].

This transmission dependence provides a natural explanation for the displaced maxima in C3. It can also make an individual order difficult to resolve because contributions from many channels with different transmissions overlap. The missing fourth-order maximum therefore does not imply that fourth-order MAR is absent. It means only that this order does not produce a separate extremum in the measured *dI–dV*. I nevertheless cannot reproduce the detailed combination of peak positions, amplitudes, and the unresolved fourth order with any of the tested transmission distributions. The comparison thus does not support a unique pincode, but the offset between maxima and threshold lines is not by itself evidence against conventional MAR.

I then test whether the zero-voltage and resistive branches can be described within a common Josephson-MAR model. I add a separate current from Andreev bound states, but it does not improve the description, while overdamped and time dependent RCSJ variants produce low-voltage states and plateaus that are not observed experimentally. I treat the measured switching boundary sep-

arately and use effective distributions with two or three channel groups, which yields somewhat better agreement with the stable resistive branches. The fitted noninteger channel weights, however, cannot be interpreted as a literal pincode.

Despite these improvements, none of the tested models provides a convincing description of the complete *I*–*V* characteristic. In particular, they do not reproduce its sharp, step-like shape and instead connect the subgap structures too smoothly. Phenomenological corrections improve some transitions locally but do not provide a physical explanation. I therefore do not attempt a more detailed interpretation of this characteristic.

Phenomenological Replica Description

Because no microscopic model reproduces the static *I*–*V* characteristic convincingly, I use the measured characteristic directly as input for a phenomenological description of the irradiated transport. The complete measured map is shown in Appendix Fig. F.18. I describe replicas originating from the quasiparticle branch and the static subgap structure using the threshold weighted PAMAR procedure introduced in Sec. 4.2.2. This approach preserves the measured step-like line shape instead of replacing it with a simulated MAR *I*–*V* characteristic.

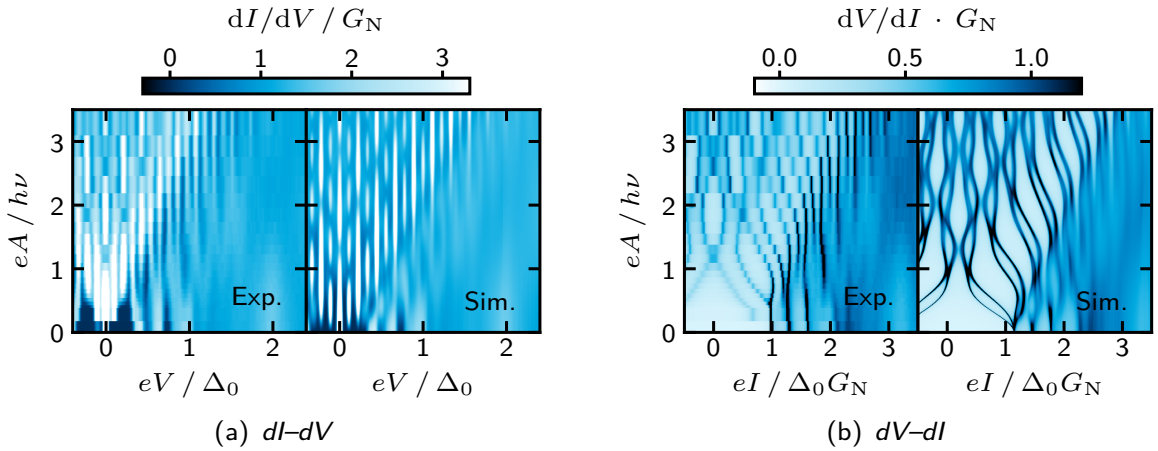
The replicas originating at zero voltage require an additional contribution. I parameterize their positions as

$$V_{n,p} = \frac{n}{p}\,\frac{h\nu}{2e}\,, \tag{4.27}$$

where $p = 1$ gives the conventional Cooper pair spacing $h\nu/(2e)$ and $p = 2$ generates the trajectories halfway between these positions. This regular spacing resembles the PAMAR pattern, but the anchor points distinguish the two responses. Each PAMAR fan emerges from its corresponding finite-bias MAR threshold, whereas both families in Eq. (4.27) emerge from the zero-voltage contribution. In a coherent Josephson picture, the $p = 2$ family could be associated with fractional Shapiro steps produced by a nonsinusoidal current–phase relation. The corresponding single-harmonic estimate for the step width is given by Eq. (2.81). I instead extend the incoherent replica construction used above. This extension is empirical. In particular, the $p = 2$ term behaves mathematically like an effective $4e$ contribution, but it does not by itself establish coherent transfer of two Cooper pairs or any other microscopic mechanism.

Following Sec. 4.2.2, I integrate the static zero-voltage conductance feature to obtain $I_0(V)$. I then construct the irradiated contribution as

$$I(V,A) = \sum_{p=1}^{2} \sum_{n=-\infty}^{\infty} J_n^2\left(\frac{p(2e)A}{h\nu}\right) I_p\left(V + \frac{nh\nu}{p(2e)}\right)\,, \tag{4.28}$$



(a) *dI–dV* (b) *dV–dI*

Figure 4.41. Measured and simulated microwave response of C3 at 19.3 GHz. Each panel compares the measured data on the left with the phenomenological replica calculation on the right. (a) The *dI–dV* maps show replicas of the quasiparticle branch and the static subgap structure. (b) The corresponding *dV–dI* maps emphasize the sharp current biased transitions. The calculation combines threshold weighted PAMAR with the two-component Cooper pair replica model and uses the measured static *I–V* characteristic as input. It reproduces the dominant trajectories, while differences remain around the sharpest low-voltage features.

with

$$I_p(V) = w_p\, I_0(V)\,, \qquad \sum_{p=1}^{2} w_p = 1\,. \tag{4.29}$$

I select the relative weights by comparing the calculated maps and amplitude cuts with the measured data. I obtain the closest qualitative agreement for

$$\{w_1, w_2\} = \left\{\tfrac{3}{5}, \tfrac{2}{5}\right\}\,. \tag{4.30}$$

I do not extract these weights through an independent microscopic fit. They should be interpreted only as empirical spectral weights of the two replica components.

Figure 4.41 compares the resulting calculation with the measured derivative maps. The combined construction reproduces the principal finite-bias and low-voltage trajectories in both representations. It also retains the sharp structure of the measured static *I–V*, although differences remain in the detailed line shapes and spectral weights.

Figure 4.42 compares amplitude cuts at the integer positions $V_{n,p}$. Both components contribute at these voltages because $V_{n,1} = V_{2n,2}$. The cuts therefore contain the nth replica of the $p = 1$ contribution and the $2n$th replica of the $p = 2$ contribution. Their superimposed squared-Bessel amplitude dependence follows the measured redistribution more closely than the ideal Shapiro response.

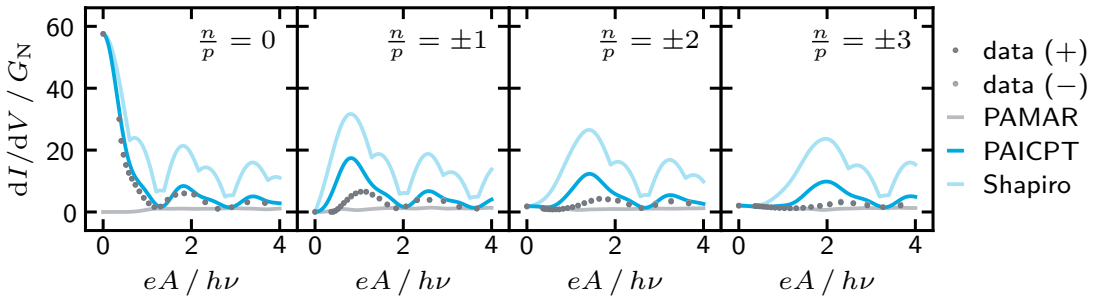


Figure 4.42. Amplitude dependence at the integer replica positions $V_{n,p}$ of C3 at 19.3 GHz. The panels compare the measured positive- and negative-bias branches with the PAMAR background, the two-component phenomenological PAICPT construction, and an ideal Shapiro response. At each integer position, the $p = 1$ replica of order n overlaps the $p = 2$ replica of order $2n$. Their combined squared-Bessel amplitude dependence follows the measured redistribution more closely than the ideal Shapiro limit but does not determine the microscopic origin of the $p = 2$ component.

The intermediate trajectories in the full maps provide complementary evidence that a separate $p = 2$ component is required.

I include the complete measured and reconstructed transport maps in Appendix Figs. F.18 and F.19 in Sec. F.6. They show the same comparison as heat maps, surfaces, and waterfall cuts over the full calibrated amplitude range.

Taken together, the intermediate trajectories, the overlapping $p = 1$ and $p = 2$ contributions at the integer positions, and their squared-Bessel-like amplitude dependence consistently point toward a subharmonic PAICPT response. The present phenomenological construction does not provide a precise microscopic model, and it does not establish whether the $p = 2$ component originates from incoherent correlated Cooper pair transfer or from another mechanism. Nevertheless, conventional PAICPT has been reported only with the usual $h\nu/(2e)$ replica spacing [30]. To the best of my knowledge, a subharmonic PAICPT response containing the additional $p = 2$ structure observed here has not been reported previously. The experimental observation is therefore a new result in its own right, independent of the unresolved microscopic interpretation.

5 Conclusion and Outlook

In this thesis, I systematically map microwave driven superconducting transport from the tunnel limit to atomic contacts with nearly perfect channel transmission. Across this range, the same experimental strategy connects static junction characterization with complete microwave amplitude maps. This progression resolves conventional photon assisted tunneling with unusually high resolution and establishes a unified phenomenological framework for photon assisted transport across finite to high transmission channels. Threshold weighted photon assisted multiple Andreev reflection (PAMAR) accounts for the finite-bias sidebands, and photon assisted incoherent Cooper pair tunneling (PAICPT) accounts for the coexisting zero-bias replicas. Along the way, it reveals two previously unreported responses, a tunable microwave induced rectification and a subharmonic PAICPT response at half the conventional replica spacing.

Static transport determines the superconducting energy scales and, for atomic contacts, the mesoscopic pincode. Microwave irradiation then introduces a second spectroscopic dimension. The voltage positions of the resulting structures fix their energy thresholds and transferred charges, while their evolution with drive amplitude distinguishes processes that occupy the same voltage range. This combination transforms complex transport maps into a systematic comparison of quasiparticle, Cooper pair, and MAR transport.

Stable mechanically controllable break junctions and a common measurement and analysis chain make the comparison possible across all regimes. In situ calibration converts the hardware dependent microwave output into the local ac voltage experienced by each junction, so the maps can be compared on the physical drive coordinate of the transport models.

From Tunnel Barriers to Atomic Contacts

The first tunnel junction establishes the high-resolution tunnel reference. Its static transport fits the SIS BCS model, and I reproduce its complete microwave driven evolution with the Tien-Gordon model, achieving exceptional agreement. Individual photon assisted replicas remain visually resolved and can be followed continuously to at least order 25. The measured ladder extends substantially beyond the superconducting gap while retaining the predicted redistribution of spectral weight across the full amplitude range. To my knowledge, no previous

SIS amplitude map has continuously resolved quasiparticle replicas to a higher order. This result demonstrates quantitative Tien-Gordon transport deep into the multiphoton regime and sets the reference against which the more complex junctions can be understood.

A second, nominally similar tunnel junction reveals the first new phenomenon. I observe microwave induced rectification with a local current asymmetry reaching up to 25 %. Dc bias, microwave amplitude, and frequency jointly control its magnitude and polarity, switching it on with increasing drive and reversing its direction across the investigated interval. The response is nonhysteretic and turns the junction into a microwave controlled diode-like element. Comparison with the bias symmetric Tien-Gordon response isolates the additional asymmetric contribution and establishes a regime beyond uniform ac driving. Static magnetic field maps show no unexpected features and the transport stays unchanged across a wide range of gate voltages, supporting a microwave induced origin over a static flux sensitive or Coulomb blockaded state. The narrow drive frequency at which the effect appears also matches an electromagnetic mode of the sample environment identified independently in an earlier transmission measurement of the same cryostat. A resonantly driven Josephson weak link, consistent with the conventional switching current of a comparable junction, provides a plausible explanation for the drive threshold, frequency sensitivity, and polarity reversal.

Moving from the tunnel regime to atomic contacts introduces discrete channels of finite and nearly perfect transmission, and with them the question of how to separate overlapping transport processes when several channels contribute at once. Static characterization along a complete opening trace shows that the extracted pincodes evolve continuously within each conductance plateau and reorganize at abrupt jumps, revealing contact changes that stay hidden in the normal-state conductance alone. Under microwave irradiation, the same channels support quasiparticle, MAR, and Cooper pair processes that overlap in the same voltage range, so the static pincode alone can no longer explain the driven maps. To separate these processes, I decompose the static MAR response deterministically around its known energy thresholds and apply charge dependent photon dressing to each resulting order. Combined with PAICPT, this framework reproduces the dominant trajectories and their spectral weight evolution across complete amplitude maps, without adjusting parameters to fit individual driven features.

To test the generality of this description, I move beyond the single contact where I develop it. Testing it across contacts with substantially different pincodes, from moderate transmission to a nearly ballistic leading channel, shows that threshold weighted PAMAR captures the dominant finite-bias response throughout this range, while the squared-Bessel amplitude evolution of the replicas originating from zero bias supports their assignment to PAICPT rather than to a coherent Shapiro response. The model therefore does more than reproduce a collection of lines. It assigns overlapping structures consistently to their energy

thresholds, effective charge orders, and static transport origins, across contacts that share little beyond the same measurement and analysis chain.

At the highest conductance investigated, a second new phenomenon emerges, a subharmonic PAICPT response at half the conventional replica spacing. To my knowledge, no earlier measurement has reported this response. In this contact, the microwave maps contain trajectories running halfway between the conventional Cooper pair replicas, positions that no combination of the established single-pair replicas can reach. A second, distinct replica family at half the conventional spacing is therefore required to describe them. Within the phenomenological construction, this spacing matches the effective transfer of two Cooper pairs, placing the response one step beyond the incoherent single-pair transport that describes the other contacts. Their squared-Bessel-like amplitude evolution is consistent with this assignment, but does not by itself establish a simultaneous, correlated two-pair transfer or identify the microscopic process behind it. The replica pattern is a new result in its own right, independent of this unresolved microscopic interpretation. It marks the point where the controlled few-channel description reaches its limit, opening a new region in which higher charge correlations, phase dynamics, and the electromagnetic environment can be investigated together.

From Microwave Response to Transport Processes

Microwave maps become interpretable transport data only when combined with three complementary coordinates. Feature spacing alone is insufficient when quasiparticle, MAR, and Cooper pair structures overlap. Static transport supplies the channel composition and energy thresholds, microwave spacing determines the transferred charge, and amplitude evolution distinguishes incoherent tunneling from coherent phase locking. Together, these coordinates separate coexisting processes by their static origin, charge, and coherence without selecting isolated features from the driven data.

This perspective also clarifies when a macroscopic junction description is sufficient and when microscopic charge processes must be resolved. In the tunnel reference, a single driven voltage and the static I–V characteristic capture the response over an exceptionally wide range. At finite transmission, the individual channels and voltage dependent MAR charge orders become essential. The electromagnetic environment adds another active degree of freedom by setting the local drive, enabling incoherent Cooper pair transfer, and coupling phase dynamics back into quasiparticle transport. The rectification in the second tunnel junction and the subharmonic response in the highest-conductance atomic contact appear precisely where these additional degrees of freedom become visible. They therefore extend the systematic progression rather than standing as unrelated anomalies.

The distinction between circuit parameters and microscopic transport has direct implications for superconducting quantum circuits. Their Josephson junctions are commonly represented by a few circuit quantities, although dissipation and microwave response originate from charge transport within a local electromagnetic environment. The measurements here show that this environment can set the junction drive, redistribute quasiparticle and Cooper pair transport, and generate frequency dependent responses absent from static characterization. Dedicated test structures based on the same spectroscopy could probe microwave delivery, resonances of the sample holder and wiring, subgap transport, and unintended high-transmission channels. They would not measure qubit coherence directly, but could reveal how junction fabrication and the surrounding microwave circuit influence microscopic dissipative processes that limit circuit reproducibility.

The underlying approach is not restricted to superconducting qubits or metallic atomic contacts. Hybrid nanowires, quantum dots, molecular junctions, superconducting detectors, and driven single-charge devices all support regimes in which several charge transfer processes coexist under periodic drive. The same three coordinates apply there too, provided static transport independently fixes the relevant energy thresholds, as the pincode does here, so the microwave spacing and amplitude evolution can be compared against a prediction rather than fitted freely. That condition can fail, for example when the relevant thresholds are not independently accessible or several channels remain unresolved. A different characterization step would then have to substitute for the pincode used here.

The framework remains phenomenological, but independently characterized static transport limits its physical assignments. It bridges the transparent Tien-Gordon limit and a microscopic nonequilibrium treatment, providing a benchmark for determining which features require coherence, occupation dynamics, or environmental backaction.

Future Research

The rectification found in the second tunnel junction motivates a dedicated follow-up experiment rather than a side test. A deliberately fabricated series of tunnel junctions with systematically varied circuit asymmetry, for example in sample position within the shielding environment, lead impedance, or electrical path length, would turn the uncontrolled asymmetry of this sample into a controlled parameter. It would then map how the drive threshold, frequency sensitivity, and polarity reversal scale with this parameter. Combined with the frequency and amplitude sweeps already established here, this would test whether the effect scales as expected for a resonantly driven Josephson weak link, and could turn a favorable asymmetry into a deliberately designed microwave recti-

fier. This step needs no new measurement technique and only variations on the existing sample fabrication, making it a natural first project on its own.

The next step is to embed the junction in a deliberately designed electromagnetic environment. A calibrated impedance, resonator, or superconducting loop would turn microwave amplitude, phase, and mode structure into controlled variables, extending the asymmetry series above with an independent check of the resonantly driven weak link picture. The same control addresses a broader question in the atomic contacts, separating incoherent Cooper pair transfer from Josephson phase locking by making the environment an experimental variable rather than a fixed backdrop. A tunable weak link embedded in a resonator could additionally correlate charge transfer and photon exchange with resonator loss, quasiparticle dynamics, or qubit relaxation, so that the environment becomes a controlled quantum degree of freedom rather than an uncontrolled boundary condition.

The subharmonic response, observed in a single contact reached after reclosing the break junction, calls for the same systematic mapping already applied across the rest of the few-channel regime, and reaches further into open theoretical territory. A series of highly transmissive contacts, tuned deliberately into and across the conductance range where the subharmonic replicas appear, would establish how the effect depends on transmission, channel number, and drive amplitude. Finite-frequency current noise would add an independent, charge sensitive probe alongside the amplitude maps, since a genuinely correlated two-pair transfer and two uncorrelated single-pair events at the same average current would leave distinct signatures in the noise. Microscopic Floquet or Keldysh calculations can build directly on the extracted pincodes, calibrated drives, and charge order assignments to determine which structures follow from charge quantization and static thresholds and which require coherent interference, nonequilibrium occupations, or environmental backaction, settling in particular whether the subharmonic response reflects higher charge correlations or a more conventional process appearing at an unexpected spacing.

A superconducting single-electron transistor provides the natural next platform beyond this. It adds charging energy and gate induced charge to the controls established here. The connection between MAR charge order and energy threshold offers a way to organize its dense stability diagrams. Gate–bias–amplitude maps could follow how a process transferring multiple charges crosses a Coulomb threshold, exchanges photons, and changes the island charge, while phase control distinguishes coherent Josephson cycles from incoherent quasiparticle and MAR sequences. This extends the present framework toward the joint study of superconducting coherence, nonequilibrium dynamics, and charge fluctuations.

Final Remarks

Through systematic microwave amplitude spectroscopy, I connect superconducting transport across widely different transmission regimes. I resolve the Tien-Gordon response with exceptional dynamic range, relate finite-transmission structures to their energy thresholds and effective charges, and reveal microwave controlled rectification and a subharmonic response at the boundaries of these descriptions. These results establish a basis for experiments with controlled phase, impedance, and island charge.

Superconducting quantum technologies ultimately depend on controlling how charge and phase respond to electromagnetic fields at individual junctions. By translating these microscopic responses into experimentally distinguishable signatures, I contribute to the understanding needed to make superconducting circuits more controllable, reproducible, and scalable.

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A. Supplementary Material and Software

The research software, source files, and supplementary digital material associated with this dissertation are available through the GitHub repositories listed below.

`schmampf/dissertation`
LaTeX source of this dissertation.

`schmampf/dissertation-supplementary`
Supplementary video material, animations, interactive figures, and other digital content accompanying this dissertation will be published in this repository. It serves as the persistent entry point for material that cannot be represented adequately in the printed document.

`schmampf/dissertation-figures`
Source code and processing scripts used to generate figures for this dissertation.

`schmampf/superconductivity`
Analysis, simulation, fitting, and visualization tools for superconducting transport experiments.

`schmampf/p5control-bluefors-evaluation`
Setup specific evaluation code for measurements performed with the Bluefors MCBJ experiment.

`mawenzy/p5control-bluefors`
Setup specific measurement control layer for the Bluefors MCBJ experiment, developed primarily by me on top of `p5control`.

`mawenzy/p5control`
Instrument control and data acquisition framework developed by Florian Kayatz under my supervision.

B. Diffusion and Coherence Lengths

Static disorder and impurities randomize the quasiparticle momentum through elastic scattering. We characterize this process by the elastic mean free path ℓ_{qp}. Disorder produces normal-state momentum relaxation in the aluminum films, permits diffusion that remains phase coherent over finite distances, and reduces the superconducting coherence length from its clean-limit value. These effects characterize propagation in the electrodes. They do not imply that the atomic constriction itself is diffusive. This appendix develops these distinctions and collects the length estimates used for the short-junction approximation in Sec. 2.3.3.

Normal-State Diffusion

Impurities, structural disorder, and phonons randomize the quasiparticle momentum. We characterize elastic transport by the mean free path ℓ_{qp}. On length scales much larger than ℓ_{qp}, repeated scattering produces diffusion. For an isotropic three-dimensional conductor, the diffusion constant is

$$D = v_{\mathrm{F}} \ell_{\mathrm{qp}} / 3 \,. \tag{B.1}$$

The Einstein relation then connects diffusion to the normal-state conductivity,

$$\sigma_{\mathrm{N}} = e^2 N_0 D \,. \tag{B.2}$$

Here N_0 denotes the total density of states including both spin directions. Under the assumptions of an isotropic parabolic band, this expression is equivalent to the Drude conductivity. The electric field creates a small imbalance in the quasiparticle distribution, while elastic scattering limits the drift. The local response is $\vec{j} = \sigma_{\mathrm{N}} \vec{E}$. This description assumes that the field, current density, and material parameters vary slowly on the scale of ℓ_{qp}. [37, 118]

For a homogeneous wire of length L and cross-sectional area A, integration of the local relation gives Ohm's law,

$$I = G_{\mathrm{N}} V \,, \qquad\qquad G_{\mathrm{N}} = \sigma_{\mathrm{N}} A / L \,. \tag{B.3}$$

The measured normal-state conductance therefore combines material transport parameters with sample geometry. This local description provides the material basis for discussing coherent propagation over a finite conductor. [119]

The parameters v_{F}, N_0, ℓ_{qp}, and D summarize the low-energy normal-state properties needed in this thesis. They do not specify a spectral lifetime or de-

termine whether phase coherence survives across a finite device. That question requires comparing the conductor size with the relevant transport lengths.

Phase Coherent Diffusive Transport

In the diffusive mesoscopic regime, the elastic mean free path remains much shorter than the system size, $\ell_{\mathrm{qp}} \ll L$. Quasiparticles undergo many scattering events that randomize momentum while traversing the conductor. If $L \lesssim \ell_\phi$, phase coherence persists over many such events and interference between multiple paths becomes observable.

For diffusive motion, the phase coherence length is conveniently expressed in terms of the diffusion constant D and a dephasing time t_ϕ,

$$\ell_\phi = \sqrt{D t_\phi}\,. \tag{B.4}$$

At finite temperature, thermal averaging over an energy window $\sim k_{\mathrm{B}}T$ further reduces interference. A common characteristic thermal length is

$$\ell_T = \sqrt{\hbar D / k_{\mathrm{B}} T}\,, \tag{B.5}$$

such that coherent effects are most pronounced when $L \lesssim \min(\ell_\phi, \ell_T)$. Numerical prefactors in the definition of ℓ_T depend on the geometry and convention. [120, 121]

On average, transport still follows the bulk drift-diffusion picture with the conductivity and diffusion constant introduced above. Quantum mechanically, coherent superposition of scattering amplitudes adds corrections, most prominently weak localization and universal conductance fluctuations. In low-dimensional or strongly disordered systems, these effects can become strong and may lead to Anderson localization. [120 - 124]

In the present work, this regime serves as a reference rather than the primary contact model. The aluminum electrodes can support diffusive propagation, while the atomic-scale constrictions remain shorter than the relevant scattering lengths and are therefore described as ballistic scatterers by their transmission eigenvalues.

Superconducting Length Scales in Aluminum

The short-junction approximation requires the weak-link length L to remain much smaller than the relevant superconducting coherence length, $L \ll \xi$. The clean-limit BCS coherence length of aluminum is

$$\xi_{\mathrm{clean},0} = \frac{\hbar v_{\mathrm{F}}}{\pi \Delta_0} \approx 2.4\,\mu\mathrm{m}\,. \tag{B.6}$$

For a diffusive aluminum film, the corresponding zero-temperature decay length is

$$\xi_{\mathrm{dirty},0} = \sqrt{\frac{\hbar D}{\Delta_0}} \approx 67\text{-}111\,\mathrm{nm}\,. \tag{B.7}$$

The dirty-limit estimate is film dependent. Steinberg *et al.* reported $\ell_{\mathrm{qp}} = 1.8\text{-}5.0\,\mathrm{nm}$ for several thin aluminum films, yielding the range above [67].

These are representative bulk and film estimates rather than device specific coherence lengths. Even the smaller dirty-limit coherence length greatly exceeds the length of an atomic-scale constriction, placing the contacts considered here in the short-junction limit [16]. The coherence lengths characterize superconducting propagation in clean or diffusive material. They do not determine whether transport through the constriction itself is ballistic or diffusive.

In the short-junction limit, quasiparticles acquire only a negligible energy dependent propagation phase while traversing the weak link. Its normal-state scattering properties may therefore be treated as energy independent over the gap. Each channel is characterized by one transmission eigenvalue τ_i, and its ABS spectrum depends only on τ_i, the gap, and the phase difference, as expressed by Eq. (2.67).

In summary, the aluminum electrodes may lie in the dirty limit while the atomic constriction remains a short ballistic scatterer. Electrode disorder determines the relevant superconducting coherence length, but it does not place the weak link in the diffusive short-junction, or KO1, limit. That limit applies only when the weak link itself is diffusive.

C. Practical Notes on Sample Fabrication

This appendix collects practical notes from the laboratory that complement the conceptual description in Sec. 3.1. It is not intended as a fully standardized protocol. Instead, it documents the small but critical steps, common pitfalls, and empirically useful practices that improve the reliability of sample preparation.

Wafer Preparation

I start with a bronze wafer with a thickness of 300 µm, which is covered by a protective foil. To remove this foil without damaging the wafer, I cool the wafer in liquid nitrogen and gently peel off the brittle fragments. This method proves effective and largely nondestructive. I then polish the wafer using a polishing head made of sewn cotton cloth attached to a drill and aluminum oxide from a white polishing compound block as the abrasive. I soak the polishing head with ethanol or IPA and fix the wafer to a smooth milled steel block using double-sided adhesive tape[1], taking care that no tape protrudes beyond the wafer edge. During polishing, I move the polishing head from top to bottom so that debris generated in the process is less likely to settle back onto the wafer surface. I rotate the steel block periodically because the lower part of the wafer is generally easier to polish uniformly. I continue until the entire surface has a homogeneous mirror-like finish. To detach the wafer from the steel block, I cool it again in liquid nitrogen. I then rinse the wafer first with acetone and subsequently with IPA, ensuring that the acetone is removed completely before switching solvents. Finally, I dry the wafer with nitrogen gas while the surface is still wet with IPA to avoid residue formation. [125]

An alternative polishing method uses sandpaper with progressively finer grit on a glass slide, as described by Patrick Raif [89]. It is better suited to reducing long-range surface variations, but it is also considerably more time-consuming and is therefore not used routinely.

Next, I prepare the spin coater, the polyamic acid (PA), and the vacuum oven. The PA is preportioned into small crimp seal vials to minimize moisture exposure and reduce the time required for warming. I remove one vial from the freezer and allow it to reach room temperature while preheating the convection oven to 135 °C. I line the inside of the spin coater with aluminum foil and verify that the vacuum oven is operational and open.

To remove adsorbed water, I place the wafer on a hot plate at 100 °C for at least one minute. For reliable adhesion to the 1 inch chuck, I apply a layer of

[1] Tesa, Universal.

parafilm to the chuck and bend it securely around the chuck edge. I then make five small holes in the parafilm using a syringe, one in the center and one near the edge in each cardinal direction. I allow the wafer to cool to room temperature before centering it on the chuck and only then switch on the vacuum pump. Before dispensing the PA, I run the spin coating program once to confirm that the wafer is held securely. This check is particularly important for larger wafers, for which insufficient adhesion is not always immediately apparent.

I blow the wafer with nitrogen to remove residual dust and pour the PA onto the wafer until roughly 90 % of the surface is covered, taking care to avoid trapped air bubbles. If bubbles remain, I move or burst them carefully with a syringe without scratching the wafer. The spin coating program consists of an initial spread step at 300 rpm for 30 s, followed by a final spin step at 5000 rpm for 90 s. Immediately afterward, I transfer the wafer to the convection oven for a 5 min soft bake at 135 °C. I then perform the hard bake in the vacuum oven using a programmed cycle that ramps to 400 °C, holds this temperature for 30 min, and cools back to room temperature over several hours. Detailed instructions and the corresponding temperature profile are given in Ref. [89]. In practice, starting the bake immediately after spin coating improves reproducibility.

I apply the EL11 and A4 resist layers using the same general workflow, but with different spin and bake parameters[2]. Aluminum foil inside the spin coater is unnecessary because these resists can be removed easily with acetone. I check the age of the resist before use because prolonged storage can noticeably degrade its performance. I apply EL11 first and perform the corresponding soft bake. After applying the A4 layer, the subsequent A4 bake also hard-bakes the underlying EL11 layer. The relevant process parameters are listed in Table C.2.

I then cut the wafer into pieces with dimensions of 3×8 mm using a custom sheet shear equipped with 3 mm and 18 mm spacers. Before cutting, I clean the cutting edges thoroughly with IPA to reduce contamination and maintain clean cuts. Samples taken from the outer wafer region are more likely to show thickness variations in the PI and resist layers. I therefore prefer pieces from the central region of the wafer for reproducible fabrication. Edge related thickness variations can affect the dose required during electron beam lithography. After cutting, I store the samples in a chip tray[3] for safe handling.

Electron Beam Lithography

I perform electron beam lithography (EBL) using the `Crossbeam` system. I begin by measuring the beam current with the jig containing the Faraday cup. A position list in the Zeiss software helps me locate the Faraday cup quickly. I keep the working distance as close as possible to 5.00 mm. Because the beam

[2] EL11 (MMA(8.5)MAA) & 950 A4 (PMMA)
[3] Entegris, custom configuration.

current requires some time to stabilize, I either wait before measuring or return later and repeat the check.

Accurate horizontal alignment of the sample is essential for exposure. I use the alignment tools in the focused ion beam (FIB) toolbox to simplify this step. I zoom out to the maximum field of view and use the upper-right corner together with the leftmost sample edge for the initial alignment. This is usually sufficient to correct the dominant angular misalignment. Additional refinement within `Elphy Plus` is possible, but in practice it rarely improves the final angular alignment once the sample is mounted in the cryostat.

Next, I focus the beam in the upper-right corner of the sample. I use the wobble function to center the beam in the aperture and minimize astigmatism with the x- and y-stigmators. Once the beam is well focused, I check the current again at the Faraday cup. For the 30 µm aperture in high-current mode, the beam current is about 0.5 nA. I use this setting for the small high-resolution structures. For the 120 µm aperture, which I use for the 1000 µm writing fields, the exact current is less critical because the larger leads tolerate moderate underexposure. In practice, slight underexposure in these larger fields reduces the writing time without compromising the result. The beam current for the 120 µm aperture in high-current mode is typically around 5 nA. I perform a final focusing step at a central position along the upper sample edge, using a particle on the resist as a focusing aid. At this stage, I confirm once more that the working distance is 5.00(1) mm and that the beam shift is set to zero.

Before starting the exposure, I note the current time. Because the design layout includes a timestamp, this provides a convenient way to match laboratory notes to the physical sample. If the measured beam current differs from earlier sessions, I recalculate the settling time for the 100 µm field accordingly. The position list executed in `Elphy Plus` then locates the writing fields automatically, applies the corresponding settings, and writes the design. I enforce stitching between adjacent writing fields by overlapping their nominal positions by 100 µm.

In practice, it is advantageous to keep the written design as simple as possible and avoid an unnecessarily large number of polygons. This reduces writing overhead and makes the exposure procedure more robust. I also avoid contamination dots as focusing markers. Instead, the combination of high-current mode and the 30 µm aperture proves sufficient for reliable focusing because it provides a comparatively large depth of focus. The drawback is that very low exposure doses are no longer readily accessible, but this is not a practical limitation for the structures used here.

I repeat the full alignment, focusing, and writing sequence for the remaining samples until I have completed a batch of roughly four to five samples.

If several weeks have passed since the previous EBL session, I verify the alignment between the 100 µm and 1000 µm writing fields before exposing the first sample. I expose one 100 µm field together with a neighboring 1000 µm field in the upper-right corner of a sample. The exposed resist appears visibly different from the unexposed surroundings and can therefore be used to judge the rela-

tive alignment. I avoid prolonged observation because it could unintentionally expose the resist and reduce its contrast. If the fields are misaligned, I adjust the settings accordingly. For example, if the 100 µm field is shifted to the left, I correct U by $-\Delta X$. If it is shifted upward, I correct V by $-\Delta Y$. I repeat the test until the alignment is satisfactory.

After exposure, I develop the samples. I prepare a crimp seal vial containing a mixture of one part MIBK and three parts IPA and one additional vial of pure IPA for each sample. I immerse the sample in the MIBK/IPA developer for 25 s while gently swirling the vial to maintain uniform development. I then transfer the sample immediately into pure IPA and continue swirling. I reuse the developer for all samples in the same batch but use fresh IPA for each sample. I leave the sample in IPA for 60 s, remove it, and carefully dry it with a stream of nitrogen. Throughout the process, I hold the sample securely with tweezers to avoid dropping it while waiting or transferring it between vials. Finally, I inspect the developed structures under a light microscope to verify that the pattern is complete and free of obvious defects.

Shadow Evaporation

To begin shadow evaporation, I place the sample holder on a hot plate at 100 °C for a few minutes to remove adsorbed water. This step noticeably improves the vacuum and adhesion. I carefully check the sample orientation to ensure that the shadow evaporation proceeds in the intended direction. I then load the sample into the load lock and pump it overnight, or longer if possible, before starting the deposition sequence.

Once the vacuum is satisfactory, I transfer the sample holder into the main chamber for the first evaporation. Using electron beam evaporation, I deposit 60 nm of aluminum at an angle of 4°. I heat the aluminum gradually over approximately 5 - 10 min to avoid stressing the crucible and promote stable evaporation. I use one to three aluminum pellets per evaporation to ensure that the crucible contains sufficient material.

After the first evaporation, I transfer the sample back into the load lock and close the pumping line. I introduce pure oxygen until the pressure reaches 3.0(2) mbar and oxidize the sample for 3 min. To terminate the oxidation, I reopen the pumping line. I thoroughly evacuate the oxygen feed line to recover a vacuum on the order of 1×10^{-6} mbar afterward.

Once the load lock vacuum has recovered, I transfer the sample back into the main chamber for the second evaporation. I deposit 100 nm of aluminum at an angle of 34° to complete the shadow evaporation sequence.

A smooth aluminum surface is desirable for forming a well-defined AlO_x layer. In practice, however, the deposited aluminum often shows a large number of dimples, about 100 nm in diameter. Several measures are tested in an attempt to reduce them, but no decisive remedy is identified. There is some indication

that higher evaporation rates, particularly values above 4 Å s^{-1}, may reduce the number of dimples, although the effect is not statistically convincing. Contrary to earlier expectations, a higher pressure or a shorter time between the last vacuum break and evaporation occasionally appears to improve the surface. One possible interpretation is that a very clean sample mainly suppresses the desorption of water or other contaminants during deposition, whereas less favorable chamber conditions may influence nucleation in a way that incidentally smooths the film. Because the observed differences remain small, the issue is not pursued further.

I perform lift-off by placing the sample in a crimp seal vial filled with pure acetone and heating it on a hot plate at about 60 °C for 1 - 3 h. Afterward, I use a pipette to agitate the acetone carefully until all visible metal flakes have detached. I rinse the sample with IPA and dry it with nitrogen.

Reactive Ion Etching (RIE)

For the penultimate fabrication step, I use the `PlasmaPro 80` ICP RIE system from Oxford Instruments. Although this tool is not designed specifically for highly isotropic etching, I operate it in a regime that gives sufficiently isotropic results by combining low table RF power with elevated pressure. I first use an ignition step to stabilize the plasma before switching to the less stable regime that favors isotropic etching. The sample temperature is particularly important for obtaining the desired undercut. A table temperature of 100 °C proves to be a good compromise because it produces a sufficient undercut while keeping the etch slow enough to control the depth reliably. The process parameters that give the best results are listed in Table C.1.

I use a laser interferometer to monitor the etched depth. For this purpose, I choose a spot on the sample without metal structures where the reflected intensity is as large as possible without reaching saturation. During the etch, the signal follows an approximately cosine-like oscillation that can be used to estimate the removed thickness. The required number of periods m depends on the laser wavelength $\lambda = 632.8\,\mathrm{nm}$, the target depth $d = 500\,\mathrm{nm}$, and the refractive index[4] $n = 1.81$. For the present process, the target number of periods is

$$m = \frac{2 \cdot n \cdot d}{\lambda} = \frac{2 \cdot 1.81 \cdot 500\,\mathrm{nm}}{632.8\,\mathrm{nm}} = 2.86\,. \tag{C.1}$$

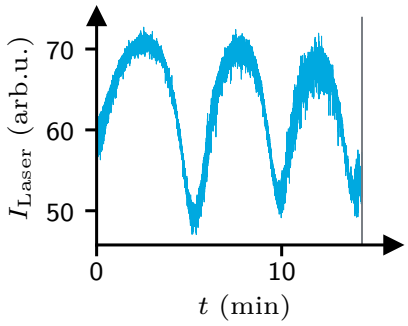


Figure C.1. Laser intensity recorded during plasma etching. I terminate the process after about 2.8 periods at the target etch depth.

[4] Fujifilm Electronic Materials U.S.A., Inc., `Polyamic Acid Durimide 100`

Table C.1. Process parameters for RIE, `mcbj sset v5 @ 100C`.

Parameter	Stabilization	Ignition	Etching
Process time	5 min	5 s	30 min
Table heater	100 °C	-	-
Oxygen flow	50 sccm	-	-
Pressure	100 mTorr	ramp	250 mTorr
Table RF			
Power Demand	-	20(5) W	3(5) W
Max Reflected Power	-	5 W	2 W
Tolerance Time	-	7 s	-
Min dc bias	-	1 V	-
AMU (C1, C2)	-	(41 %, 23 %)	-
ICP RF			
Power Demand	-	400(50) W	-
Max Reflected Power	-	100 W	-
Tolerance Time	-	10 s	-
AMU (C1, C2)	-	(64 %, 39 %)	-

An example of the recorded laser intensity during etching is shown in Fig. C.1.

Because the etch rate varies noticeably from sample to sample, I etch each sample individually.

Electrical Contacting

Electrical contacting is one of the most delicate steps in the entire fabrication sequence. Many otherwise successful samples are lost at this stage, so patience and a calm working environment are more important than speed. In practice, I obtain better results by contacting fewer samples carefully rather than rushing several samples in one session.

I use a setup with stable hand positioning, appropriate lighting, and a comfortable bench height. Sitting close to the sample while keeping both hands supported makes the process significantly easier. I also minimize avoidable sources of hand tremor before starting.

To fix the sample, I use adhesive strips[5] on a copper plate. This arrangement keeps the sample in place while still allowing me to reposition the plate. I prepare 90 µm insulated copper wires soldered to post connectors in advance. I strip the

[5] Tesa, tesafilm standard.

wire ends and twist the bias line wires. I keep the wires as short as practical, typically 4 - 5 cm, and insulate the rear part of each post connector with varnish[6].

I position the wire ends on the sample pads with the help of modeling clay, as shown in Fig. C.2. Then, using a sharpened toothpick, I apply a small amount of silver paste to connect each wire to its pad. The viscosity of the silver paste is critical. If it is too fluid, shorts could form. If it is too viscous, it tends to remain on the toothpick instead of wetting the pad properly. In practice, paste collected from the lid after shaking the bottle works well. I transfer only enough paste to wet the pad. Afterward, I apply two-component epoxy as strain relief for the wires. Again, the viscosity is important. If I use the epoxy too soon after mixing, it continues to flow and could spread into unwanted areas. Because the interaction between epoxy and silver paste is not fully understood, I keep the two materials separate as much as possible. I test the epoxy on small trial dots and wait until it reaches a suitable consistency before applying it to the sample. I glue the two sides of the sample separately because the useful working window is fairly short.

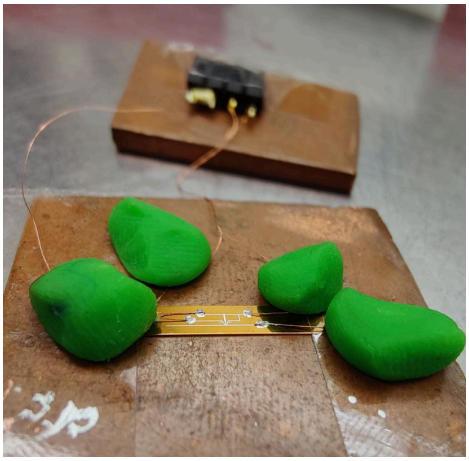


Figure C.2. Example of a contacted sample, before adding strain relief.

After mounting the sample in the cryostat, I scratch the shorting bridge with an engraving tool. I ensure proper grounding during this step and while closing the radiation shields by wearing a grounding bracelet. In dry conditions, I also increase the ambient humidity to reduce the risk of electrostatic discharge.

[6] Oxford Instruments, C5-101, GE Low Temperature Varnish.

Summary of Key Process Parameters

Table C.2. Key process parameters used for sample preparation.

Sacrificial layer	Durimide 115A
Spinning	spread cycle: 300 rpm for 30 s
	spin cycle: 5000 rpm for 90 s
	ramp time: 3 s, `Program 3`
Soft baking	135 °C for 5 min in convection oven
Hard baking	400 °C for 30 min in vacuum oven
Spacing layer	EL11 (MMA(8.5)MAA)
Spinning	spread cycle: 500 rpm for 5 s
	spin cycle: 2500 rpm for 90 s
	ramp time: 3 s, `Program 1`
Soft baking	150 °C for 1 min on hot plate
Electron resist	950 A4 (PMMA)
Spinning	spread cycle: 500 rpm for 5 s
	spin cycle: 5000 rpm for 60 s
	ramp time: 3 s, `Program 2`
Hard baking	170 °C for 30 min in convection oven
Exposure	`Crossbeam` by Zeiss, `Elphy Plus` by Raith
Settings	10 kV acceleration voltage, 5 mm working distance,
Aperture	30 µm and 120 µm aperture, high current,
	155 µC cm^{-2} area dose
Development	25 s in 1:3 MIBK:IPA
	60 s in IPA
Shadow evaporation	First layer: 60 nm Al at 4°
	Oxidation: 3.0(2) mbar O_2 for 3 min
	Second layer: 100 nm Al at 34°
Lift-off	1 - 3 h in acetone at 60 °C
Reactive ion etching	`PlasmaPro 80` ICP RIE by Oxford Instruments
	`mcbj sset v5 @ 100C`, see Table C.1
Electrical contacting	90 µm insulated copper wire, silver paste as
	electrical contact and epoxy as strain relief

D. Cryogenic Environment

The measurements are carried out in a Bluefors LD400 dry dilution refrigerator. The cryostat must remove heat from room temperature to the millikelvin regime and thermalize the wiring and sample environment. The relevant cooling chain is therefore described here in operational order. The dilution cycle uses a circulating ^{3}He / ^{4}He mixture as its working fluid. The pulse tube provides closed-cycle precooling and brings this mixture into the few-kelvin regime. The Joule-Thomson stage then condenses the precooled mixture. The dilution unit supplies the continuous cooling power at the mixing chamber, enabling mK temperatures.

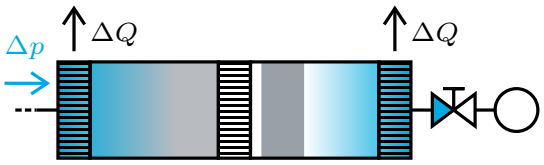
Pulse Tube Cryocooler

The first active cooling stage is a two-stage Gifford–McMahon (GM) type pulse tube. It is a closed-cycle helium refrigerator. An external room-temperature compressor and rotary valve generate a pressure oscillation, while the cold head contains no mechanical displacer, unlike, e.g., a Stirling cooler. The pulse tube removes the dominant heat load, cools the radiation shields, provides thermal anchors for wiring, and brings the helium mixture to the few-kelvin range. The pressure oscillation and rotary valve also generate mechanical vibration.

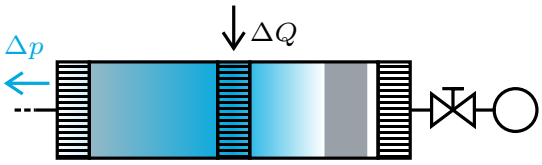
Figure D.1 shows the basic operation of a single-stage orifice pulse tube, following Refs. [126, 127]. The compressor drives helium back and forth through the aftercooler, regenerator, cold heat exchanger, pulse-tube volume, warm heat exchanger, and the flow impedance to the buffer volume. The gas does not circulate through the cold head in one direction. It oscillates. The regenerator stores heat during one part of the cycle and returns it during another. In addition to this cyclic exchange, the regenerator itself must sustain a steep temperature gradient along its own length, from the warm end down to the cold end, without conducting that heat back across.

Cooling is produced by the phase relation between pressure and gas displacement. In a conventional GM cooler, this phase relation is imposed by a cold mechanical displacer. In a pulse tube, the gas column itself acts as a virtual piston. The warm-end impedance and buffer volume delay the flow relative to the pressure oscillation. With the correct timing, gas rejects heat at the warm heat exchanger and absorbs heat at the cold heat exchanger. The useful cooling can therefore be understood as an enthalpy flow through an oscillating gas column.

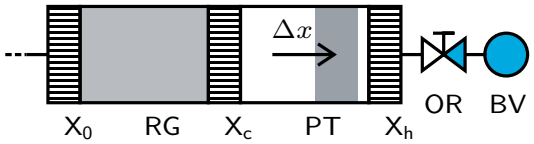
The LD400 uses two cascaded pulse tube stages, as sketched in Fig. D.2. The first stage removes heat at intermediate temperature and provides a thermal



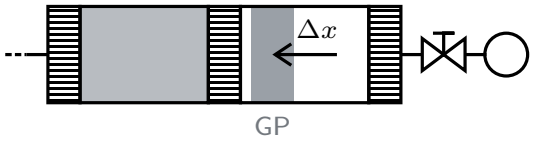
(a) Pressure Δp enters from the left. Heat ΔQ is rejected at the warm end. The high-pressure front is delayed by the orifice.



(c) The returning gas absorbs heat at the cold heat exchanger and leaves to the left.



(b) The virtual gas piston is displaced, while the buffer volume delays the gas flow.



(d) The virtual gas piston moves back toward equilibrium before the next cycle begins.

Figure D.1. Schematic of pulse-tube cooling. The stage consists of an aftercooler (X_0), regenerator (RG), cold heat exchanger (X_c), pulse tube (PT), warm heat exchanger (X_h), virtual gas piston (GP), orifice (OR), and buffer volume (BV). High-pressure regions are indicated in blue (■). Adapted from Ref. [126].

anchor near 40 K. The second stage is precooled by the first and reaches the few-kelvin range. These stages also affect the cooldown time and the heat load that must later be handled by the dilution unit.

The pulse-tube system is driven by an oil lubricated room-temperature helium compressor.[1] Low-pressure helium returning from the cold head is compressed, cooled, purified, and supplied again as high-pressure helium. The oil improves sealing and absorbs and transports heat internally, while the cooling water heat exchanger ultimately removes it. The oil must not reach the cold head. Oil separators, coalescing filters, and adsorbers therefore remove oil droplets, mist, and residual contaminants before the helium returns to the cryostat. This purification chain is essential because impurities can freeze out at cryogenic temperatures and degrade the regenerator or block small flow impedances.

Compressor maintenance becomes relevant during this work. I replace the adsorber, and the compressor heat exchanger requires repeated repairs due to leaks. One practical cause is the cooling water supply. The compressor is cooled with water from Lake Constance, which is very soft and can corrode copper cooling water circuits if it is not chemically conditioned. Dirt particles and shell debris in insufficiently filtered lake water can additionally abrade the piping [128]. Eventually, a leak between the cooling water circuit and the helium circuit leads to replacement of the compressor package. This history is included because compressor degradation or helium contamination can reduce pulse-tube performance before a problem is visible in the sample temperature.

[1] Cryomech, `CPA1110W2FNCE-21770300001`.

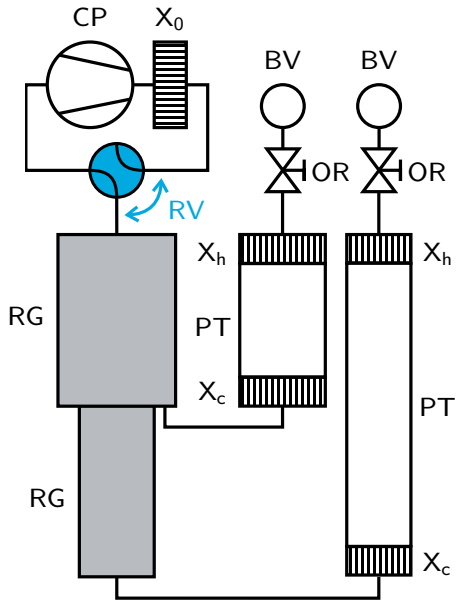


Figure D.2. Schematic of a two-stage Gifford–McMahon-type pulse tube. The rotary valve (RV) alternately connects the cold head to the high- and low-pressure sides of the compressor (CP). The first stage provides an intermediate thermal anchor, while the second stage is precooled by the first stage. Adapted from Ref. [127].

The cryostat frame is supported by pneumatic air springs at the four corners, operated at about 6 bar, and the suspended assembly has a large mass. This arrangement reduces the transmission of building and compressor vibrations to the suspended assembly. I do not measure the vibration coupling to the junction independently. In this configuration, contacts can remain stable on the scale of weeks as long as the cryostat is not mechanically disturbed. A small fraction of contacts still shows atomic rearrangements during individual measurements. I therefore do not attribute these rearrangements specifically to cryostat vibration.

Joule-Thomson Cooling

After pulse tube precooling, the circulating ^{3}He / ^{4}He mixture must be condensed before it can feed the dilution unit. The mixture cannot simply be expanded from room temperature into the cold part of the cryostat, because the incoming gas would carry too much enthalpy. It is first thermalized at the pulse tube stages and by lower-temperature return flows. It then passes through a flow impedance. This is the Joule-Thomson stage of the refrigerator [127, 129].

Joule-Thomson cooling is the temperature change of a real gas during throttling at approximately constant enthalpy H,

$$H = U + pV\,, \qquad \Delta H \approx 0\,, \tag{D.1}$$

with internal energy U, pressure p, and volume V. During throttling through a narrow restriction, no useful mechanical work is extracted and ideally no heat is exchanged within the impedance. The gas therefore changes pressure without changing its enthalpy appreciably.

For a small pressure drop, the corresponding temperature change can be written as

$$\Delta T \approx \mu_{\mathrm{JT}}\, \Delta p\,, \tag{D.2}$$

where μ_{JT} is the Joule-Thomson coefficient, which is given by

$$\mu_{\mathrm{JT}} = \left(\frac{\partial T}{\partial p}\right)_H = \frac{V}{C_{\mathrm{p}}}(\alpha T - 1)\,. \tag{D.3}$$

Here V is the molar volume, C_{p} is the molar heat capacity at constant pressure, and α is the volumetric thermal expansion coefficient. The Joule-Thomson coefficient is the slope of an isenthalpic path in the temperature–pressure plane. Its sign determines whether a pressure drop cools or heats the gas. For an ideal gas, enthalpy depends only on temperature, so an isenthalpic expansion would not change the temperature. The Joule-Thomson temperature change therefore arises from deviations from ideal-gas behavior.

The sign of μ_{JT} changes at the inversion temperature, as shown in Fig. D.3 for ^{4}He. Helium has an unusually low inversion temperature, so room-temperature helium would not be cooled by Joule-Thomson expansion. The exact values differ for the circulating ^{3}He / ^{4}He mixture, but the practical requirement is the same. The gas must be precooled before throttling can be used for condensation. The pulse tube brings the mixture into the thermodynamic regime where expansion removes heat instead of adding it.

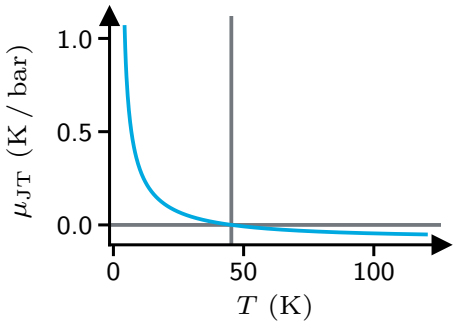
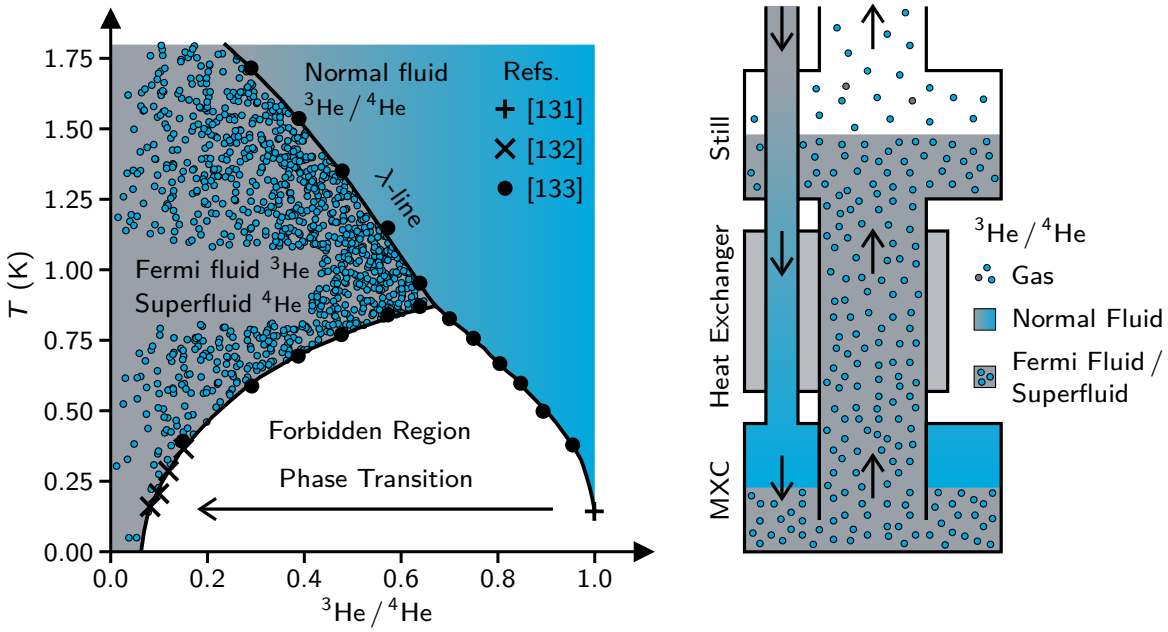


Figure D.3. Joule-Thomson coefficient of ^{4}He as a function of temperature. The illustrative curve is calculated for a pressure of 1 bar using CoolProp [130] and shows the change of sign at the inversion temperature.

In the dry dilution refrigerator, the incoming high-pressure mixture is cooled stepwise before it reaches the Joule-Thomson impedance. It is anchored at the pulse tube stages and exchanges heat with colder return flows in the lower-temperature part of the dilution unit. After this precooling, the pressure drop across the impedance lowers the temperature further and condenses the mixture. The Joule-Thomson stage thus connects pulse tube precooling to dilution refrigeration by converting a precooled, pressurized gas flow into a cold liquid mixture for continuous circulation.

Initial condensation and regular circulation have different failure modes. During condensation, the mixture line is operated at overpressure to drive gas through the impedance and condense it. Leaks can therefore lose mixture to the vacuum space or laboratory environment. During regular circulation, by contrast, the condensing line is usually below atmospheric pressure. Small leaks are then more likely to draw residual gas or contamination into the circulation path. This is dangerous because the Joule-Thomson impedance is narrow and can become clogged by gases or moisture that freeze out at cryogenic temperatures.



(a) Phase diagram of liquid ^{3}He / ^{4}He mixtures. The white region is forbidden for a homogeneous liquid, because mixtures in this range lower their free energy by separating into a ^{3}He-rich concentrated phase and a ^{4}He-rich dilute phase. The λ-line marks the superfluid transition of the ^{4}He-rich phase. Adapted from Ref. [134].

(b) Dilution unit circulation. The incoming ^{3}He-rich stream is precooled, enters the mixing chamber (MXC), and cools it as ^{3}He crosses into the dilute phase. The still evaporates ^{3}He and closes the cycle.

Figure D.4. Dilution refrigeration with a ^{3}He / ^{4}He mixture. Together, the phase diagram and circulation schematic connect the finite low-temperature ^{3}He solubility of the dilute phase to continuous cooling of the mixing chamber. The blue shading uses the same mixture composition encoding in both panels, including the intentional gradient along the condensing line. In the dilute Fermi fluid, the density of the ^{3}He symbols within the ^{4}He-rich background indicates the local ^{3}He concentration.

For this reason, the mixture handling line is passed through a liquid nitrogen trap before condensation or circulation.

Once the mixture has condensed and circulation is established, the same impedance remains part of the continuous flow path. The pressure drop and counterflow heat exchangers supply cold liquid mixture to the lower-temperature stages, while the actual millikelvin cooling is produced in the mixing chamber.

Dilution Refrigerator

The dilution unit is governed by the low-temperature phase diagram of the helium mixture rather than by a further gas expansion step. Figure D.4(a) shows that a homogeneous liquid is stable only outside the phase separation boundary. The white "Forbidden Region" is therefore not an impossible total mixture composition. It is a miscibility gap. If the average ^{3}He concentration and temperature

place the mixture inside this region, the homogeneous liquid has a higher free energy than two coexisting liquids. The mixture therefore separates into a ^{3}He-rich concentrated phase and a ^{4}He-rich dilute phase with compositions on the phase boundary at the same temperature [129, 134].

This phase separation is useful because the dilute phase does not become pure ^{4}He as the temperature approaches zero. Instead, it retains a finite ^{3}He molar concentration of about 6.6 %, while the concentrated phase approaches nearly pure ^{3}He. The mixing chamber exploits exactly this property. Its phase boundary separates the upper concentrated phase from the lower dilute phase. This arrangement results from their density difference, with the less dense ^{3}He-rich phase floating above the denser ^{4}He-rich phase in Earth's gravitational field. When ^{3}He atoms cross from the concentrated phase into the dilute phase, they enter a state with larger enthalpy. The required energy is taken from the mixing chamber, so the transfer cools the sample stage. The refrigerator can keep cooling only if this transfer is driven continuously. Without circulation, the phases would approach equilibrium and the cooling power would vanish.

The circulation path in Fig. D.4(b) shows how this non-equilibrium condition is maintained. The incoming ^{3}He-rich stream is precooled by the outgoing cold dilute stream in the counterflow heat exchangers and enters the mixing chamber on the concentrated side of the phase boundary. After crossing into the dilute phase, the ^{3}He is carried upward with the dilute mixture toward the still. At the still temperature, ^{3}He has a much higher vapor pressure than ^{4}He, so controlled heating of the still preferentially evaporates ^{3}He. The pumps remove this ^{3}He-rich gas, the room-temperature gas handling system cleans it, and the condensing line returns it to the cryostat. The cooling power is therefore set mainly by the ^{3}He circulation rate and by the heat exchanger efficiency. The heat exchangers determine how much of the outgoing cooling is recovered before the incoming stream reaches the mixing chamber [127, 129].

Typical preparation and cooldown values are useful diagnostics for later operation. Before cooldown, the cryostat vacuum space is pumped for up to 24 h and reaches pressures of about 6×10^{-3} mbar. During cryogenic operation, the insulating vacuum measurement reaches about 2×10^{-6} mbar. A full cooldown to the few-kelvin regime, including the magnet, takes about 36 h. Under typical operating conditions, the nominal 40 K stage stabilizes near 47 K, while the 4 K stage reaches about 3.2 K. Once the system is at 4 K, condensation of the mixture takes about 1 h, and the subsequent cooldown from 4 K to base temperature takes about 3 h.

The LD400 system is specified with a guaranteed base temperature of 10 mK and an expected base temperature of 8 mK. The specified cooling powers at the mixing chamber flange are 14 µW at 20 mK, 400 µW at 100 mK, and 575 µW at 120 mK [135]. These values are manufacturer specifications for the cryostat. I do not remeasure them for the installed experiment. With the installed MCBJ experiment, wiring, filters, thermometers, and microwave infrastructure, the sample and mixing chamber stages can reach about 30 mK, while the still is

operated at about 1 K. A normal automated warm-up with the Bluefors warm-up script takes about 2.5 - 3 d.

I do not optimize the ^{3}He circulation rate for the absolute lowest mixing chamber temperature during this work. The limiting factor for the transport measurements is not the final few millikelvin of lattice temperature. The setup can be operated comfortably between about 30 mK and 40 mK, and the MCBJ can be moved step by step without leaving this temperature regime. The largest temperature excursions occur during microwave irradiation. Depending on the applied microwave power and coupling path, the sample stage temperature can rise to several hundred millikelvin. In extreme cases it approaches 1 K. Improving the last part of the base temperature performance is therefore less relevant than controlling microwave heating and maintaining a well-thermalized, well-filtered electrical environment. These conditions are more important for obtaining sharp dI–dV spectra. The measured sample stage temperature should therefore not be identified automatically with the electronic temperature relevant for the junction, which can be higher because of residual radiation, imperfect line thermalization, bias dissipation, or microwave induced heating. Earlier optimization of the same cryostat and MCBJ platform by Martin Prestel reached temperatures below 30 mK before the modifications described here [94].

Together, these stages define the thermal environment of the experiment. The pulse tube provides the dominant cooling power at intermediate temperatures, the Joule-Thomson stage condenses the helium mixture and enables continuous circulation, and the dilution unit provides the millikelvin cooling power at the mixing chamber. The sample, filters, thermometers, and wiring are thermally anchored to the corresponding cryostat stages. The cryostat therefore does not only set the lattice temperature of the device. It also determines how well electrical lines are thermalized, how much external radiation reaches the contact, and how much mechanical motion is transmitted to the break junction. These limits directly affect the sharpness of the dI–dV spectra and the reproducibility of contact tuning.

E. Supplementary Analysis of TB2

For completeness, Fig. E.1 shows branch resolved line cuts corresponding to the comparison maps in Fig. 4.20. Figures E.2 and E.3 provide the corresponding detailed measured transport landscape. The corresponding simulated transport landscape highlights the deviations from the BCS SIS TB behavior.

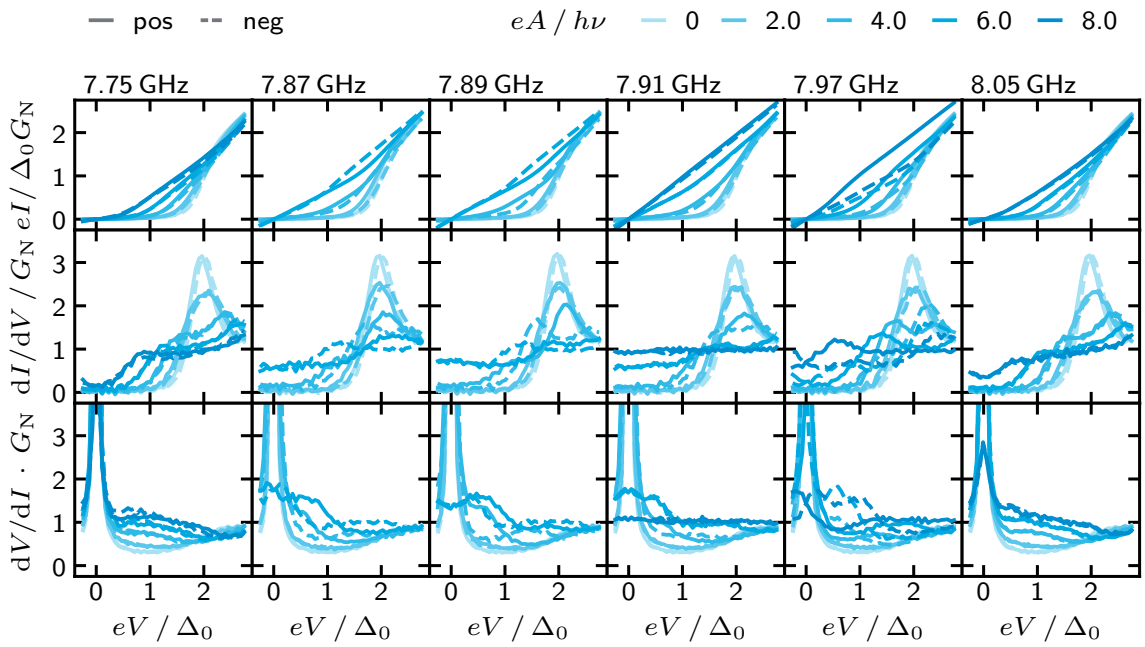
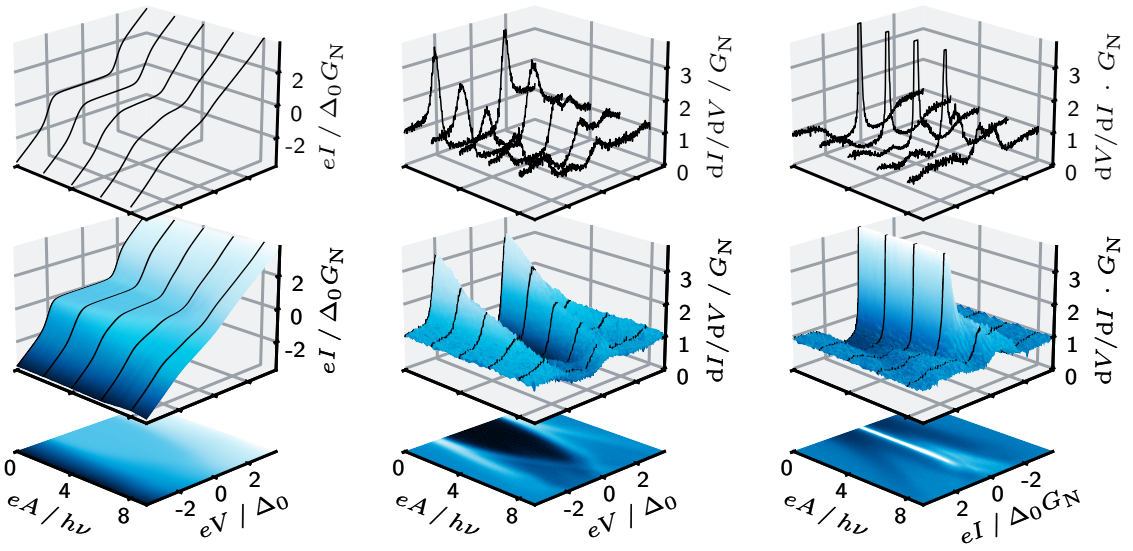
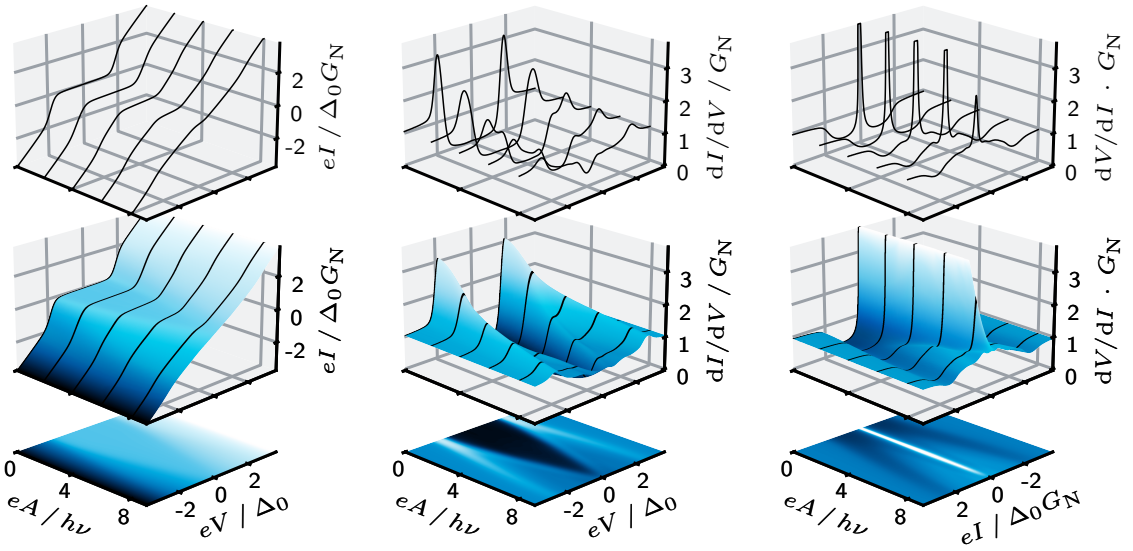


Figure E.1. Comparison of the positive- and negative-bias transport branches of TB2 at selected frequencies and calibrated microwave amplitudes. The columns correspond to the six frequencies shown in Fig. 4.20, and the rows show the normalized *I–V*, *dI–dV*, and *dV–dI* responses. Solid lines denote the positive-bias branch, while dashed lines denote the negative-bias branch reflected about the origin. The color encodes the normalized calibrated amplitude $eA_{\mathrm{cal}}/h\nu$. The increasing separation of the two branches at intermediate frequencies reveals the onset and frequency dependent reversal of the rectification.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure E.2. Calibrated transport landscape of TB2 at 7.97 GHz. The three columns show the *I–V*, *dI–dV*, and *dV–dI* responses as selected traces, surfaces, and false-color maps. The vertical axis uses the effective calibrated amplitude defined in Eq. (4.9).



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure E.3. Symmetric SIS plus Tien-Gordon reference calculation at 7.97 GHz. The simulation shows the response expected from bias symmetric photon assisted quasiparticle tunneling and provides a model reference for the asymmetric measurement. I calculate the temperature over the amplitude axis with the fit I obtain from Eq. (E.1).

Coupling and Heating for TB2

I use the following diagnostic relations to compare microwave coupling and heating across the investigated frequencies:

$$\begin{aligned} A_{\mathrm{fit}} &= \eta_{A_{\mathrm{out}}}^{(A_{\mathrm{fit}})}\, A_{\mathrm{out}}\,, \\ T_{\mathrm{fit}} &= \min\Big[T_{\mathrm{c}},\, \max\Big(T_{\mathrm{base}},\, \eta_{A_{\mathrm{out}}}^{(T_{\mathrm{fit}})}\, A_{\mathrm{out}} + T_{\mathrm{off}}\Big)\Big]\,, \\ T_{\mathrm{bath}} &= \eta_{A_{\mathrm{out}}}^{(T_{\mathrm{bath}})}\, A_{\mathrm{out}} + \mathrm{const.}\,, \\ T_{\mathrm{fit}} &= \min\Big[T_{\mathrm{c}},\, \max\Big(T_{\mathrm{base}},\, \eta_{A_{\mathrm{fit}}}^{(T_{\mathrm{fit}})}\, A_{\mathrm{fit}} + T_{\mathrm{off}}\Big)\Big]\,, \\ T_{\mathrm{bath}} &= \eta_{A_{\mathrm{fit}}}^{(T_{\mathrm{bath}})}\, A_{\mathrm{fit}} + \mathrm{const.}\,, \\ T_{\mathrm{fit}} &= \min\Big[T_{\mathrm{c}},\, \max\Big(T_{\mathrm{base}},\, \eta_{T_{\mathrm{bath}}}^{(T_{\mathrm{fit}})}\, T_{\mathrm{bath}} + T_{\mathrm{off}}\Big)\Big]\,. \end{aligned} \tag{E.1}$$

The fitted-temperature data form empirical lower and upper clusters near

$$T_{\mathrm{base}} \approx 0.47(1)\,\mathrm{K}\,, \qquad\qquad T_{\mathrm{c}} \approx 1.29(1)\,\mathrm{K}\,. \tag{E.2}$$

I use these values as the bounds in the clipped temperature relations. The coefficients η quantify correlations between the respective coordinates and do not by themselves identify a causal coupling mechanism.

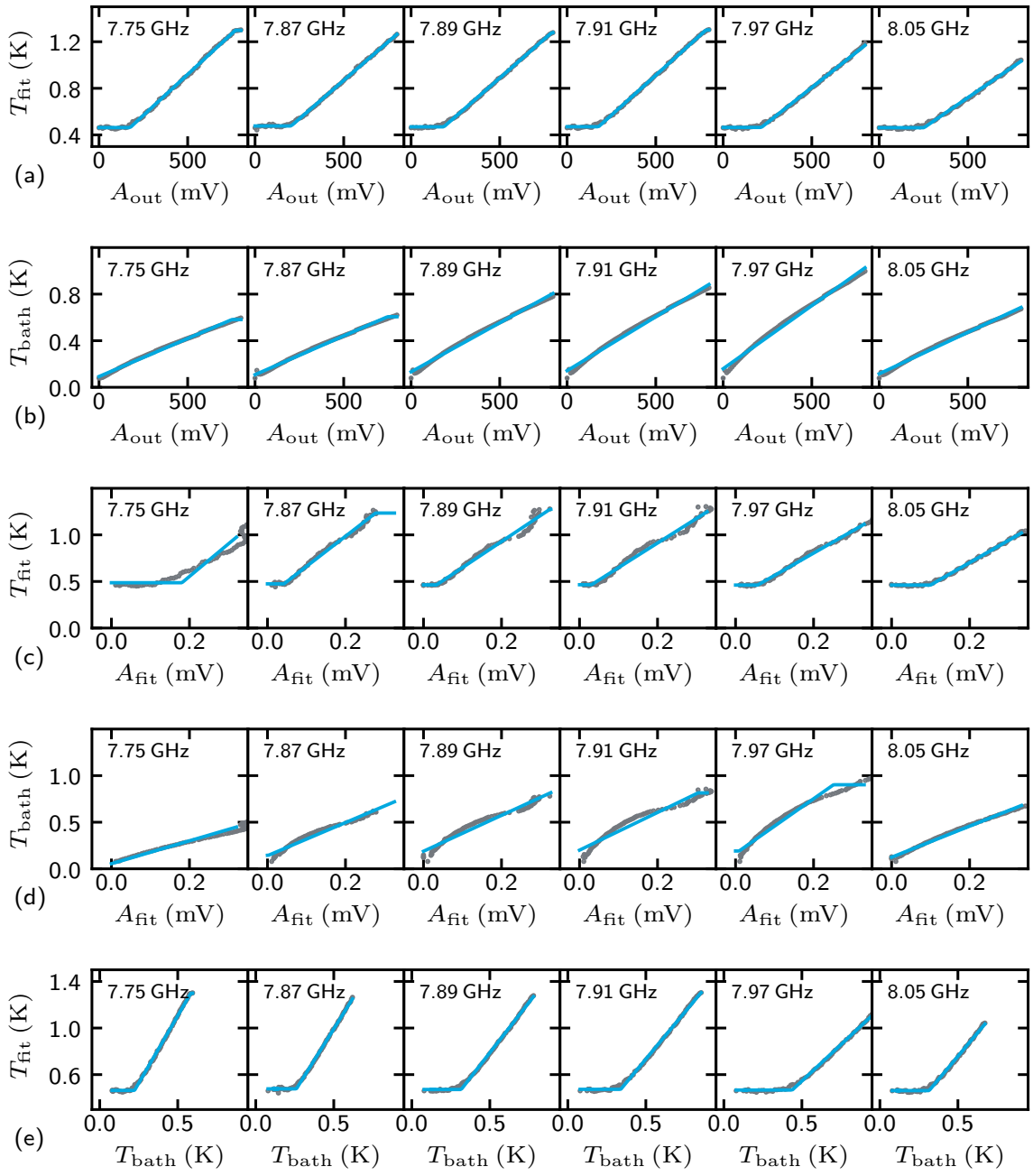


Figure E.4. Coupling and heating diagnostics for TB2 at the selected microwave frequencies. Panels (a) and (b) show the fitted transport temperature and measured bath temperature as functions of the nominal source amplitude. Panels (c) and (d) show the same temperatures as functions of the effective fitted microwave amplitude. Panel (e) compares the fitted transport temperature directly with the measured bath temperature. The clipped fits in panels (a), (c), and (e) use the empirical bounds T_{base} and T_{c}, while the linear slopes in panels (b) and (d) quantify microwave induced bath heating. These relations characterize correlations between the control quantities but do not establish a causal mechanism for the rectification.

F. Supplementary: Atomic Contacts

This appendix complements the atomic contact measurements in Sec. 4.2 with characterization data and extended model comparisons that support the main analysis. I first characterize the intact constriction before breaking. Its hysteretic zero-field response and magnetic field dependence establish the initial superconducting device state, which is distinct from the subsequently selected contact configurations C0 - C3. I then introduce C0 in the tunnel regime. Static MAR fitting, the temperature dependence measured on the separate contact C0′, and microwave irradiation establish the superconducting energy scales and recover conventional photon assisted quasiparticle tunneling while exposing the first weak finite-transmission correction.

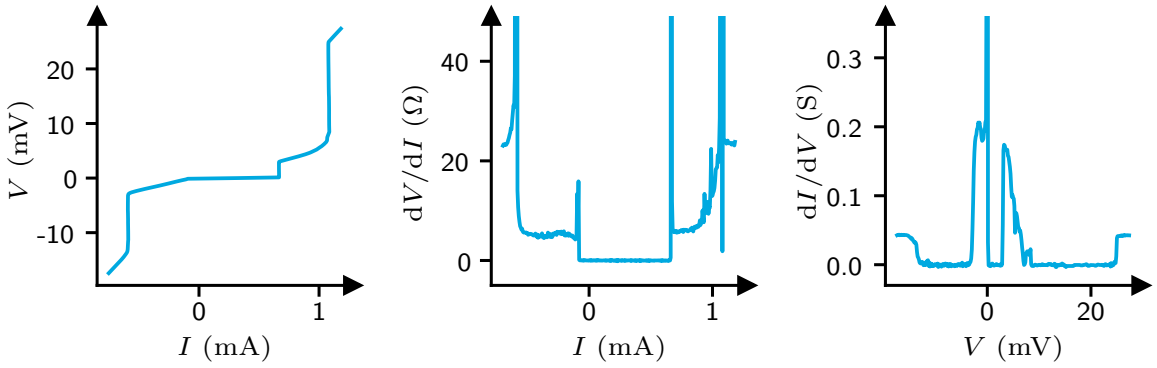
The following magnetic field study compares C0 - C3. It establishes a common empirical suppression scale near $28\,\mathrm{mT}$ and documents the configuration dependent suppression of the superconducting features. I then provide representative channel resolved MAR fits and the complete measured and reconstructed transport maps underlying the mesoscopic pincode analysis of the contact opening trace.

The final sections collect the extended microwave results. Tables summarize the static pincodes and coupling parameters of the measured configurations, while the C1 and C2 figures provide the charge order decomposition, model selection residuals, and complete measured and simulated photon assisted transport maps. The appendix closes with the full measured and phenomenological maps of the high-conductance configuration C3. Together, these results document the parameter extraction, fit quality, model limits, and field dependence that support the more compact comparisons in the main text.

F.1 Characterization of the Unbroken Device

Before opening the break junction, I characterize the intact constriction in its unbroken state. This measurement establishes the electrical response of the device before mechanical adjustment and provides a reference for the subsequent contact configurations. I do not include the unbroken device in the C0 - C3 series because its low resistance and hysteretic switching response place it in a distinct transport regime. The same device is denoted sample S2b in the detailed characterization by Raif [47].

Figure F.1 shows the characteristic hysteresis of the intact constriction. On increasing the current magnitude, the device leaves the low-voltage branch at the switching current I_{sw}. On the return sweep, it reenters this branch at the lower



(a) *V–I* (b) *dV–dI* (c) *dI–dV*

Figure F.1. Zero-field transport of the unbroken device. The *V–I* characteristic exhibits hysteretic switching from the low-voltage branch at $I_{\mathrm{sw}} = 1.075(5)\,\mathrm{mA}$ and retrapping at $I_{\mathrm{rt}} = 0.595(5)\,\mathrm{mA}$. The corresponding *dV–dI* and *dI–dV* traces emphasize the switching transitions and the finite slope of the low-voltage branch.

retrapping current I_{rt}. Linear fits to the low-voltage and normal branches give the resistances R_{s} and R_{N}, respectively. The extracted zero-field parameters are

$$I_{\mathrm{sw}} = 1.075(5)\,\mathrm{mA}\,, \qquad I_{\mathrm{rt}} = 0.595(5)\,\mathrm{mA}\,, \tag{F.1}$$

$$R_{\mathrm{s}} = 0.3387(2)\,\Omega\,, \qquad R_{\mathrm{N}} = 23.151(2)\,\Omega\,. \tag{F.2}$$

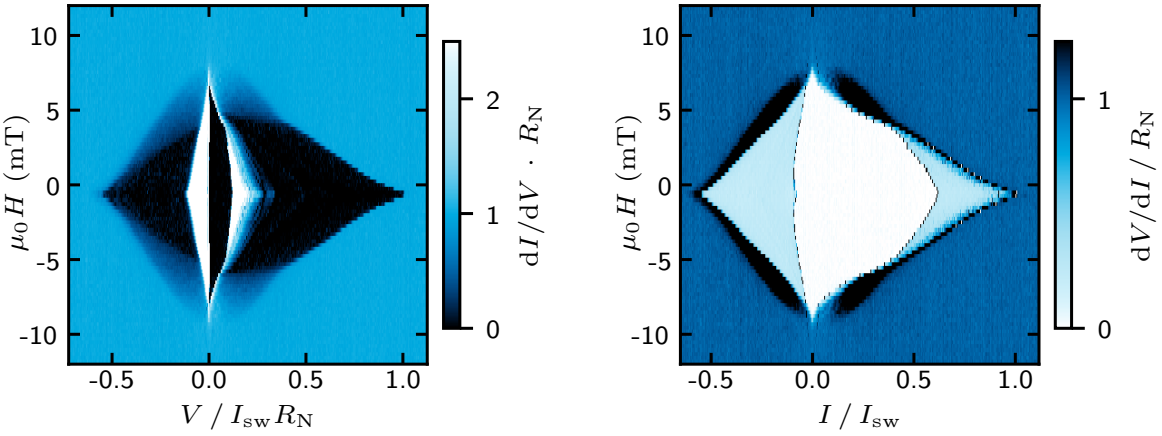
The large separation between R_{s} and R_{N} identifies a well-resolved superconducting branch. The inequality $I_{\mathrm{rt}} < I_{\mathrm{sw}}$ quantifies the hysteresis. These values characterize the complete intact constriction and its measurement environment. I do not use them as parameters for the channel resolved transport models of the opened junction.

I next apply a magnetic field perpendicular to the sample plane. Figure F.2 shows that the switching region narrows for both field polarities and is centered at a small field offset. I characterize this evolution by

$$\mu_0 H_{\mathrm{c}} = 9.9(1)\,\mathrm{mT}\,, \qquad \mu_0 H_{\mathrm{off}} = 0.5(1)\,\mathrm{mT}\,. \tag{F.3}$$

Here, $\mu_0 H_{\mathrm{c}}$ is the field scale at which the switching structure is no longer resolved and $\mu_0 H_{\mathrm{off}}$ is the center of the measured field response. I treat $\mu_0 H_{\mathrm{c}}$ as an empirical device parameter. The measurement does not distinguish orbital, geometric, and microscopic pair breaking contributions.

Figure F.3 resolves the same positive-field evolution in the complete transport traces. The low-voltage branch shortens with increasing field, while the switching and retrapping transitions move toward lower current magnitude. The complementary *dI–dV* and *dV–dI* peaks follow these boundaries and disappear as the field approaches H_{c}. The agreement between the individual traces and

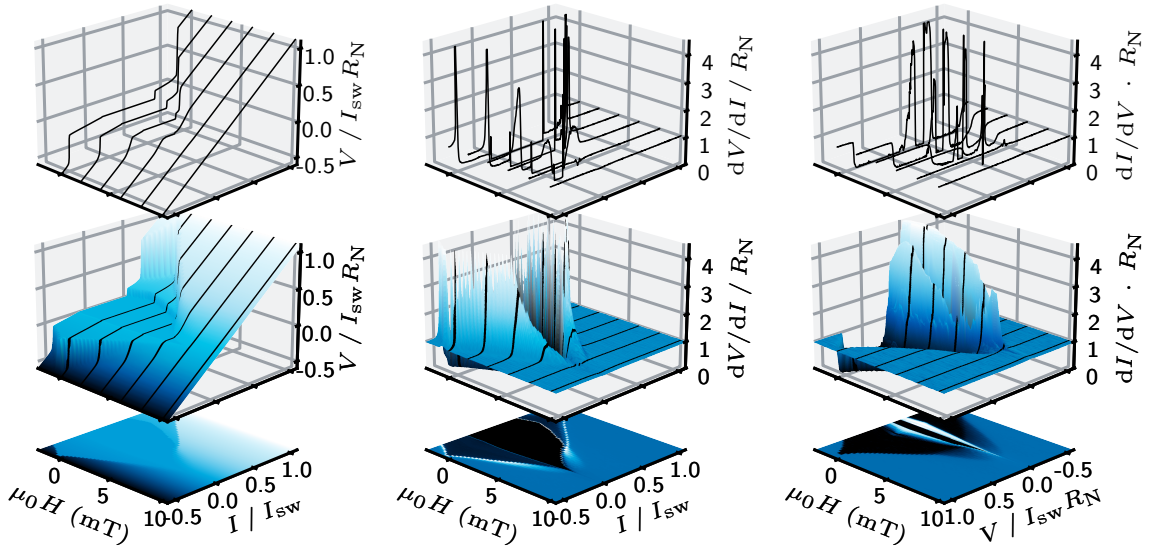


(a) *dI–dV* (b) *dV–dI*

Figure F.2. Magnetic-field dependence of the unbroken device. The normalized *dI–dV* and *dV–dI* maps show the hysteretic switching region for both field polarities. The region narrows as $|\mu_0 H - \mu_0 H_{\mathrm{off}}|$ increases and disappears at an empirical critical-field scale of $\mu_0 H_{\mathrm{c}} = 9.9(1)\,\mathrm{mT}$. The small offset $\mu_0 H_{\mathrm{off}} = 0.5(1)\,\mathrm{mT}$ describes the displacement of the response along the field axis.

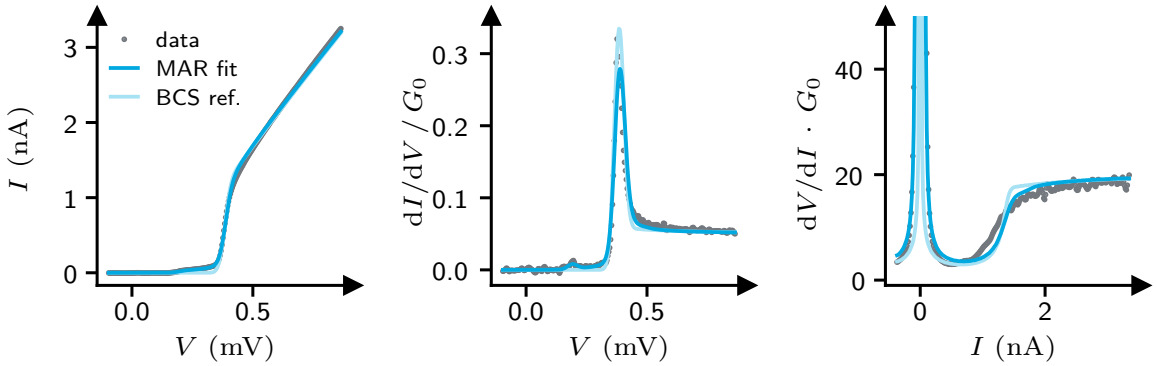
the maps in Fig. F.2 shows that the extracted field scale is not tied to a single selected sweep.

The unbroken constriction therefore behaves as a hysteretic superconducting weak link whose switching response is continuously suppressed by magnetic field. This characterization establishes the initial device state, but it does not determine the transport parameters of the junction after breaking. I therefore treat C0 - C3 as separate configurations in the following sections.



(a) *V–I* (b) *dV–dI* (c) *dI–dV*

Figure F.3. Positive-field transport map of the unbroken device. Each panel shows the measured traces as waterfall cuts, a continuous surface, and a projected heat map. The *V–I* map shows that the low-voltage branch and its hysteretic switching interval contract as the perpendicular field increases. The *dV–dI* and *dI–dV* peaks trace the same switching boundaries and merge into the normal-state response near $H/H_c = 1$. Current, voltage, field, differential resistance, and differential conductance are normalized by I_{sw}, $I_{sw}R_N$, H_c, R_N, and $1/R_N$, respectively.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.4. Static transport of C0 in the tunnel regime. The multichannel MAR fit reproduces the measured *I–V*, *dI–dV*, and *dV–dI* characteristics using the transmission eigenvalues in Eq. (F.4). The first-order BCS reference captures the dominant quasiparticle response but does not describe the weak $m = 2$ MAR onset at $|eV| = \Delta_0$.

F.2 Characterization of the Tunnel Regime

I next open the break junction and select a low-conductance configuration denoted C0. This adjustment changes the device from the low-resistance, hysteretic weak link characterized above into a tunnel regime junction. The static MAR fit shown in Fig. F.4 gives

$$\{\tau_i\} = \{0.045, 0.001, 0.001, 0.001, 0.001, 0.001\}\,, \quad G_\mathrm{N} = 0.050\,G_0\,. \tag{F.4}$$

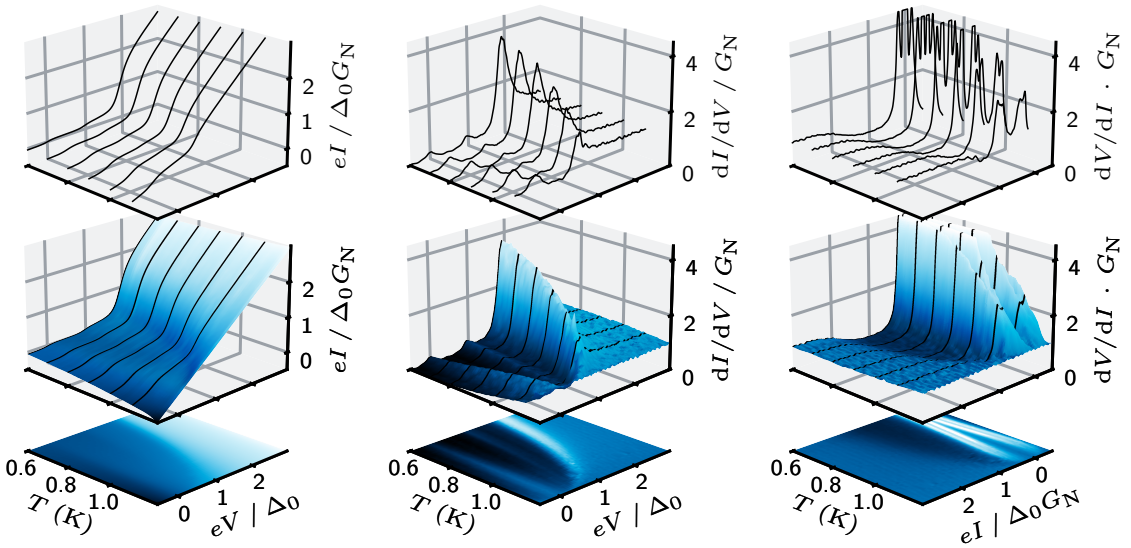
Transport is therefore dominated by one weakly transmitting channel, and the first-order quasiparticle current provides the minimal description. C0 serves as a tunnel regime reference between the oxide tunnel junctions discussed earlier in Sec. 4.1 and the finite-transmission contacts analyzed in Sec. 4.2.2.

The temperature sweep in Fig. F.5 tests whether the dominant spectral structure follows the superconducting electrodes. At low temperature, the differential conductance is suppressed inside the gap and shows a pronounced quasiparticle onset near 2Δ. Increasing the temperature shifts spectral weight toward lower bias and weakens the onset as superconductivity is suppressed. The temperature dependence of C0′ and the magnetic field characterization of C0 in Sec. F.3 give

$$\Delta_0 = 192.0(5)\,\mu\mathrm{eV}\,, \qquad T_\mathrm{c} = 1.21(1)\,\mathrm{K}\,, \qquad \mu_0 H_\mathrm{c} = 28(1)\,\mathrm{mT}\,. \tag{F.5}$$

Using the extracted gap for the weak-coupling relation gives $T_\mathrm{c} = 1.24\,\mathrm{K}$. This value is close to the measured $1.21(1)\,\mathrm{K}$ and confirms that the temperature evolution of C0′ is consistent with the extracted superconducting gap.

I then irradiate C0 through the stripline at $\nu = 15.0\,\mathrm{GHz}$. Figure F.6 shows replicas of the quasiparticle onset whose spacing and redistribution with drive



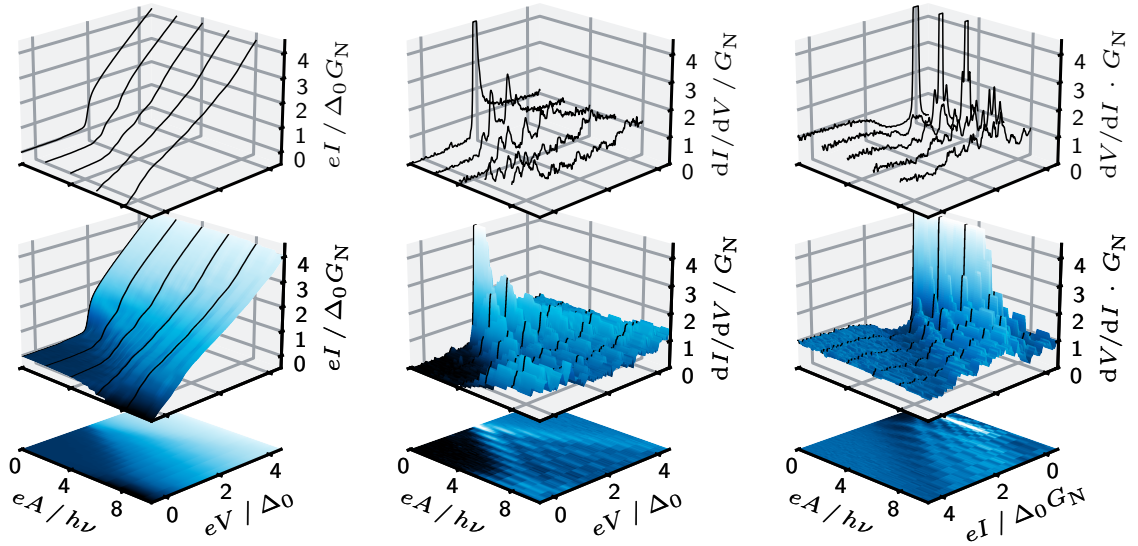
(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.5. Temperature dependence of the separate tunnel regime contact C0′. This contact has a slightly higher conductance than C0, as listed in Table F.1. Thermal broadening progressively fills the subgap region and smears the quasiparticle onset as T approaches T_c. The three transport representations retain the complete evolution from the gapped low-temperature characteristic toward the normal-state response.

amplitude follow Tien-Gordon photon assisted tunneling. I calibrate the effective junction amplitude with the same procedure used earlier in Sec. 4.1.

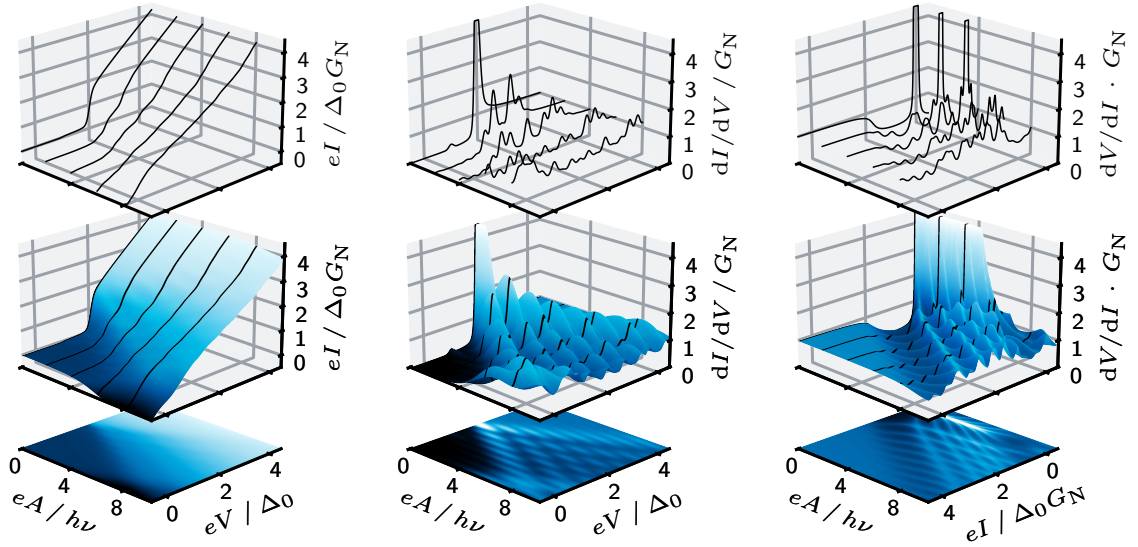
PAT simulations reproduce the principal quasiparticle sidebands and their transfer of spectral weight over the measured amplitude range. A weak measured subgap structure near the $n = 2$ MAR threshold is absent from the first-order tunnel calculation. This deviation is consistent with the onset of Andreev reflection from the leading channel. I do not resolve a separate photon assisted MAR replica pattern in this contact, so the data do not justify a quantitative PAMAR analysis. The comparison consequently establishes the useful scope of the tunnel reference. Tien-Gordon dressing describes the dominant irradiated quasiparticle transport, while the remaining subgap signal marks the beginning of finite-transmission corrections.

The tunnel regime measurements therefore provide a control for the later finite-transmission analysis. They verify the superconducting energy scales through independent field and temperature sweeps, and they recover conventional photon assisted quasiparticle tunneling under irradiation. At the same time, the weak subgap deviation shows why the tunnel model must be replaced by a channel resolved MAR description as the contact transmission increases.



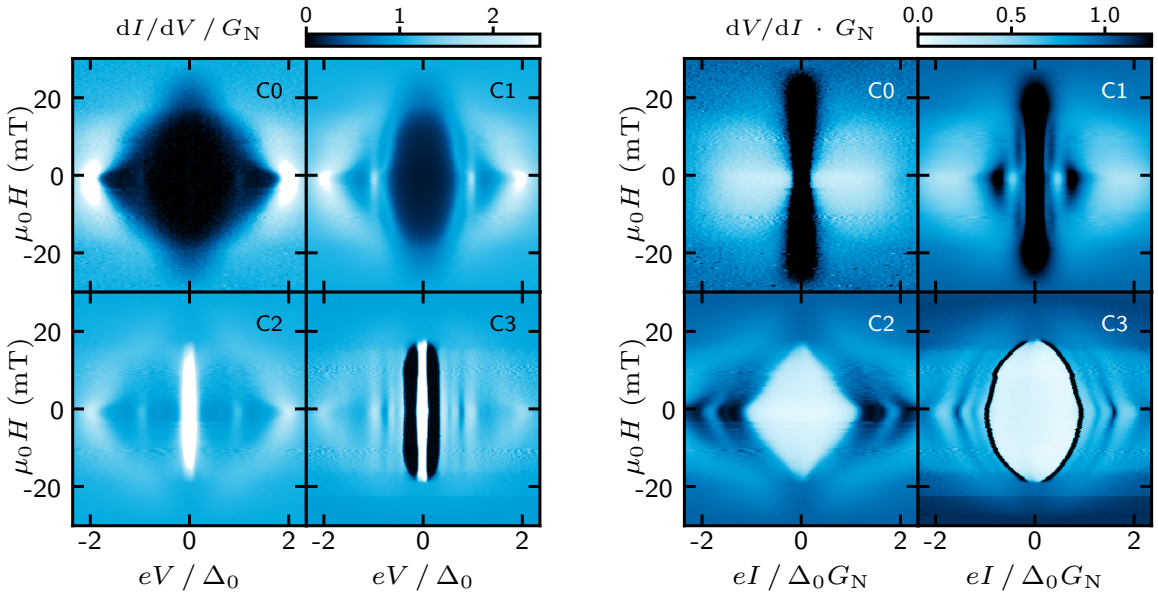
(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.6. Calibrated microwave amplitude map of C0 under stripline irradiation at 15.0 GHz. Photon assisted replicas emerge from the quasiparticle onset and redistribute spectral weight as $eA/h\nu$ increases. A weaker subgap feature near $|eV| = \Delta$ indicates the onset of Andreev reflection.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.7. Simulated microwave amplitude map of C0 under stripline irradiation at 15.0 GHz according to PAT. The calculation reproduces the dominant quasiparticle sidebands in Fig. F.6, but it does not contain the weak Andreev contribution visible in the measured subgap response.



(a) *dI–dV* (b) *dV–dI*

Figure F.8. Magnetic field dependent transport of C0 - C3 for a field perpendicular to the sample plane. The maps show the normalized *dI–dV* as a function of eV/Δ_0 and the normalized *dV–dI* as a function of $eI/(\Delta_0 G_\mathrm{N})$. Each panel is labeled by its contact configuration, and both field polarities are shown.

F.3 Magnetic Field Dependent Characterization

I measure C0 - C3 as functions of magnetic field perpendicular to the sample plane. Figure F.8 compares the resulting *dI–dV* and *dV–dI* maps. The center of the superconducting response occurs at a small configuration dependent offset $\mu_0 H_\mathrm{off} = 0.5\text{-}1.5\,\mathrm{mT}$. I therefore characterize the suppression relative to the effective field $H - H_\mathrm{off}$. I attribute this small offset operationally to the residual field of the magnet and cryostat, without an independent calibration of its origin.

Starting from H_off, the superconducting transport features weaken with increasing field magnitude. In C0, the quasiparticle onset moves toward zero voltage and the subgap region fills as the gap closes. In C1 - C3, the supercurrent contribution and subharmonic gap structure are progressively suppressed. This evolution is consistent with earlier measurements on aluminum atomic contacts, in which a perpendicular magnetic field acts as a pair breaking perturbation, rounds the MAR current steps, suppresses the excess current, and shifts the conductance maxima to lower voltage before the normal-state response is recovered [116]. Although the visible structures depend on the contact regime, they disappear at a relatively consistent field scale in both *dI–dV* and *dV–dI*. I de-

fine the empirical critical field as the field at which no resolved superconducting transport features remain. This criterion gives the common value

$$\mu_0 H_\mathrm{c} = 28.0(5)\,\mathrm{mT}\,. \tag{F.6}$$

The uncertainty is given by the field resolution.

Particularly noisy regions occur around $\mu_0 H \approx \pm 10\,\mathrm{mT}$ in the *I–V* characteristics. The noise likely originates from internal processes in the magnet power supply or its controller.

A distinct abrupt change occurs in C3 for $\mu_0 H \lesssim -20\,\mathrm{mT}$. In contrast to the continuously suppressed contribution, this transition changes the background conductance after the superconducting features have largely disappeared. I therefore attribute it tentatively to a reformation of the atomic contact. The data recorded before and after this event do not necessarily represent the same microscopic contact configuration. I consequently determine the superconducting suppression field from the continuous evolution outside this reformation event and do not interpret the conductance jump as superconducting hysteresis.

In addition to the abrupt conductance change in C3 discussed above, the zero-voltage feature and the subgap structure (SGS) appear to be suppressed at different field scales. The SGS disappears first, reducing the overall subgap conductance, while the supercurrent persists to a higher field.

In C2 and C3, the MAR structure remains visible up to comparatively large fields in *dI–dV*. This apparent robustness does not imply that the electrodes remain unaffected by the field. The corresponding *dV–dI* maps show that the low-resistance current region contracts continuously from much lower fields. The field therefore modifies the current-carrying state before the MAR features become unresolvable. This gradual evolution is consistent with flux penetration and orbital pair breaking in dirty thin-film aluminum. Such films can enter the type-II regime even though bulk aluminum is type I [136]. The present transport maps neither resolve individual vortices nor determine the Ginzburg–Landau parameter independently. They are therefore consistent with, but do not establish, a type-II mixed state in the electrodes.

The supercurrent contribution exhibits a pronounced difference between opposite bias sweep directions. This behavior is consistent with slightly different switching and retrapping currents, $I_\mathrm{sw} > I_\mathrm{rt}$. It may indicate weakly hysteretic junction dynamics, but it does not by itself establish a departure from the overdamped limit.

The similar suppression fields are consistent with a dominant field scale set by the common aluminum electrodes rather than by the individual channel transmissions. This conclusion remains empirical because I do not model the electrode geometry, orbital depairing, or microscopic pair breaking. The value in Eq. (F.6) therefore denotes the disappearance of resolved superconducting transport features and should not be identified directly with an intrinsic bulk critical field.

F.4 The Mesoscopic Pincode

This section collects representative channel resolved MAR fits and the complete measured and reconstructed transport maps supporting the mesoscopic pincode analysis in Sec. 4.2.1. Figure F.11 illustrates the fit quality for selected contact states, while Figs. F.9 and F.10 provide the global comparison across the complete opening trace.

F.5 Photon Assisted Transport

This section documents the full set of atomic contacts included in the photon assisted transport analysis and provides the detailed model decompositions and transport maps supporting Sec. 4.2.2. Tables F.1 and F.2 summarize the static MAR parameters and microwave coupling values used across this set, including C0, C0′, and C3 from Secs. F.2 and 4.2.3.

As in the mesoscopic pincode analysis, I assume the following minimum temperature and Dynes parameter:

$$T = 0\,\mathrm{K} \qquad\qquad \gamma = 1\,\mathrm{neV}. \tag{F.7}$$

For the pincodes of the irradiated junctions, the gap and voltage noise parameters remain free and cluster around

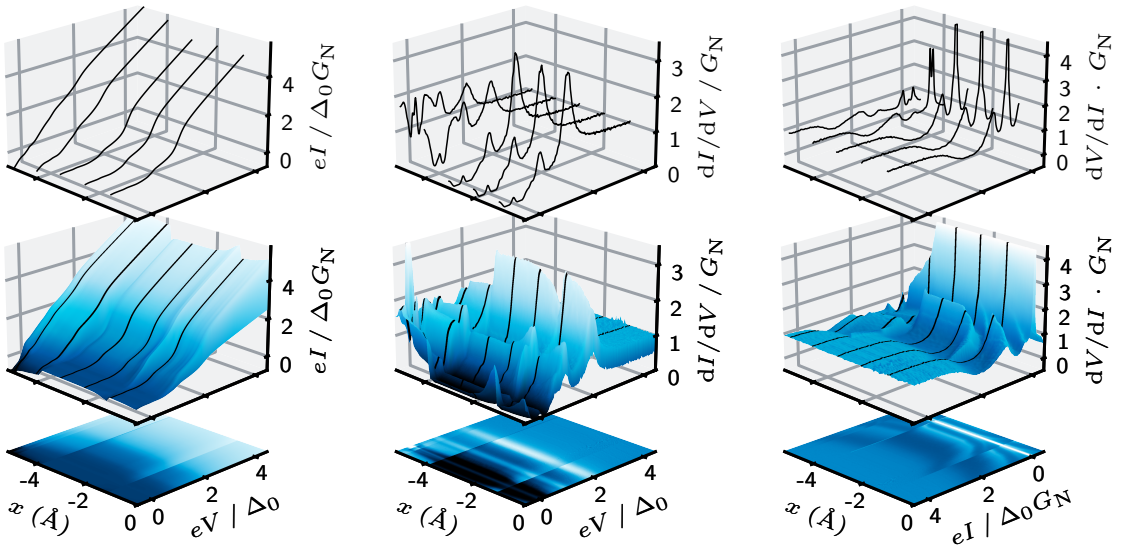
$$\Delta_0 = 190.4(8)\,\mu\mathrm{eV}\,, \qquad\qquad \sigma_V = 10(4)\,\mu\mathrm{V}\,. \tag{F.8}$$

The compact figures in the main text establish the model choices using two representative contacts. Figure F.12 resolves the threshold weighted PAMAR contributions at fixed amplitudes, while Fig. F.13 provides the complete model selection test by comparing the four combinations of the two MAR constructions and the two Cooper pair responses. The remaining figures retain the complete voltage, amplitude, and representation dependence of the data.

The measured and simulated maps for C1 are shown in Figs. F.14 and F.15. Figures F.16 and F.17 provide the corresponding PAMAR-only validation for C2 and expose the different low-voltage amplitude evolution discussed in the main text.

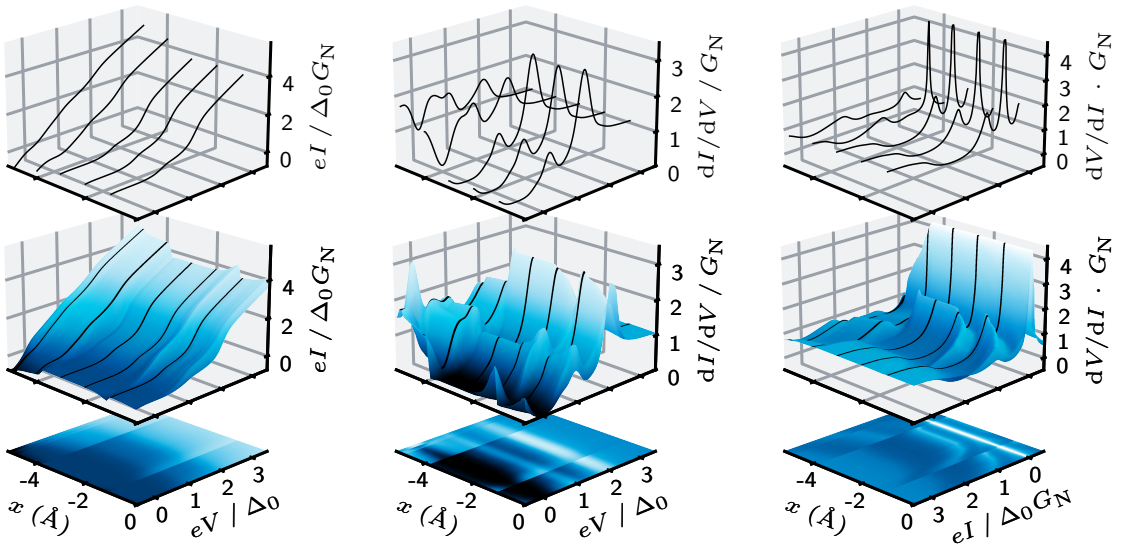
F.6 Subharmonic Transport at High Conductance

Figures F.18 and F.19 show the full transport maps for C3. I discuss these data in Sec. 4.2.3.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.9. Measured transport map during contact opening. The *I–V*, *dI–dV*, and *dV–dI* representations show the evolving MAR structure as a function of the approximate relative displacement x. Abrupt spectral changes separate successive contact configurations.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

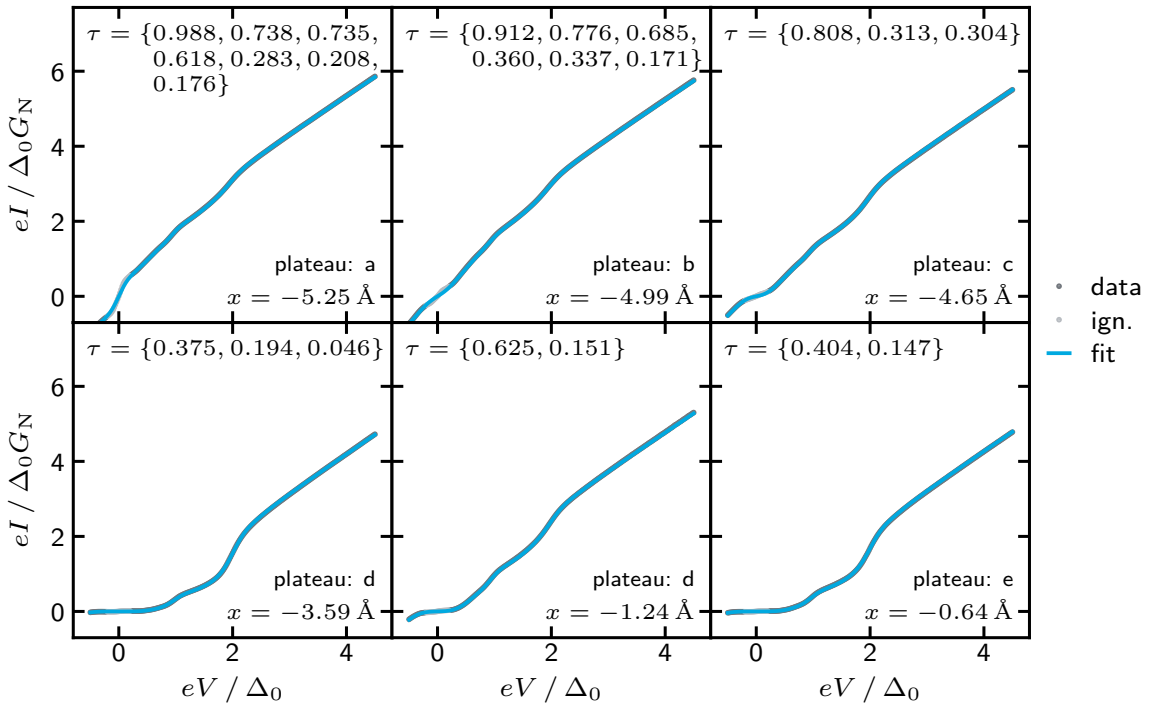
Figure F.10. Transport map reconstructed from the fitted pincodes. The calculation uses the position dependent transmissions, $T = \gamma = 0$, and the common parameters in Eq. (4.14). It reproduces the principal displacement dependent structures of Fig. F.9.

Table F.1. Static MAR parameters and mesoscopic pincodes of the atomic contacts included in the photon assisted transport analysis.

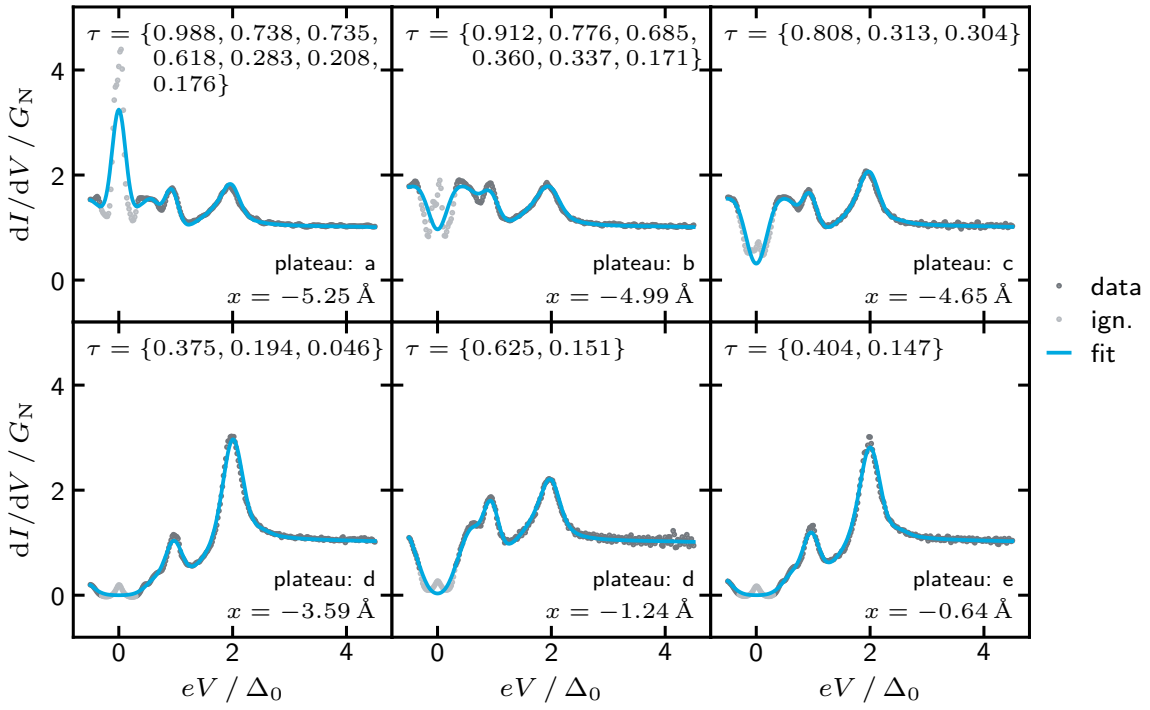
C	$G_{\mathrm{N}}\,(G_0)$	$\Delta_0\,(\mu\mathrm{eV})$	$\sigma_V\,(\mu\mathrm{V})$	$\{\tau_i\}$
C0	0.050	192.0	19.5	{0.045, 0.001, 0.001, 0.001, 0.001, 0.001}
C0′	0.133	189.0	25.5	{0.118, 0.008, 0.004, 0.001, 0.001, 0.001}
	0.794	190.5	12.5	{0.682, 0.112}
	0.926	190.5	9.0	{0.767, 0.159}
C1	1.757	189.5	8.0	{0.517, 0.399, 0.332, 0.305, 0.204}
C2	1.987	191.0	11.5	{0.997, 0.440, 0.184, 0.184, 0.182}
	1.992	190.5	10.5	{0.780, 0.672, 0.266, 0.263, 0.011}
	2.035	189.5	4.5	{0.988, 0.655, 0.272, 0.120}
	2.819	190.5	8.0	{0.834, 0.747, 0.444, 0.383, 0.269, 0.142}
	2.822	190.5	8.0	{0.846, 0.735, 0.504, 0.289, 0.284, 0.164}

Table F.2. Microwave coupling parameters of the atomic contact measurements included in the photon assisted transport analysis.

C	$G_{\mathrm{N}}\,(G_0)$	$\nu\,(\mathrm{GHz})$	$\eta_{\mathrm{antenna}}\ (10^{-3})$	$\eta_{\mathrm{stripline}}\ (10^{-3})$
C0	0.050	15.0	-	5.803(1)
	0.794	7.8	9.680(19)	8.403(9)
		15.0	6.556(13)	7.085(13)
		19.3	5.677(15)	7.763(18)
	0.926	15.0	-	7.785(12)
C1	1.757	15.0	6.643(17)	6.179(16)
C2	1.987	7.8	-	6.079(7)
		15.0	-	5.230(12)
		19.3	-	5.421(15)
	1.992	7.8	10.580(10)	7.120(7)
		15.0	6.465(11)	6.939(11)
		19.3	5.758(19)	7.521(22)
	2.035	7.8	9.188(11)	6.148(6)
		15.0	5.146(7)	5.293(6)
		19.3	3.633(16)	4.924(14)
	2.819	7.8	-	7.645(11)
		15.0	-	7.345(16)
		19.3	-	7.472(17)
	2.822	7.8	-	5.409(3)
		15.0	-	5.081(6)
		19.3	-	5.100(7)
C3	29.00	7.8	8.108(4)	6.566(3)
		15.0	4.365(3)	4.384(3)
		19.3	4.200(9)	5.682(10)



(a) *I–V*



(b) *dI–dV*

Figure F.11. Representative channel resolved MAR fits during contact opening. The measured *I–V* characteristics and corresponding *dI–dV* spectra are compared with the best-fit multichannel model. Each trace is labeled by its relative displacement x and extracted mesoscopic pincode $\{\tau_i\}$. The derivative spectra are obtained from the same *I–V* data and are not independent fit observables.

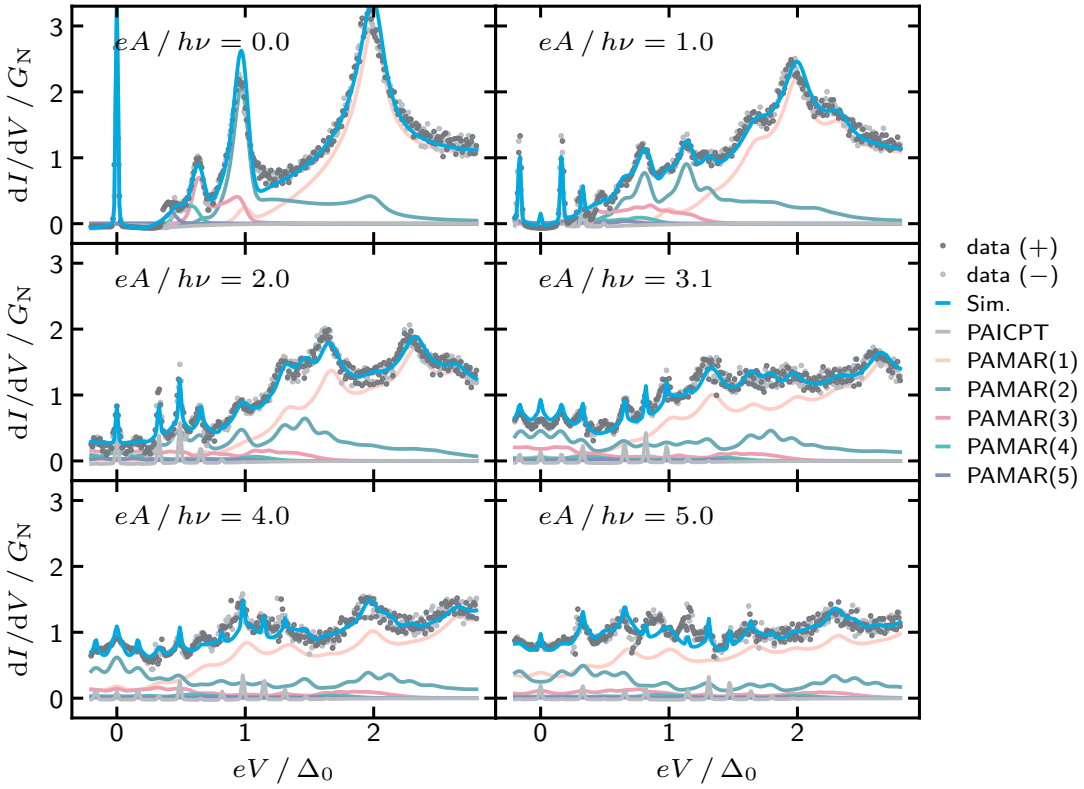
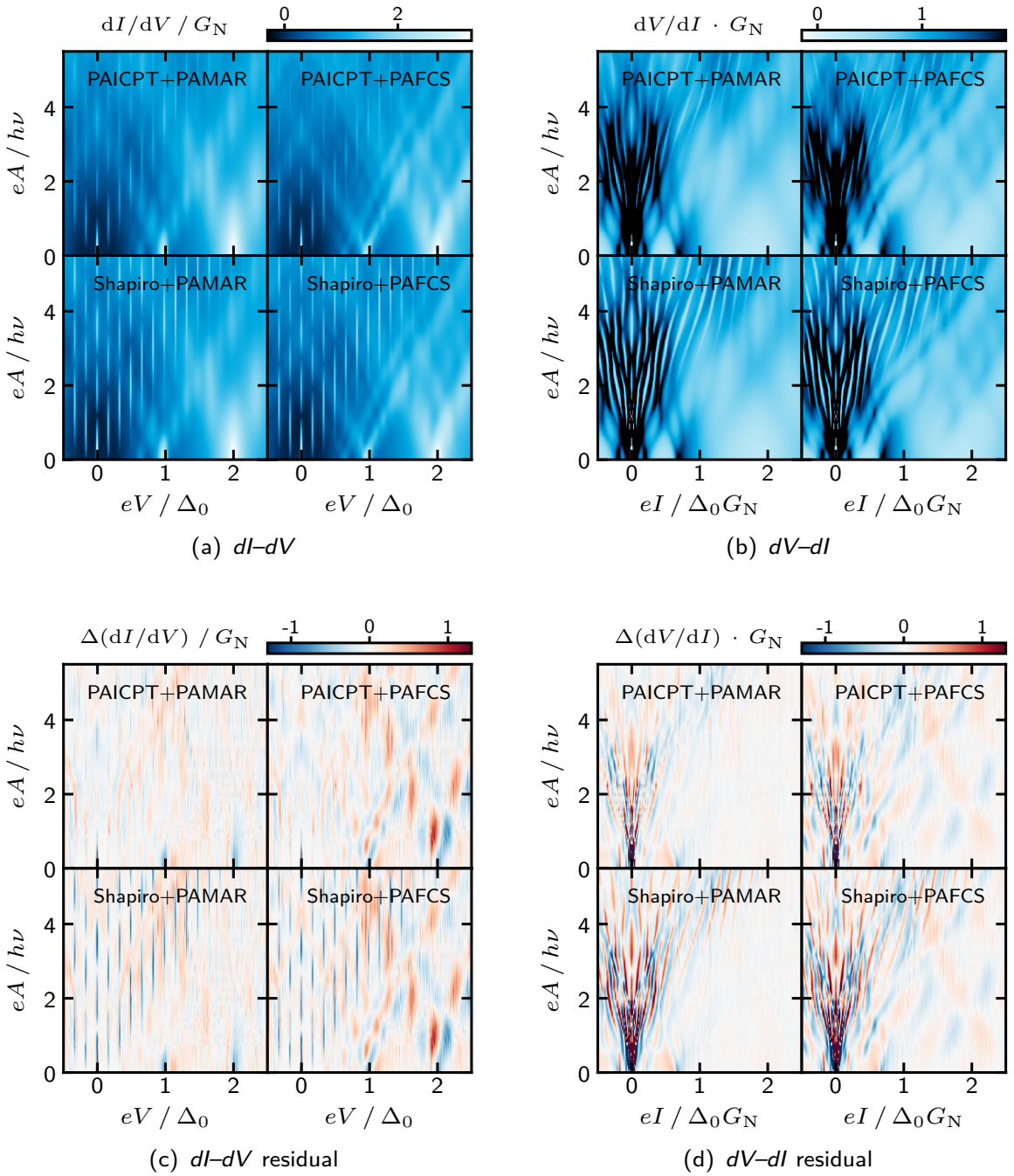


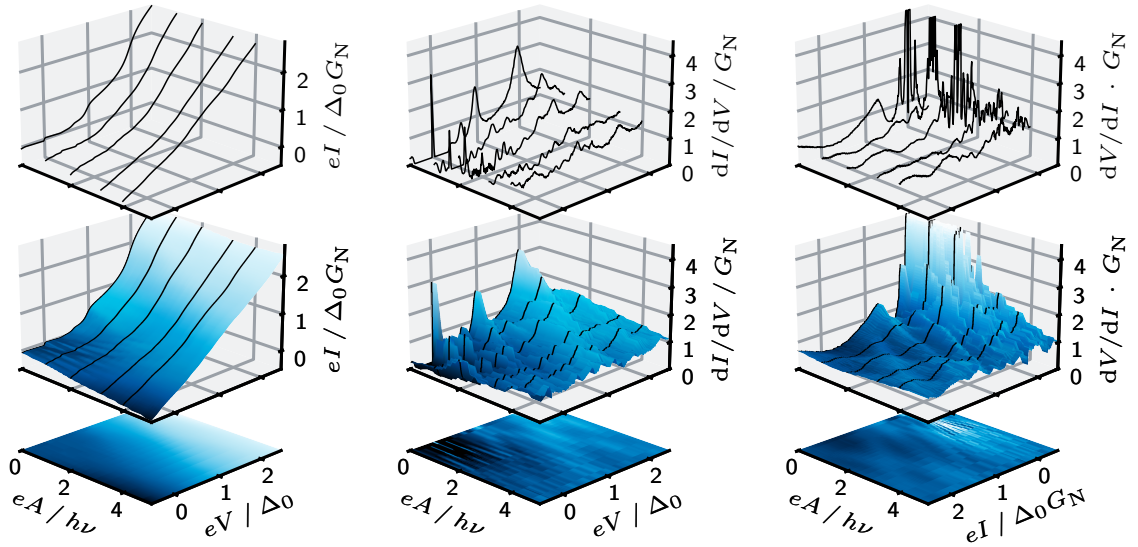
Figure F.12. Charge order decomposition of voltage cuts through the microwave driven *dI–dV* of C1 at 15.0 GHz. The six panels show increasing reduced drive amplitude from $eA/h\nu = 0$ to 5. The measured positive- and negative-bias branches are compared with the composite simulation, its PAICPT contribution, the summed threshold weighted PAMAR response, and the individual PAMAR components $m = 1$ to 5. The dominant charge order shifts toward larger m as the relevant subgap threshold approaches zero bias.



(a) *dI–dV* (b) *dV–dI*

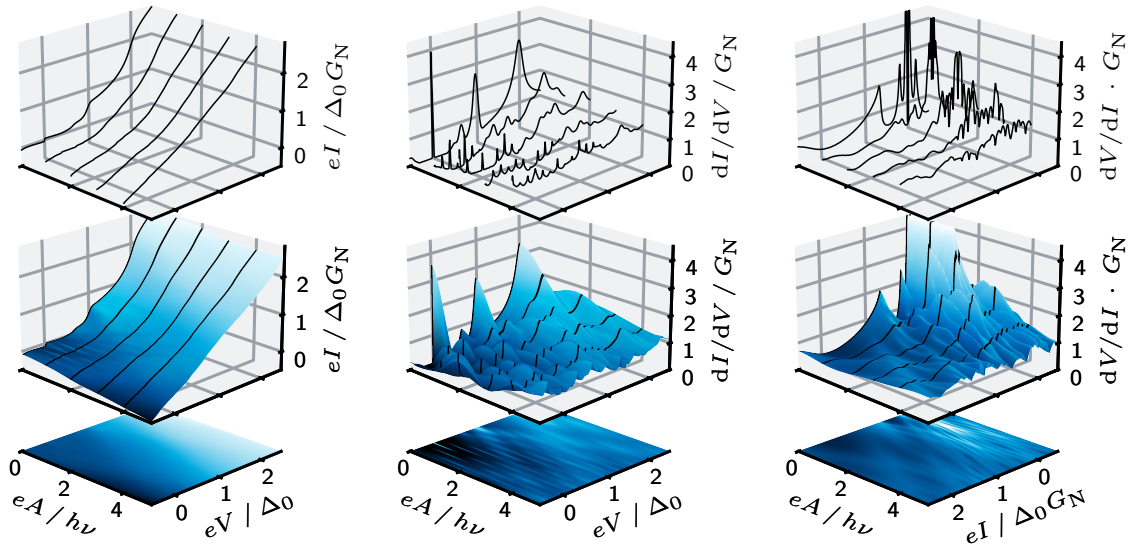
(c) *dI–dV* residual (d) *dV–dI* residual

Figure F.13. Comparison of four photon assisted models for C1 at 15.0 GHz. The calculated (a) *dI–dV* and (b) *dV–dI* maps combine either PAICPT or an ideal coherent Shapiro response with either threshold weighted PAMAR or direct dressing of the FCS components. Panels (c) and (d) show the corresponding residuals relative to the measurement. The combination of PAICPT and threshold weighted PAMAR gives the least structured residual in both representations. Direct FCS dressing leaves pronounced deviations along the MAR sidebands, while the coherent Shapiro model does not reproduce the amplitude evolution of the replicas originating from zero bias. All four calculations use the same calibrated drive amplitude and static contact parameters.



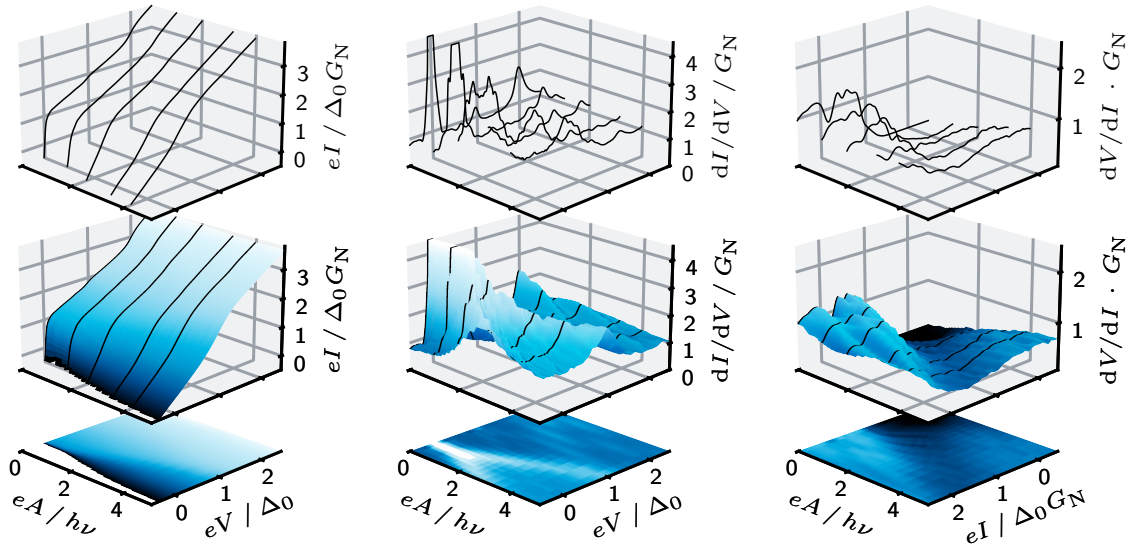
(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.14. Calibrated microwave amplitude map of C1 under stripline irradiation at 15.0 GHz. The *I–V*, *dI–dV*, and *dV–dI* columns display the same measurement as heat maps, surfaces, and representative waterfall cuts. Photon assisted replicas emerge from the quasiparticle and MAR thresholds, while narrow features originating from the zero-bias peak disperse across the reduced amplitude axis $eA/h\nu$.



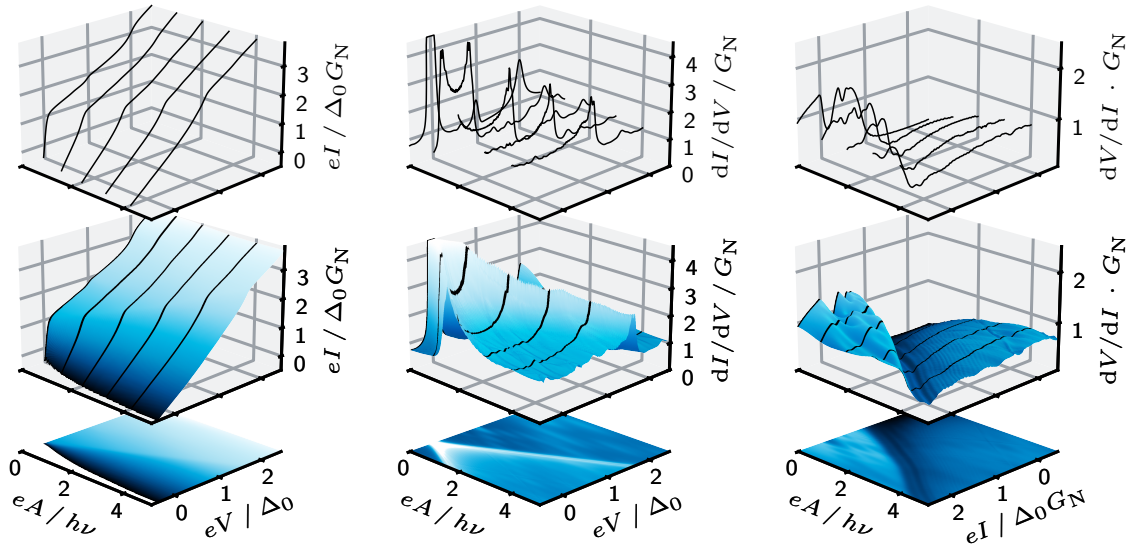
(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.15. Simulated microwave amplitude map of C1 at 15.0 GHz. The calculation combines threshold weighted PAMAR with PAICPT using the calibrated drive amplitude and the static transport parameters of C1. The three representations reproduce the principal sideband trajectories and spectral weight redistribution observed in Fig. F.14, while smaller line shape differences remain at high drive.



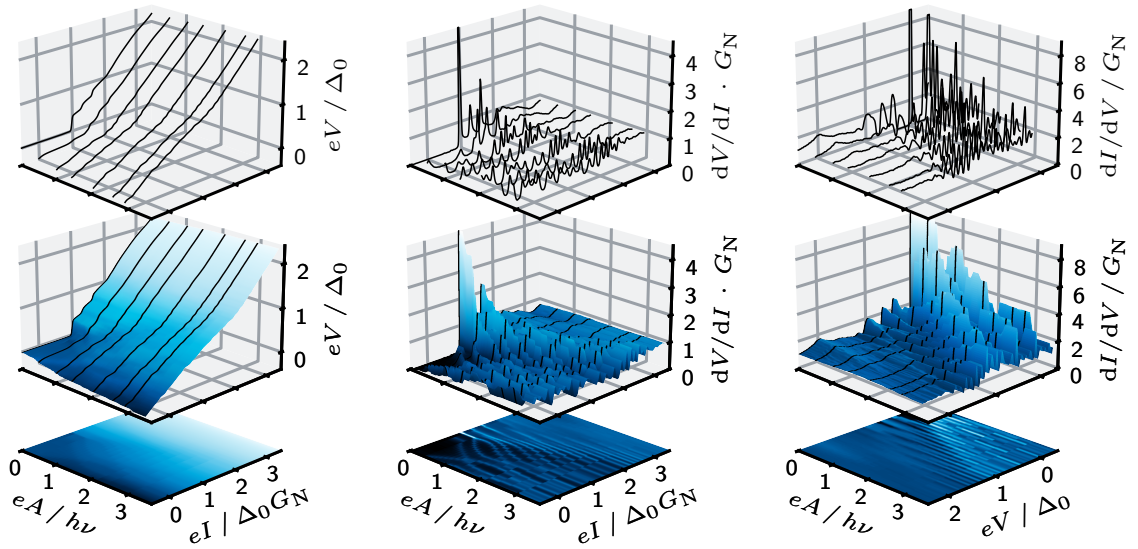
(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.16. Calibrated microwave amplitude map of C2 under stripline irradiation at 15.0 GHz. The *I–V*, *dI–dV*, and *dV–dI* representations show the strong subgap response of the nearly ballistic leading channel. The measured low-voltage features exhibit an amplitude dependence that is not fully captured by the PAMAR-only simulation in Fig. F.17.



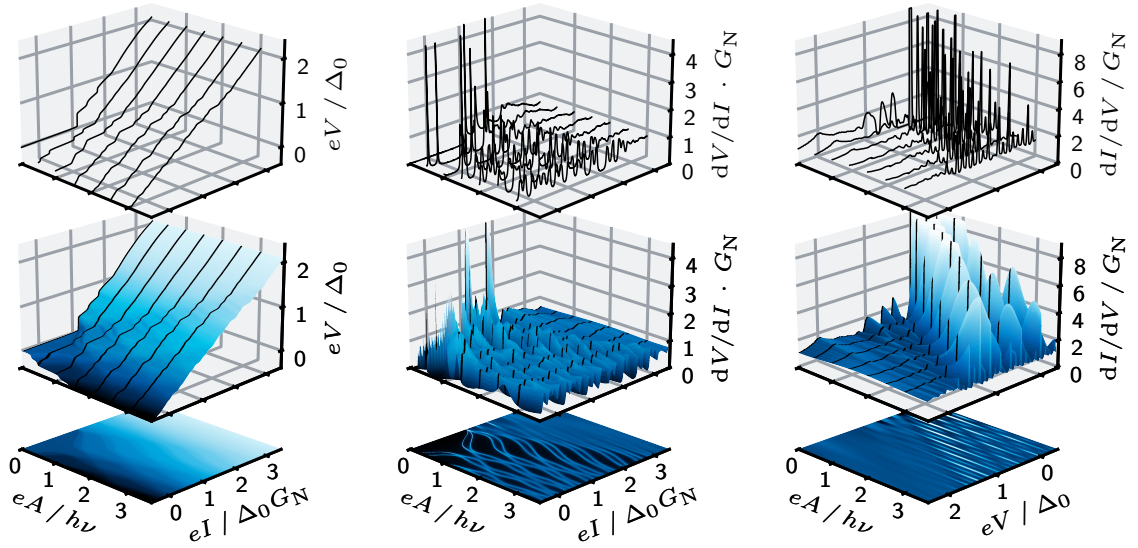
(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.17. PAMAR-only microwave amplitude map of C2 at 15.0 GHz. The calculation uses the calibrated drive amplitude and the simulated static MAR current calculated with the pincode in Eq. (4.25). It reproduces the dominant quasiparticle and MAR structure over the full bias and amplitude range but assigns too much of the zero-voltage response to MAR and therefore misses the measured low-voltage amplitude evolution in Fig. F.16.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.18. Calibrated microwave amplitude map of C3 under antenna irradiation at 19.3 GHz. The *I–V*, *dI–dV*, and *dV–dI* columns show the measured response as waterfall cuts, surfaces, and heat maps. Increasing the drive produces replicas of the quasiparticle branch and the static subgap structure, together with additional closely spaced features originating near zero voltage.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure F.19. Phenomenological microwave amplitude map of C3 at 19.3 GHz. The calculation reproduces the principal replica trajectories and their amplitude evolution, while differences remain in the sharpness and spectral weight of the low-voltage features observed in Fig. F.18.

G. SSN Hybrid Devices

This chapter presents transport measurements on two hybrid devices in an SSN configuration. Here, SSN denotes the sequence of a superconducting lead, a superconducting island, and a normal conducting lead. Each device combines a fixed Al–AlO_x–Cu tunnel barrier with an Al mechanically controllable break junction. Opening the break junction creates the second junction and geometrically defines the superconducting island between the two contacts. Although SET1 and SET2 share the same nominal design and are fabricated in the same production batch, they develop distinct transport regimes after opening. SET1 forms a gate controlled single-electron transistor, whereas SET2 retains a high-conductance contact and exhibits a gate independent zero-bias anomaly.

I fabricate and measure SET1, shown in Fig. G.1, and SET2. Both devices come from the same production batch and use the same nominal design and process parameters. The Al layer is 40 nm thick, and the Cu top layer is approximately 80 nm thick. I perform the oxidation at 4.9(1) mbar for 2:30 min. The SSN architecture and the specific design used for these devices originate from the preceding work of Laura Sobral Rey [88], who introduces me to the design at the beginning of this work. I use this design without adaptation. The fabrication principles and general design considerations are described in Sec. 3.1.

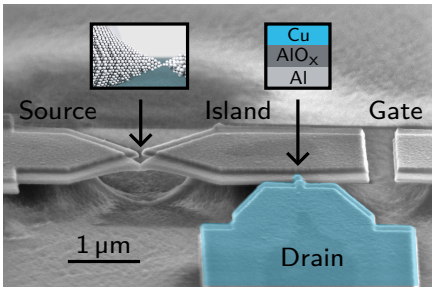


Figure G.1. SEM image of SET1. The overlays identify the break junction, island, fixed tunnel barrier, and gate electrodes.

Each device provides only one mechanically formed contact configuration suitable for the measurements reported here and fails afterward. Repeated contact configurations and systematic comparisons across contact parameters are therefore not available. The comparison establishes two experimentally observed outcomes of the nominal design, but it does not determine their statistical prevalence or reproducibility across devices.

I perform these measurements on the Scheer II setup, which had been developed and operated in the preceding work of Thomas Lorenz, Susanne Sprenger, and Laura Sobral Rey. Detailed descriptions of the platform are given in their theses [85, 86, 88]. Laura Sobral Rey introduces me to the operation of the cryostat, after which I operate the setup independently for these measurements. Before the measurement campaign presented here, I repair the cryogenic system with valuable technical input from Thomas Lorenz.

I perform the complete numerical evaluation and model fitting presented here. This includes the SIN reference fits, the orthodox normal-state SET fit, the

analysis of its structured residuals, and the dynamical Coulomb blockade fit. The data treatment follows the procedure described in Sec. 3.3 and uses the Python based analysis pipeline that I develop. The physical interpretation benefits from discussions with Lukas Kammermeier.

Unless stated otherwise, the parenthetical uncertainties of directly fitted parameters are standard errors obtained from the least squares covariance matrix. They quantify the local statistical uncertainty within the respective model and do not include systematic uncertainty from calibration, contact drift, or model choice. I estimate the bath temperature uncertainties as 1 mK.

Section G.1 first establishes the intact tunnel barrier of SET1 and then examines the normal and superconducting transport after opening the break junction. In the normal-state SET, the gate–bias response reveals Coulomb diamonds, and an orthodox SET fit provides the minimal electrostatic description. I additionally report the structured residuals that remain after this fit. In the superconducting SSN SET, I report gate dependent superconducting switching as a separate experimental outcome.

Section G.2 presents SET2 as a contrasting outcome of the same device concept. After partially opening the break junction, I do not resolve any gate modulation. Instead, the normal-state response is consistent with dynamical Coulomb blockade, and the superconducting response develops two conductance regimes separated by a crossover. The analysis compares the independently extracted resistance and energy scales while keeping the microscopic origin of their correspondence open.

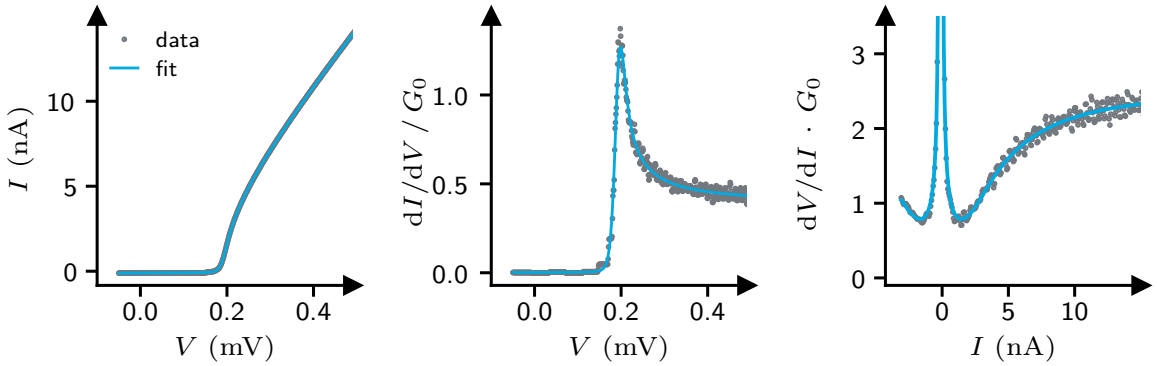
G.1 Static Coulomb Blockade

In this section, I use the term static Coulomb blockade for charging effects described by discrete island charge states and orthodox sequential tunneling, in contrast to environmental dynamical Coulomb blockade.

I first characterize SET1 with the mechanically controlled break junction in its unbroken state. The closed metallic constriction has a much lower resistance than the oxide tunnel barrier, so the measured voltage drops predominantly across the Al–AlO_x–Cu junction. The device therefore provides an SIN tunnel junction reference before I open the break junction to form the superconducting island.

In the superconducting state, the current is suppressed below the quasiparticle threshold and increases sharply near $eV = \Delta_0$, as expected for an SIN junction. I fit the complete *I*–*V* characteristic with the BCS density of states including Dynes broadening. The fit yields the normal-state resistance and conductance

$$R_{\mathrm{TB}} = 32.381(3)\,\mathrm{k\Omega}\,, \qquad G_{\mathrm{TB}} = 0.39857(4)\,G_0\,. \tag{G.1}$$



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure G.2. Static transport of SET1 before opening the break junction. The panels show the measured *I–V*, *dI–dV*, and *dV–dI* responses together with the SIN BCS Dynes fit. The fit describes the quasiparticle onset and the conductance above the gap.

The same fit gives the superconducting gap, fitted electronic temperature, and Dynes broadening

$$\Delta_0 = 196.106(9)\,\mu\mathrm{eV}\,, \qquad T_{\mathrm{fit}} = 68.1(5)\,\mathrm{mK}\,, \qquad \gamma = 0.86(3)\,\mu\mathrm{eV}\,. \tag{G.2}$$

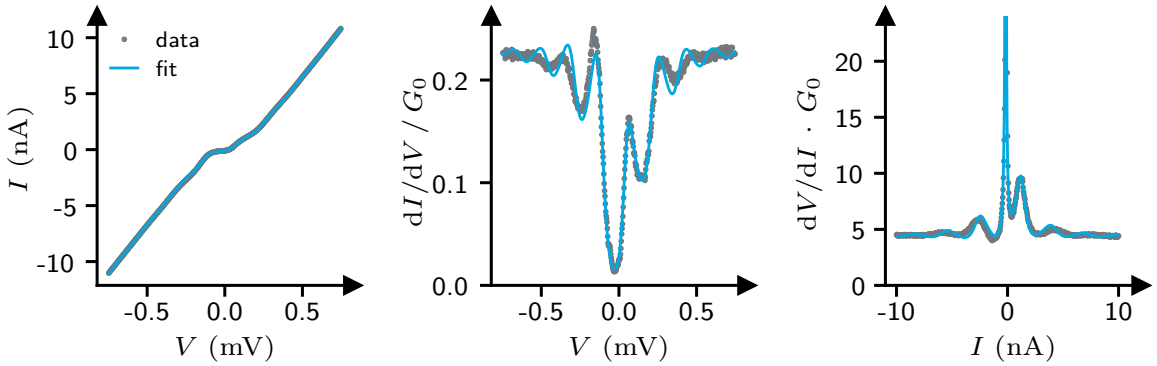
The fitted temperature is higher than the measured bath temperature, $T_{\mathrm{bath}} = 52(1)\,\mathrm{mK}$. Such a difference is expected because the bath thermometer does not directly measure the electronic temperature of the junction, and residual electrical noise or imperfect thermalization can broaden the transport response. I therefore treat T_{fit} as an effective electronic temperature.

The fit reproduces both the quasiparticle onset and the conductance peak in Fig. G.2. The extracted gap provides the superconducting energy scale used below, while R_{TB} characterizes the fixed oxide barrier before I open the break junction to convert the device into an SET.

In addition, I measure the device after suppressing superconductivity. The normal-state *I–V* characteristic shows no resolvable zero-bias anomaly. I therefore find no evidence for appreciable dynamical Coulomb blockade in the unbroken device.

Normal-State Single-Electron Transistor

Opening the break junction separates the constriction and the fixed oxide tunnel barrier by a small island. I suppress superconductivity to first characterize this three-terminal device in the normal state. A representative *I–V* curve is shown in Fig. G.3. It shows a reduced low bias conductance compared with the approximately linear response at larger bias.



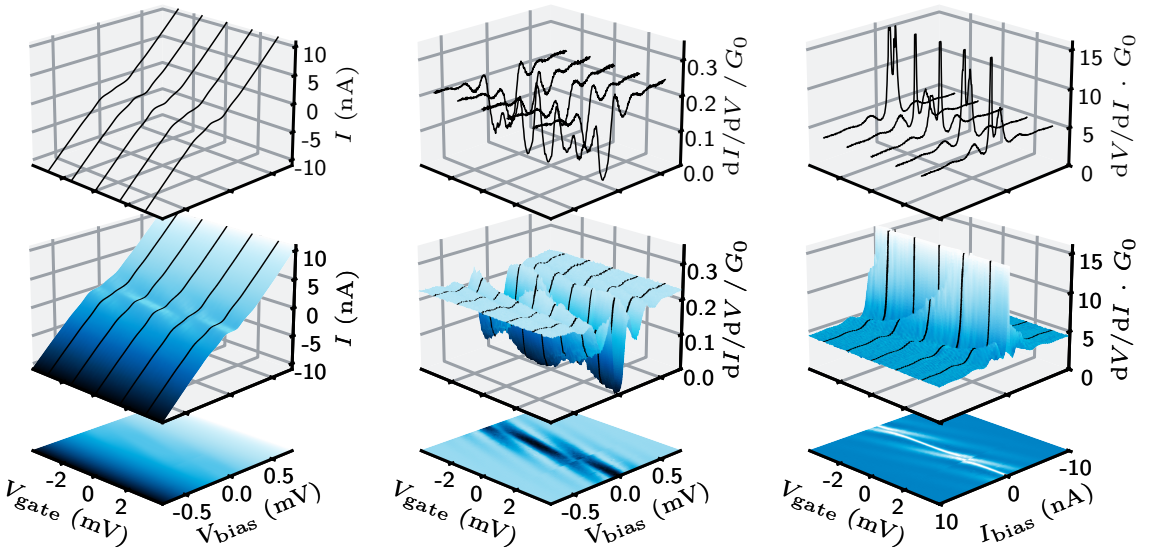
(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure G.3. Normal-state transport of SET1 after opening the break junction at $V_{\mathrm{gate}} = 0\,\mathrm{mV}$. The measured *I–V*, *dI–dV*, and *dV–dI* responses are compared with the orthodox SET fit. The suppressed low bias conductance is the local signature of Coulomb blockade.

Sweeping the gate voltage reveals that this suppression is periodic and bounded by diagonal conductance onsets, as shown in Fig. G.4. The two families of Coulomb diamond edges have different absolute slopes in the gate–bias plane, as expected for unequal junction capacitances. Above $V_{\mathrm{gate}} \approx 1.7\,\mathrm{mV}$, the measured pattern changes abruptly, as seen more clearly in Fig. G.6. This discontinuity is consistent with an atomic rearrangement of the mechanically formed break junction during the sweep, although the conductance does not provide an independent signature of such a change. The normal-state data therefore establish gate controlled Coulomb blockade and an asymmetric SET geometry before I consider superconducting transport, but the complete map does not represent a single unchanged configuration.

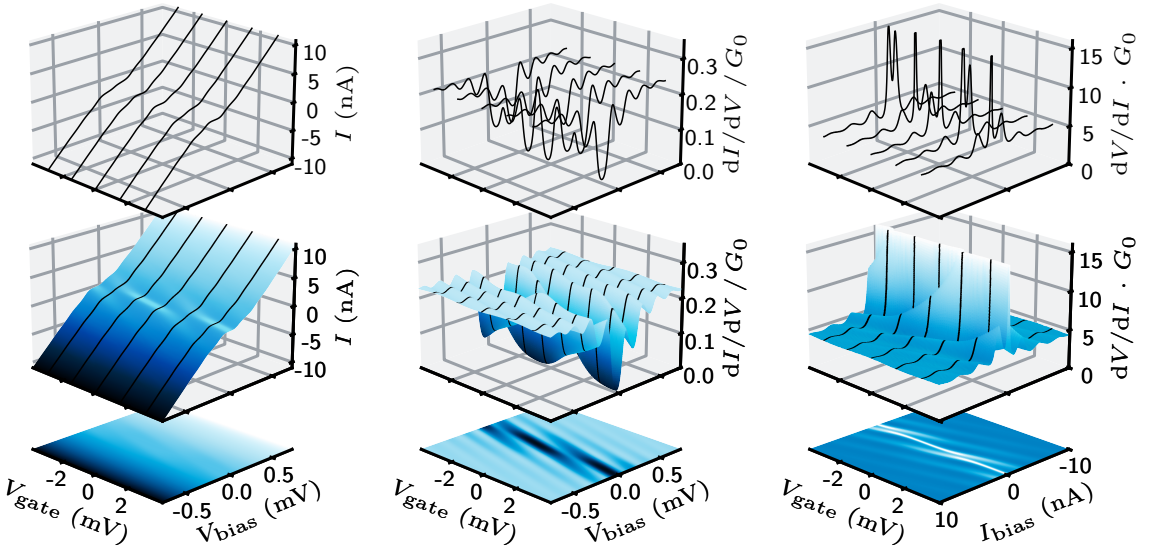
I fit the measured *I–V* gate map below the discontinuity with the orthodox sequential tunneling model introduced in the theory chapter. I fit all bias and gate points at $V_{\mathrm{gate}} < 1.7\,\mathrm{mV}$ simultaneously with a common set of circuit parameters rather than by fitting individual gate traces separately. The island energy and normal-state tunneling rates follow Eqs. (2.111) and (2.112). At each bias and gate voltage, I solve the stationary master equation in Eq. (2.113) and evaluate the current from Eq. (2.114). I optimize the circuit parameters with the nonlinear least squares implementation in SciPy using a soft L1 loss to reduce the influence of isolated outliers.

The intact SIN measurement already determines the resistance of the oxide tunnel barrier. I therefore hold this resistance fixed and fit the complete normal-state transport map simultaneously to extract the remaining junction resistances and capacitances, the gate coupling, the offset charge, and the effective electronic temperature.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure G.4. Measured normal-state gate–bias transport of SET1 after opening the break junction. The panels show the *I–V*, *dI–dV*, and *dV–dI* responses as selected traces, surfaces, and false-color maps. The periodic low conductance regions form Coulomb diamonds whose two edge families have different absolute slopes. The abrupt change during the gate sweep is consistent with an atomic rearrangement of the break junction.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure G.5. Orthodox SET fit to the stable region of the normal-state transport landscape in Fig. G.4. The panels show the calculated *I–V*, *dI–dV*, and *dV–dI* responses in the same representation as the measurement. The region beyond the discontinuity is excluded from the optimization.

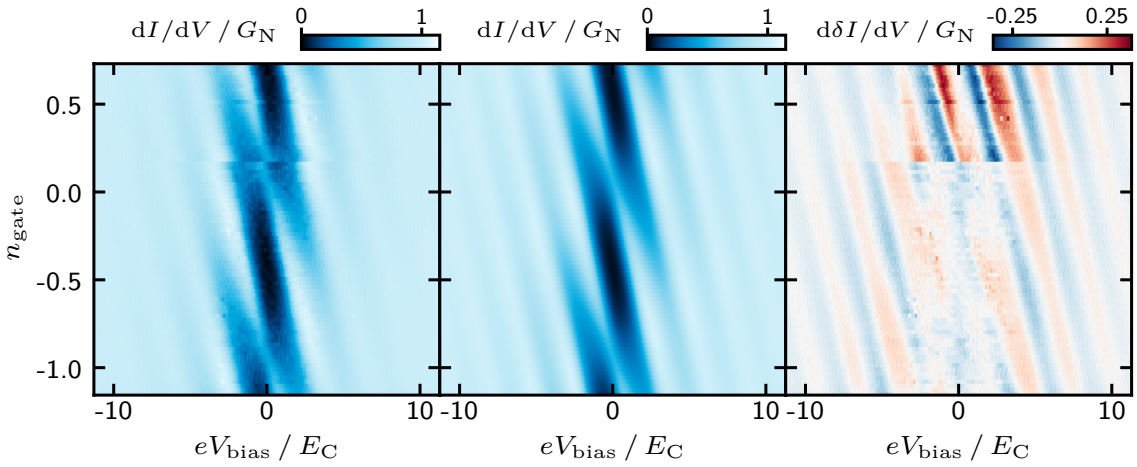


Figure G.6. Comparison of the measured and fitted normal-state differential conductance. The left and center panels show the experimental map and the orthodox SET result, while the right panel shows the differential conductance of the current residual $\delta I = I_\mathrm{exp} - I_\mathrm{fit}$. Only the stable region is included in the fit. Data above the fit boundary are shown as a visual reference for the discontinuity and are not interpreted as fit residuals.

The individual junction resistances and conductances, together with the resulting totals, are

$$\begin{aligned} R_\mathrm{TB} &= 32.38\,\mathrm{k\Omega}\,, & R_\mathrm{BJ} &= 24.15(4)\,\mathrm{k\Omega}\,, & R_\mathrm{N} &= 56.53(4)\,\mathrm{k\Omega}\,, \\ G_\mathrm{TB} &= 0.3985\,G_0\,, & G_\mathrm{BJ} &= 0.5344(9)\,G_0\,, & G_\mathrm{N} &= 0.2283(2)\,G_0\,. \end{aligned} \tag{G.3}$$

The resistance of the opened break junction is comparable to that of the oxide tunnel barrier. Both junctions therefore contribute appreciably to the measured voltage drop. From this point onward, $R_\mathrm{N} = R_\mathrm{TB} + R_\mathrm{BJ}$ and $G_\mathrm{N} = 1/R_\mathrm{N}$ denote the total normal-state resistance and conductance of the complete SET, rather than those of the intact SIN reference.

The fit further yields the junction and gate capacitances

$$C_\mathrm{TB} = 957(3)\,\mathrm{aF}\,, \qquad C_\mathrm{BJ} = 214(3)\,\mathrm{aF}\,, \qquad C_\mathrm{gate} = 37.3(2)\,\mathrm{aF}\,, \tag{G.4}$$

and therefore the total island capacitance and charging energy

$$C_\Sigma = 1.208(4)\,\mathrm{fF}\,, \qquad E_\mathrm{C} = 66.3(2)\,\mu\mathrm{eV}\,. \tag{G.5}$$

The fit also yields an effective temperature. Expressed as the thermal energy $E_\mathrm{T} = k_\mathrm{B} T_\mathrm{fit}$, it is

$$T_\mathrm{fit} = 132(2)\,\mathrm{mK}\,, \qquad E_\mathrm{T} = 11.4(2)\,\mu\mathrm{eV}\,. \tag{G.6}$$

The charging energy is therefore about six times larger than the thermal energy, $E_\mathrm{C}/E_\mathrm{T} \approx 5.8$. This separation is consistent with the clearly resolved Coulomb blockade. As in the SIN reference measurement above, the fitted temperature is

higher than the measured bath temperature, $T_{\mathrm{bath}} = 97(1)\,\mathrm{mK}$. I again interpret this fit parameter as an effective electronic temperature rather than the direct cryostat reading.

The substantially different junction capacitances quantify the asymmetry between the two families of Coulomb diamond edges. The oxide tunnel barrier supplies most of the total island capacitance, while the gate capacitance sets the conversion from applied gate voltage to induced charge,

$$n_{\mathrm{gate}} = C_{\mathrm{gate}} V_{\mathrm{gate}}/e - n_{\mathrm{gate},0}\,, \qquad n_{\mathrm{gate},0} = 0.213(3)\,. \tag{G.7}$$

Overall, the calculated maps in Fig. G.5 reproduce the dominant gate period, asymmetric blockade boundaries, and normal-state current scale of the stable break junction configuration.

Within the fitted region, the residual in Fig. G.6 is structured rather than uniformly distributed. Its diagonal features follow approximately the same directions as the measured conductance onsets. I show the data above the fit boundary only as a visual reference and do not interpret them as fit residuals. I report the structures in the fitted region without assigning a microscopic origin here.

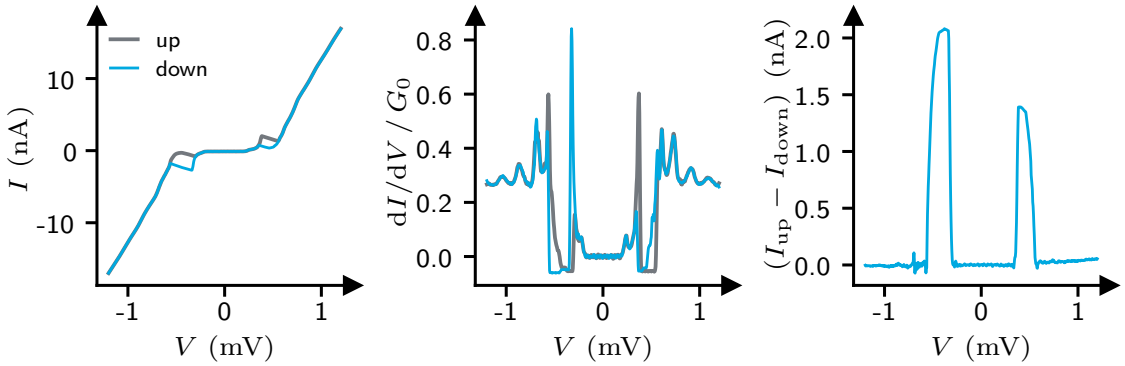
I use the orthodox model as a minimal electrostatic description. Its validity in the tunnel limit is marginal because $R_{\mathrm{BJ}} = 24.15(4)\,\mathrm{k\Omega}$ is slightly below the resistance quantum $R_{\mathrm{K}} = h/e^2$. The normal-state resistance alone does not determine the individual channel transmissions of the break junction, and the present data do not provide its mesoscopic pincode. Higher-order and transmission dependent processes could therefore have contributed to the structured residuals. I consequently use the fit to quantify the dominant charging scale, gate coupling, and junction asymmetry, but I do not interpret the remaining structures microscopically.

SSN Single-Electron Transistor

The normal-state measurement establishes the electrostatic calibration of the island. I use this reference to examine how superconductivity changes the low-voltage transport and how the resulting features depend on the gate induced charge.

After the normal-state characterization, I remove the magnetic field and measure the same device in the superconducting state. Figure G.7 compares increasing and decreasing voltage sweeps. The two traces differ around the low-voltage branch and show abrupt transitions whose positions depend on sweep direction. I refer to this difference as hysteresis without assigning it to a particular dynamical model.

I next repeat the measurement as a function of gate voltage. Figures G.10 and G.11 show that both the low-voltage response and the abrupt transitions vary periodically with the gate. I use the electrostatic calibration from the nor-



(a) *I–V* (b) *dI–dV* (c) Sweep difference

Figure G.7. Superconducting transport of the broken device for increasing and decreasing voltage sweeps. Panels (a) and (b) show the *I–V* and *dI–dV* responses, while panel (c) shows the current difference $I_{\uparrow} - I_{\downarrow}$ between the two sweep directions. The low-voltage response is sweep direction dependent and contains abrupt transitions, but no separate retrapping transition can be distinguished from the broader sweep dependent response.

mal SET fit for the gate coordinate. I scale the superconducting bias by the gap, the current by the corresponding normal-state current scale, and the differential conductance by the total normal-state conductance. This follows the normalization used for the normal SET, with the charging energy replaced by the superconducting gap as the relevant bias scale.

The high-bias superconducting conductance does not coincide exactly with G_{N}. This mismatch could result from a contact rearrangement between the measurements or from superconducting transport contributions not captured by the normal-state conductance scale. I therefore retain G_{N} only as a common normalization and do not interpret the mismatch quantitatively.

To quantify these boundaries, I operationally define V_{sw}^{+} as the positive-bias discontinuity in the up sweep and V_{sw}^{-} as the negative-bias discontinuity in the down sweep. I do not assign the other sweep direction dependent features to retrapping because they do not provide additional boundaries that could be identified with comparable confidence. Figure G.8 shows the extracted switching voltages together with the measured differential conductance landscape.

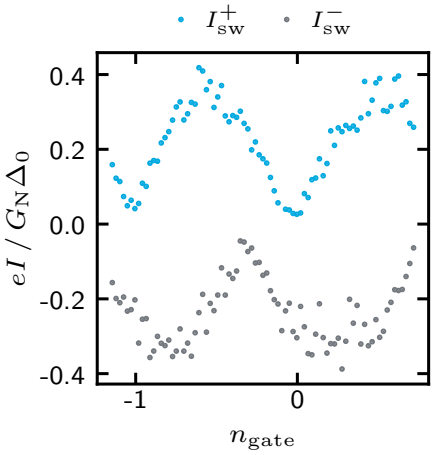


Figure G.9. Positive and negative switching currents obtained from the current at V_{sw}^{+} in the up sweep and at V_{sw}^{-} in the down sweep, respectively. Both magnitudes vary periodically with n_{gate}.

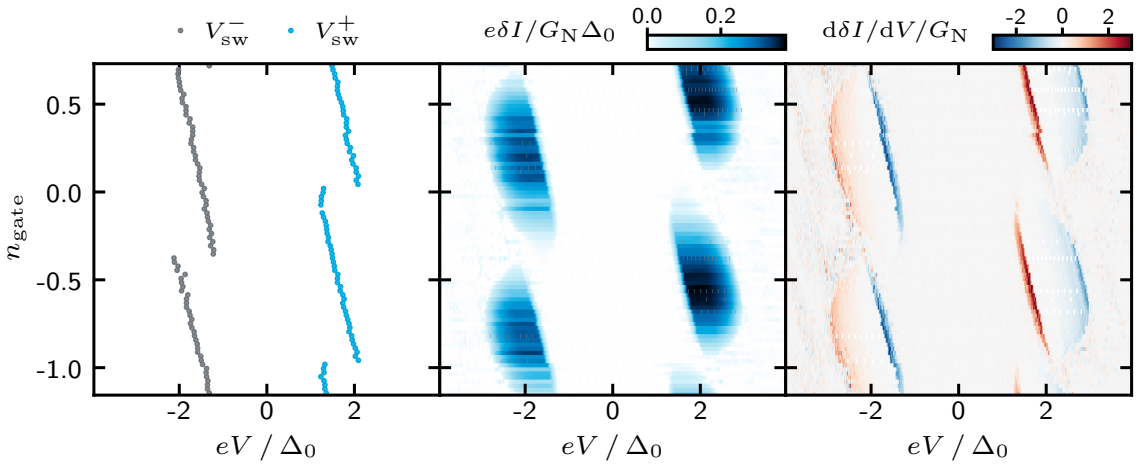


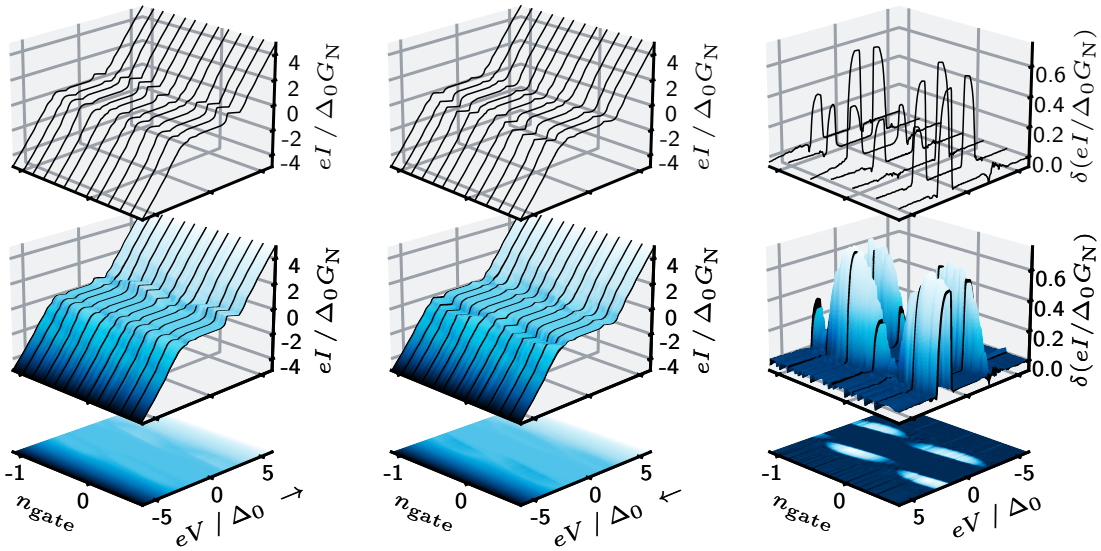
Figure G.8. Gate dependent sweep difference and switching voltages in the superconducting state. The left panel shows the positive-bias discontinuity $V_{\rm sw}^+$ extracted from the up sweep and the negative-bias discontinuity $V_{\rm sw}^-$ extracted from the down sweep. The center panel shows the current difference $\delta I = I_\uparrow - I_\downarrow$, while the right panel shows the corresponding differential conductance difference. The axes use the normalized coordinates $n_{\rm gate}$ and $eV_{\rm bias}/\Delta_0$. The extracted switching voltages follow the abrupt boundaries of the sweep dependent region.

I evaluate the current at $V_{\rm sw}^+$ in the up sweep and at $V_{\rm sw}^-$ in the down sweep to obtain $I_{\rm sw}^+$ and $I_{\rm sw}^-$, shown in Fig. G.9. Their magnitudes are approximately symmetric between the two bias polarities and vary periodically with gate induced charge. Lukas Kammermeier and I identify this gate dependence during the analysis. The data therefore establish gate dependent switching in the superconducting SSN device. I associate the discontinuities with switching of the superconducting break junction, but the present measurement does not determine the microscopic origin of their gate dependence or of the observed hysteresis.

Together, these gate dependent maps show that both the switching boundaries and the low-voltage hysteretic response follow the same periodic gate dependence identified above, establishing the superconducting SSN-SET as a gate tunable two-terminal device whose microscopic switching mechanism remains open.

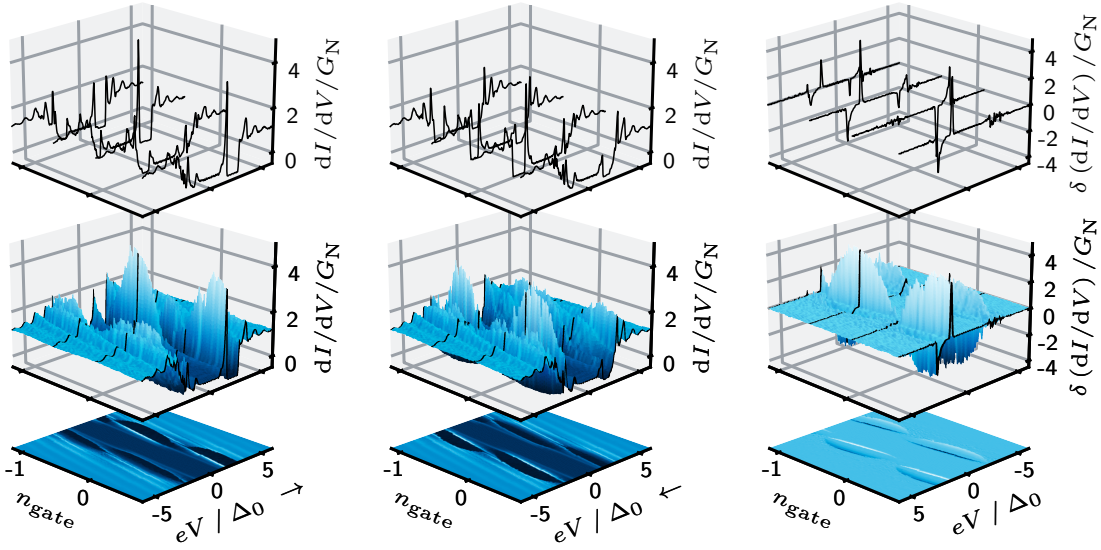
G.2 Dynamical Coulomb Blockade

SET2 provides a contrasting outcome to the gate controlled transport of SET1. I first characterize the device before opening the mechanically controlled break junction. I then partially open the junction while retaining a contact configuration with a conductance of several G_0. Gate sweeps of the intact superconducting device and of the partially opened device in both the normal and superconducting states show no resolvable modulation. These measurements cover the same applied gate voltage range in which SET1 shows periodic transport. I therefore



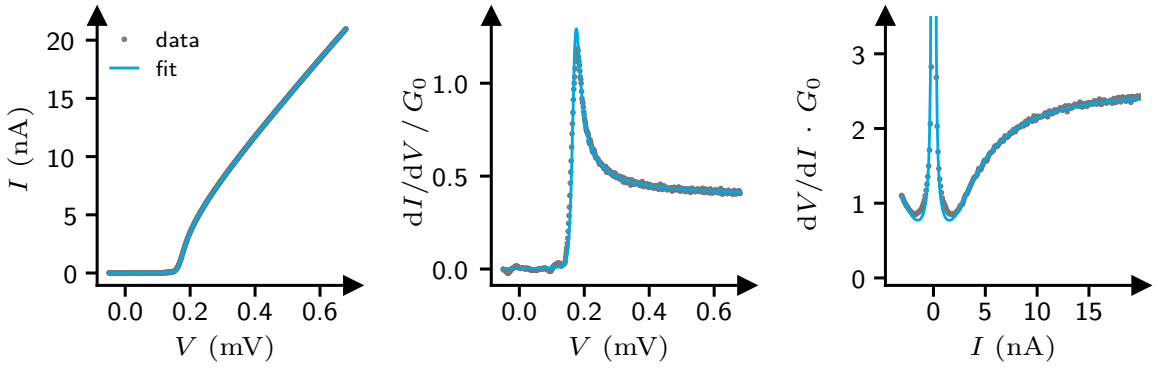
(a) Up sweep (b) Down sweep (c) Sweep difference

Figure G.10. Gate dependence of the superconducting current recorded during voltage sweeps. Panels (a) and (b) show the current for the up and down sweeps, respectively, while panel (c) shows their difference. Each panel presents selected traces, a surface, and a false-color map. The low-voltage response and its abrupt boundaries vary periodically with the gate induced charge. Bias is normalized by Δ_0, and current by $G_N\Delta_0/e$.



(a) Up sweep (b) Down sweep (c) Sweep difference

Figure G.11. Gate dependence of the differential conductance. Panels (a) and (b) show $\mathrm{d}I/\mathrm{d}V$ for the up and down sweeps, respectively, while panel (c) shows their difference. The same normalized bias and gate coordinates as in Fig. G.10 are used. The conductance maps independently resolve the periodic displacement of the abrupt positive- and negative-bias boundaries.



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure G.12. Superconducting transport of SET2 before opening the MCBJ. The measured *I–V*, *dI–dV*, and *dV–dI* responses are compared with an SIN BCS Dynes fit. The fit reproduces the quasiparticle onset and establishes the normal-state resistance and superconducting parameters of the fixed tunnel barrier.

find no evidence that opening SET2 produces a gate controlled island. Instead, the partially opened configuration develops a gate independent zero-bias anomaly consistent with dynamical Coulomb blockade.

SIN Tunnel Barrier

Before I open the break junction, its resistance is much smaller than that of the fixed Al–AlO_x–Cu tunnel barrier. The device therefore behaves approximately as a single SIN tunnel junction. Figure G.12 shows its superconducting transport measured at $T_{\mathrm{bath}} = 36(1)\,\mathrm{mK}$ together with a fit using the BCS density of states with Dynes broadening. The fit yields the normal-state resistance and conductance

$$R_{\mathrm{TB}} = 32.379(2)\,\mathrm{k\Omega}\,, \qquad G_{\mathrm{TB}} = 0.39859(3)\,G_0\,, \tag{G.8}$$

and the superconducting parameters

$$\Delta_0 = 195.99(2)\,\mu\mathrm{eV}\,, \quad T_{\mathrm{fit}} = 59.5(8)\,\mathrm{mK}\,, \quad \gamma = 1.48(4)\,\mu\mathrm{eV}\,. \tag{G.9}$$

After I suppress superconductivity, the intact device shows no resolvable deviation from a linear normal-state response and no zero-bias anomaly within the measurement noise. This measurement establishes the fixed tunnel barrier as a reference before I change the mechanical configuration.

Normal-State Dynamical Coulomb Blockade

I then partially open the MCBJ. Unlike SET1, the resulting configuration does not show periodic gate modulation. Instead, its normal-state conductance measured at $T_{\mathrm{bath}} = 70(1)\,\mathrm{mK}$ shows a gate independent suppression around zero bias, as shown in Fig. G.13. This behavior motivates a DCB fit. I fit the complete measured *I–V* range using the DCB formulation of *P(E)* Theory from Sec. 2.4.2 and the same effective parallel-RC environment ansatz used in Sec. 4.2.2. The normal-state resistance, effective resistance, effective capacitance, and effective temperature are free parameters in a single simultaneous fit. The fit reproduces the voltage dependent conductance with

$$R_{\mathrm{N}} = 34.43(3)\,\mathrm{k\Omega}\,, \qquad G_{\mathrm{N}} = 0.37488(3)\,G_0\,, \tag{G.10}$$

and the effective environmental parameters

$$R_{\mathrm{eff}} = 0.446(3)\,\mathrm{k\Omega}\,, \qquad C_{\mathrm{eff}} = 4.36(5)\,\mathrm{fF}\,, \qquad T_{\mathrm{eff}} = 77(6)\,\mathrm{mK}\,. \tag{G.11}$$

These parameters define the RC cutoff energy

$$E_{\mathrm{RC}} = \frac{\hbar}{R_{\mathrm{eff}} C_{\mathrm{eff}}} = 339(5)\,\mu\mathrm{eV}\,. \tag{G.12}$$

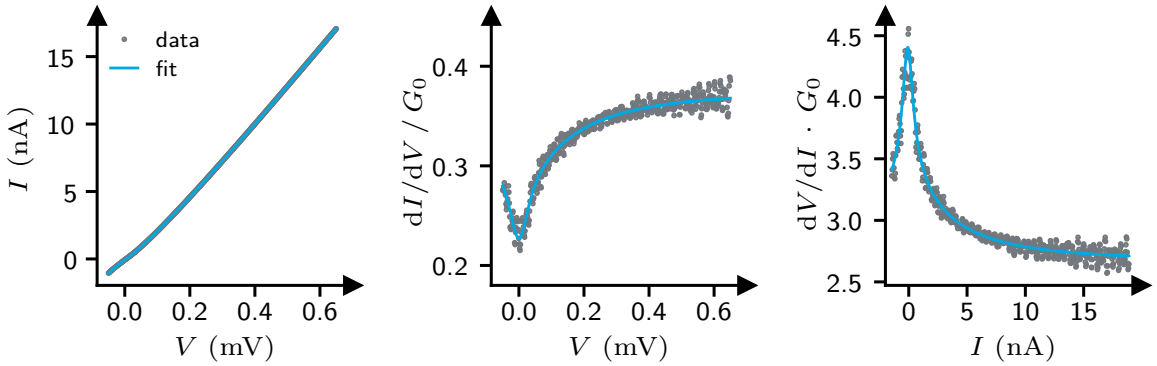
The contrast with the linear response of the intact device links the anomaly to the mechanically modified configuration. Together with the absence of gate modulation, the effective RC fit supports an environmental DCB interpretation rather than static Coulomb blockade of a gate controlled island. I treat R_{eff}, C_{eff}, and the resulting E_{RC} as phenomenological parameters because the measurement does not identify them with individual circuit elements.

The measured bath temperature remains stable during the measurement, and I resolve no corresponding cryostat temperature drift. No temperature series or independent characterization of the effective environment is available. The RC fit therefore does not uniquely distinguish DCB from other contact or density of states effects, and stable bath temperature does not exclude electronic heating. I consequently use DCB as an effective phenomenological interpretation of the gate independent anomaly rather than as a uniquely established microscopic mechanism.

Subtracting the intact tunnel barrier resistance from the normal-state resistance of the partially opened configuration gives an estimate for the normal-state break junction resistance,

$$R_{\mathrm{BJ}} = 2.05(3)\,\mathrm{k\Omega}\,, \qquad G_{\mathrm{BJ}} = 6.30(9)\,G_0\,. \tag{G.13}$$

This estimate shows that the MCBJ remains a high-conductance multichannel contact rather than entering the tunnel regime. Its conductance is compatible



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure G.13. Normal-state transport of SET2 after partially opening the MCBJ. The measured *I–V*, *dI–dV*, and *dV–dI* responses are compared with the dynamical Coulomb blockade fit for an effective RC environment. The gate independent zero-bias conductance suppression contrasts with the linear normal-state response of the intact device.

with an atomic-scale metallic constriction, although the transport measurement does not determine the contact geometry or number of atoms.

Superconducting Crossover

Directly after the normal-state measurement, I restore superconductivity and measure the device again at $T_{\mathrm{bath}} = 81(1)\,\mathrm{mK}$ without mechanically adjusting the MCBJ. The two measurements therefore probe the same nominal contact configuration back to back. In the superconducting state, however, a single SIN BCS Dynes model does not reproduce the complete voltage range, as shown in Fig. G.14. I consequently analyze a low bias region (LB), $|V| < 0.50\,\mathrm{mV}$, and a high bias region (HB), $|V| > 0.55\,\mathrm{mV}$, separately.

In the LB region, the response resembles that of an SIN tunnel junction. A Dynes fit yields

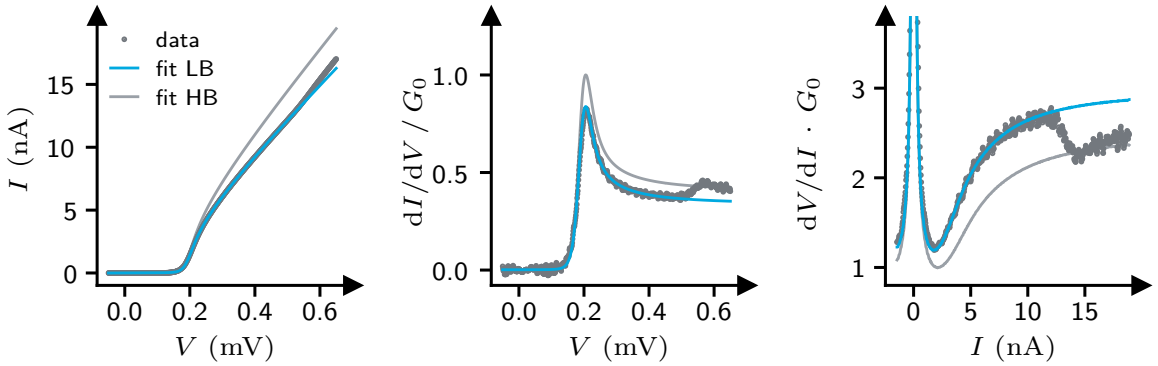
$$R_{\mathrm{N}}^{\mathrm{LB}} = 38.34(2)\,\mathrm{k\Omega}\,, \qquad G_{\mathrm{N}}^{\mathrm{LB}} = 0.33657(2)\,G_0\,, \tag{G.14}$$

and the superconducting parameters

$$\Delta_0 = 195.94(6)\,\mu\mathrm{eV}\,, \qquad T_{\mathrm{fit}} = 121(2)\,\mathrm{mK}\,, \qquad \gamma = 0.88(9)\,\mu\mathrm{eV}\,. \tag{G.15}$$

In the HB region, the measured differential conductance approaches a larger value. I fit the *dI–dV* response in this region while keeping Δ_0, T_{fit}, and γ fixed to the values extracted from the LB fit. The remaining conductance scale is

$$R_{\mathrm{N}}^{\mathrm{HB}} = 32.13(2)\,\mathrm{k\Omega}\,, \qquad G_{\mathrm{N}}^{\mathrm{HB}} = 0.40172(3)\,G_0\,. \tag{G.16}$$



(a) *I–V* (b) *dI–dV* (c) *dV–dI*

Figure G.14. Superconducting transport of the partially opened SET2 configuration measured directly after the normal-state response in Fig. G.13. The panels compare the measured *I–V*, *dI–dV*, and *dV–dI* responses with the LB Dynes fit and the HB reference calculated with the same superconducting parameters. The derivative responses resolve the crossover from the LB conductance scale toward the HB scale.

The HB conductance is approximately 19% larger than the conductance inferred from the LB response. Its resistance differs from the intact tunnel barrier resistance by only

$$\left|R_{\mathrm{N}}^{\mathrm{HB}} - R_{\mathrm{TB}}\right| = 0.25(2)\,\mathrm{k\Omega}\,, \tag{G.17}$$

corresponding to a relative difference of $< 1\%$. Thus, the HB resistance closely matches the independently measured tunnel barrier resistance.

The derivative panels resolve the crossover between the LB and HB responses at

$$V_{\mathrm{x}} = 0.52(3)\,\mathrm{mV}\,, \qquad I_{\mathrm{x}} = 13(1)\,\mathrm{nA}\,. \tag{G.18}$$

I identify the crossover interval from the visible changes in *dI–dV* and *dV–dI* without using the RC energy scale. I approximate the uncertainties by the half-widths of these intervals. They therefore quantify the crossover width rather than statistical fit uncertainty.

The crossover provides an additional comparison with the normal-state DCB fit. Its bias energy and the sum of the superconducting gap and RC cutoff energy are

$$eV_{\mathrm{x}} = 520(30)\,\mu\mathrm{eV}\,, \qquad \Delta_0 + E_{\mathrm{RC}} = 535(5)\,\mu\mathrm{eV}\,. \tag{G.19}$$

The two scales agree within the estimated crossover width. This correspondence is consistent with a crossover from an environmentally modified quasiparticle response toward the bare SIN response once the excess bias energy becomes comparable to the RC cutoff energy. Because the effective RC environment

produces a gradual energy dependence rather than a sharp threshold, I do not identify this agreement as a derived onset condition.

Expressed as resistance, the separation between the two fitted scales is

$$\Delta R_{\mathrm{N}} = R_{\mathrm{N}}^{\mathrm{LB}} - R_{\mathrm{N}}^{\mathrm{HB}} = 6.21(3)\,\mathrm{k\Omega}\,. \tag{G.20}$$

This value is approximately 4 % below $h/(4e^2) \approx 6.45\,\mathrm{k\Omega}$. This scale can be written as the resistance quantum for charge-$2e$ transfer, but it also equals the resistance of two parallel spin-degenerate unit-transmission electron channels. The present treatment within *P(E)* Theory does not predict an additive resistance of this magnitude, so I retain the proximity as a numerical observation without assigning either interpretation.

SET2 thus shows a gate independent zero-bias anomaly rather than the static gate controlled blockade observed for SET1. The normal-state response is consistent with DCB in an effective RC environment. In the superconducting state, the HB resistance agrees with the independently measured tunnel barrier resistance, while the LB Dynes fit yields a larger effective resistance. The crossover energy agrees with $\Delta_0 + E_{\mathrm{RC}}$ within its estimated width, while the difference between the fitted resistance scales is close to R_{Q}^{2e}. The first correspondence is consistent with environmental modification of the quasiparticle response. The second does not follow directly from the model used here. I therefore report both without assigning a microscopic mechanism to the LB–HB crossover.

Taken together, the two devices show that the same nominal SSN design accesses distinct transport regimes when I open the MCBJ to different contact conductances. SET1 samples the weak to intermediate coupling regime. Its normal-state response is described at a minimal level by orthodox SET theory, and it shows gate dependent switching in the superconducting state. SET2 remains in the strong coupling regime and develops a gate independent zero-bias anomaly consistent with an effective DCB description. This contrast is consistent with the broader coupling regime behavior reported for the same SSN architecture by Laura Sobral Rey *et al.* [87, 88]. The measurements do not identify the microscopic contact properties responsible for these different outcomes. Because each device provides only one usable configuration, the comparison establishes the observed regimes but not a systematic transition between them.

Acknowledgments

First and foremost, I thank Elke Scheer for giving me the opportunity to conduct this research under her supervision. Her scientific guidance, the many discussions we had, and the freedom and resources she provided shaped this project throughout. Her support in the laboratory and her encouragement to present my work at conferences allowed me to engage deeply with mesoscopic superconductivity.

While writing the theory part of this thesis, I came to appreciate how the different parts of the field come together in this project. These are microscopic quasiparticle tunneling, macroscopic Josephson dynamics, mesoscopic transport through atomic contacts, and stochastic energy exchange with the electromagnetic environment. All these perspectives ultimately meet in microwave driven superconducting devices, such as atomic contacts and single-electron transistors. I am very grateful that this project allowed me to work on a system at the heart of mesoscopic superconductivity.

I thank Carlos Cuevas for his theoretical insight. He helped me interpret the measurements carefully and avoid overinterpretation. The numerical codes he provided allowed me to explore and compare the models considered in this thesis.

I thank Patrick Raif, whose practical project and master's thesis I had the pleasure of supervising over almost two years. Working out experimental plans and solutions together sharpened my own thinking, and I learned a great deal from our exchanges. His careful sample fabrication and data analysis supported several of the atomic contact measurements I present in this thesis, and his diligence made supervising him particularly rewarding.

I thank my longtime friend Florian Kayatz, whose early development of the instrument control framework I supervised. He structured and documented that code with great care and a strong emphasis on reproducibility, providing a foundation of high quality that I subsequently extended into the measurement software used throughout this thesis. I also thank David Capalbo and Lazkin Habo for their motivated engagement and their willingness to help out in the lab and with simulations.

The weekly discussions at the "Meso Supra" meetings contributed greatly to my scientific development. I thank all participants for our discussions about mesoscopic superconductivity and laboratory work. I particularly appreciated the contributions of Laura Sobral Rey and Lukas Kammermeier concerning SSETs. I thank Roman Hartmann for his help with many practical challenges in the laboratory, especially those involving cryogenic equipment. I also thank Annika Zuschlag for her support with sample fabrication.

I thank my predecessors Susanne and Thomas Sprenger, as well as Martin Prestel, for sharing their experience from earlier projects. They repeatedly took

the time to discuss previous experiments and unresolved questions with me. Thomas's dedication to laboratory work left a lasting impression on me.

I thank Ralf Sieber and Jürgen Maier for their patient assistance with the helium supply and for performing several last-minute open-heart surgeries on my equipment. I thank Ansgar Fischer and Ingmar Jäger from the scientific workshop for their expertise, patience, and careful realization of many technical modifications. I also thank Sabine Wallrabe for her administrative support and willingness to help.

I thank Ronja Fischer-Süßlin, Alfredo Spuri, Jennifer Koch, Roman Hartmann, and Valentin Barth for making daily life in the "party office" enjoyable and for the many conversations beyond research.

I thank Daniel Kazenwadel and Jacob Holder for their friendship, honest feedback, and advice. I am also grateful that they have been my companions in physics since the beginning of my studies. Our late-night conversations about physics and life accompanied me throughout this journey. Over the years, we helped each other overcome many obstacles [137].

Finally, I thank Katrine Aschenbrenner for supporting me through a long and difficult period. She gave me the space and clarity to rediscover my love for physics and complete this thesis.

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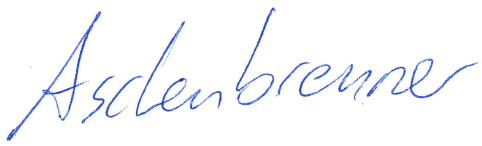
Konstanz, 24.08.2026 

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